



**GREEN
CLIMATE
FUND**

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Provisional agenda item 11

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27 January 2025

Consideration of funding proposals – Addendum VI

Funding proposal package for FP258

Summary

This addendum contains the following six parts:

- a) A funding proposal titled "Multi-country Project Advancing Early Warnings for All (EW4All)";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Secretariat's assessment;
- d) Independent Technical Advisory Panel's assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- f) Gender documentation.

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Funding Proposal

Project/Programme title: *Multi-country Project Advancing Early Warnings for All (EW4All)*

Country(ies): *Antigua and Barbuda | Chad | Cambodia | Ecuador | Ethiopia | Fiji | Somalia*

Accredited Entity: *United Nations Development Programme (UNDP)*

Date of first submission: *[2024/09/20]*

Date of current submission *[2025/01/23]*

Version number *[V.11]*



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A. PROGRAMME SUMMARY			
A.1. Project or programme	Programme	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. Not applicable		
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.		
		GCF contribution	Co-financers' contribution ¹
	Mitigation total	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Energy generation and access	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Low-emission transport	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Buildings, cities, industries and appliances	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Forestry and land use	<u>Enter number</u> %	<u>Enter number</u> %
	Adaptation total	<u>Enter number</u> %	<u>Enter number</u> %
	☒ Most vulnerable people and communities	85 %	85 %
	☒ Health and well-being, and food and water security	5 %	5 %
	☒ Infrastructure and built environment	5 %	5 %
	☒ Ecosystems and ecosystem services	5 %	5 %
A.5. Expected mitigation outcome	Not Applicable	A.6. Expected adaptation outcome	Indicate total number of direct and indirect beneficiaries

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

		<i>Indicate number of direct beneficiaries</i> Cambodia: 514,098 Antigua and Barbuda: 91,500 Chad: 1,200,000 Ecuador: 513,900 Ethiopia: 16,000,000 Fiji: 416,282 Somalia: 7,600,000		<i>Indicate number of indirect beneficiaries</i> Cambodia: 2,153,028 Antigua and Barbuda: 114,000 Chad: 5,500,000 Ecuador: 14,000,000 Ethiopia: 21,600,000 Fiji: 488,308 Somalia: 7,800,000	
		<i>Indicate % of direct beneficiaries vis-à-vis total population</i> Cambodia: 2.91% Antigua and Barbuda: 100% Chad: 6% Ecuador: 2.6% Ethiopia: 12.12% Fiji: 46% Somalia: 40.9%		<i>Indicate % of indirect beneficiaries vis-à-vis total population</i> Cambodia: 12.21% Antigua and Barbuda: N/A² Chad: 28% Ecuador: 72% Ethiopia: 15.9% Fiji: 54% Somalia: 42%	
A.7. Total financing (GCF + co-finance ³)	114,586,309 USD	A.9. Project size	Medium (Upto USD 250 million)		
A.8. Total GCF funding requested	103,246,722 USD For multi-country proposals, please fill out annex 17.				
A.10. Financial instrument(s) requested for the GCF funding	<i>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</i> ☒ Grant 103,246,722 ☐ Equity <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>				
A.11. Implementation period	5 years (60 months)	A.12. Total lifespan	30 years (360 months)		

² Indirect beneficiaries for Antigua and Barbuda are considered the significant number of tourists that visit the country during the hurricane season.

³ Refer to the Policy of Co-financing of the GCF.

A.13. Expected date of AE internal approval	9/20/2024	A.14. ESS category	Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category. C
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☐ No ☐	A.18. Is this FP included in the country programme?	Yes ☐ No ☐
A.19. Complementarity and coherence	Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1. Yes ☒ No ☐		
A.20. Executing Entity information	United Nations Development Programme (UNDP)		

A.21. Executive summary

- 1 This Multi-country Project (MCP) proposal is part of a global initiative to advance Early Warnings for All (EW4All) as announced by the United Nations Secretary General in March 2022⁴ and articulated in the EW4All Executive Action Plan (EAP).⁵ The project will improve the production, dissemination and effective use of Multi-Hazard Early Warning Systems (MHEWS), aiming to reach the most vulnerable communities in the targeted seven countries, namely Antigua and Barbuda, Cambodia, Chad, Ecuador, Ethiopia, Fiji, and Somalia. National projects tailored to each participating country's needs will establish Early Warning Systems (EWS) infrastructure, set up systems for its long term operation, support local capacities in the generation and dissemination of early warnings products and help institute relevant policies and governance mechanisms for end-to end MHEWS. This MCP aims to accelerate the delivery of EAP priority actions by stimulating the design and funding of multiple regional delivery mechanisms to support the financial and technical needs of the targeted countries.
- 2 The project will follow the approach of the ADB CRPP project⁶. An EW4All Global Funding Proposal (FP) serves to provide details on the global program and financing architecture, its theory of change, alignment with the GCF investment criteria, and other relevant information common to all participating countries. Eligibility criteria for the selection of interventions, activities, sites and beneficiaries are provided in Appendix 3.

Climate change problem

- 3 Rising annual global near-surface temperatures associated with anthropogenic climate change have already exceeded 1.5 °C above pre-industrial levels,⁷ exacerbating disaster risk triggered by different types of hazardous events. Societies and the natural systems on which they depend are unable to cope with and adapt to the effects of climate change, some of which are irreversible.⁸ The intensity, geographic extent and frequency of both slow and rapid onset disasters has increased, exacerbated by changes in the weather and climate, including increases in temperature (leading to heat-waves, damages to agriculture and human health) and the intensity of rainfall, storms and cyclones (causing floods, landslides and coastal inundation), as well as decreases in rainfall (leading to agricultural and hydrological droughts).
- 4 These changes in climate increase the risks faced by communities and the livelihoods on which they depend. They create food and water insecurity for millions of people by reducing production from agriculture and natural resources and disrupting ecosystem services.⁹ These impacts are reflected within National Determined Contributions (NDCs) which identify freshwater resources, food production and nutritional security and terrestrial and wetland ecosystems as the top priority areas and sectors. National Adaptation Plans (NAPs)¹⁰ submitted by 48 countries identified drought, floods, heat waves, increasing temperatures, sea level rise, and land and forest degradation as critical impacts of climate change, with cyclones, typhoons and storm surges major concerns in most of the SIDS. The impacts of these hazards are numerous and include reduced crop yield, soil degradation, animal diseases, loss of livestock, reduction of water supplies, salinization of water resources, human migration, ecosystem and biodiversity loss, increased coastal erosion, damage to infrastructure

⁴ World Meteorological Organisation (WMO), "Statement of Antonio Guterres, Secretary-General of the United Nations," March 2022.

⁵ The UN Global Early Warning Initiative for the Implementation of Climate Adaptation, "Early Warnings for All - Executive Action Plan 2023-2027" (World Meteorological Organization (WMO), 2023).

⁶ FP215: Community Resilience Partnership Program Multiple Countries: Asia and Pacific Region | Asian Development Bank | Decision B.37/12, 24 November 2023

⁷ WMO. 'State of the Climate 2024-Update for COP29'. World Meteorological Organization, 2024. <https://library.wmo.int/idurl/4/69075>.

⁸ David I. Armstrong McKay et al., "Exceeding 1.5°C Global Warming Could Trigger Multiple Climate Tipping Points," *Science*, September 2022.

⁹ Valérie Masson-Delmotte et al., "Global Warming of 1.5 °C," An IPCC Special Report on the Impacts of Global Warming of 1.5°C above Pre-Industrial Levels and Related Global Greenhouse Gas Emission Pathways, in the Context of Strengthening the Global Response to the Threat of Climate Change, Sustainable Development, and Efforts to Eradicate Poverty, 2019.

¹⁰ UNFCCC LDC Expert Group. "National Adaptation Plans 2023: Progress in the Formulation and Implementation of NAPs". Bonn, Germany: United Nations Framework Convention on Climate Change, 2023.

and increases in forest fires. They further highlight the need for establishing national MHEWS, including community-based systems, or sector specific systems ¹¹.

- 5 EWS(EWS) to monitor climate and weather variability and change are a critical investment that can warn against impending shocks and enhance climate resilience of governments and communities by providing critical information to guide risk management, including anticipatory action and make informed long-term decisions that help avoid otherwise crippling damages and losses to communities and livelihoods. Early warnings furthermore save lives and assets and can be worth more than ten times the investments and can also mitigate millions of cases of morbidity (cholera, dengue, diarrhea, malaria, etc.). Twenty-four hours warning of a coming storm or heat wave can cut the ensuing damage by 30%, while spending USD 800 million on such systems in developing countries could avoid losses of \$3–16 billion per year.¹²

Key climate change induced risks in participating countries

A summary of key climate change risks in each of the seven participating countries in the MCP is provided below. Antigua and Barbuda

- 6 Ranked as the 5th highest in the World Risk Index in 2023, Antigua and Barbuda is highly exposed to an array of baseline hazards, including hurricanes, extreme rainfall, droughts, and rising sea levels. Hurricanes are considered the country's main climate hazard, having faced 15 hurricanes and 14 tropical storms between 1995 and 2021. By 2030, the general building stock in Antigua and Barbuda, using the Representative Concentration Pathway (RCP) 8.5, will experience direct losses from storm surges and wind in the amount of USD 451M for a historic event, USD 4.03 billion for a 1:100-year event, and USD 9.4 billion for a maximum credible event¹³. Additionally, the islands endured 14 drought episodes between 2000 and 2020, resulting in droughts being considered the 2nd major climate threat.¹⁴
- 7 Rising sea levels and increasing land and sea surface temperatures are expected in the future. The low-lying geography of Antigua and Barbuda makes it particularly susceptible to land loss due to sea-level rise, as projected under all future scenarios assessed by the Intergovernmental Panel on Climate Change (IPCC). Moreover, hurricane intensity is anticipated to increase by 18% for category 4 and 5 hurricanes over the next 30 years, based on General Circulation Models. Under the RCP 4.5 scenario, models predict nine drought years for the period 2040–2069, representing an 8% increase.

Chad

- 8 Chad faces severe environmental degradation, and climate change is aggravating already challenging climatic conditions by increasing the frequency and magnitude of extreme events. The current and projected warming is increasing evapotranspiration, and while rainfall projections for the region are very uncertain, they suggest that extreme rainfall will increase while overall precipitation decreases. As a result, both floods and longer droughts are very likely¹⁵.
- 9 The food production and livelihood systems in Chad are highly dependent on climatic factors, particularly rainfall patterns and distribution. Extreme heat, heavy rainfall and flooding pose risks to crops, health, farmers' and herders' livelihoods, food security and farm workers' productivity. Since 1995, Chad has been ranked at the bottom of the ND-GAIN Country Index (the country ranked 184th from 1995-2004 and 185th from 2005-2021) due to a combination of high poverty, frequent conflicts, and the risk of both droughts and floods. This underlines the country's great need for investment and innovations to improve readiness, and a great urgency for action.

Ecuador

¹¹ UNFCCC, "Report of the Conference of the Parties Serving as the Meeting of the Parties to the Paris Agreement on Its Third Session, Held in Glasgow from 31 October to 13 November 2021 Addendum Part Two: Action Taken by the Conference of the Parties Serving as the Meeting of the Parties to the Paris Agreement at Its Third Session," March 2022.

¹² Global Commission on Adaptation. 'Adapt Now: A Global Call for Leadership on Climate Resilience', September 2019.

¹³ GoAB 2022. Adaptation Communication for Antigua and Barbuda

¹⁴ GoAB 2022. Adaptation Communication for Antigua and Barbuda.

¹⁵ <https://dicf.unepgrid.ch/chad/climate-change>

- 10 In Ecuador, projections foresee changes in the frequency and magnitude of precipitation and temperature, both for average and extreme values¹⁶. Climate change is project to increase melting of glacier cover, increase variations in sea surface temperature, cause spatial and temporal changes precipitation regimes, increase flood-prone areas, intensify droughts, reduce water quantity and quality, expand the range of disease-transmitting vectors (dengue and malaria), and increase loss of biodiversity¹⁷. The increased intensity and frequency of extreme and slow-onset climatic events is projected to cause economic and non-economic losses and damages to goods and services¹⁸.

Fiji

- 11 Daily maximum temperatures in Fiji have increased at the rate of about 0.2°C per decade. There has been a clear shift towards more years being warmer than average during the warm/wet season, especially since the 2000s. Fiji is becoming hotter, with more intense temperatures occurring more frequently. An increase in the frequency and intensity of extreme rainfall events is also projected. The 1-in-20-year daily rainfall is projected to increase by approximately 5-7 mm by 2030 and by 36 mm by 2090. Whilst climate projections suggest a decrease in the total number of cyclones, their severity is expected to increase¹⁹. This is expected to result in increased floods, flash floods and landslides. Coupled with projected changes in sea level rise, increased storm surges and coastal erosion is expected. The increased temperatures are expected to contribute to drought and together with changes in precipitation result in accelerated ecosystem degradation leading to bleached coral reefs and saline water intrusion into aquifers.

Ethiopia

- 12 Projections²⁰ indicate that temperatures in Ethiopia will rise by 1.5-2.5°C by mid-century under RCP4.5 and up to 3-4°C under RCP8.5. Rainfall will become more erratic, with prolonged dry spells worsening drought conditions, wild land and forest fires, and intensified rainfall events leading to severe flooding, land and mudslides. Significant impacts of climate change in Ethiopia include reduced water availability, frequent crop failures, and heightened food insecurity. South-West Ethiopia is facing devastating floods causing widespread destruction of homes, public infrastructure and affecting essential services like water, sanitation, and hygiene.
- 13 In July 2024, torrential rains in the Gambella region triggered floods and landslides severely affecting 143,000 people and causing more than 2,300 fatalities. Wildfires have ravaged ecosystems, including humid forests and protected areas. The International Organization for Migration (IOM) Displacement Tracing Matrix DTM, 2024 reports that drought in the Somali region is causing the highest number of displacements caused by climate-induced events (17% of Internally Displaced Persons (IDPs)), followed by floods, landslides and fires (9% of IDPs).

Cambodia

- 14 Cambodia faces a high level of disaster risk, ranking 58th out of 191 countries in the 2024 INFORM Risk Index. The country is especially vulnerable to flood and drought events, with about 80% of its land in the Mekong River and Tonle Sap regions. Underlying vulnerabilities such as economic and environ-mental challenges further heighten the risk from climate hazards for disadvantaged populations.
- 15 Climate change is projected to exacerbate flooding and drought risks and decrease Cambodia's GDP by as much as 10% by 2050 compared to projected GDP in 2050 in the absence of climate change²¹. Agriculture, infrastructure, forestry, human health, and coastal areas are most affected by climate change. The observed impacts of climate change include

¹⁶ Ministry of Environment, Water and Ecological Transition of Ecuador (MAATE). (2022). Fourth National Communication to the United Nations Framework Convention on Climate Change. MAATE, Quito, Ecuador

¹⁷ ibid

¹⁸ ibid

¹⁹ CSIRO and SPREP. 'Current and Future Climate for Fiji: Enhanced "NextGen" Projections Technical Report'. CSIRO Technical Report. Melbourne, Australia: Commonwealth Scientific and Industrial Research Organisation (CSIRO) and Secretariat of the Pacific Regional Environment Programme (SPREP), 2021. <https://publications.csiro.au/publications/publication/PIcsi:EP2021-2149>.

²⁰ <https://climateknowledgeportal.worldbank.org/country/ethiopia/climate-data-projections>

²¹ UNDRR. 'Disaster Risk Reduction in Cambodia: Status Report 2019'. Bangkok, Thailand, United Nations Office for Disaster Risk Reduction (UNDRR), Regional Office for Asia and the Pacific, 2019. <https://www.undrr.org/publication/disaster-risk-reduction-cambodia-status-report-2019>.

more intense rains leading to floods, delayed rainy season and longer dry seasons affecting rain-fed and irrigated crops, unexpected dry periods during the rainy season, and untimely rains which affect agricultural planning. Dams and poor river management exacerbate climate change related impacts, increasing vulnerabilities and risks, especially near waterways such as the Mekong River.

Somalia

- 16 Critical climate change impacts in Somalia are drought and flood.²² Drought affects nomadic and agro-pastoral communities who are more vulnerable to prolonged dry seasons and consequent water scarcity and reduced pasture. Droughts in Somalia result in reduced livestock production and agricultural productivity, livestock loss, food insecurity, and ultimately, famine and the loss of human life.²³ Somalia experiences regular river floods and flash flooding that result in human casualties and major impacts on the economy and damage to infrastructure. Riverine flooding typically occurs along the Juba and Shabelle Rivers in southern Somalia and flash floods, in the north along intermittent watercourses.²⁴ Intense rainfall events and subsequent floods have become more frequent, washing away fertile soil and causing widespread devastation.

Proposed Interventions

- 17 The project interventions, delivered through two Outputs, intend to assist vulnerable countries in making continuous strides towards meeting the EW4All vision. This will be achieved by supporting concrete investments in the seven countries in Output 2, and by establishing a sustainable architecture at the global and regional levels to continuously support more countries in promoting country-level actions towards EW4All (Output 1).
- 18 The project addresses critical barriers that prevent the effective generation, dissemination, communication, and use of early warnings for climate change adaptation, preparedness and response to warnings, through two broad outputs tailored to specific country needs.
- 19 Output 1 aims to accelerate and scale up national access to global and regional technical and operational support for countries to develop comprehensive MHEWS and respective capabilities. Recognizing the need for continuous and significant efforts to achieve the goal on universal access to early warnings, enabling conditions will be enhanced globally and regionally to promote country-level actions toward a global coordination and support architecture for EW4All that will be sustained beyond the project implementation period. The output will be led by the World Meteorological Organization (WMO) and United Nations Office for Disaster Risk Reduction (UNDRR).
- 20 Output 2 addresses barriers that prevent the implementation of a comprehensive and end-to-end MHEWS in the seven countries including: 1) Gaps in policies, inter-agency coordination and lack of coherent strategies; 2) Inadequate technical and scientific capacities; 3) Missing observations particularly in SIDS and Least Developed Countries (LDCs); and 4) Barriers that prevent effective local actions. The activities will mirror the four pillars of the EW4All EAP and will be implemented through the national governments with support from UNDP, the EW4All Pillar leads (WMO, UNDRR, ITU, IFRC) and their in-country national offices and other partner agencies where relevant.
- 21 The two outputs will work in tandem to accelerate the delivery of the EW4All EAP and provide long-term and uninterrupted support to critical baseline capacities on which MHEWS are built. These are: i) robust scientific information and analytics on weather, water and climate hazards and vulnerabilities within National Meteorological and Hydrological Services (NMHSs), National Disaster Risk Management Authorities (NDMAs) and relevant sectors to support greater and more diverse understanding of disaster risk and its underlying factors, with greater Impact Based Forecasting (IbF) to inform decision-making and planning; ii) effective internal coordination among various government entities that are responsible for different elements of managing a national EWS; and iii) a long-term financing plan to maintain the necessary hydrometeorological infrastructure, and community level capacities to receive, understand and respond to EWS thus enhancing adaptive capacities at national levels to protect lives and livelihoods.

²² GFDRR 2024, Somalia Coastal Flood: <https://thinkhazard.org/en/report/226-somalia/CF>

²³ National Drought Plan For Somalia, 2020.

https://www.unccd.int/sites/default/files/country_profile_documents/FINAL%20NATIONAL%20DROUGHT%20PLAN%20FOR%20SOMALIA%28final%29%2016%20Dec%202020%28%20PDF%20version%29.pdf

²⁴ FAO 2022. Somalia water and land information management. Food and Agriculture Organization. <http://www.faoswalim.org>

22 Summary of proposed interventions in participating countries

- 23 To be able to address the climate change induced and exacerbated risks highlighted above, through MHEWS and associated early actions/responses ahead of an anticipated weather/climate event, each country will need to develop and strengthen their existing MHEWS, strategies, policies and protocols to mobilize resources and people. This MCP will address these country-specific requirements, which are given below and based on extensive consultations with national stakeholders (government, NGOs and CSOs), as well as the EW4ALL Pillar leads (UNDRR, WMO, ITU, IFRC).

Antigua and Barbuda

- 24 This project will update and align disaster management legislation and policies with international standards, including modernizing the Disaster Management Act, integrating disaster risk reduction (DRR) into sectoral policies, and strengthening EWS for pandemics. The nationally defined EW4All implementation plan emphasizes improving national strategic plans for emergency management agencies and promoting data-sharing collaborations. Enhancing public awareness and understanding of hazards and disaster risk, including exposure and vulnerability, through a comprehensive communication strategy is also proposed. An online risk data platform will be created which is accessible to stakeholders and the public. Hazard maps will be updated along with exposure and vulnerability models and climate change analyses for critical sectors covering both hazards and risks across multiple scenarios will be conducted. The project will incorporate indigenous knowledge and practices into the national EWS and improve disaster risk knowledge in schools.
- 25 The project will strengthen the Antigua and Barbuda Meteorological Services (ABMS) by improving infrastructure for real-time monitoring of hydro-meteorological data, for hydrological and marine monitoring and for routine operations and broadcasting activities. It will improve telecommunications and broadcasting infrastructure, conduct country assessments on network infrastructure resilience, and set up communication campaigns to raise awareness of EWS. Under the project, a National Emergency Telecommunication Plan (NETP) will be developed and adoption of the Common Alert Protocol (CAP) will be supported. The project will implement networks configured to respond to assistive technologies for people with disabilities. It will also support the adoption of the Caribbean Disaster Emergency Management Agency (CDEMA) Model MHEWS Policy 2020, and integrating CAP procedures and impact-based forecasting (IbF) into local response systems. Finally, the project will enhance community disaster response teams, providing training and equipment for first responders, and will support the government to adopt a dedicated financing mechanisms and structured early action measures for pre-impact actions based on the needs of vulnerable groups.

Cambodia

- 26 The project proposes to improve and update platforms for data integration by enhancing data management protocols and security and generating risk knowledge at the national level. It will develop a comprehensive approach to disaster management, focusing on human resource development in EWS, particularly for women and those with disabilities. Additionally, early warning dissemination strategies and platforms will be developed to ensure timely and accurate information reaches all stakeholders.
- 27 At the sub-national level, the project will develop DRR capacities and implement community-based EWS for floods and lightning to ensure access to improved weather information. It will provide training for community members in emergency preparedness and response, focusing on the most vulnerable populations using the government's Shock Resistant Social Protection frameworks. By strengthening local capacities and integrating community-focused strategies, the project aims to build a more resilient and prepared Cambodia at all levels.

Chad

- 28 The intervention in Chad will cover the four pillars of EW4ALL and address underlying institutional and regulatory barriers through a cross-pillar approach to governance and coordination issues, considering the critical need for data interoperability and holistic, multi-hazard knowledge generation and management. It will complement existing programmes including the Systematic Observations Financing Facility (SOFF) Investment Funding and will build on community based early warning initiatives by bringing in innovations allowing the dissemination of intelligible and actionable alerts, risk education and preparedness information to the public.

- 29 The intervention will promote a people-centered approach by strengthening risk knowledge, gender analysis, and community-based research to identify and act on trust and perception issues. It will support local and devolved authorities responsible for crisis management while integrating anticipatory action into response frameworks. Significant stakeholder engagement, and preparatory work on institutional and regulatory frameworks, will be needed to ensure that coordination and governance approaches are set in motion to support these activities and outcomes.

Ecuador

- 30 Strengthened governance and coordination will be established through a robust regulatory and institutional framework for MHEWS. The project will support the establishment of a functional regulatory framework for MHEWSs and the development of a National Framework for Climate Services (NFCs), including coordination with National Gender and Development (GAD) initiatives. A National Climate Finance Framework will be developed to ensure sustainable climate services and create a comprehensive climate data management strategy with standardized data frameworks, including a climate data centre with a user interface. Governance capacities will be enhanced through the definition of Standard Operating Procedures (SOPs) for each hazard. A harmonized Monitoring, Evaluation, Accountability, and Learning (MEAL) strategy will ensure effective tracking of project outcomes and integration of lessons learned.
- 31 Disaster risk knowledge and observations will be enhanced through developing methodologies for risk assessments and implementing these assessments to create a robust data collection and risk database. Specific actions include conducting hazard and risk assessments and establishing a Multi-Hazard Risk Platform for the dissemination of risk information. Observations and monitoring networks will be strengthened by rehabilitating and integrating existing monitoring systems and improving data management and telemetry capabilities. Strengthening MHEWS forecasting capacity through improving access to, and use of Numerical Weather Prediction (NWP) models will focus on extending forecasting capacity, enhancing now-casting, improving assimilation processes, and adopting ensemble forecasting techniques. IbF will be developed for specific hazards such as floods, landslides, droughts, and extreme temperatures, including pilot projects to test these forecasts.
- 32 Warning dissemination systems will be strengthened through mapping stakeholders and focal points for receiving and disseminating alerts, establishing a multi-level and inter-institutional communication protocol, and generating binding documents to ensure effective communication. Permanent operational mechanisms will be set up for warning dissemination and activation of necessary actions. The project will enhance message dissemination by identifying priority alert mechanisms, implementing cell broadcasting, strengthening alternative communication means, and developing a warning platform using CAP. The project will analyze existing warning content, standardize communication bulletins, and generate alerts and notices in various languages with an inclusive approach. Raising awareness through educational communication strategies and drills at national and local levels will also be emphasized, along with integrating Indigenous Knowledge Systems (IKSs) into MHEWS Preparedness protocols will be developed by assessing regulatory frameworks and standardizing methodologies for Community Emergency Plans and supporting their implementation. The project aims to strengthen disaster response capacities by assessing and enhancing municipal disaster management units, including improving technical capacities and equipment. Anticipatory action protocols will be developed, including reviewing and implementing existing protocols for El Niño and droughts.

Ethiopia

- 33 The project will address gaps in the implementation of each of the four EW4All pillars, identified in the Ethiopia Impact-Based Multi-Hazard Early Warning Early Action National Roadmap 2023-2030, complementing existing programmes and projects and contributing to the EW4All objectives. The project will support and ensure coordination with national Pillar leads and stakeholders, and relevant UN, non-governmental organizations (NGO) international organizations and civil society organizations (CSO) supporting the EW4All initiative in Ethiopia.
- 34 A national disaster risk database will be created, coordinated by the Ethiopian Disaster Risk Management Commission (EDRMC), with contributions from all sector Agencies/Ministries to collect and develop multi-hazard data and risk analyses and strengthening the Woredas risk profiles in priority areas with compilation and analysis of hazard, exposure and vulnerability data. The Ethiopia Meteorological Institute (EMI) will upgrade observing stations and coverage in East and South-East areas of the country, improve data sharing, and develop specific gender-based EWS focusing on women, elderly and children.

- 35 Media communications channels will be developed according to types of audience, including vulnerability and gender-based issues. The project will build capacities of communications specialists to address media, and of journalists to communicate about disaster risk, produce media content about disaster risk, identify mitigation measures and anticipatory actions and put in place an Emergency Call short code. Disaster responses will be coordinated jointly by the Ethiopian Red Cross Society (ERCS) and EDRMC who will develop multi-hazard Anticipatory Action plans linked to a national financing mechanism (funding reserve) at national and local levels. This will include inter-sectoral/ministerial agreement on multi-hazard alert thresholds, development of community-based early-warning systems in priority areas, stockpiling and preposition of essential emergency stock at national level and in identified priority areas (food and non-food), in cooperation with private sectors and mainstream Anticipatory Action as part of emergency preparedness and response planning, integrated into social protection mechanisms.

Fiji

- 36 The project will strengthen governance and coordination through development and implementation of protocols and relevant mechanisms, including protocols and SOPs for data sharing among MHEWS agencies to enhance collaboration and legal frameworks. It will establish interagency working group for data exchange, analyze laws related to data exchange, establish Memoranda of Understanding (MoUs) between authorities for early warnings, and create protocols for data sharing. The project will additionally enhance coordination among stakeholders in developing MHEWS plans and establish collaboration mechanisms among government, non-government, and community levels. Training workshops for national and sub-national trainers will be undertaken along with strengthening of community disaster management committees.
- 37 A robust tracking system for assessing hazard impacts and future risks will be established. It will expand the necessary location-specific and targeted forecasting to all areas are affected by hazards. The project will address challenges in accessing and sharing disaster data by improving infrastructure and data utilization. It will standardize risk assessment methodologies and parameters in Fiji, including disaster data; creating a centralized database system to store risk information across sectors and disaggregated by gender, age, and disability; and improving infrastructure to collect and analyze data for climate projection applications. The project will improve the expansion of river gauges to un-gauged locations. The project will upgrade observation networks, including Automatic Weather Station (AWS) installation, spares, and training. It will upgrade TB3 and AWS data loggers to utilize historical climate data for enhancing EWS and expand location-specific predictions and develop a scientifically based drought risk map for Fiji. A drought monitoring system for health and food security will be established. It will strengthen the use of AI and a centralized database for forecasting and early warning.
- 38 The project will support the implementation of cell broadcasts and help establish a regulatory framework for mobile manufacturers to support it, specifically focusing on automated systems and mobile coverage in rural areas and for fishermen at sea. The project will activate the National Emergency Telecom Cluster, develop SoPs/contingency plans, automate EWS communication dissemination and improve warnings through mass messaging, utilize traditional knowledge and enhance national and regional communications. Communities based response plans will be developed to respond to early warnings based on previous successful implementations, focusing on how they access and act on received messages. Traditional and local knowledge will be incorporated into the local level EWS, as well as reviewing and integrating anticipatory actions into national and community-based disaster management plans.

Somalia

- 39 Somalia's EW4ALL project activities contribute directly to the realisation of the proposed activities under Somalia's National Roadmap for Implementation of EW4All 2024-2027. The project will develop a comprehensive Risk Assessment Framework including hazard risk assessment methodology harmonised with regional approaches and international framework-based targets and indicators. Using the framework, hazard risk assessments will be undertaken for national and local levels and applied/interpreted and translated for critical considerations of key vulnerability factors. A database of disaster impact data and loss and damage tracking is also to be established. The project will develop disaster risk knowledge materials and communication for national and local levels, which include maps of multiple hazards that identify specific geographic areas, sectors and groups, tailored for EWS and DRR plans and local risk management plans. Climate risk summaries of key vulnerability factors will be developed which are accessible for national and local levels.

The project will support building capacity and coordination for applying improved disaster risk knowledge at national and local levels through key training and enhancing public awareness and education through a national campaign to empower communities.

- 40 The project will also strengthen the harmonization of early warning data collection, analysis and exchange between federal line ministries, sectorial offices and regional bodies, as well as improving data automation, modelling and forecasting of early warning. It will undertake activities to improve the interoperability and early warning information exchange between sectors. The project will invest in accessible nationwide warning dissemination and communication. These will include early warning communication strategies and SOPs being established for early warning information, communication and dissemination governance, functions, roles and responsibilities, impact-based forecast and warning services (IBFWS), and CAP. It will establish comprehensive infrastructure networks and early warning dissemination services and equipment which cover simulation exercises, CAP adoption, telecommunication service provider agreements, automated system setups and protocols for AA and communication strategies and feedback mechanisms, amongst others.
- 41 The project will emphasize the building of local preparedness capacities. The project will support inclusive, people-centred EW4All by bolstering local networks to receive and disseminate information and last-mile connectivity. Finally, the project will enhance preparedness for timely responses and early actions by improving response and recovery; disaster preparedness measures for early actions. This includes developing participatory, gender-responsive disaster preparedness measures and plans for district levels, as well as engaging policy, legislation and strategies to integrate Anticipatory Action and preparedness, forecast based plans and protocols/EAPs and SOPs.

Climate results/benefits

- 42 The MCP will establish and strengthen the role of MHEWS in informing decision as a foundation for risk informed development processes. It is aligned with the Agenda for Sustainable Development,²⁵ specifically SDG 13 and SDGs 1, 2, 9, 11, 12 and 15. The project will reduce disaster risk and the impacts of shocks and hazard events by supporting national strategic guidelines and frameworks towards achievement of the Sendai Framework for Disaster Risk Reduction 2015-2030²⁶, and create a regional support framework as well as direct action in selected countries to address the four priorities of the Sendai call for action.²⁷ It contributes specifically to target G of the Sendai Framework for Disaster Risk Reduction.²⁸ The project contributes to the priorities identified in Article 8 of the Paris Agreement²⁹ by supporting the following areas of cooperation: i) EWS; ii) emergency preparedness; iii) slow onset events; iv) events that may involve irreversible and permanent loss and damage; v) comprehensive risk assessment and management; and vi) resilience of communities, livelihoods and ecosystems.
- 43 In sum, the project supports achieving global targets in reducing exposure to climate changes impacts, with a strong focus on those most vulnerable. It will lay the foundations for further projects and the supporting mechanisms that are needed by countries in establishing, operating and expanding the coverage of effective, end-to-end MHEWS. Regional and global facilities and networks for technical and financial support will be strengthened. Implementation strategies and approaches from past projects and initiatives will feed into effective and pragmatic project design, which address the unique constraints under which NMHS, NDMA and other stakeholders operate. This will specifically include the challenges and contexts of SIDS and LDCs, those with limited infrastructure or in fragile and conflict affected states.

²⁵ United Nations, "Transforming Our World: The 2030 Agenda for Sustainable Development," 2015.

²⁶ UN, "Sendai Framework for Disaster Risk Reduction 2015 - 2030" (United Nations, 2015); UNFCCC, "The Cancun Adaptation Framework" (Cancun, 2010); International Strategy for Disaster Reduction, "Hyogo Framework for Action 2005-2015: Building the Resilience of Nations and Communities to Disasters" (United Nations, 2007); World Conference on Natural Disaster Reduction, "Yokohama Strategy and Plan of Action for a Safer World: Guidelines for Natural Disaster Prevention, Preparedness and Mitigation" (Yokohama, Japan: United Nations, 1994).

²⁷ UNISDR, "Build Back Better in Recovery, Rehabilitation and Reconstruction," 2017.

²⁸ UNDRR and WMO, "Global Status of Multi-Hazard Early Warning System: Target G," 2022.

²⁹ United Nations Framework Convention on Climate Change, "Paris Agreement," 2015.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context

44 Climate hazards and associated risks (impacts, exposures and vulnerabilities)

45 Approximately half the global population of 3.3 to 3.6 billion people live in countries or regions that are highly sensitive to climate change³⁰. Most live in LDCs, SIDS and Africa with limited MHEWS and are unable to undertake anticipatory action and risk reducing measures at scale.³¹ The acceleration of climate risks is outpacing the financial flows towards adaptation, the requirements of which are estimated to be 5-10 times higher and will increase to USD 160-340 billion by 2030 and USD 315-565 billion by 2050.³² Human populations will be exposed to the highest magnitude of one to three climate hazards concurrently by the turn of the century. Some coastal regions will face up to six different hazards concurrently.³³

46 The **Latin America and the Caribbean (LAC)** region has seen the strongest warming trend of about 0.2 °C per decade on record, warming faster than global averages with temperatures projected to rise by 1.5 to 2.5 °C by 2050. Prolonged droughts have affected multiple sectors, including agriculture, energy, transportation, and water supply. Droughts combined with higher temperatures and lower humidity have triggered wildfires in the region. Sea levels have risen at a higher rate than the global mean and tropical storms such as Fiona, Lisa and Ian caused extensive damage as did floods and landslides triggered by heavy rain which also caused hundreds of deaths.³⁴ Climate change-driven impacts and risks are amplified for small islands, where more severe and frequent hurricanes compound the impacts of sea level rise, storm surge and coastal erosion, flooding and salinisation of water bodies and aquifers, threatening coastal communities and infrastructure.³⁵ In Central and South America, changes in the timing and magnitude of precipitation and extreme temperatures are affecting agricultural production among subsistence farmers. This includes more frequent long dry spells, reduced precipitation and altered rainfall during the start and end of the rainy season. Warming temperatures have further increased the transmission of vector borne diseases.³⁶

47 In Antigua and Barbuda these regional changes lead to (see Appendix 1 for further details):

- Increases in the frequency of high intensity (category 4 and 5 hurricanes), leading to increases in storm surge, coastal flooding and high winds;
- Increases in drought intensity, reducing freshwater in groundwater and increasing saline intrusion;
- Rising sea levels lead to increases in coastal flooding, exacerbated by low-lying geography.

48 In Ecuador climate risks include (see Appendix 1 for further details):

- Increases in rainfall and temperature extremes which increase the risk of flooding, landslides and mudflows compounded with volcanic ash;
- Intensification of droughts reduce water quantity and quality;
- Increasing temperatures expand the range of disease-transmitting vectors (dengue and malaria).

³⁰ The UN Global Early Warning Initiative for the Implementation of Climate Adaptation, "Early Warnings for All - Executive Action Plan 2023-2027" (World Meteorological Organization (WMO), 2023).

³¹ IFRC, "IFRC Operational Framework for Anticipatory Action 2021-2025," 2022.

³² UNEP, "Adaptation Gap Report 2022: Too Little, Too Slow. Climate Adaptation Failure Puts World at Risk," 2022.

³³ Camilo Mora et al., "Broad Threat to Humanity from Cumulative Climate Hazards Intensified by Greenhouse Gas Emissions," *Nature Climate Change* 8, no. 12 (December 2018): 1062–71, <https://doi.org/10.1038/s41558-018-0315-6>.

³⁴ World Meteorological Organisation (WMO), *State of the Climate in Latin America and the Caribbean 2022*, WMO-No. 1322 (Geneva, Switzerland: WMO, 2023).

³⁵ Intergovernmental Panel on Climate Change (IPCC), ed., "Small Islands," in *Climate Change 2022 – Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge: Cambridge University Press, 2023), 2043–2122, <https://doi.org/10.1017/9781009325844.017>.

³⁶ Castellanos, Edwin J., Maria Fernanda Lemos, Laura Astigarraga, Noemí Chacón, Nicolás Cuví, Christian Huggel, Liliana Miranda, et al. 'Central and South America'. In *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, edited by Hans-Otto Pörtner, Debra Cynthia Roberts, Melinda M. B. Tignor, Elvira S. Poloczanska, Katja Mintenbeck, Andrés Alegria, Marlies Craig, et al. Cambridge University Press, 2022.

- 49 In the **South-West Pacific**, TCs, floods, droughts, sea level rise and ocean acidification have had significant impacts on people, economies and ecosystems. The region is influenced by both the El Niño-Southern Oscillation (ENSO) and the Indian Ocean Dipole (IOD)³⁷. Most of the South-West Pacific shows upper-ocean (0–700 m) warming since 1993, with rates exceeding 2–3 times the global average in the tropical Pacific. Sea-level rise in the eastern Indian Ocean, western tropical Pacific, and south and north mid-latitude Pacific are also higher than the global mean. Observed sea surface temperatures have increased in tropical western Pacific Ocean and Rainfall has decreased in parts of the Pacific islands. This drying trend is expected to continue in the coming decades, except in parts of western and equatorial Pacific. Heavy rainfall events will increase in the western tropical Pacific and the higher evapotranspiration is expected to amplify or partially offset the effect of decreases or increases in rainfall respectively, resulting in increased aridity in parts of the Pacific under the 2°C global warming and above scenario. The intensity and frequency of TCs will increase and, together with sea level rise will contribute to storm-surges leading to extreme coastal water levels in most regions in the Pacific islands³⁸.
- 50 In Fiji these regional changes lead to (see Appendix 1 for further details):
- Current warming of 0.2°C per decade is expected to accelerate, reaching 1.3-2.5°C above pre-industrial temperatures by 2070;
 - Greatest impacts are due to typhoons (F\$500 million in annual expected losses – 5% GDP), the most intense of which are expected to increase in frequency due to climate change;
 - Increases in rainfall intensity will increase the likelihood of 1/100 yr floods with expected losses of F\$ 2 billion;
 - Drought and coastal salinization due to SLR and storm surge affects crop production, leading to food imports.
- 51 Climate hazards in **sub-Saharan Africa** include floods, droughts, storms and landslides. Floods have caused widespread damage to infrastructure and crops and have contributed to waterborne diseases. Droughts have resulted in crop failures leading to food and water insecurity while storms have damaged infrastructure and crops and lead to loss of lives. Landslides, which are linked to heavy rain have also contributed to deaths and damages to infrastructure and crops. Most African countries will face increasingly high temperature climates, with increased mean and extreme temperatures, including increased heat waves as a consequence of climate change. Increased droughts are expected over large parts of southern Africa, decreased precipitation in Northern Africa at 2°C global warming. The intensity of heavy rainfall is also expected to increase, increasing exposure to pluvial and riverine flooding across all but north and southwestern Africa³⁹.
- 52 In Chad these regional changes lead to (see Appendix 1 for further details):
- Successive droughts have affected the food security and livelihoods of households across large swathes of the country, with 1.5 million people affected in 1981-85, and 2.4 million people affected in 2009-2010. The most recent drought in 2022 affected 2.1 million people;
 - Floods (riverine and pluvial) rendered 121,088 people homeless. These disasters led to a total of 10,556 recorded deaths, of which 67% were a result of viral or bacterial disease epidemics such as cholera, meningitis and measles;
 - Climate change is expected to increase temperatures 2.1°C to 4.3°C (very likely range) by 2080, with more very hot days leading to greater incidences of heat stress and reduced agricultural labour production;
 - More variable rainfall is likely to lead to great extremes of drought and flood, with reductions in yields of maize and other crops.
- 53 In Ethiopia climate risks include (see Appendix 1 for further details):

³⁷ World Meteorological Organisation (WMO), *State of the Climate in South-West Pacific*, WMO-No. 1324 (Geneva, Switzerland: WMO, 2023).

³⁸ Valérie Masson-Delmotte et al., eds. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press, 2021. <https://doi.org/10.1017/9781009157896>.

³⁹ Christopher H. Trisos et al., 'Africa'. In *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, edited by Hans-Otto Pörtner et al. Cambridge University Press, 2022.

- Droughts cause substantial economic damage, with a 1/5 yr return period leading to losses exceeding US\$1 billion, and a 1-in-100 year drought event resulting in losses of over US\$3 billion;
- Flooding also has a severe impact on GDP, with a flood event with a 1/25 yr return period causing a 3.3% deviation in GDP baselines, and a 1/100 yr flood leading to a 7% deviation;
- Heat stress results in significant losses due to its impact on labour productivity and livestock yields. Infrastructure damage, particularly to roads and bridges, can incur annual costs of up to US\$755 million;
- Climate change is expected to increase the damages from all three risks due to increases in temperatures and extreme rainfall, coupled with seasonally dependent reductions in average rainfall.

54 **Somalia** is considered to have the least capacity to respond and recover from extreme weather events.⁴⁰ Climate risks for Somalia include (see Appendix 1 for further details):

- The frequency and intensity of heavy precipitation events are projected to increase with potential impacts of flooding and soil erosion.⁴¹ The higher rainfall and runoff expected across the various climate scenarios, are likely to lead to increased flash flooding.⁴² In 2023 around 2.48 million people were affected, with over 1.2 million displaced.⁴³ The 2023 floods also resulted in the loss of 188 lives.⁴⁴
- Increased frequency, persistence and intensity of droughts in Somalia are likely.⁴⁵ Climate change projections suggest that the duration, frequency and intensity of droughts will increase by the end of the 21st Century under all scenarios and that drought areas will increase by 16 per cent, 36 per cent and 54 per cent under increasing Representative Concentration Pathways (RCPs) (2.6, 4.6, 8.5 respectively).⁴⁶ Prolonged dry seasons result in water scarcity, reduced grazing land, reduced livestock production, livestock perishes, reduced agricultural productivity, food insecurity, and ultimately, famine and the loss of human life.⁴⁷ During 2015 – 2022, 6.1 million people were affected by drought, and 760,000 - 1.2 million people were displaced by drought.⁴⁸
- Daily temperature maximums have also increased over the last two decades⁴⁹ and are projected to continue over the next decade.⁵⁰ The very southern areas of Somalia are projected to experience extreme heat, and the southern and some parts of the northern areas are projected to experience very high heat.⁵¹

⁴⁰ ND-GAIN Index Country Rankings, Somalia. <https://gain-new.crc.nd.edu/country/somalia> [accessed 7 July 2024]

⁴¹ Seneviratne, S. I. & Zhang, X. (2021) Chapter 11: Weather and Climate Extreme Events in a Changing Climate [MassonDelmotte, V., P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J. B. R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekci, R. Yu & B. Zhou. Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change In Press, Cambridge University Press. https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Chapter_11.pdf.

⁴² Government of Somalia. (2021). National Water Resource Strategy 2021–2025. Ministry of Energy and Water Resources. <https://www.climate-laws.org/geographies/somalia/policies/national-water-resource-strategy-2021-2025>

⁴³ SODMA 2024. Deyr Floods Somalia 2023 Rapid Post-Disaster Needs Assessment: United Nations (UN), United Nations Development Programme (UNDP), the World Bank (WB), the European Union (EU) and Government of Somalia, https://www.undp.org/sites/g/files/zskgke326/files/2024-06/somalia_deyrfloods_rpdna_executive_summary_07062024.pdf

⁴⁴ SODMA 2024. Deyr Floods Somalia 2023 Rapid Post-Disaster Needs Assessment: United Nations (UN), United Nations Development Programme (UNDP), the World Bank (WB), the European Union (EU) and Government of Somalia, https://www.undp.org/sites/g/files/zskgke326/files/2024-06/somalia_deyrfloods_rpdna_executive_summary_07062024.pdf

⁴⁵ Somalia: Drought Situation Report No.2 (As of 21 December 2021), <https://www.unocha.org/publications/report/somalia/somalia-drought-situation-report-no2-21-december-2021>

⁴⁶ Haile, G.G., Tang, Q., Hosseini-Moghari, S., Liu, X., Gebremicael, T.G., Leng, G., Kebede, A., Xu, X., & Yun, X. (2020). 'Projected impacts of climate change on drought patterns over East Africa', *Earth's Future*, 8(7). <https://doi.org/10.1029/2020EF001502>

⁴⁷ National Drought Plan For Somalia, 2020. https://www.unccd.int/sites/default/files/country_profile_documents/FINAL%20NATIONAL%20DROUGHT%20PLAN%20FOR%20SOMALIA%28final%29%2016%20Dec%202020%28%20PDF%20version%29.pdf

⁴⁸ IOM, OCHA, UNHCR 2022 Somalia: Drought and Famine Displacement Monitoring Dashboard (September 2022), <https://reliefweb.int/report/somalia/somalia-drought-and-famine-displacement-monitoring-dashboard-september-2022>

⁴⁹ World Bank 2021, Climate Change Knowledge Portal: Somalia, II – Variability and Change in Variability <https://climateknowledgeportal.worldbank.org/country/somalia/trends-variability-historical>

⁵⁰ World Bank 2021, Climate Change Knowledge Portal: Somalia, I – Trends within Variability <https://climateknowledgeportal.worldbank.org/country/somalia/trends-variability-projections>

⁵¹ World Bank 2021, Climate Change Knowledge Portal: Somalia, High-level Summary: Compound Heat Risk <https://climateknowledgeportal.worldbank.org/country/somalia/heat-risk>

- 55 In Asia, surface temperature has increased in the past century with large increases and extremes in West and Central Asia, resulting in an increasing trend of growing-season length across the continent and causing strong, more frequent and longer heat-waves in South and East Asia. There is a decreasing trend of the South Asian summer monsoon precipitation during the second half of the 20th century. On the other hand decreased precipitation and increased evapotranspiration in West and Central Asia have contributed to drought conditions and decreased surface runoff⁵².
- 56 Climate projections suggest rising temperatures increase the likelihood of the threat of heat-waves across Asia, droughts in arid and semiarid areas of West, Central and South Asia, floods in monsoon regions in South, Southeast and East Asia, and glacier melting in the Hindu Kush Himalaya region. A greater warming in North Asia, including the Tibetan Plateau, Central Asia and West Asia are also projected. The highest levels of warming for extremely hot days are expected to occur in West and Central Asia with increased dryness of land. Elevation-dependent warming is expected to continue and leading to glaciers shrinking at varying rates among regions, Temperature rise will be strongest in winter in most regions, while it will be the strongest on summer in the northern part of West Asia and some parts of South Asia where a desert climate prevails. The wet-bulb temperature, a measure of heat stress, is likely to approach critical health thresholds in West and South Asia under the RCP4.5 scenario and in East Asia, under the RCP8.5 scenario. The occurrence of extreme heat-waves is likely to increase in Asia with a sizeable part of South Asia experiencing heat stress conditions in the future. A large percentage increase in annual precipitation is projected in South and North Asia, but decrease over the northwest part of the Arabian Peninsula and increase over its southern part. Both heavy and intense precipitation are projected to intensify and become more frequent in South, Southeast and East Asia. Monsoon land precipitation is likely to increase in East, Southeast and South Asia due to increasing moisture convergence by elevated temperature⁵³. There will be a large increase in flood frequency in monsoon regions in Southeast Asia, Peninsular India, eastern Africa and the northern half of the Andes⁵⁴ which without further mitigation efforts, will lead to continued loss of lives and infrastructure⁵⁵. On the other hand, the warming in higher latitudes in the Himalayas is likely to adversely affect glacial melt, increasing the likelihood of glacial lake outbursts and significantly affecting water resources over time⁵⁶.
- 57 In Cambodia these regional changes have resulted in the following (see Appendix 1 for further details):
- Over the past 25 years riverine and flash floods have caused the most deaths, with droughts and lightning causing fewer deaths but higher damages;
 - Rising temperatures and decreases in average rainfall during the dry season are expected to increase the demand for irrigation for most crops, including rice;
 - Increases in extreme rainfall during the monsoon are likely to increase riverine and flash flood risks;
 - Whilst it is not clear if lightning risk will change, the availability of more moisture, energy and a deeper convective monsoon under climate change may potentially increase the risk.

MHEWS as an adaptation solution to address the Impacts of climate change amongst the most vulnerable

- 58 The adverse impacts of climate change and resulting risks often cascade across sectors and regions through multiple pathways that impact human health, water stress, food systems, economic resilience, infrastructure, and security, as well as ecosystem structure, species ranges and phenology.⁵⁷ Insecurity in fragile and conflict affected states, including conflicts over rights and access to natural resources and water particularly for pastoral communities, raises the risks due

⁵² R Shaw et al. 'Asia'. In *Climate Change 2022 – Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, edited by Intergovernmental Panel on Climate Change (IPCC), 1457–1580. Cambridge: Cambridge University Press, 2023. <https://doi.org/10.1017/9781009325844.012>.

⁵³ *ibid*

⁵⁴ Yukiko Hirabayashi et al., 'Global Flood Risk under Climate Change'. *Nature Climate Change* 3, no. 9 (September 2013): 816–21. <https://doi.org/10.1038/nclimate1911>.

⁵⁵ Oppenheimer, Michael, and Amro Abd-Elgawad. 'Sea Level Rise and Implications for Low Lying Islands, Coasts and Communities 2'. *Notes* 23 (2018). https://www.academia.edu/download/78648590/SROCC_FOD_Ch04_Final.pdf.

⁵⁶ Wester, Philippus, Arabinda Mishra, Aditi Mukherji, and Arun Bhakta Shrestha, eds. *The Hindu Kush Himalaya Assessment: Mountains, Climate Change, Sustainability and People*. Springer Nature, 2019. <https://doi.org/10.1007/978-3-319-92288-1>.

⁵⁷ Hans-O. Pörtner et al., *Climate Change 2022: Impacts, Adaptation and Vulnerability* (IPCC Geneva, Switzerland:, 2022).

to climate hazards and the need for humanitarian and development assistance.⁵⁸ Furthermore the expansion, shifts and intensification of climate hazards due to climate change, leads to the exposure of rural communities and rapidly growing informal urban settlements, to both higher existing risks and novel risks they have never previously experienced.⁵⁹ The case of small islands provides a useful example where geopolitical, environmental, socioeconomic and cultural factors are interconnected and are likely to increase exposure and vulnerability to the impact of climate change induced sea level rise, increasingly intense cyclones, coastal flooding and coastal erosion⁶⁰.

- 59 Climate change has a disproportional impact on the poor and vulnerable who lack assets and access to information and resources and are therefore less able to anticipate, adapt or mitigate these impacts.⁶¹ Often these communities are reliant on livelihoods linked to climate sensitive sectors such as agriculture, or are located in regions more exposed to climate risks, including on small islands, along coastlines and in flood and land-slide prone regions. Many live in unplanned settlements and shanty-towns and lack basic facilities and infrastructure. Furthermore, vulnerable groups within communities, including women, women-headed households,⁶² households with infants, children and elderly, people with disabilities,⁶³ often lack the means to receive information and respond to climate related disasters.

Well-functioning MHEWS allow people, institutions and businesses to prepare for and respond to multiple hazards (see Table 1), including those occurring simultaneously or whose impact accumulates over time, and their potential interrelated impacts.⁶⁴

- 60 Disaster impacts in countries with high risk and high poverty can negatively affect welfare and cause major setbacks in human and economic development, with long-term consequences⁶⁵ to resilience building and well-being. MHEWS are a proven, effective and feasible climate change adaptation measure that saves lives and provides at least a tenfold return on investment⁶⁶. The socio-economic benefits of weather prediction are approximately USD 160 billion per year and improvements in forecasting and EWS could provide additional annual benefits of US\$30 billion⁶⁷. Economic assessments in several countries have found strong positive cost-benefit ratios for investment in weather and climate services between 1:4 to 1:36 from reduction in weather-related economic losses, reduced humanitarian relief costs and maladaptive risk-coping strategies⁶⁸.

⁵⁸ C. Mbow et al., et al. 'Food Security'. In *Climate Change and Land: IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse Gas Fluxes in Terrestrial Ecosystems*, edited by Intergovernmental Panel on Climate Change (IPCC), 437–550. Cambridge: Cambridge University Press, 2022. <https://doi.org/10.1017/9781009157988.007>.

⁵⁹ *ibid.*

⁶⁰ M. Mycoo et.al., 'Small Islands'. In *Climate Change 2022 – Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, edited by Intergovernmental Panel on Climate Change (IPCC), 2043–2122. Cambridge: Cambridge University Press, 2023. <https://doi.org/10.1017/9781009325844.017>.

⁶¹ Aleksandra Kosanic et al., "An Inclusive Future: Disabled Populations in the Context of Climate and Environmental Change," *Current Opinion in Environmental Sustainability* 55 (April 2022): 101159, <https://doi.org/10.1016/j.cosust.2022.101159>.

⁶² Yvonne Corcoran-Nantes and Snigdha Roy, "Gender, Climate Change, and Sustainable Development in Bangladesh," ed. Janet McIntyre-Mills, Norma Romm, and Yvonne Corcoran-Nantes, *Balancing Individualism and Collectivism*, Contemporary Systems Thinking (Cham: Springer International Publishing, 2018), http://link.springer.com/10.1007/978-3-319-58014-2_8; Debashish Sarker Dev and Jaime A. Manalo IV, "Gender and Adaptive Capacity in Climate Change Scholarship of Developing Countries: A Systematic Review of Literature," *Climate and Development*, 2023, 1–12; Houria Djoudi, Maria Brockhaus, and Bruno Locatelli, "Once There Was a Lake: Vulnerability to Environmental Changes in Northern Mali," *Regional Environmental Change* 13, no. 3 (June 2013): 493–508, <https://doi.org/10.1007/s10113-011-0262-5>; Alicea Garcia, Petra Tschakert, and Nana Afia Karikari, "Less Able: How Gendered Subjectivities Warp Climate Change Adaptation in Ghana's Central Region," *Gender, Place & Culture* 27, no. 11 (November 2020): 1602–27, <https://doi.org/10.1080/0966369X.2020.1786017>.

⁶³ Paul S. Davies and Jeffrey Hemmeter, "Supplemental Security Income Recipients Affected by Hurricanes Katrina and Rita: An Analysis of Two Years of Administrative Data," *Population and Environment* 31, no. 1-3 (January 2010): 87–120, <https://doi.org/10.1007/s11111-009-0093-1>.

⁶⁴ WMO, "Climate Services for Supporting Climate Change Adaptation Supplement to the Technical Guidelines for The National Adaptation Plan Process," Provisional Edition, 2021.

⁶⁵ United Nations Office for Disaster Risk Reduction (UNDRR). 'Words into Action: A Guide to Multi-Hazard EWS'. Geneva, 2023. <https://www.undrr.org/media/87690/download?startDownload=20240523>.

⁶⁶ The UN Global Early Warning Initiative for the Implementation of Climate Adaptation. 'Early Warnings for All - Executive Action Plan 2023-2027'. World Meteorological Organization (WMO), 2023.

⁶⁷ SOFF. 'The Value of Surface-Based Meteorological Observation Data: Costs and Benefits of the Global Basic Observing Network'. Information Brief. Systematic Observations Financing Facility, October 2020.

⁶⁸ WMO. 'Valuing Weather and Climate: Economic Assessment of Meteorological and Hydrological Services'. World Meteorological Organization, 2015.

Table 1: Summary of how hydrometeorological observations and MHEWS can help in adapting, mitigating and responding to climate change induced hazards

Hazard	Hydrometeorological Observations and indices	Type of Early Warning	Adaptation, Disaster Preparedness and Response
Drought	Standardized Precipitation Index (SPI), Palmer Drought Severity Index (PDSI), soil moisture content	Drought monitoring, seasonal forecasts, anticipatory/early action triggers, IbfS	Water conservation, drought-resistant crop varieties, emergency response plans
Heat Wave	Temperature and humidity indices (e.g., Heat Index, Wet Bulb Globe Temperature), urban heat island intensity	Heat wave warnings, heat health watch/warning systems	Urban green spaces, cooling centers, public awareness campaigns, emergency medical services
Shortened Growing Season	Growing Degree Days (GDD)	Phenological models, pasture availability assessments	Crop variety selection, planting date adjustments, agricultural extension services, crop insurance
Flood	River stage, discharge, rainfall intensity, soil moisture	Flood forecasting, flood warning systems, flood inundation mapping	Floodplain zoning, evacuation plans, flood insurance
Flash Flood	Real-time rainfall intensity, radar-estimated rainfall, streamflow	Flash flood warning systems, nowcasting	Emergency response plans, public education, land use planning, infrastructure resilience
Extreme Precipitation Event	Rainfall intensity, duration, return period	Extreme precipitation forecasting, landslide monitoring	Storm water management systems, evacuation plans, infrastructure resilience
Rainfall Variability	Precipitation time series, trend analysis	Climate change projections, seasonal precipitation forecasts	Water storage infrastructure, diversified cropping systems, agricultural extension services
Thunderstorm	Radar reflectivity, Doppler velocity, lightning detection	Severe thunderstorm warnings, nowcasting	Lightning protection, public awareness campaigns, emergency response plans, building codes
Tropical Cyclone	Satellite imagery, wind speed, atmospheric pressure	Tropical cyclone track and intensity forecasts, storm surge models	Coastal zone management, evacuation plans, building codes
Storm Surge	Tide gauge data, wind speed, atmospheric pressure	Storm surge forecasting, warning systems	Coastal protection measures, evacuation plans, disaster recovery planning
Sea Level Rise	Tide gauge data, satellite altimetry, coastal inundation models	Sea level rise projections, coastal vulnerability assessments	Coastal retreat, infrastructure elevation, ecosystem-based adaptation,

Scenario in the absence of proposed interventions

- 61 Global warming is on track to cross the 1.5 °C threshold, even if greenhouse gas emissions are substantially reduced in the near-term.⁶⁹ This can have unprecedented and potentially catastrophic outcomes on societal wellbeing.⁷⁰ Local to regional changes in the intensity and frequency of weather and climate extremes generally scale with global warming and new evidence strengthens the conclusion that even relatively small incremental increases in global warming cause significant changes in extremes on the global scale and for large regions.⁷¹
- 62 About 13 million people lose their lives annually due to climate-related impacts on health through pollution, extreme weather events, forced displacement, food insecurity and pressures on mental health.⁷² By 2050, the world economy may lose up to 14% (USD 23 trillion) compared to what world GDP is projected to be in 2050, on account of climate

⁶⁹ Amy McNally, "FLDAS Noah Land Surface Model L4 Global Monthly 0.1 X 0.1 Degree (MERRA-2 and CHIRPS)," 2018, <https://doi.org/10.5067/5NHC22T9375G>.

⁷⁰ Luke Kemp et al., "Climate Endgame: Exploring Catastrophic Climate Change Scenarios," *Proceedings of the National Academy of Sciences* 119, no. 34 (August 2022): e2108146119, <https://doi.org/10.1073/pnas.2108146119>.

⁷¹ Sonia I. Seneviratne et al., "Weather and Climate Extreme Events in a Changing Climate," 2021.

⁷² WHO, "Health and Climate Change," COP24 Special Report (World Health Organisation, 2018).

change, unless decisive action on reducing greenhouse emissions is taken.⁷³ The IPCC report highlights the acute food insecurity, malnutrition, displacement and insecurity faced by communities in LDCs, who are disproportionately affected due to poverty, governance challenges, limited access to basic resources and high levels of climate-sensitive livelihoods.⁷⁴

- 63 Differences in regional climate between the present day and warming between 1.5 °C and 2 °C are expected to be widespread across temperature increases and extremes, heavy precipitation events, significant increases in runoff and flood hazard. There is an increased probability of extreme drought, precipitation deficits, and risks associated with water availability. Global mean sea level rise between 0.04 – 0.16 m is projected along with an overall increase in multiple and compounded risks.^{75, 76} There is a 66% likelihood that the annual average near-surface global temperature between 2023 and 2027 will be more than 1.5 °C above pre-industrial levels for at least one year and it is possible that we have already exceeded the 1.5 °C levels.⁷⁷ This highlights the need for global coverage of MHEWS and response systems which allow preparedness activities and preventative and anticipatory action to attenuate the impacts of climate-induced shocks.⁷⁸

Goal outputs and activities

- 64 The objective of this MCP is to accelerate the design and implementation of the EW4All initiative through coordinated global support involving all the key actors across the early warnings value chain. This project, through its country interventions, will target communities whose livelihoods depend on climate sensitive sectors, those with greater physical exposure to climate hazards, with clear emphasis on women, poor and vulnerable groups within these communities. It will lead to the enhanced production of actionable multi-hazard early warnings and IbFs to inform and trigger preparedness, early and anticipatory actions. It will strengthen multiple channels of communication to ensure timely and effective dissemination of these warnings and information to the last mile. The project will also prepare disaster management and supporting agencies at national and the sub-national level as well as communities to make effective use of these early warnings and advisories for early and anticipatory action which allows them to adapt to, avoid and facilitate recovery from climate change hazards. Finally it will address barriers to mobilizing financial resources that prevent many countries from translating early warnings into on-ground action.
- 65 The underlying assumption for the theory of change of the project is: **IF** generation, delivery and adoption of multi-hazard early warning information is improved for effective early and anticipatory action, **THEN** avoidance of and adaptation to climate induced natural hazards will be enhanced and the loss of lives and disruption of livelihoods, services and productive assets will be reduced **BECAUSE** of improved, effective and timely disaster mitigation and response by national and sub-national governments, communities and businesses supported by multilateral agencies and private entities.

Barriers to establishing robust and effective end-to-end MHEWSs

- 66 The GCF Sector Guidance note on Climate Information and EWS identifies key barriers for effective EWS ranging from a lack of enabling environments, coordinated interventions, well-managed hydrometeorological operations, to budgetary constraints and resource mobilization.⁷⁹ The effective development, generation and use of MHEWS is challenged by the absence of consistent, reliable and robust sources of technical, human and financial support to and within countries. In

⁷³ Swiss Re Institute, "The Economics of Climate Change: No Action Not an Option," 2021.

⁷⁴ Hans-Otto Pörtner et al., eds., "Summary for Policymakers," in *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2022); Hans-Otto Pörtner et al., "IPCC Special Report on the Ocean and Cryosphere in a Changing Climate," *Ipcc Intergovernmental Panel on Climate Change: Geneva, Switzerland* 1, no. 3 (2019).

⁷⁵ Masson-Delmotte, Valérie, Panmao Zhai, Anna Pirani, Sarah L. Connors, Clotilde Péan, Sophie Berger, Nada Caud, et al., eds. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press, 2021. <https://doi.org/10.1017/9781009157896>; Pörtner, Hans-O., Debra C. Roberts, Helen Adams, Carolina Adler, Paulina Aldunce, Elham Ali, Rawshan Ara Begum, Richard Betts, Rachel Bezner Kerr, and Robbert Biesbroek. *Climate Change 2022: Impacts, Adaptation and Vulnerability*. IPCC Geneva, Switzerland; 2022.

⁷⁶ Ove Hoegh Guldberg et al., "Impacts of 1.5 C Global Warming on Natural and Human Systems" (Intergovernmental Panel on Climate Change, 2018).

⁷⁷ McCulloch et al., "300 Years of Sclerosponge Thermometry Shows Global Warming Has Exceeded 1.5°C," 2024.; 'European State of the Climate 2023 | Copernicus'. Accessed 23 May 2024. <https://climate.copernicus.eu/esotc/2023>.

⁷⁸ World Meteorological Organisation (WMO), "Statement of Antonio Guterres, Secretary-General of the United Nations," March 2022.

⁷⁹ GCF, "Climate Information and EWSs Sectoral Guide" (Yeonsu: Green Climate Fund, September 2022).

this regard, it is important to strengthen the internal mechanisms and capacities by which institutions at the national level develop and communicate early warning messages, and how these messages are received, understood and utilized by local responders, communities and populations. These barriers often limit countries' capacities to provide climate information services in a manner that effectively reaches all vulnerable communities, economic sectors and livelihoods. Some of the root causes and barriers associated with these aspects of MHEWS and that need to be addressed are described below.

Poor strategic planning for long term financial support to MHEWS

- 67 There is insufficient capital at the global, country and community levels to establish and maintain the observation networks and equipment needed to generate meaningful weather, climate and hydrological forecasts. Critical socio-economic observations are also difficult to gather in real time because suitable crowd sourcing and other technologies are underutilized. These observations are needed to improve our understanding of multi-hazard exposure and vulnerability and hence risks faced by people and communities. Resources at national, sub-national, and community levels are not enough to translate forecasts into warnings, share these warnings effectively, and prepare communities, especially those at risk and vulnerable, to use and respond to them through mitigative actions that have been proven effective.
- 68 Limited financial resources and human/institutional capacity to establish and maintain observation networks has led to gaps in data needed to meet the requirements of the WMO Integrated Global Observing System (WIGOS) and the Global Basic Observing Network (GBON) systems. These systems are used to generate early warning products at the regional and global levels. The SOFF has been established to address this resource constraint in SIDS and LDCs, and this project will ensure that investments through SOFF are leveraged and coordinated through additional GCF investments. Many countries face similar challenges in operating and maintaining their hydro-meteorological and communication networks.⁸⁰ Limited awareness about the value of hydrometeorological and communication networks in reducing climate risks constrains opportunities to raise necessary public budgets. This often leads to the deterioration of existing installations or the use of equipment that fails to meet required standards, creating challenges in quality control and interoperability.
- 69 Efficiency and efficacy of preparedness and early response efforts is constrained by lack of forecast-based planning and financial resources at the grassroots and municipal levels. This is critical as local governments are often mandated with maintaining local EWS, particularly last mile access. The inadequacy of mechanisms through which necessary funds and other resources can be rapidly released before and during emergencies is a significant barrier to early response and anticipatory action to early warnings needed to mitigate disasters. There is a lack of financing to put in place and maintain the mechanisms, systems and capacities needed for anticipatory action. Funding is also needed for the actual implementation of anticipatory action, including within national systems⁸¹.
- 70 Many countries lack a coherent and long-term financing strategy for establishing, rehabilitating and/or maintaining their soft and hard infrastructure for early warning services. In some countries, it has become challenging to effectively coordinate available resources to ensure end-to-end systems. Even fewer have the fiscal frameworks to allow for funding of EWS, particularly those that are maintained by local actors. Such strategies should explore raising resources from private sources. They need to enhance the capacities and resources to develop comprehensive and compelling proposals that help unlock funding for projects to set up EWS and address different national priorities and vulnerabilities and contexts. The absence of a comprehensive gap analysis at the national level prevents these countries from benefiting from the global and regional support architecture and facilities, such as GCF or the Global Environment Facility (GEF), or from global financing campaigns which targets donors across different sectors.

⁸⁰ Alliance for Hydromet Development, "Country Hydromet Diagnostics," n.d., accessed April 1, 2024; WMO. 'Hydromet Gap Report 2021'. Alliance for Hydromet Development, 2021.

⁸¹ Also referred to as "build funding" and "fuel funding", respectively.

Limited global, regional and national support for country level actions for a comprehensive MHEWS

- 71 Technical, scientific and institutional barriers prevent the generation and development of reliable, meaningful and actionable weather and climate information, as well as its effective communication and use by people, government, communities and businesses. Absence of such information contributes to the low level of awareness and appreciation of the value of MHEWS and its associated components, including among policy makers. Many LDCs and SIDS fail to leverage regional facilities and partnerships for necessary and sustained technical assistance due to limited human capital and capacities. Consequently, they are excluded from the knowledge exchange, peer-to-peer learning, coordination and collaboration with key initiatives and frameworks which provide contextual operational support across a wide range of hazards. This is a barrier to access technical support and information services from regional climate centres, regional centres of excellence including national and regional academic institutions, initiatives and projects.⁸²

Inadequate policies and strategies for inter-agency coordination, data sharing and governance for preparedness and response

- 72 Inconsistencies in policies, institutional structure and mechanisms that govern the collection and sharing of hydrometeorological, vulnerability and exposure data can pose a barrier to implementation of international agreements and frameworks for EWS and DRR.⁸³ National policies often fall short of a comprehensive framework to facilitate private financing and fail to provide suitable structures to strengthen inclusive, people-centred approaches to early warnings. Capacity gaps exist in transformational planning for such approaches which would address each country's unique "paradigm shift" needs. There are barriers within regions and countries which prevent the effective use of available climate information and EWS (CIEWS) by line agencies for different sectors, and those responsible for national disaster response, relief and recovery, particularly civil protection organizations at the district and community levels.
- 73 In most countries, ministries or departments with independent administrative structures and procedures are involved in the generation, analysis and use of climate information. Meteorological and hydrological services often reside in different ministries or departments from those whose expertise is needed in co-developing advisories and provide the sectoral expertise, for example health and agriculture. Inadequate and inefficient mechanisms for data sharing and coordinated action between these agencies prevents the development of targeted and Ibf⁸⁴ and scenarios for triggers to inform early and Anticipatory Action⁸⁵ or Forecast based Financing⁸⁶ for relevant sectors and areas⁸⁷. Furthermore, the Information and Communication Technology Ministry and telecommunication regulatory authority and private telecommunication companies are not actively involved or consulted in the design of communications strategies which includes data transmission from observation stations, dissemination of early warnings and advisories and obtaining feedback from targeted stakeholders. Insights from behavioral science about how end-users respond to early warnings are also neither contextualized nor institutionalized in national EWS. Their exclusion results in missed opportunities to take advantage of the growth in networks and services.
- 74 The NMHSs often do not have staff or facilities below the provincial or district level, de-linking the generation and use of CIEWS. The inadequacies and deficiencies in SoPs is a barrier to the dissemination of CIEW using public and private communications infrastructure. SoPs are often lacking in the development of easily understood, targeted, tailored and

⁸² These include the IGAD Climate Prediction and Applications Centre (ICPAC), Fully Optimized User Centric Climate Services Value Chain for Southern Africa (FOCUS-Africa, Caribbean Community Climate Change Centre (CCCC), Pacific Community (SPC), the EU-horizon project and the Mediterranean and pan-European forecast and EWS against natural hazards (MedEWSa).

⁸³ Jonathan Raikes et al., "The Influence of International Agreements on Disaster Risk Reduction," *International Journal of Disaster Risk Reduction* 76 (June 2022): 102999, <https://doi.org/10.1016/j.ijdr.2022.102999>.

⁸⁴ Melanie Harrowsmith et al., "The Future of Forecasts: Impact-Based Forecasting for Early Action" (International Federation of Red Cross and Red Crescent Societies; UK Met Office; World Meteorological Organisation, 2020); Rochelle Campbell, Daniel Beardsley, and Sezin Tokar, "Impact-Based Forecasting and Warning: Weather Ready Nations" (November 2018); World Meteorological Organization (WMO) and World Meteorological Organization (WMO), *WMO Guidelines on Multi-hazard Impact-based Forecast and Warning Services*, WMO (Geneva: WMO, 2015).

⁸⁵ IFRC, "IFRC Operational Framework for Anticipatory Action 2021-2025," 2022.

⁸⁶ WFP, "Forecast-Based Financing - Anticipatory Actions for Food Security," *Sdg Integration* (<https://sdgintegration.undp.org/forecast-based-financing-anticipatory-actions-food-security>, n.d.), accessed February 10, 2023.

⁸⁷ UNDRR, "Introductory Guide for Anticipatory Finance", 2024, <https://www.undrr.org/publication/anticipatory-finance-introductory-guide>

actionable CIEWS for different audiences. Sub-national agencies are often excluded from the dissemination or the use of CIEWS. There are very few sub-national agencies where procedures exist for the use of CIEWS by communities, for who they are intended, this is particularly the case in un-planned settlements and remote areas where national services may not be available. Institutional structures at the settlement, village or community level are often weak, have limited roles in governance and weak linkages with government agencies. Consequently, the mechanisms needed to effectively prepare for disasters and to implement mitigation and adaptive action at the grassroots are not enabled and/or do not exist.

Inadequate technical and scientific capacities

- 75 NMHS in developing countries, face a number of technical and scientific challenges and constraints in the generation and delivery of timely, targeted and contextual CIEWS. The transition to IbF, Anticipatory Action / Forecast Based Financing increase the requirement for higher resolution spatio-temporal data, technical and scientific expertise as well as facilities for statistical downscaling, and the ability to develop models and interpreting model outcomes to build scenarios and triggers to activate early and anticipatory action, including through scalable social protection systems⁸⁸.
- 76 NMHS often have insufficient training and capacities required to develop and run models locally. There is therefore a need to develop in-house skills in these areas, but also, to have the parallel development of CIEWS through regional and global support arrangements. There is an urgent need for regional centres and relevant academic and research institutions to support weather and seasonal forecasts as much as possible through provision of data to NHMS and other agencies involved in the early warning/early action value chain, who in turn need to use those data in conjunction with other forecasts available through global producing centres, for e.g. the UK Met Office (UKMO), the European Centre for Medium-Range Weather Forecast (ECMWF), and the National Oceanic and Atmospheric Administration (NOAA).
- 77 Many countries have a shortage of staff with field data collection, analytical and modelling skills required for the generation of IbF to inform anticipatory and early action. In many countries the subject course for hydrometeorology is not existent, resulting in a systemic barrier in the development of skilled professionals to support met services, much less at a sub national level. The localised nature of CIEWS, creates a greater need to develop national and sub-national systems as they have better access to local information and contexts. Agencies need to be in a position to conduct local level monitoring and data gathering on factors that are important for impact-forecasting, such as the changing dynamics of hazard exposure, social vulnerabilities and disaggregated distribution of assets, access to services and infrastructure at risk. While regional centres have a role in making first order estimates, it is national institutions that must undertake more detailed risk mapping and assessments based on both historic loss and damage data as well as risk data that enables outlook and foresight on the possible impacts of climate change. The number of countries that have accessible, understandable, usable, and relevant disaster risk information and risk assessments available at the national and local levels represents the lowest of all Sendai Framework Target G indicators. In large part this is a result of a lack of field capacities within hydrometeorological systems as well as those related to data processing through new technologies. For example the field data required for IbF include multidisciplinary vulnerability assessments and tracking of bio-physical and hydrologic variables such as those pertaining to vegetation (land use and cover), soil properties (saturated conductivity, erosivity) and hydrology (flow-generation/accumulation and discharge). Processing and visualising this data require programming skills and use of programming languages and tools. Skills are also required in analysing spatial data and hydrologic modelling using tools/software and cloud-based tools which leverage large data infrastructure without the costs associated with setting up such systems locally. Capacities to link forecasts with hydrologic models and the use of Artificial Intelligence (AI) tools are also lacking. It is these new innovative technologies that paradoxically would allow met services to address in a more a cost-efficient manner existing gaps in CIEWS coverage and in some cases can provide solutions to data gaps.
- 78 Gaps in observational data are particularly severe and persistent in SIDS and LDCs and even in middle income economies. This is a bottleneck at the start of the EWS value chain about which the international community has been concerned

⁸⁸ UNDRR, "Analysis of Barriers to Financing and Operationalizing Linkages between Anticipatory Action and Social Protection Systems", 2023, <https://www.undrr.org/publication/analysis-barriers-financing-and-operationalizing-linkages-between-anticipatory-action>

about for decades. Many attempts have been made to address the problem. However, substantial investments in observing systems have not translated into increased observational data sharing or in addressing data gaps effectively.

- 79 The lack of technical and financial capacities at a national level has acted as a barrier to adoption of innovative digital technologies. These technologies can enable a transformational shift towards solutions that address the lack of consistent and reliable field monitoring such as the use of IoT based sensors to autonomously measure and transmit data using mobile networks. Generation of timely early warnings, particularly in remote areas, requires use of sensors and systems which are autonomous and "off-grid". Many of these, such as water level recorders, have dual channels for real-time warnings to local communities as well as transmission of data to NMHS for further analysis and interpretation. Nonetheless, operation and maintenance (O&M) of these systems can present challenges for NMHS who have a shortage of ground personnel and for communities without the necessary technical skills.

- 80 Effective local actions based on MHEWS are prevented by a number of barriers.

Ineffective or absence of dissemination to the last mile

- 81 Many countries have significant gaps in telecommunication networks which exclude some of the most climate vulnerable communities in rural areas, especially nomadic and migratory pastoral groups and forest dwelling communities. Access to early warnings and advisories can also be a serious constraint among poor rural communities who may not own phones or have electricity to power radios. Insufficient capacities for translation, contextual rephrasing and packaging of early warning information and advisories for multiple channels can prevent end-users from receiving such information. Most MHEWS are developed through a centralized process. In large and diverse countries, this fail to sufficiently address local contexts and tailor risk communications to diverse audiences, and tend to use technical terms in major national languages. They are not adequately designed around end-users, are not people-centred and warning messages are often not timely, actionable or understood by local actors. Warning messages and advisories usually lack the local context and are not tailored to the different needs of targeted communities, including the elderly, persons with disabilities and health impairments, female-headed households, ethnic minorities and IDPs, and vulnerable children. The design and content of early warnings often fail to address the differential needs of end-users arising from differential access to information, resources and representation. This prevents the most vulnerable members of communities to access early warnings or take appropriate action.

Insufficient national and local level disaster preparedness and anticipatory action planning

- 82 There is a major gap in terms of national and/or local plans setting out anticipatory action and development and use of triggers which determine when to act, roles of different actors in the early warning value chain, to define the relevant and most appropriate and effective types of anticipatory actions for these actors. Under one third of countries globally have reported on the percentage of local governments that have plans to act on early warnings (Sendai Framework Monitor, Target G indicator G4), suggesting they may not have these plans. Furthermore, the percentage of local governments that have a plan to act is very low even among the countries that have reported on this indicator. Few governments have integrated anticipatory action into their national systems, although their number is growing.
- 83 Equitable and meaningful decision-making opportunities and roles for women in the design and management of MHEWS and related areas are significantly limited.⁸⁹ There are gender gaps in access to information, technology and digital services. For example, in South Asia and Sub-Saharan Africa where, more than half a million women are excluded from the mobile connection and the gender gap scores are 41% and 37%, respectively.⁹⁰ Inadequate resources at the local institution level often prevent women from leading preparedness and coordinating effective inter-agency responses to follow up warning messages with local action. Barriers that limit effective use of CIEWS by first responders and communities include local socio-cultural and economic contexts, infrastructure and services that determine the ability of communities to adopt measures that mitigate the impact of climate-induced disasters.⁹¹ First responders and communities include technical limitations in accessing CIEWS, their contextual interpretation, particularly in the case if

⁸⁹ UNDRR, "Inclusive and Accessible Multi-Hazard Early-Warning Systems: Learning from Women-Led Early-Warning Systems in the Pacific," 2022.

⁹⁰ GSMA, "The Mobile Gender Gap Report 2022," *Mobile for Development*, June 2022.

⁹¹ Pörtner et al., *Climate Change 2022*, 2022.

indigenous and minority populations, and inadequate procedures and plans to take anticipatory action or to mitigate impacts of disasters. There is almost a complete absence of information on dissemination and evacuation procedures that consider people with visual or hearing impairments, physical disabilities. In the few cases where evacuation procedures and facilities exist, they are male-centred and designed for persons without disabilities.

B.2 (a). Theory of change narrative and diagram

- 84 Addressing the barriers to effective end-to-end MHEWS require engagements at different levels of organisation, governance systems and across multiple institutions. The project will need to operate at different scales and leverage new and innovative technologies while ensuring the interventions engage meaningfully with local communities, their own knowledge, understanding and coping strategies for climate related disasters.
- 85 The project is structured so that Output 1 provides support at the regional and global levels to set up, scale up and accelerate technical and operational support to countries with the ambition to enhance their MHEWS coverage and capacities. This output will catalyze regional and global collaboration among actors active in EWS support, including through regular coordination, joint action, and exchanges to reduce duplication, save scarce resources, and better leverage the comparative advantage of diverse actors to deliver shared value to countries aligned to the EW4All call to action. The output will strengthen strategic planning for long term financing for maintaining such a support architecture to avoid fragmented, discontinuous, and uncoordinated support to countries in improving end-to-end MHEWS.
- 86 Output 2 ties together the four key components of the EW4All EAP by addressing barriers in observations, risk knowledge, dissemination and disaster preparedness and response at the country level. Activities under Output 2 address gaps at the national level, these are designed to ensure effective use of MHEWS for disaster preparedness and response. Hence the investments in observations will be guided by the vulnerability analysis and specifically seek to facilitate development of impact based and actionable early warnings for early and anticipatory action by national actors on the ground. Engagements with the governments will also seek strengthening of institutional mechanisms and governance for more effective preparedness and response to attenuate the scale of recovery needs. Dissemination systems will be improved to ensure usable multi-hazard risk information is transferred to end-users, including women and vulnerable groups within communities, along with guidance on effective mitigation actions to reduce impacts (loss and damage). Finally, the work with communities will ensure disaster planning and preparedness as well as community-based systems that hybridize new technologies and local knowledge.
- 87 The project will improve technical support mechanisms and financial flows to meet the needs of countries (particularly LDCs and SIDS) to establish, operate and maintain their national end-to-end MHEWS. To do so, it will establish regional and national collaborations, multiple funding instruments and complementary mechanisms that maximize country ownership and impact. The project will facilitate national investment planning into the MHEWS value chain, including the development of national roadmaps, investment pipelines, and approaches to sustain national EWS through state and private sector resources, and innovative finance. This will help address gaps across the continuum of country-specific technical and financial support needs for end-to-end MHEWS, translating country visions into concrete investments.
- 88 The project has two broad outputs and associated activities which address the barriers described above. These outputs and activities are mutually supportive and reinforcing (Figure 1).

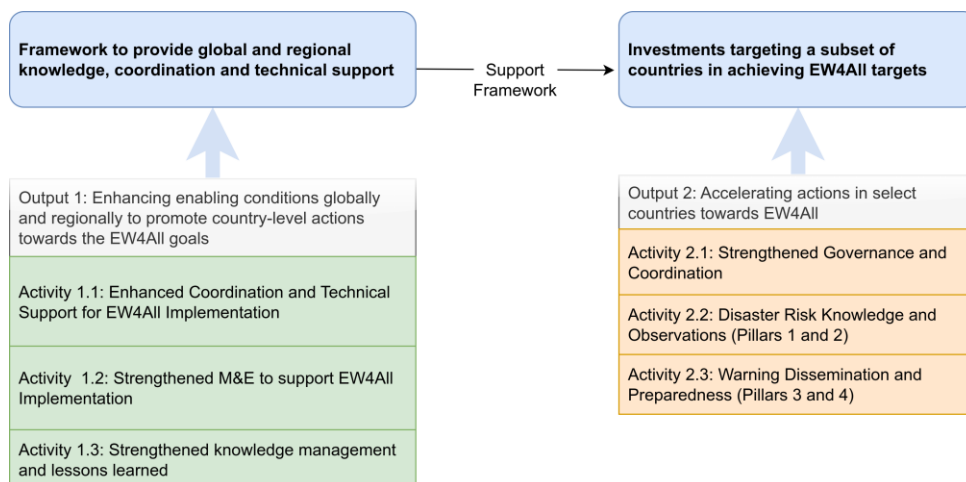


Figure 1: Interactions between the two proposed outputs

89

- 90 Output 1: Enhancing enabling conditions, both globally and regionally, to promote country-level actions towards the EW4All goals. This output will address barriers pertaining to both support for country level actions as well as gaps in global and regional policies, standards and inter-agency coordination to scale up coherent national strategies for end-to-end MHEWS. The output will improve existing global support architecture to scale-up access to technical expertise, operational support, collaboration and knowledge sharing and the existing global support architecture and will be led by WMO and UNDRR in collaboration with global and regional organizations.
- 91 Output 2: Accelerating actions towards EW4All goals, will implement national level projects and priority investments listed under the four pillars in the EAP, based on the context of each participating country. This will be led by UNDP, WMO, UNDRR, ITU, IFRC, in collaboration with other key partners.

B

IMPACT: PARADIGM SHIFT

IF generation, delivery and use of multi-hazard early warning information is improved for effective early and anticipatory action, THEN it will enhance avoidance of and adaptation to climate induced natural hazards and reduce the loss of lives, disruption of livelihoods BECAUSE it will result in improved, effective and timely disaster mitigation and response from national governments supported by multilateral agencies and private entities.

OUTCOMES and CO-BENEFITS

1: Sustained support for global coordination and support architecture for EW4All

Co-Benefits

1. Improved livelihoods and food security of communities dependent on climate-sensitive natural resources.
2. Improved management and conservation of the environment and critical ecosystems.
3. Better design and planning of settlements and infrastructure.

2: End-to-end EWS that is inclusive and builds community resilience to climate hazards

OUTPUTS

1. Enhancing enabling conditions globally and regionally to promote country-level actions towards the EW4All goals

2. Accelerating actions in select countries towards EW4All

PROJECT ACTIVITIES

- 1.1: Enhanced Coordination and Technical Support for EW4All Implementation
- 1.2: Strengthened M&E to support EW4All Implementation
- 1.3: Strengthened knowledge management and lessons learned

- 2.1: Strengthened Governance and Coordination
- 2.2: Disaster risk knowledge and observations
- 2.3: Warning dissemination and preparedness

BARRIERS, RISKS

- Barrier 1.1: Limited global, regional and national support for country level actions for a comprehensive EW4All
- Barrier 1.2: Inadequate policies and strategies for inter-agency coordination and governance for preparedness and response
- Barrier 1.3: Poor strategic planning for long term financial support to MHEWS

- Barrier 2.1: Inadequate technical and scientific capacities in the generation and delivery of timely, targeted and contextual CIEWS
- Barrier 2.2: Insufficient disaster risk knowledge for effective management and response
- Barrier 2.3: Ineffective or absence of dissemination to the last mile
- Barrier 2.4: Insufficient local level disaster preparedness and planning

ASSUMPTIONS

1. Participating countries are able to sustain O&M of established hydromet infrastructure and communication networks
2. NMHS and NDMA retain technical staff trained and capacitated in developing forecasts, advisories, and development of participatory anticipatory action plans
3. International funding organisations are willing to invest in the EW4All and related initiatives
4. Regional technical centres of WMO and other pillar/partner agencies have the capacity and resources to provide necessary technical backstopping to countries
5. Agencies involved in the EWS value chain, including government, multilateral, non-government, private sector agree to coordinate their activities

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☒	☒
Outcome 2	☐	☐	☐	☐	☒	☒	☒	☒
Outcome 3	☐	☐	☐	☐				

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1	☒	☒	☒	☒	☒	☐
Co-benefit 2	☒	☐	☐	☐	☐	☐
Co-benefit 3	☐	☒	☒	☐	☐	☐

B.3. Project/programme description

- 92 This project addresses the four key elements of MHEWS, i) disaster risk knowledge; ii) detection, observation, monitoring, analysis and forecasting of hazards; iii) warning dissemination mechanisms and communication; and iv) preparedness and response capabilities.⁹²
- 93 Effective MHEWS depend on coordination with and across different sectors and at multiple levels and require feedback mechanisms which allow for their improvement.⁹³ The involvement of both national and local institutions and communities in dissemination of warnings and response is a critical component of successful MHEWS.⁹⁴
- 94 Addressing persistent gaps in establishing and managing an end-to-end MHEWS at the country level is a key requirement. The proposal recognises that there are multiple on-going initiatives and mechanisms that aim to enhance countries' climate information and EWS. These programmes and initiatives in the participating countries is summarized in Table 2 in the appendices. The proposal also recognizes that an effective end-to-end MHEWS can only be built on timely and long-term support for critical baseline capacities such as: i) robust scientific risk information and analytics on weather, hydrology, environmental conditions, climate hazards and socio-economic, infrastructure and other dynamic vulnerabilities; ii) optimised internal coordination among various government entities that are responsible for managing different elements of a national EWS value chain; and iii) a long-term financing plan to maintain the necessary hydrometeorological infrastructure, and community level capacities to receive, understand and respond to EWS through risk-informed preparedness and response actions. These baseline capacities in many countries, especially in LDCs and SIDS, can be built only through long-term support; the project will therefore strengthen the existing architecture and create the environment in which countries can better utilise timely, long-term, uninterrupted support.

⁹² UNDRR, "EWS"; The UN Global Early Warning Initiative for the Implementation of Climate Adaptation, "Early Warnings for All - Executive Action Plan 2023-2027," 2023.

⁹³ UNDRR and WMO, "Global Status of Multi-Hazard EWSs: Target G," 2022.

⁹⁴ Matthew L. Collins and Naim Kapucu, "EWSs and Disaster Preparedness and Response in Local Government," *Disaster Prevention and Management: An International Journal*, 2008.

95 Additionally, there is recognition that the scale and speed required to deliver universal early warning coverage by 2027 requires an “all-hands-on-deck” approach which utilises support available from all partners. Achieving this will require sustained assistance to maintain effectiveness and sustainability. Furthermore, the support architecture described above cannot be solely dependent on a single project or programme and this project will include activities in which a long-term financing and management strategy is developed.

96 Targeted countries will be provided global and regional assistance to mobilise technological resources for upgrading and sustaining their MHEWS. A comprehensive assessment of climate-related risks required improvements in capacities for early action, which includes local level disaster planning, will be conducted in all countries. National policies, procedures and mechanisms will be instituted to coordinate and collaborate between NMHS, NDMA, government ministries (including agriculture, water and health) and non-government agencies. The capacities of critical agencies and institutions at all levels along the early warnings/Anticipatory Action value chain will be enhanced.

GCF proceeds will be used to provide technical assistance to the beneficiaries or finance limited improvements to existing infrastructure / equipment purchases and there will be no transfers of cash or funds to the beneficiaries.

Component/Output 1: Enhancing enabling conditions globally and regionally to promote country-level actions towards the EW4All goals

97 Output 1 aims to scale up and accelerate global and regional support for national efforts to increase Multi-Hazard EWS (MHEWS) capacity, coverage and performance. Three key activities are envisaged: Activity 1.1 focuses on enhancing global and regional coordination and providing coordinated technical support for the implementation of the EW4All initiative; Activity 1.2 emphasizes the strengthening of monitoring and evaluation (M&E) processes to support EW4All implementation; and Activity 1.3 aims to reinforce knowledge management, evidence from national practice, and the dissemination of lessons learned to inform national progress in EWS. Together, these outputs are designed to improve the effectiveness and reach of EWS, ensuring that more communities are better prepared for multiple hazards.

98 The output will support the global and regional coordination and support architecture for EW4All that will be sustained beyond the project implementation period. Activities through this output will help the EW4All Pillar leads build on the Climate Risk and Early Warning Systems Initiative (CREWS), SOFF, and similar financing and coordination mechanisms to: i) achieve substantive synergies with other partners providing similar support towards the EW4All target; and ii) to help unlock required funding for sustaining EW4All support architecture to 2027 and beyond. The latter will target conventional and emerging donors such as bilateral donors, multilateral financing institutions, and philanthropists. A long-term financing strategy will be developed which is informed by detailed needs assessment from LDCs and SIDS, including fragile and conflict-affected states. EW4All pillar leads are simultaneously engaged in global advocacy for EWS finance with the G20 Presidency, the ‘Friends of EW4All’, a coalition of 17 major bilateral donors, MDBs, the private sector, and non-traditional donors to unlock funding for national EWS. These efforts will also include finance gaps in the 7 countries. The private sector will be prioritized as part of EW4All financing, with several entities already actively engaged in EW4All through provision of technical services, pro bono cloud bandwidth, and satellite mapping, e.g., Microsoft, Google, Insurance Development Forum, and representation of the high-level EW4All Advisory Panel.

99 The output will augment on-going support for synthesis of knowledge, experience and lessons by providing operational and technical assistance aligned with international standards and best practices towards the effective implementation of the EW4All initiative. This will involve the development, dissemination and application of comprehensive normative guidance, tools, and methodologies across all four pillars of EWS and at the inter-pillar level. Specifically, this includes the development of detailed guidelines, strategic orientation documents, and AI-based tools designed to improve processes along the entire early warning value chain and the provision of tailored technical assistance to strengthen national EWS capabilities. National and regional DAEs, as well as sectoral representatives and associated ministries will be included in the knowledge exchange to ensure they can contribute to discussions on practically implementing MHEWS within different contexts and sharing innovations, lessons, and evidence of national EWS effectiveness.

100 In addition, this output will support the integration of EWS into national priorities, roadmaps, and financing strategies, including governance and legal frameworks. Technical support will also be provided to facilitate simulations and testing of national EWS operational effectiveness, develop coordinated funding requests, strengthen the global EW4All M&E framework and develop a knowledge management system to document and disseminate key learnings. Collectively,

these global and regional activities aim to establish robust, sustainable, and effective national EWS globally, ensuring that countries are well-prepared to respond to and mitigate the impacts of disasters.

- 101 This component will be led by EW4All Pillar lead agencies and will aim to support enabling conditions at both the global, regional and national levels, so that countries receive relevant and timely assistance to meet the universal coverage target of EW4All. Funds from the GCF will flow from the UNDP to the respective Pillar leads who will be the responsible parties of these activities.

Activity 1.1: Enhanced Coordination and Technical Support for EW4All Implementation

- 102 Activity 1.1 focuses on enhancing coordination and providing comprehensive joint technical support to ensure the effective implementation of the EW4All initiative. This involves the development and application of normative guidance, tools, and methodologies that span across all four pillars of EWS and integrate these efforts at the inter-pillar level. The four Pillar lead agencies will collaborate to create detailed and standardized guidelines, training materials, and strategic orientation documents. The goal is to provide countries and partners with the necessary resources to align their national EWS initiatives with international standards and best practices. This includes integrating EWS into national development and adaptation plans, roadmaps, and financing strategies, as well as supporting governance and legal frameworks for EWS. Furthermore, the development and application of AI-based tools will be emphasized, ensuring that advanced technologies are used responsibly to address gaps and improve the effectiveness of the entire early warning value chain. Activity 1.1 further aims to accelerate harmonized technical support to countries by providing operational assistance and facilitating simulations and testing of national EWS operational effectiveness. Capacity-building efforts will also be undertaken to enable DAEs to fast-track access to GCF financing for early warning initiatives.

Sub- Activity 1.1.1: Develop normative guidance and tools across all four pillars & at the interpillar level

- 103 This sub-activity will result in the development of pillar-specific and joint inter-pillar guidelines, tools, training curricula and methodologies, building on what exists and accounting for variations in regional and national contexts. Strategic Orientation and Policy Guidance will be developed and provided to countries and partners in thematic areas such as: improved access and harmonization of climate and disaster data and information for application in national EWS; global standardization of loss and damage accounting for rapid and slow-onset hazard events for inform national tracking; standardized EWS gap analyses and national EWS roadmaps and financial frameworks; global standards and diagnostics of national data ecosystems related to risk, climate and disaster loss open-data sharing for Ibf; national climate and risk assessment capacity strengthening; and minimum standards for national climate and disaster risk hotspot mapping, multi-channel warning dissemination and risk communications; and preparedness and response protocols.
- 104 This sub-activity will also focus on advancing the application of AI-based tools within the framework of EWS, offering guidance and support to countries on the responsible and effective use of AI across the entire MHEWS value chain. This effort aims to address existing gaps and emerging needs, leveraging the expertise and resources of the AI for EW4All sub-group. AI is a key output identified in the EW4All EAP. The AI for EW4All sub-group will be leveraged to explore AI use cases that align with the four pillars of EW4All, providing practical examples and identifying opportunities to enhance the initiative's impact. This sub-activity will include the development of an EW4All Toolkit, which will include tools, models, applications, and resources designed to enhance EWS. To ensure broad accessibility, a searchable webpage will be designed to host the Toolkit, enabling countries to effectively leverage these resources for managing risk, communication, and preparedness. This sub-activity will also involve cultivating strategic partnerships and collaborating with stakeholders globally to support the implementation of AI in the EW4All initiative. Additionally, it will focus on developing and delivering capacity-building and awareness-raising sessions. These sessions will demonstrate how AI can significantly enhance the effectiveness of EWS, improve the flow of critical information during emergency response and preparedness, and support the achievement of outcomes across all four EW4All Pillars.
- 105 84% of persons with disabilities reported not having a personal preparedness plan for disasters; 56% (3,549) reported not being aware of or not having access to disaster risk information in accessible formats in their communities⁹⁵. The findings

⁹⁵ According to the 2023 Global Survey on Persons with Disabilities and Disasters: <https://www.unwater.org/news/undrr-2023-global-survey-persons-disabilities-and-disasters>

underscore the urgent need to invest in accessible risk communication, tailored for the specific needs of persons with disabilities. The global normative guidance will provide a comprehensive framework to ensure that all individuals, including persons with disabilities, have equal access to vital information to protect themselves from disasters. The guidelines will explore how inclusive communication strategies should cater to diverse needs, covering content and messaging, accessible formats and inclusion of persons with disabilities in the development of risk communication strategies and products. Regional broadcasting unions will be strengthened to support their national members to: build operational resilience through SOPs and emergency broadcasting plans; implement CAP (in partnership with ITU); strengthen editorial capacity to contribute to early warning chain ahead of disasters and cover DRR topics in times of peace in order to build a culture of preparedness in the context of national EWS. Where relevant, these tools will be validated and suitably tailored to the context of the seven participating countries to inform the activities under Activity 2.1 and 2.2.

- 106 In addition to this, WMO will undertake a review of its guidance across hazard types and update as needed to enable the use of multi-model ensemble and probabilistic forecast to issue warnings and communicate uncertainty. As part of the review and update emerging technologies and the role of AI will be addressed. Work will also be done to ensure that the guidance is packaged and disseminated with WMO Members to enable their uptake and use.

Sub-Activity 1.1.2: Accelerate Technical Support to Countries for EWS implementation

- 107 Building on delivery of normative guidance and tools articulated in the preceding activity, Activity 1.1.2 focuses on accelerating technical support for countries implementing EWS, focusing on providing operational and technical assistance for the development and implementation of EW4All national roadmaps, supporting countries to test the capacities of the MHEWS, and support the financing elements that underpin MHEWS systems and capacities. Efforts will be focused on scaling up and enhancing the work of the Tropical Cyclone Programme (TCP), the Severe Weather Forecasting Programme (SWFP), IbF and Warning (IBFW), and the CAP. WMO's Regional Associations (RAs), working groups and expert teams will be engaged to reach more countries with the afore mentioned capacity development programmes to address a broader range of hazards, and provide better support to WMO Members.
- 108 Under this sub-activity operational and technical assistance will be provided to countries, ensuring alignment with international standards and best practices outlined in sub-Activity 1.1.1. Pillar leads, WMO Regional Climate Centres (RCCs), together with other global, regional and national agencies will co-design and implement analytic frameworks and models for national adoption to scale-up use of data related to critical multi-hazard exposure, vulnerability, loss and damage, and effective mitigation action to provide the basis of impact-based forecast warning services (IbWFS). Alignment will also be sought with existing regional and global initiatives around early warning early action. This will include providing and helping to design systems for climate and multi-hazard risk data sharing, archiving and access and open-source software and scripts for data cleaning, analysis, visualisation and modelling. Regional centres will also be responsible for developing models which can be used by the NMHS together with the NDMA's, for IbWFS, scenario building and triggers to inform early/anticipatory actions. Wherever feasible, the regional centres will explore co-production of Anticipatory Action protocols involving the NMHS, NDMA's together with research agencies and grassroots organisations. The capacity of WMO, UNDRR, ITU and IFRC will be augmented to enhance the quality of support to countries via Output 2. In particular, the existing WMO supportive frameworks such as SWFP, TCP, CAP, IbFW and RAs will be expanded geographically to provide real time operational support to more countries and also cover a wider range of hazards to match country needs. These activities will be closely coordinated with SOFF investments which directly support WMO corporate priorities and frameworks.
- 109 The sub-activity will support the integration of EWS into national priorities, roadmaps, and financing strategies. Technical support and coordination across all pillars of the EAP which relate to governance and legal frameworks for EWS and early warning dissemination will be provided to participating governments. A sustainability strategy for the global support architecture will be developed by implementing global financing campaigns and consultations with various partners and donors that will take place throughout the implementation of the project. The sub-activity will also develop methods to facilitate simulations and testing of national EWS operational effectiveness. It will provide capacity-building and support to DAEs to fast-track access to GCF financing for EWS strengthening.

- 110 A global strategy will be developed to unlock financing at the global level to deepen or expand support to countries towards the EW4All target. Countries will be supported in sourcing financing from international financial institutions, international organisations and vertical funds and corporate donors seeking to support climate adaptation, such as those participating in the [UN Global Compact](#). Countries will also be facilitated in supporting grant financing for women-led and women only initiatives.

Activity 1.2: Strengthened M&E to support EW4All Implementation

- 111 Enhancements to the global M&E framework for EW4All will be implemented and rolled out, including updates and expanded metrics available on the dedicated dashboard to track progress of the initiative globally and at country levels across all four pillars. Technical assistance will be provided to the 7 participating countries to improve national capacity for reporting on EWS performance based on a global reporting framework. The framework will include annual reporting on progress on the EW4All initiative at the UNFCCC COPs. Work will also be supported on the development of an index capturing EWS effectiveness and country capacity. The activity will also support countries on Sendai Monitor reporting, particularly target G. Efforts will focus on: development and implementation of comprehensive tools to evaluate, continuously monitor, and report on national EWS capacities, development and implementation of the EW4All maturity index system for benchmarking global progress, strengthening and consolidating the EW4All M&E framework across pillars, considerate of interpillar elements, updating and validating pillar assessments and score cards to regularly track country capacities, and integration of country data into the EW4All dashboard and maintain the dashboard, including the creation of a database.

Sub-Activity 1.1.1: Operationalising the EW4All maturity index

- 112 This sub-activity will develop and implement the EW4All maturity index system for benchmarking global progress across all pillars and at the inter-pillar level. It will update and validate assessments and score cards to track country progress across the EWS value chain, and provide diagnostics support to countries on monitoring EWS effectiveness. This sub-activity will also enable WMO to complete the rapid assessment of Pillar 2 for new countries entering the initiative. WMO will also track progress on completed assessments and ensure that country score cards are updated regularly. Moreover, WMO will undertake the task of updating the methodology for CHD as needed. These activities will strengthen global information and reporting systems on MHEWS capacities for informed action.

Sub-Activity 1.1.2: Strengthen global EW4All M&E and reporting to enable regular progress updates

- 113 This sub-activity will strengthen and consolidate the EW4All monitoring and evaluation framework across pillars, considerate of inter-pillar elements, It will conduct data collection and related monitoring actions on field and develop and publish studies, reports, other knowledge materials to showcase outcomes of framework monitoring, including annual Global Status of MHEWS reports.

Sub-Activity 1.1.3: Regular update of EW4All Dashboard

- 114 Sub-Activity 1.1.3 will improve integration of EW4All implementation progress and country capacities data into the EW4All dashboard. It will improve analytical and visual functions and maintain the dashboard. It will ensure country-level tracking and updates to the global database⁹⁶ respecting measurable improvements in national capacity in areas such as risk assessment, risk data use and governance for EWS, application of hazard-specific exposure and vulnerability data to improve IbF, and progress on avoiding and minimizing loss and damage through EWS. In addition the sub-activity will ensure regular reporting by participating countries on the Sendai Monitor, particularly for target G reporting.

Activity 1.3: Strengthened knowledge management and lessons learned

- 115 The activity will support knowledge exchange among countries at different stages of implementation of the EW4All EAP across all Pillars and implementing partners. Countries will be assisted in developing plans to improve their MHEWS and Anticipatory Action approaches and systems, ensuring that each approach takes a people-centric view with ensured last-mile communications and including all stakeholders (DAEs, humanitarians, International Financial Institutions (IFIs) and

⁹⁶ <https://earlywarningsforall.org/site/early-warnings-all/early-warnings-all-dashboard>

sectoral representatives – both government and NGOs). This will be linked to the SOFF community of practice and will bring together peer advisors and implementing entities to document lessons learned.

Sub-Activity 1.3.1: Development of knowledge management framework that guides learning and exchange

- 116 This sub-activity will develop a knowledge management framework to systematically capture, document, and disseminate lessons learned. It will establish processes and protocols for knowledge sharing among stakeholders and develop a repository that is open access and online for the dissemination of knowledge products. The sub-activity will contribute to the UN global knowledge bank for EWS to improve easy access by country-level actors to robust tools, guidance, peer-learning, and reliable technical guidance from multiple UN and non-UN sources on themes cutting across EWS.

Sub-Activity 1.3.2: Documentation and dissemination of good practices and lessons learned

- 117 The identification and documentation of good practices, challenges, innovations and solutions encountered in early warning programming will be done under this sub-activity. Post-disaster forensics will be conducted to learn lessons related to EWS. Lessons learned on EWS functioning will be integrated in post-disaster needs assessments (PDNA), and the reinforcement of functional EWS will be included in Recovery Frameworks. The sub-activity will also disseminate early warning knowledge products, in support of peer-to-peer learning and continuous improvement and informed decision-making.
- 118 Under this sub-activity, WMO will develop and maintain an enhanced section on the WMO website which centralizes information, tools, and guidance materials dedicated to providing resources for the implementation of the EW4All initiatives.

Sub-Activity 1.3.3: Development of global and regional technical expert working groups

- 119 This sub-activity will establish global technical expert groups and communities of practice composed of national representatives to advance collective action in core thematic areas such as open data as a digital public good for EWS, mobile EWS, early and anticipatory action and finance, preparedness and response planning. It will facilitate the sharing of implementation expertise and good practices, serve as a platform for capacity building and technical support aligned with Activity 1.1.2, and augment opportunities for national experts in LDCs and SIDS to influence global dialogue on EWS.
- 120 The Cell Broadcast Expert Group provides the platform for technical experts to share information on topics such as public warning systems architecture and regulation, cell broadcast implementation, multi-channel deployment, satellite adaptation, and standardization for handset and chip manufacturers, to assist countries in their deployment; as well as to create awareness and share challenges and solutions in international events and workshops.

Component/Output 2: Accelerating actions in select countries towards EW4All

- 121 This component will roll out national level investment activities in seven countries to address country and local level needs to achieve the EW4All target. All seven of these countries are from the list of the 30 focal countries identified under the global EW4All initiative for 2023⁹⁷. Concrete investments will focus on EWS hardware and software, risk knowledge, disseminating and communicating early warnings and preparedness for local-level actions that reduce vulnerability of communities. These will be based on a comprehensive analysis of capacities and gaps in each country.
- 122 Concerted efforts will be made to close gender gaps in access to, control over, and use of MHEWS technology, communications and services. This includes considerations of needs and challenges facing rural women, ethnic minorities, men and women who are illiterate, people with disabilities, elderly and households with infants and children. It will support women's equitable decision-making opportunities, including in design and management of EWS areas, laws, policies and operational procedures. Lessons from other countries such as [Cambodia](#) will be incorporated to ensure women play meaningful roles in MHEWS and early action. In a country like Somalia, for example, women and girls have low levels of understanding about early warning messages and generally have lower access to information than men. The

⁹⁷ UNDRR. 'Global Status of Multi-Hazard Early Warning Systems 2024'. United Nations Office for Disaster Risk Reduction, 4 December 2023. <https://doi.org/10.18356/9789213587058>.

project is targeting for 60% of EWs to specifically target women and girls. This will include by taking into account the constraints they face in accessing and understanding information and building on opportunities. Working with women's groups and other outreach opportunities that target women, for example, will be key. If women and girls are able to access EWs in a way that is understandable and meaningful, this will be transformational as it will mean they can take adequate measures to protect themselves in times of emergency. Overcoming literacy barriers, traditional roles of women and lower access to technology are among the challenges that will be encountered. The project mainstreams gender and all activities essentially have a gender lens which is documented in the country-level GAAPs. In Cambodia, for example, project activities focused on training of local-level stakeholders on MHEWS will specifically target women. For example, the project will include the participation of women in capacity development related to impact-based forecasting and engage in technical areas sessions to build their capacity and representation within the Ministry of Water Resources and Meteorology. Moreover, the project targets that 70% of training sessions on MHEWS, regardless of participant's gender, include gender-sensitive content on the differential impacts of hazards and effective communication strategies for diverse groups. UNDP Cambodia implemented an LDCF financed project on EWS earlier where women in community-level leadership positions, such as school directors, members of district disaster committees, commune chiefs, etc, have been trained as gender champions in their respective communities and they played a big part in bringing other women to participate in project activities while the role they played empowered themselves (see: <https://undpcambodia.exposure.co/>). The GCF project will take these lessons in the project implementation.

- 123 Each country has unique opportunities and barriers to implementing effective EWS, and the activities and sub-activities described here will be based on the eligibility criteria listed in Appendix 3. However, there are some common needs related to the barriers mentioned within the project's theory of change, particularly as they relate to capacity and finance gaps, CIEWS coverage particularly in remote areas and lack of access to innovative solutions. This component will fulfil those needs and leverage available opportunities by executing time bound investment activities and allowing for measuring the impacts on ground. The implementation of the national investment activities will be supported by WMO, UNDRR, ITU, IFRC, and other partner agencies, depending on the capacity of the host government and in-country presence and capacities of these agencies. The UNDP and its respective Country Offices will provide technical and administrative support, coordination and oversight to the implementation of the Output. All activities will coordinate with and be designed to complement existing work being undertaken by DAEs, NGOs (IFRC, Care, Christian Aid etc.) and other multilateral (World Bank, GFDRR, FAO, WFP) and bilateral (GIZ, FCDO, SIDA, NORAD etc.) donors. This Output is designed to mirror the structure of the EAP so that activities within this Output will be fully aligned to it.
- 124 The implementation of the seven country projects will be supported by a global team of experts and a global Project Management Unit (PMU). The international expertise will be drawn from the Pillar Leads and other partner agencies and expertise will span the technical elements of all the four pillars and will include social and environmental safeguards, gender, inclusion and disability experts who guide and participate in the implementation of project activities in the seven countries. Most of the international experts will be hired on a part-time basis and will support activities identified and approved by the national governments for which local expertise is not available. This arrangement will ensure access for all countries to a wide range of highly qualified and experienced experts on a cost-sharing and therefore, cost-effective basis.

Activity 2.1: Strengthened Governance and Coordination

- 125 The activity will focus on legislation and policies governing the exchange of information to facilitate coordinated dissemination of warnings between different actors. It will establish clear functions, roles and responsibilities of agencies in the national warning dissemination process with associated updating of the WMO Register of Alerting Authorities (RAA). Alert dissemination and feedback channels will be expanded to reach end users with targeted and actionable information by leveraging community-based institutions, civil society and faith-based organisations. Considerations of differential access to information and resources by women and vulnerable groups will be included in the dissemination strategy. Partnerships with intermediaries will be established and training provided to local authorities and relevant agencies for interpretation and dissemination of alerts to ensure delivery of clear and actionable warnings to end users. This will include extension officers and other government staff in line departments and journalists.

126 This activity will also attempt to establish or strengthen existing local Early Warning Committees (EWC) or equivalent grassroots organisations with representatives from elected/traditional leaders who are linked to govt. development or DRR structures wherever feasible given the country context. Meaningful involvement and representation of women and vulnerable groups, including persons with disabilities and their representative organizations (OPDs) in these institutions will be ensured. Each EWC will seek to be a registered entity under the relevant national provisions and will seek to have a formal structure of office bearers and a bank account, based on the existing governance structure and institutional arrangements. A clear coordination and reporting structure will be instituted between sub-national agencies from the lowest level of governance up to the district level. In all countries except Chad and Somalia, each EWC will have direct access to focal persons in the relevant government agency at the district, NMHS and NDMA. In Chad and Somalia organizations fulfilling these roles will be engaged. Capacities of these local institutions, from district down to village level will be enhanced through regular engagements and training programmes that strengthen institutional structures and procedures.

Sub-Activity 2.1.1: Strengthened coordination mechanisms

- 127 The sub-activity will advocate for enabling policies, laws and strategies for effective early warning, early action and preparedness. It will support the development of a strategy and action plan and associated legislation to mainstream early warning and anticipatory action, while ensuring alignment with best practices and a supportive framework and legislation to mainstream early warnings and anticipatory action from national to local scales. This will be done through a comprehensive review of existing disaster risk management (DRM) policies, laws and strategies as well as climate plans. The establishment of national liability frameworks that hold stakeholders accountable for not acting on warnings will be explored as mechanism for strong legal incentives for accountability⁹⁸. The sub-activity will be disability inclusive, consulting Organizations of Persons with Disabilities (OPDs) in the development, implementation and monitoring of emergency-related laws and policies, to ensure meaningful participation of persons with disabilities to comprehensively understand their needs in humanitarian emergencies.
- 128 The lead government agency of each country in coordination with the NDA, is expected to use existing interministerial platforms, particularly those established for the NDC and NAP coordination, to inform the long-term sustainability options, including financial incentives, for maintaining the hydrometeorological network..
- 129 Data sharing between and within agencies at the national and sub-national levels will be strengthened and a formal delivery mechanism will be provided for government agencies by providing technical assistance, equipment, and other support to overcome the barrier that is preventing effective data sharing between national and local authorities, first responders and other stakeholders involved in early warning on the ground and to better align and activate their respective response plans to reduce the impact of the hazard (Sub-Activity 2.3.3). Targeted groups include national and local authorities, businesses, communities, National Red Cross Red Crescent Societies, NGOs, the UN, and community groups. Mechanisms that will be used include MOUs between relevant agencies combined with standard operating procedures for user access to data hosted on servers. Data integration frameworks and systems which utilize application programming interfaces (APIs) to access data across multiple servicers may additionally be required in countries and agencies where such systems are not currently operational.
- 130 This activity will additionally ensure policy and financial support for mainstreaming of early earning, early action and preparedness in government frameworks, systems and plans based on local needs and priorities and using local knowledge. Committed financing will be critical to anticipatory action and response in the event of a crisis. The success of the work in this activity hinges on committed financing for disaster preparedness and anticipatory actions ahead of a crisis. The activity will embed the principles of anticipatory action into social assistance through enhancing shock-responsive social protection system at the grassroots level.

⁹⁸ <https://www.undrr.org/news/one-step-closer-legal-protection-disasters-and-resolution-79/128> of the UN General Assembly: <https://undocs.org/Home/Mobile?FinalSymbol=A/79/476>

Sub-Activity 2.1.2: Enhanced governance capacities

- 131 This will include: i) a clear assignment of roles and responsibilities at the national and subnational level for services; ii) strategic plan and organizational structure; iii) periodic process to assess and evaluate its services; iv) coordination platform(s) – such as the NFCs – established; v) inter-institutional arrangements are in place for data sharing, interoperability of systems, co-design of services; vi) comprehensive plans are in place for communication, competency framework, cooperation, studies, quality system and financial resources. Development and updating of national plans, strategies and legislation for NMHS and DRM agencies in each country will lead to forecasting and warning services being installed for all priority hazards and meeting Sendai Target G Indicator 4: All countries' local governments have a plan to act on early warnings.
- 132 Strengthened governance, collaboration and inclusion. Governance arrangements, particularly for disaster preparedness and response, will be strengthened to include clear provisions, protocols, roles and responsibilities for early warning and early action, and will include an emphasis on inclusion of women, persons with disabilities and vulnerable groups within communities. Collaboration will be strengthened between grassroots organisations including government agencies, traditional and religious leaders, local schools and health centres and civil society organisations.
- Capacity building and creation of enabling conditions for better integration of relevant ministries and entities concerned with gender and disability, with the work of NDMA and NMHS in order to more efficiently use existing capacity within government structures.
 - Raising awareness on crucial intersections of gender inequality, disability social exclusion with success of MHEWS uptake and use by ensuring the inclusion of gender and inclusion topics in every capacity development activity to be carried out.
 - Strengthened collaboration between key ministries, academia and the private sector, including at the community level by supporting countries in defining their roles, responsibilities and identify ways of working which prioritise collective action between the different agencies and institutions.
- 133 The sub-activity will strengthen risk knowledge capability in these institutions by combining local, traditional, Indigenous, generational and scientific knowledge that can enable resilience under a range of future risk scenarios. Building on existing approaches, it will strengthen locally led, people-centred and citizen science-based approaches. It will use innovative approaches to leverage new and existing technologies to expand risk knowledge capability at all scales, including continuous monitoring / crowdsourcing of important vulnerability metrics as a basis for responding to and anticipating emerging crises and disasters.

Sub-Activity 2.1.3: Monitoring, Evaluation and Learning

- 134 Sub-Activity 2.1.3 ensures continuous monitoring of other project activities and sub-activities based on quantitative and where possible spatially explicit indicators and means of verification to track project progress and support adaptive management decisions. This is expected to result in improved monitoring, evaluation and learning systems integrated within project and national structures to provide up to date status of activities and emerging risks, as well as promoting adaptive approaches to the implementation of MHEWS. It will promoting institutional arrangements to ensure capacity and resources (e.g. staff, budget, etc.) to implement, monitor, and report gender-responsive actions and will enhance capacities of countries to monitor the availability and effectiveness of EWS, to use this to update their approaches and to report better on Sendai Framework Target G and custom indicators. Specific actions will be taken to set up a formal mechanism for feedback on the EWS from end-users and communities. This will leverage mobile phone based technologies (surveys, forms) where available combined with regular consultations for feedback from focal persons in communities, specifically including representatives of women and vulnerable groups. Specific actions to be taken for monitoring and assessing project impact are described in the M&E plan (Annex 11 pg. 3). A baseline survey will be administered in the initial phase of the project with additional surveys at the mid-point and at the end of the project.

135 This activity will require assessments and data collection for which digital technologies⁹⁹ will be leveraged where possible. The activity will also undertake disability inclusive surveys to assess the efficacy of critical project interventions, for example, measuring the socio-economic benefits from the use of climate products and advisories at the level of institutions and communities. The activity will establish formal linkages with academia and institutions at the national and regional level for collaborating on the impact assessments and also for reviews and evaluations of the products themselves for their incremental improvement and development. The proposed actions under Sub-Activity 2.1.3 include: i) M&E survey design, implementation and analysis of results for feedback into project reporting, management and assessing project impacts. This will be split into monitoring visits undertaken at quarterly intervals and disability inclusive assessment surveys done annually; ii) Product evaluation which will involve accuracy assessments through field measurements and validation with target stakeholders, both institutions and communities, with disaggregated analysis which captures feedback on EWS and its impacts on women, men, youth and vulnerable groups such as elderly and people with disabilities; iii) M&E and assessing the reach and effect of communication and dissemination strategies. This will cover: a) monitoring of data sharing and integration protocols at the institutional level, specifically between agencies involved in observations and those engaged in development of sector specific products using this data; b) monitoring of reach and c) effectiveness across different channels of dissemination and across both institutions and communities. The latter including the different vulnerable group; iv) M&E of the gender strategy, assessing the gender indicators for reporting on a quarterly basis and updating the strategy on an annual basis; and v) M&E of environmental and social safeguards as per the country specific safeguard plans and strategies on a quarterly basis. This will specifically include any safeguards put in place for FCAS and for inclusion of marginalized and disadvantaged groups.

Activity 2.2: Disaster risk knowledge and observations

136 This activity will result in the systematic collection of risk data, information and analysis leading to improved awareness about hazards, exposure populations, services, infrastructure, vulnerabilities of communities, and improved loss and damage accounting.

137 Existing data collection and repository systems in countries will be strengthened for increased data production, aggregation and use of risk information to support IbFs. This will include mainstreaming disability-inclusive data collection tools, including the Washington Group Short Set of Disability Questions and the UNICEF-Washington Group Child Functioning Module, as well as data related to risks and barriers for persons with disabilities. Relevant authorities such as NDMA, NMHS and other stakeholders, including sector ministries, and sub-national governments, will benefit from capacity building. The countries will strengthen their disaster loss accounting systems and monitoring of Sendai Target G Indicator 5:1 All countries have accessible, understandable, usable and relevant disaster risk information and assessment available to people at national and local levels. The activity will also establish collaborations with relevant academic institutions to develop effective communication strategies for contextual and actionable alerts to end users.

Sub-Activity 2.2.1: Develop disaster risk knowledge systems

138 The sub-activity will increase production and access to risk information and effective application of risk information at national, sub-national and local levels. It will result in improved capability of each country to produce quality, timely and contextually relevant risk information for EWS which integrates community knowledge through a comprehensive stakeholder engagement plan. This will result in improved generation of national, sub-national and local risk information to a minimum defined level of vulnerability, exposure and risk across major climate-related hazards. Capacities of relevant agencies in data access, metrics and analysis to support decision-making will be built to facilitate access to updated risk information in standardised, inter-operable formats. Finally, the sub-activity will build institutional capacities to incorporate risk knowledge into EWS. This will result in high resolution vulnerability (hazard, exposure) datasets for each participating country and high resolution, forecast linked models for specific hazards (floods, dry-spells, heatwaves, drought, thunderstorms, landslides, glacial lake outburst floods).

⁹⁹ Mobile ICT based Kobo/ODK, Interactive Voice Response, interactive mobile quizzes using key-entry and UMTS.

Sub-Activity 2.2.2: Broaden observation and monitoring networks

- 139 This sub-activity will focus on the strengthening and/or establishment of meteorological, hydrological and climate forecast and monitoring systems for each participating country. This sub-activity will leverage SOFF investments which support capacity and infrastructure of countries to achieve the GBON and ensure that weather and climate data is shared internationally in order to improve NWP products which improve the skill of forecasts. Capacities of NMHS will be further enhanced to address gaps in hydrometeorological data acquisition beyond GBON, through institutional and staff capacity development, installation of hardware and software leading to improved observations for developing forecasts which in turn feed into the generation of accurate and timely early warnings. Additionally, the NMHS will adopt a quality management system for regular review and reporting on the accuracy and timeliness of their services leading to compliance with GBON. The activities will build on implementation plans of existing programmes such as the Flash Flood Guidance System (FFGS) or the Hydrological Status and Outlook Systems ([HydroSOS](#)) initiative.
- 140 The sub-activity will result in enhanced national capacities for observations and data exchange by enhancing data sharing and information exchange infrastructure to support EWS. Implementation of the WMO Information System 2.0 (WIS 2.0) will be accelerated to enable timely and sustained data access and develop technical capabilities, while ensuring there is not duplication of investments of SOFF in WIS 2.0 implementation in some countries. This will be coupled with policy advocacy and development to enable sharing of critical data required for early warnings, based on the WMO Unified Data Policy. Existing efforts on observation, monitoring and forecasting for EWS led by multilateral agencies will be optimised through data sharing and sharing of forecasting products.
- 141 The sub-activity will enhance data sharing between actors involved in the generation of IbFs. It will improve both infrastructure and develop local capacities to better interpret forecasts for effective use by communities and people.
- 142 A gap analysis on early warning needs will be conducted at national and regional levels to expand the WMO cataloguing of hazardous events initiative (WMO-CHE) to all weather and climate driven hazards and their monitoring requirements; the catalogue will also be deployed to track loss and damage¹⁰⁰. Timely access to and use of in-situ observing networks, satellite observation, radar, lightning detection and other advanced technologies will be used to build detection and forecasting capabilities. It will close gaps in ocean and cryosphere data in hot spots such as coastal areas and high mountains and will accelerate the implementation of the WMO action plan for hydrology by establishing regional hydrological status systems.
- 143 An assessment of interoperability of different EWSs will be done to identify and address gaps in communication and data exchange. This will ensure seamless data integration and communication between the different components of the EWS in the country as well as with external systems.

Sub-Activity 2.2.3: Strengthen MHEWS forecasting capacity

- 144 The sub-activity will strengthen capabilities for data processing, forecasting and analysis. The network of Regional Specialised Meteorological Centres (RSMCs) and associated National Meteorological and Hydrological Centres will be supported to meet region specific needs for severe weather forecasting, flood and flash-flood forecasting, sand and dust storm forecasting, marine weather forecasts, high resolution weather predictions, now-casting, TCs and IbF. Additionally, WMO supportive frameworks (WIGOS, WIS and WMO Integrated Processing and Prediction System (WIPPS)) will be expanded to provide real-time operational support to monitor a wide range of hazards.

Activity 2.3: Warning dissemination, preparedness and response

- 145 This Activity, corresponding to the third and fourth pillar of the EAP, seeks to ensure that clear, understandable and actionable warning messages reach everyone at risk, in line with the Sendai Targets¹⁰¹ G, Indicator G-3¹⁰², and Indicator G-6¹⁰³. This will be achieved by developing and taking advantage of last mile connectivity and by establishing multi-

¹⁰⁰ <https://www.undrr.org/building-risk-knowledge/disaster-losses-and-damages-tracking-system-dldt>

¹⁰¹ UN, "Sendai Framework for DRR 2015-2030," 2015.

¹⁰² All people will be covered and can receive early warning information through local governments or through national dissemination mechanisms

¹⁰³ Preparedness to Respond translates early warnings into life-saving actions that reduce losses and harm from climate shocks

channel EWS using radio, TV, the Internet, mobile and satellite networks, sirens etc. Early warning messages will be developed in a multitude of accessible formats and languages, including for persons with disabilities in their full diversity, illiterate persons, etc. Information will be disseminated in a manner which ensures that risks and warnings are understood, are clear and actionable by their target audience. For persons with disabilities, risk and warning information will be (i) disseminated (i) in accessible formats, including oral, print, sign language, easy-to-read/plain language, text-to-speech, and (ii) via Organisations of Persons with Disabilities and to locations where persons with disabilities reside, to ensure that information about what is happening and how to keep safe must be easy for persons with disabilities to understand.¹⁰⁴ The activity will also ensure that participating countries have updated registries of authorities responsible for alerting communities and are issuing warnings that are compliant with the CAP¹⁰⁵ whilst being understandable in culturally appropriate formats and media and being cognizant of data protection and privacy concerns.

- 146 The activity will support a spectrum of actions, including integrating clear protocols for early warning and early action in disaster management/ preparedness plans. It will support actions based on early warnings before disasters hit, such as evacuation, strengthening of infrastructure, pre-positioning of essential stocks and supplies. This activity will strengthen the preparedness capacities, systems and procedures of local governments, responders and vulnerable communities through training and equipping them to respond based on integrated community and national EWS and communications. The activity will include development of a protocol for the prevention and reduction of risks for persons with disabilities, which will be validated by persons with disabilities and OPDs.

Sub-Activity 2.3.1: Strengthen early warning dissemination systems

- 147 The Sub-Activity will lead to expanded and strengthened communication network and services by supporting the implementation of multi-channel systems for communicating and disseminating early warnings and alerts to the last mile. Infrastructure and network services will be strategically strengthened by targeting the most efficient, trusted and preferred communication channels, with at least one channel as backup where feasible. Geo-located mobile early warning services based on cell-broadcast or location-based SMS systems will be implemented and strengthened by working with mobile network operators, ICT regulators in countries and the GSM Association (GSMA), the global association of mobile network operators. Best practices for data privacy will be put in place. Countries will receive technical assistance to set up and implement mobile EWS. This will be further strengthened by supporting governments to institute appropriate regulatory approaches and policies that mandate the use of geo-located alerts using satellite in areas where people are at risk, but mobile networks do not exist (e.g. in remote mountainous areas). Here, alternative networks, in particular, satellite-based and radio networks will be exploited as the primary dissemination channel. Funds will be identified to pay for these systems where governments do not have resources to establish a basic satellite/mobile EWS. Assessments will be carried out to understand how best to reach communities at risk and innovative and emerging technologies will be used to develop "client profiles" to scale up targeted dissemination of actionable early warnings. This activity proposes country piloting of the Disaster Connectivity Map and visualization tool to assess connectivity levels and disaster risks using AI, helping visualise local connectivity, identify risk areas, and enhance national assessments and communication strategies. Digital Connectivity Mapping (DCM) enables analysis of telecoms infrastructure vulnerability to extreme weather events and other hazards, enabling national authorities to take prevention action. Digital connectivity is essential for early warning systems, and DCM visually demonstrates weaknesses that will disable message disseminations in the event of shocks. This information is produced by ITU and will be shared through a very detailed/granular map of the entire country to show exactly where people are within/outside of digital networks. This map is shared with the telecommunication regulatory authority, for feedback and comments. This information on how and where people have connectivity is critical to understand how many people can be reached via digital networks (such as on mobile phones) but also to highlight areas where people at risk may be outside connectivity. This will allow

¹⁰⁴ Human Rights Council (2015), Thematic Study on the Rights of Persons with Disabilities under article 11 of the Convention on the Rights of Persons with Disabilities, on situations of risk and humanitarian emergencies. Report of the Office of the High Commissioner for Human Rights, A/HRC/31/30.

¹⁰⁵ WMO, "Guidelines for Implementation of Common Alerting Protocol (CAP)-Enabled Emergency Alerting" (Geneva, Switzerland, 2013).

governments and operators to take informed policy decisions (for example, to expand networks, or to set up alternative communication channels such as siren, radio etc to reach those at risk). For more information [see here](#).

- 148 At the community level, the activity will promote and support inclusive, people-centered co-development of gender responsive, disability inclusive actionable early warnings by targeted communities in accessible formats and local languages using and extending approaches such as the Participatory Integrated Climate Services for Agriculture (PICSA).¹⁰⁶ Existing community-based systems including Indigenous Knowledge Systems (IKS) and flood and drought warning systems set up through other agencies such as the IFRC will be leveraged and feedback mechanisms from end users, including women and vulnerable groups, will be established to facilitate improvements in the development and delivery of early warning services. Innovative systems that leverage mobile communications and Internet of Things (IoT) will be used to set up low-latency autonomous systems (described in Activity 2.3.3) with local communities. Communities will be supported in accessing and interpreting forecasts and weather data to better manage their risks.

Sub-Activity 2.3.2: Develop adaptation and preparedness protocols

- 149 Actions under this sub-activity will lead to comprehensive procedures and protocols for climate change adaptation and preparedness, with emphasis on supporting communities most at risk in each country. Its actions will build on the strengthened coordination mechanisms and governance (Activity 2.1) and better disaster knowledge and observations (Activity 2.2). Actions under this sub-activity will include: i) Communication protocols and identification of focal persons for each interest group, including a disability protocol incorporating accessible communication formats; ii) Protocols for graduated warning levels to have heightened awareness during periods of increased risk exposure; iii) Protocols for evacuation during disasters; and 4) Protocols for relief, extension and health services to deliver appropriate services to communities.

Sub-Activity 2.3.3: Strengthen disaster adaptation and response capacity

- 150 This activity will result in the development of disability inclusive disaster preparedness/contingency plans to ensure effective response, including provisions for early warning and early/ anticipatory action. Capacities for preparedness based on risk-informed and impact-based forecasts will be ensured at the local levels to enable local authorities, first responders and communities to act quickly and effectively based on the early warning alerts. Training of local governments and first responders including agencies such as National Red Cross or Red Crescent Societies will be done and their preparedness capacities, systems and procedures will be strengthened based on the integrated community and national EWS. A National Early Warning Service training programme will be developed and implemented to enhance capability, develop a sustainable knowledge base, and ensure the training programme is disability inclusive in line with the Inter-Agency Standing Committee Guidelines on Inclusion of Persons with Disabilities in Humanitarian Action.¹⁰⁷ Institutional and operational level capacities will be built by scaling up investments in preparedness, cash systems, supply chains and logistics, as well as enhancing accountability of local responders.
- 151 Communities will be engaged in developing holistic early warning and anticipatory action in local disasters management, DRR and climate change adaptation plans based on feasibility studies, climate information and local knowledge. These will include evacuation routes and plans for future construction and retrofitting of small infrastructure for emergencies. Specific attention will be given to the needs of women, persons with disabilities and vulnerable groups, specifically building awareness of the increased risk of gender-based violence during emergencies and need for accessibility of facilities to persons with disabilities, including physical, visual and hearing impairments. As part of CIEWS, communities will be trained on O&M of early warning devices such as water level recorders, rain-gauges and anemometers. Support and oversight may also be provided for small scale infrastructure for emergencies such as installing water storage tanks, minor improvements to existing access paths, and small repairs or improvements to buildings (such as waterproofing, installing toilets). Existing preparedness systems and operational protocols should reflect the roles, needs and rights of

¹⁰⁶ Center for Independent Evaluations (CIE), "Participatory Integrated Climate Services for Agriculture (PICSA) Under Saving Lives and Protecting Agriculture-Based Livelihoods in Malawi: Scaling up the Use of Modernized Climate Information and EWSs (M-CLIMES) Project," Final Evaluation Report, March 2021.

¹⁰⁷ Inter-Agency Standing Committee (2019), Guidelines on Inclusion of Persons with Disabilities in Humanitarian Action.

underrepresented and disadvantaged groups, including women, children, elderly, persons with disabilities, people on the move, minorities, etc. Inclusivity will be central to the approaches adopted in the project across all activities.

Country based project description and strategy summary

- 152 A summary of climate hazards across critical sectors and proposed interventions for the participating seven countries is provided in **Error! Reference source not found.** in Appendix 2, Pg. 106 in this document. A description of these interventions is provided below.

Antigua and Barbuda

- 153 The project approach for Antigua and Barbuda (A&B) was developed through an extensive multi-stakeholder engagement process, which culminated in an implementation roadmap for reinforcing a people-centric MHEWS in the country. The roadmap identified several key principles which underpin the approach, focused on: i) strengthening the national governance framework; ii) data-sharing collaborations and consolidation; iii) bridging the gap between technical information and community comprehension; and iv) enhancing capacity for emergency management. Moreover, several key sectors/areas have been prioritized to fill gaps in the existing MHEWS, particularly focused on the marine environment – including ocean conditions and environmental impacts – and the introduction of an EWS for Barbuda that is not wholly reliant on already stretched services in Antigua.
- 154 From a governance perspective, interventions in A&B will focus on modernizing the disaster management policy and legislative framework, including through integration of disability issues, and integrating DRR considerations into sectoral policies – including for the built environment, land-use planning and urban development. These policy actions will be aligned with existing plans on a national and regional level and reinforced by a national strategic plan for national emergency management agencies, together providing a solid foundation for the effective functioning and sustainability of MHEWS and ensuring clarity and alignment among all involved parties. A key component of the governance support is the establishment of a Disaster Preparedness and Response Fund. This fund will provide a sustainable mechanism to channel resources toward critical disaster preparedness and response activities. The cross-cutting governance interventions will be underpinned by improved disaster risk knowledge in the country (Pillar 1), focused on creating a seamless information flow between relevant government agencies, supporting organizations and the public to improve coordination and enable more holistic developmental planning, with risk data and information readily available to all stakeholders. This will include increasing the availability of sector-specific risk data, disability inclusive risk data, knowledge and best practices for critical sectors such as tourism, energy, agriculture, education, telecommunications and health, as well as added focus the marine environment and social risks. At the heart of this effort is the development of an online risk data platform, designed to centralise and manage critical risk information across all relevant sectors, as well as undertaking comprehensive and disability inclusive hazard and risk assessments, addressing significant knowledge gaps, particularly in Barbuda.
- 155 Under Pillar 2, a core focus of interventions will be on increasing the operational capacity of the Antigua and Barbuda Meteorological Service (ABMS) – including supporting actions to direct additional budget to the services, hiring and training of technical staff, and improving the hurricane resilience of the ABMS offices through structural enhancements and the installation of backup power and communication systems. In addition, the project will improve existing hydrological observation infrastructure and monitoring systems to enhance flash flood forecast system for urban and rural areas, as well as introducing additional marine monitoring infrastructure for swells and potential environmental changes. The enhancements to these networks will include provisions for real time monitoring and analysis of hydrometeorological data, ensuring interoperability with risk data in place at ABMS through the introduction of a platform to consolidate geospatial hydrometeorological and disaster risk information.
- 156 Pillar 3 will focus on simpler warning messages that are easily understandable by the public, including persons with disabilities, ensuring that warning messages are clear, concise, and actionable in accessible formats during emergencies. This will be supported by awareness actions to build trust among communities in the warnings being presented and their associated anticipatory actions. A national emergency telecommunication plan will be developed alongside a common alert protocol that integrates multiple dissemination platforms, including cell broadcasts, radio and television broadcasting, sirens, among other traditional forms of mass warning, as well as accessible formats to reach persons with a variety of disabilities.

157 Finally, interventions under Pillar 4 the project will support the scaling of community-based EWS (CEWS) to empower local communities, including persons with disabilities, to take an active role in monitoring, disseminating, and responding to potential threats — developing specialized risk-informed response/contingency plans for extreme events based on the needs of vulnerable communities, including persons with disabilities. The plan also emphasizes enhancing community disaster response teams, providing training and equipment for first responders, as well as providing resources for rapid evacuation and shelter services, and strengthening scenario-based planning with IbF to improve response capabilities. and establishing dedicated financing mechanisms and structured early action measures for pre-impact actions based on the needs of vulnerable groups, including persons with disabilities. The anticipatory action will be closely linked with the governance support interventions – ensuring that relevant legislation is updated to accommodate the emerging functions of key response institutions.

Chad

158 The intervention in Chad will ensure that EW/EA are impact based by building capability to carry out disability-inclusive post-disaster needs assessments and loss and damages reporting, promoting longer-term adaptation planning and knowledge management. It will ensure EWS are functional, by developing a sustainable finance plan with expertise in public financial management in the context of Fragile and Conflict Affected States (FCAS). The project will promote the development of a pipeline of supporting projects, ensuring data collected is transmitted in a timely way, ensuring data governance, disability disaggregation, and mapping of location and collection of information on the support needs of persons with disabilities through consultation, and developing rights and protocols for issuing and disseminating warnings including in accessible formats for persons with disabilities. It will move from a siloed approach centred on food security, towards a multi-hazard system, considering wider adaptation challenges. It will be people centred, based on the right to information on risks, advice on preparedness measures, and warnings/advisories that are locally relevant and respond to the needs and expectations of people including women, the elderly, people with disabilities, refugees and minorities.

159 The project will develop an offer of audio recordings developed through mobile phones, conduct regular surveys and qualitative community research, and develop national capability for cell broadcasts. It will consider preferences for communication, including accessible formats for persons with disabilities, and monitor trust in existing and new communication channels being developed. It will provide end-to-end integration of early warning and early action, ensuring anticipatory action protocols are tested and adapted, providing capability to carry out simulations, lessons learned and post disaster needs assessments, supporting crisis response capacity at devolved levels, while bringing together contingency planning information with a register of responsibilities for disaster management, and defined alerting rights. The project will respond appropriately to the needs of people concerning transparency, accountability and the management of grievances, and be sensitive to conflict dynamics. Significant stakeholder engagement, and preparatory work on institutional and regulatory frameworks, will ensure that coordination and governance approaches are set in motion to support these activities and outcomes, and address critical institutional, financial and regulatory barriers.

Fiji

160 The proposed interventions in Fiji primarily focus on strengthening and enhancing the existing functions of partner organizations to address well-known priorities for improving understanding and communication and governance and coordination mechanism amongst the national stakeholders relevant to EWS in Fiji. The project aims to unify risk assessment methods in Fiji, focusing on historical data, climate models and local knowledge.

161 The project will expand or upgrade weather/climate/hydrometric observation network for EWS: Investments will be made to improve monitoring and warning for severe weather, and localized and flash flooding, and droughts. Data collected will be used to create models and forecasts of, enabling timely warnings and the development of disability inclusive evacuation plans. The project will additionally Strengthen MHEWS forecasting capacity by enhance predication accuracy through integration of AI and machine learning techniques, developing a centralized database/repository and incorporating use of data science, e.g. AI to assist in forecasting and early warning. A slow drought monitoring IbF system for food security will be established along with Drought EWS.

- 162 For warnings dissemination and preparedness, the project will support the implementation of the National Emergency Telecom Plan which includes automation of a geo-targeted, location-based messaging through the CAP and cell broadcasting. This will require a review and revision of the current licensing of the national mobile providers to ensure they upgrade their infrastructure to enable cell broadcasting. The project will also support expansion of Starlink network and maritime radio to achieve 100 % EWS coverage in Fiji. This plan will ensure inclusion of formats accessible for persons with disabilities.
- 163 The project will develop and implement community-focused communication strategies based on findings from risk perception studies, and incorporating accessible communication formats for persons with disabilities. It will improve translation and interpretation of early warning messages by incorporating message translation, caption generation, and text-to-speech capabilities in EWS messaging to enhance inclusivity for persons with disabilities and the elderly. Train-the-Trainer workshops will be run for key stakeholders, such as ground actors, government officers, Red Cross officers, and sugarcane extension officers and Technology Transfer Officer and community-level awareness campaigns will be run on how to respond to early warning information, including for persons with disabilities and OPDs. The project will evaluate the reach and effectiveness of disaster preparedness campaigns, and integrate the lessons learned into revised communications and training. Local contingency plans will be developed which are behaviourally informed, inclusive and community driven and a community-based DRM (CBDRM) Training Toolkit will be developed used to and run Train of Trainer workshops on CBDRM and early warning messages with inclusion of early actions.

Cambodia

- 164 The EW4All intervention for Cambodia provides a comprehensive plan for governance and coordination, disaster risk knowledge and coordination, and warning dissemination and preparedness in Cambodia. The plan includes developing a disability inclusive data-sharing protocol, upgrading the Cambodia Disaster Damage & Loss Information System (CamDi), and implementing disability inclusive EWS at the national and sub-national levels. It also emphasizes the need for gender, disability and inclusion considerations, as well as the establishment of river gauges and drought gauges in target provinces to address weather information needs. Furthermore, the plan focuses on raising awareness of emergency alert systems and enhancing warning dissemination through various channels, including mobile networks, online platforms, and accessible formats for persons with disabilities.
- 165 The governance and coordination aspect of the plan involves establishing a collaborative early warnings secretariat, addressing gaps in the legal and financial context, and implementing a cross-pillar approach to gender, disability and inclusion. The disaster risk knowledge and coordination section emphasize upgrading the CamDi, preparing preparedness and contingency plans in target provinces (incorporating specific needs of persons with disabilities and preparing persons with disabilities to respond effectively to warnings), and installing river gauges and drought gauges. Additionally, it highlights the development of a comprehensive human resource plan to meet early warning requirements.
- 166 The warning dissemination and preparedness component focuses on raising awareness of emergency alert systems, coordinating mobile network providers, and upgrading online presence and dissemination channels. It also entails equipping local authorities with early action capacity, implementing community-based disability inclusive EWS, and providing shock-responsive social protection for preparedness and anticipatory action in target provinces. The plan places particular emphasis on the needs of vulnerable communities, including women, persons with disabilities, children, and youth.

Ecuador

- 167 The MHEWS project proposed in Ecuador is designed to significantly enhance the country's ability to anticipate, prepare for, and respond to various climate-related hazards, such as floods, landslides, droughts and extreme temperatures. The impact of these hazards is increasing in Ecuador in the last years due to climate change. The project aims to establish a robust and integrated multi-hazard EWS that leverages advanced forecasting techniques, standardised climate data management, inclusive communication strategies, community emergency plans and anticipatory actions. The primary activities include the development of a functional legislative framework, the development of a National Framework for Climate Services (NFCS), the establishment of a comprehensive climate data centre, the implementation of national risk assessments for the relevant hazards, the improvement of the hazard forecasting, the integration of Indigenous Knowledge Systems (IKS) into early warning practices, the integration of rights of persons with disabilities into early

warning laws, policies, systems, data collection, communication strategies, risk management strategies, and protocols, the improvement of the communication strategies and the creation of gender-sensitive disaster preparedness protocols.

- 168 These activities are strategically aimed at addressing the pressing climate challenges in Ecuador. The country is highly vulnerable to natural hazards due to its geographical location in the Andean region, which is prone to extreme meteorological events, droughts, floods, and landslides. The project's focus on strengthening governance structures and enhancing institutional capacities ensures that EWS are effective and sustainable. By improving disaster risk knowledge, enhancing monitoring and forecasting capabilities, and ensuring timely and inclusive communication of warnings, the project will mitigate the adverse impacts of climate-related hazards on vulnerable communities.
- 169 Key activities of the project, such as the establishment of a National Climate Finance Framework and the enhancement of data management systems, address specific gaps identified in Ecuador's national climate strategy. These include the need for more robust and sustainable financing mechanisms for climate adaptation, as well as improved coordination among different governmental and non-governmental stakeholders involved in DRM. Furthermore, the project's emphasis on gender-sensitive and disability inclusive approaches and the inclusion of Indigenous Knowledge Systems (IKS) aligns with Ecuador's commitment to social inclusion and equitable climate adaptation. By ensuring that EWSs are accessible and responsive to the needs of women, persons with disabilities, Indigenous peoples, and other vulnerable groups, the project contributes to a more resilient and inclusive society.
- 170 In summary, the MHEWS project addresses the key challenges identified in Ecuador's national climate strategy, including gaps in institutional capacity, financing, and community preparedness. Its activities are specifically designed to strengthen the country's preparedness and resilience in the face of increasing climate variability and extreme weather events, ensuring that Ecuador is better equipped to protect its people and assets from future climate-related risks.

Ethiopia

- 171 The project interventions address critical barriers that prevent the effective generation, dissemination and use of early warnings for climate change adaptation, preparedness and response to warnings, through three broad outputs tailored to specific country needs, which are aligned with the tree activities under Output 2 of the Global FP. The project will support the collection and dissemination of multi-hazard, multi-sector, disability inclusive risk data through the development of a central National Risk Data Hub managed by EDRMC and accessible to all. This will enable the development and sharing of impact-based disaster risk profiles focusing on highly vulnerable areas especially looking at floods risk and drought. It will support the upgrade of hydro-meteorological observing stations and improve coverage in under-covered areas. An analysis on most relevant communications channels to address early warning to vulnerable populations, including accessible formats for persons with disabilities, will be undertaken and used to develop a disaster risk, early warning linked to mitigation and Anticipatory Action messaging. Early warnings systems will be operationalized at community level in priority high risk locations, linking with Anticipatory Action. An emergency call number and service will be developed.
- 172 Multi-sector line ministries will be engaged to agree on sector-specific early warning and Emergency Thresholds, and multi-hazard Anticipatory Action plans will be integrated into Preparedness Planning in highly vulnerable areas, linking with a financial support mechanism and a disability inclusive social protection scheme.
- 173 A National Risk Data Hub coordinated by EDRMC with contributions from all sector Agencies/Ministries will be formed to collect and develop multi-hazard disability inclusive data and risk analyses and undertake capacity development for expert human resources including IT and connectivity capacities to enable data sharing amongst all actors. Woreda level risk profiles in priority areas will be developed with compilation and analysis of hazard, exposure and vulnerability data. The project will improve data sharing by establishing requisite protocols and procedures and will support the integration of geo-hazards in EWS. A gender based, disability inclusive EWS focusing on elderly and children will be developed as part of this exercise which will also lead to the establishment of community-based, disability inclusive EWS in priority areas.
- 174 A survey will be undertaken to identify most relevant media communications channels according to types of audience, including vulnerability, disability inclusive and gender-based issues. This will be used to build capacities of communications specialists to address media, and to journalists to communicate about disaster risk.

Somalia

- 175 The interventions proposed for Somalia under the EW4All project fall predominately within the Country's EW4All Roadmap identified activities and the resourcing requirements for these. In collaboration with the UNDP Somalia country office, the Early Warning for All (EW4All) 2024-2027 Roadmap will be operationalised under the leadership of the Somali Disaster Management Agency (SODMA) and the National Multi-Hazard Early Warning Center (NMHEWC). Three outputs under the four pillars, corresponding to the three activities under Output 2 of the Global FP, form the interventions proposed for Somalia, and contribute directly to the realisation of Somalia's Early Warning for All (EW4All) 2024-2027 Roadmap through the identified activities and the resourcing requirements for these. The activities under the three outputs include the following:
- 176 To improve disaster risk knowledge and capacities the project will undertake the development of a comprehensive risk assessment framework and database. Disaster risk knowledge materials and communications will be developed for national and local levels, and capacity building and coordination provided for the application of improved disaster risk knowledge at national and local levels. The project will also support integrated actions for detection, monitoring, analysis and forecasts of weather and climate; harmonising the collection of early warning data, its analysis and exchange modalities among federal line ministries, sectorial offices, and regional bodies. The proposed project will support improved data automation, modelling and forecasting of early warning coherency among sectorial offices and the improved interoperability and early warning information exchange between sectors. Disaster risk knowledge activities will ensure mapping the location of persons with disabilities and gathering information on their support needs, consulting with persons with disabilities and their representative organisations.
- 177 The project will support nationwide warning dissemination and communication access by supporting the development and implementation of early warning communication strategies and standard operating procedures (SOPs) for governance, functions, roles and responsibilities for early warning information, communication and dissemination. These will include early warning communication strategies and SOPs being established for early warning information, communication and dissemination governance, functions, roles and responsibilities, impact-based forecast and warning services (IBFWS), and CAP. It will establish comprehensive infrastructure networks and early warning dissemination services and equipment which cover simulation exercises, CAP adoption, telecommunication service provider agreements, automated system setups and protocols for AA and communication strategies and feedback mechanisms, amongst others. The project will invest in accessible nationwide warning dissemination and communication. Communication strategies and SOPs will also incorporate risk and warning information in accessible formats for persons with disabilities.
- 178 The proposed project will establish comprehensive functional infrastructure networks and early warning dissemination services and equipment and ensure inclusive, people-centred EW4ALL at the local level. Finally, the project will enhance preparedness for timely responses and early actions by supporting improved response and recovery disaster preparedness measures established for early actions and the development of supporting risk management policy, legislation and strategies. This will include ensuring preparedness and response plans (i) address the needs of persons with disabilities, including through their consultation and meaningful participation, (ii) are tested, and (iii) prepared persons with disabilities to respond effectively to warnings.
- 179 The Responsible Parties for the different activities in each of the participating countries are provided in the table 3 below. These will be finalized and formalized through signed Responsible Party Agreements during the inception phase of the country projects.

Table 3: List of responsible parties for the seven participating countries disaggregated by activity:

Ecuador	Cambodia	Antigua & Barbuda	Fiji	Ethiopia	Chad	Somalia
Activity 1.1, UNDP, National Secretariat for Risk Management (SNGR),	Activity 1.1 – UNDP, WFP, Ministry of Environment (MoE)	Activity 1: Ministry of Finance, National Office of Disaster Services (NODS),	Activity 1.1: National Disaster Risk Management Office (NDRMO), UNDP	Activity 1.1: Ethiopian Disaster Risk Management Commission (EDRMC)	Activity 1.1: UNDP, UNDRR ;	Activity 1.1: UNDP, ITU, Somali Disaster Management Agency (SoDMA),

Meteorological and Hydrological Institute (INAMHI)/ Water and Ecological Transition of Ecuador (MAATE) page 20		UNDP, UNDRR, WMO				National Communications Authority (NCA)
Activity 1.2, UNDP, SNGR page 20	Activity 1.2 – UNDP, IFRC	Activity 1.2: NODS, UNDP	Activity 1.2: NDRMO, UNDP	Activity 1.2: EDRMC	Activity 1.2: UNDP, WMO	Activity 1.2 UNDP, WMO, MoEWR
Activity 1.3, UNDP, SNGR page 20	Activity 1.3 - UNDP	Activity 1.3: NODS, UNDP	Activity 1.3: NDRMO, IFRC, UNDP	Activity 2.1 EDRMC	Activity 1.3: UNDP, World Food Programme (WFP);	Activity 1.3 UNDP, UNDRR, (SoDMA)
Activity 2.1, UNDP, UNDRR, SNGR, INAMHI/MAAT E page 20	Activity 2.1 UNDP, UNDRR, IFRC, National Committee for Disaster Management (NCDM)	Activity 2.1: Ministry of Legal Affairs, NODS, UNDRR	Activity 2.1: NDRMO, UNDRR, UNDP	Activity 2.2 EDRMC	Activity 2.1: UNDP, WFP, UNDRR, IFRC	Activity 2.1 UNDP, UNDRR, SoDMA
Activity 2.2, UNDP, WMO, INAMHI/MAAT E page 20	Activity 2.2 – UNDP, People in Need (PIN), Action Aid Cambodia (AAC)	Activity 2.2. : WMO	Activity 2.2: Fiji Meteorological Service (FMS), UNDP	Activity 2.3 Ethiopian Meteorological Institute (EMI)	Activity 2.1: UNDP, WMO	Activity 2.2 UNDP, WMO, Ministry of Energy and Water Resources (MoEWR)
Activity 2.3, UNDP, WMO, INAMHI/MAAT E page 20	Activity 2.3 – UNDP, Ministry of Water Resources and Meteorology (MOWRAM), Regional Integrated Multi-Hazard EWS for Africa and Asia (RIMES)	Activity 2.3: Antigua and Barbuda Meteorological Services (ABMS), UNDP, WMO	Activity 2.3: FMS, WMO, UNDP	Activity 2.4 EMI	Activity 2.2: UNDP, WMO	Activity 2.3 UNDP, WMO, MoEWR
Activity 3.1, UNDP, ITU, page 20	Activity 3.1 – UNDP, IFRC, ITU, PIN	Activity 3.1: ITU, NODS, Telecoms Division, UNDP	Activity 3.1: NDRMO, UNDP	Activity 2.5: EMI	Activity 2.3: UNDP, WMO, WFP	Activity 3.1 UNDP, ITU, SoDMA, NCA
Activity 3.2, UNDP, IFRC, SNGR page 20	Activity 3.2 – IFRC, NCDM	Activity 3.2: IFRC, ITU, NODS	Activity 3.2: NDRMO, IFRC, UNDP	Activity 3.1: Ethiopian News Agency (ENA)	Activity 3.1: UNDP, WMO, ITU, IFRC, WFP	Activity 3.2 UNDP, ITU, SoDMA, NCA

Activity 3.3, UNDP, IFRC, SNGR page 20	Activity 3.3 UNDP, WFP, IFRC, PIN, AAC	Activity 3.3: IFRC, UNDP	Activity 3.3: NDRMO, IFRC, UNDP	Activity 3.2: ENA	Activity 3.2: UNDP, IFRC, WFP	Activity 3.3 UNDP, ITU, SoDMA, NCA
				Activity 3.3: ENA	- Activity 3.3: UNDP, WFP, IFRC	
				Activity 3.4: ENA, Ethiopian Communicatio ns Authority (ECA), ITU		
				Activity 3.5: Ethiopian Red Cross Society (ERCS), IFRC		
				Activity 3.6: ERCS, IFRC		
				Activity 3.7: ERCS, IFRC		
				Activity 3.8: ERCS, IFRC		
				Activity 3.9: ERCS, IFRC		

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B.4. Implementation arrangements

- 181 UNDP will be the Executing Entity of the EW4All-MCP and will be responsible for the overall project execution. It will collaborate with the EW4All pillar leads in the provision of technical assistance and support.
- 182 The UNDP will enter into a Funded Activity Agreement (FAA) with the GCF, on approval of the project. This will be in accordance with the Accreditation Master Agreement (AMA). UNDP's overall role as an Accredited Entity is to provide oversight and quality assurance through its Headquarter and Country Office units. This role includes: i) project preparation oversight; ii) project implementation oversight and supervision, including financial management; and iii) project completion and evaluation oversight. It also includes oversight roles in relation to reporting and knowledge-management. The 'project assurance' function of UNDP is to support the Project Board by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. Project assurance has to be independent of the Project Manager; therefore, the Project Board cannot delegate any of its assurance responsibilities to the Project Manager. A UNDP Programme Officer, or M&E Officer, typically holds the Project Assurance role on behalf of UNDP. The 'senior supplier' role of UNDP is to represent the interests of the parties which provide funding and/or technical expertise to the project (designing, developing, facilitating, procuring, implementing). The senior supplier's primary function within the Board is to provide guidance regarding the technical feasibility of the project. The senior supplier role must have the authority to commit or acquire supplier resources required. If necessary, more than one person may be required for this role. Typically, the implementing partner, UNDP and/or donor(s) would be represented under this role.

Global implementation arrangements

- 183 This multi-country project will be implemented by UNDP acting as an Accredited Entity and Executing Entity in terms of GCF standards. In line with UNDP policies and procedures for direct implementation, the Accredited Entity shall authorize seven participating UNDP Country Offices to implement the project in each of the seven countries in accordance with the AMA (the "Project Document"). For the avoidance of doubt, as the Accredited Entity will act as the Executing Entity for the implementation of the Funded Activity, the Project Document shall not be considered a Subsidiary Agreement. This EW4All global project will be implemented in multiple countries and therefore will use the multi-country project type

modality as per UNDP Policies and Procedures. It is appropriate when multiple countries are required to efficiently, effectively and accountable to joint results on equal terms, enhancing opportunities for achieving substantive and operational results. This multi-country project is considered a single project for the purposes of project management and has a unified project document that includes a joint results and resources framework describing interventions in each participating country. The funds will be transferred to countries following the Internal Control Framework/ICF and signed Financial Authorization for each country office. The global implementation arrangement will establish a Project Board as described below. The Executive will be a senior official from the UNDP. UNDP will also play the role of Senior Supplier.

- 184 Pillar Focal Points (WMO, UNDRR, ITU and IFRC) will support the Project Board through a Coordination Group and will attend the Board meetings. A formal Responsible Party agreement will be signed with each of the Pillar leads as per UNDP policy. Fund flow will be from GCF to the UNDP and from the UNDP to the respective Pillar Leads as per the activity budgets. Implementation of the global proposal will be undertaken by UNDP at global and country levels in partnership with the four Pillar Leads (WMO, UNDRR, ITU and IFRC). Activities at the global level will be undertaken by pillar leads (WMO, UNDRR, IFRC and ITU) and UNDP who are responsible parties for the global-level proposal. Relevant UNDP Programme and Operational Policies and Procedures/POPP: Financial Management and Implementation Modalities will be followed, whereas WMO, UNDRR and ITU would follow UN agency procedures. (i) At the Global and country levels, UNDP will partner with WMO, ITU, UNDRR and IFRC. UN Agency to Agency Transfer Agreement will be signed with UNDRR, WMO, and ITU (PPM_UN Agency to UN Agency Contributions Template Agreement.docx). Between IFRC and UNDP – an MOU will be signed for the project duration, following UNDP Programme and Operations Policy and Procedures: Select Responsible Parties and Grantees | United Nations Development Programme. (ii) At the country levels, UNDP will engage with Government through Standard LOA between UNDP and a Government Ministry/Institution/IGO on the implementation of a project when UNDP serves as an implementing partner | United Nations Development Programme; (iii) Partnerships with Responsible Parties and Grantees such as national NGOs will be formalized through Responsible Party Agreements. (iv) Beneficiaries will be selected through established criteria as per country-specific contexts and established criteria. All four pillar leads (UNDRR, WMO, ITU, IFRC) are headquartered in Geneva, with respective regional offices covering each of the 7 countries to deliver technical assistance, training, and capacity building based on country priorities.
- 185 As per the UNDP Programme Operation and Policy Procedure (POPP) guidelines, UNDP will be implementing the planned activities of the programme document, by following a Direct Implementation (DIM) modality. In line with the programme document and availability of funds, UNDP (HQ and the participating Country Offices) will develop and approve a Multi-Year Work Plan by following a UNDP standard work plan format - highlighting the description of activities, linking with the relevant outputs (with their respective baselines, targets and indicators), Responsible Party/UNDP, period/quarter/year of implementation, and the allocated/sources of funds, against the approved activities. Once the Multi-Year Work Plan is approved by the Project Board and the funds are available, UNDP HQ and its participating COs will follow a common transparent digital competitive procurement process at their respective levels, for the selection/hiring of the national and/or international contractors (public/private/NGOs/academia) against the planned activities and as part of the inception phase of the project.
- 186 The procurement process for the selection of contractors starts with the development of a standard Terms of Reference (TOR) for each activity (in which relevant eligibility criteria or for selection of beneficiaries and/or interventions, amongst others, would be highlighted), advertising them in the UNDP global procurement website, invitation of/receiving applications for the implementation of activities from the interested parties, reviewing and approving the recommended applications of the contractors by the UNDP procurement committees (to be formed), awarding the activities by signing of the legal contracts/agreements between UNDP and selected contractors for the implementation of activities, paying the available funds of the activity to the contractor in installment basis (per the agreement), based on the satisfactory implementation of the assigned tasks. In case where other UN agency implements activities (ITU, WMO, UNDRR), UNDP will sign a standard agreement with the UN agency, without any competitive procurement process, and the UN agencies will follow UN/UNDP procurement processes. The list of responsible parties for implementing the project activities for the global proposal are provided below, which is reflected in figure 2.

- Output 1

- Activity 1.1:
 - Sub-Activity 1.1.1: UNDRR, WMO and ITU
 - Sub-Activity 1.1.2: UNDRR, WMO, ITU and IFRC
- Activity 1.2:
 - Sub-Activity 1.2.1: WMO
 - Sub-Activity 1.2.2: UNDRR, WMO, ITU and IFRC
 - Sub-Activity 1.2.3: UNDRR
- Activity 1.3:
 - Sub-Activity 1.3.1: UNDRR
 - Sub-Activity 1.3.2: WMO, IFRC
 - Sub-Activity 1.3.3: ITU
- Output 2, all activities and sub-activities: UNDP Country Offices for Antigua and Barbuda, Cambodia, Chad, Ecuador, Ethiopia, Fiji and Somalia.

187 The Project Board is responsible for making, by consensus, management decisions when guidance is required by the Project Manager. Project Board decisions will be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition. In case a consensus cannot be reached within the Board, final decision shall rest with the UNDP Executive Coordinator. As the main governing and decision-making body of the project, the Project Board will be governed by comprehensive guidelines and the latest policies regarding the establishment and operation of Project Board following UNDP's Programme and Operations Policies and Procedures (POPP). This will be approved by all members during the first project board meeting. The terms of reference on the composition of the project board, roles and responsibilities in detail, including on quorum requirements will be outlined in the terms of references developed and shared with all members prior the first board meeting.

188 The Project Board will be comprised as following:

- a. UNDP who will hold the role of Executive and will be represented by the UNDP's Director for the Bureau of Policy Programming Support (BPPS) or his/her direct designated representative.
- b. A pillar lead representative that will hold the role of Senior Supplier on a rotation basis. Pillar lead rotation should ensure that each pillar lead holds this position at least once during the project's lifetime.

189 Project Board meetings will be held virtually, twice times a year. In addition, the members of the committee will look to meet physically every other year (three times in the five years) for a knowledge sharing meeting. Three such meetings will be held during the implementation of the seven country projects. These meetings will be attended by the EW4All Pillar Leads, GCF, UNDP and two representatives from each of the country projects, including at least one from the relevant national governments.

190 A global PMU will be constituted within UNDP for the overall project coordination, oversight and monitoring. The PMU will also provide technical support to help the country PMUs in accessing regional and international resources through networking. Perhaps most importantly, the PMU will ensure the EW4All pillar leads are kept abreast of the project progress for their inputs and support. The tentative structure of the global PMU is provided in Figure 2. It comprises of four full time staff – a Project Manager, a Finance Analyst, a Programme and Operations Associate and a Procurement Analyst. The administrative team will support the acquisition and procurement of specialised equipment during the project for specific country's needs, in an effective manner that also responds to national capacities. It will also look into developing Long-Term Purchasing Agreements that would ensure the efficient use of project resources. The PMU will be provided part-time technical support through a team of advisors who specialize in climate change adaptation, climate information and EWSs, a Gender, Disability, Social and Environmental Safeguards expert, a Communications Expert and other subject experts brought in on an as needed basis.

The Project Manager will run the project on a day-to-day basis on behalf of the Project Board within the constraints laid down by the Project Board. The Project Manager function will end when the final project terminal evaluation report and other documentation required by the GCF and UNDP have been completed and submitted to UNDP. The Project Manager is responsible for day-to-day management and decision-making for the project. The Project Manager's prime responsibility is to ensure that the project produces the results specified in the project document, to the required standard of quality and within the specified constraints of time and cost. The project Logic Framework (indicators, means of verification, assumptions) will be reviewed and the quarterly and annual plans will be refined based on the progress made in each country as well as at the regional and global level.



Figure 2: Global implementation arrangements

Country level implementation arrangements

- 191 The 7 country projects will be implemented as part of the global multi-country project following the established Direct Implementation Modality (DIM) where UNDP relevant country office acts as Executing Entity. Implementation arrangements will be vetted and agreed by the government via the relevant Nationally Designated Authority to the GCF (NDA) and other Government agencies and relevant ministries. UNDP has a country presence through established fully operational offices in each of the 7 project countries. At the country level, UNDP will sign Standard LOA between UNDP and a Government Ministry/Institution/IGO implementation of specified project activities defined by the project. UNDP will follow standards and procedures for project implementation as reflected within the SBAA signed between UNDP and each national government. as described within each existing signed country SBAA.¹⁰⁸ In the case where specific government agencies will implement certain project activities, these Agencies will act as Responsible Parties for these specific activities for the duration of the project. In this case the Responsible Party Agreement will be signed between UNDP and the Responsible Party. Following UNDP POPP guidelines: <https://popp.undp.org/procedure/responsible-parties>.

108 <https://www.uncdf.org/sbaa>

- 192 The Project Board at the national level will be comprised of UNDP as Executive, The Senior Beneficiaries are the most vulnerable communities and will be represented by the NDMA or Disaster Risk Agency as appropriate, designated by the NDA. . The Project Board will be responsible for making, by consensus, management decisions when guidance is required by the National Project Manager. Project Board decisions will be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition. In case a consensus cannot be reached within the Board, final decision shall rest with the UNDP's Resident Representative. The Project Board will meet at least once a year.
- 193 The project will be developed by UNDP under DIM modality with the support of a National Project Management Unit (PMU) that will be established in each of the countries. The PMU may rely on the support of Responsible Parties (RPs) in the implementation of specific actions and/or activities for the delivery of project objectives. RPs will be accountable to the respective UNDP country office for the implementation of specified activities and outputs of the project to achieve intended results, i outcomes, and the effective use of UNDP resources. Responsible Parties (RPs) will be identified by the NDA and UNDP based on their comparative advantages and core competence and will support the respective UNDP country office in successfully implementing the project activities.. The RPs will report to the UNDP-country office as per the terms of RP agreement that they will enter into with the UNDP country office.
- 194 The structure and staff of the National PMU will be decided by each country. The National Project Manager or National Project Coordinator will run the project on a day-to-day basis within the constraints laid down by the National Project Board. The Project Manager function will end when the final project terminal evaluation report and other documentation required by the GCF and UNDP has been completed and submitted to UNDP. The Project Manager is responsible for day-to-day management and decision-making for the project. The Project Manager's prime responsibility is to ensure that the project produces the results specified in the project document, to the required standard of quality and within the specified constraints of time and cost.
- 195 Project Support comprises stakeholders including relevant government ministries and departments at national level and local government with local offices in the districts (NMHS and NDMA and ICT ministries/regulators), the community members from targeted districts, the civil society and other local and international NGOs. Pillar Focal Points will provide technical advice to the Board through relevant subject experts, reflecting their role in each country.
- 196 Sub-national coordination of project activities will be done through relevant district agencies, and where needed, through local units of international agencies such as IFRC societies. Sub-national planning, particularly at the village level will be informed by village level institutions for social and civil protection. Community members will play a critical role in the implementation and monitoring of project activities. Extensive stakeholder engagements have been undertaken at the community level in all countries in the formulation of their respective Stakeholder Engagement Plans. These plans inform the roles, functions and responsibilities by which communities can be fully engaged in the project.
- 197 The project Logic Framework (indicators, means of verification, assumptions) will be reviewed and the quarterly and annual plans will be updated annually, based on stakeholder engagements. The stakeholders will also be engaged during the mid-term and final evaluations to assess the progress of the project and enable adaptive project management in response to the needs and priorities of the communities targeted. Community participation will be tracked through tools developed by the project management, leveraging, where possible the existing decentralised government structures to implement the project and the role and views of the community members. Community representation in the Board will be through the decentralised structure of the Implementation Partner in each country.
- Responsible Parties will be selected by UNDP based on their technical and institutional capacities, comparative advantage, cost effective and financial management to implement specific project activities and achieve intended results, compared to other potential organizations. UNDP standard rules and regulations will be followed at all times, as per POPP (<https://popp.undp.org/procedure/responsible-parties>), including the undertaking of the Harmonised Approach to Cash Transfers (HACT) assessments. Exhaustive stakeholder engagements and feasibility studies which included analysis of vulnerabilities and key priorities of stakeholders have guided the selection of sites and beneficiaries for the different project activities. The role of the Responsible Parties in each participating country will be to implement the specified project activities, manage resources, monitor and report on the results as outlined in the Responsible Party Agreement.

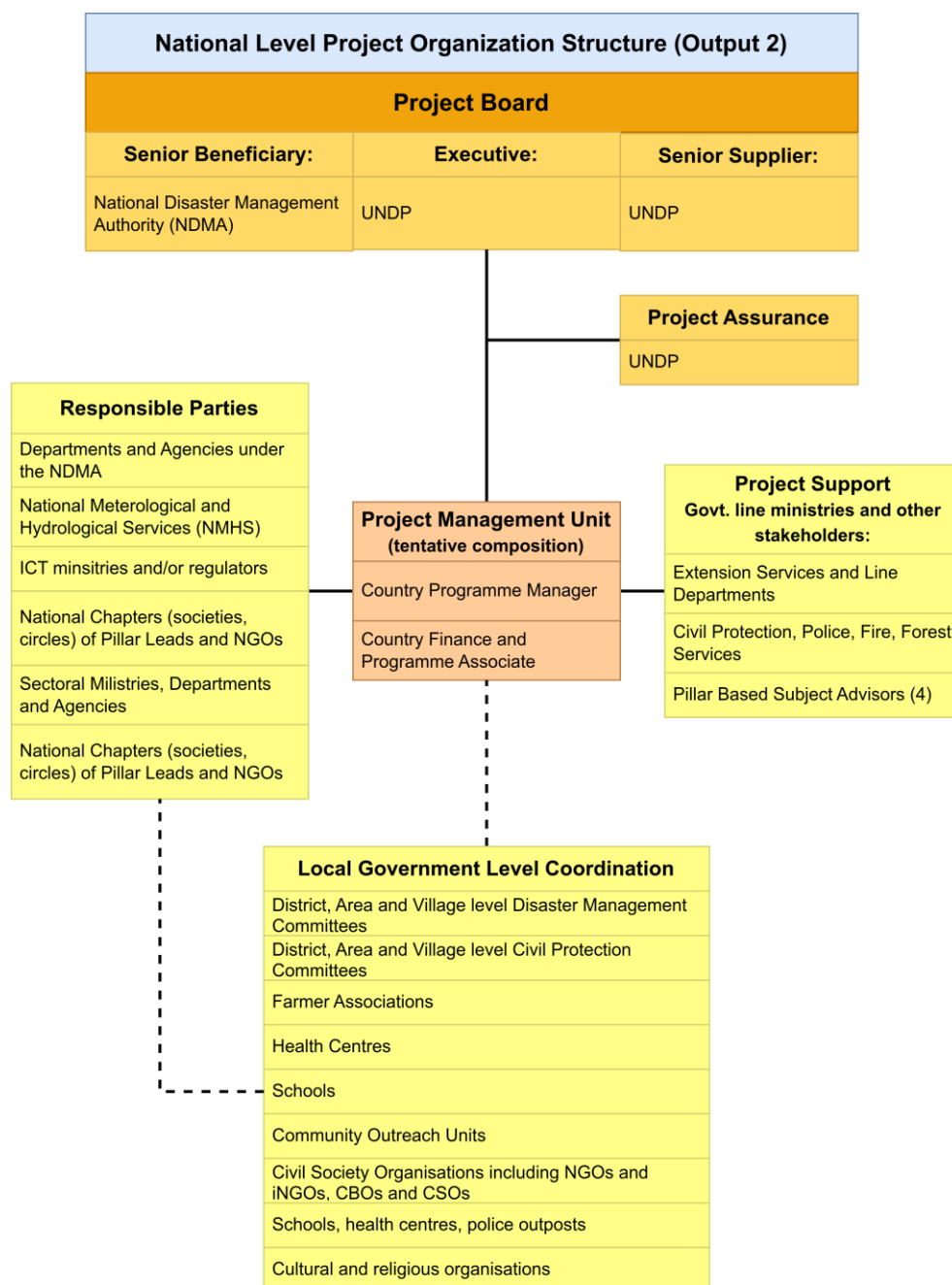


Figure 3. National project implementation arrangements

B.5. Justification for GCF funding request

198 Investments in and around MHEWS are largely public goods including the networks, capacity building at the institutional level as well as community level. While there are a few elements that are conducive for private sector engagement, in developing countries, particularly in the aviation and telecommunications sectors, such opportunities are largely untapped due to under-developed markets. Markets can also be constrained due to challenges in insecurity, access and limitations of government support and available financial services. Public investments in MHEWS have been limited and sporadic partially because the value that MHEWS can bring is not recognised. The absence of technical capacities to make

use of existing CIEWS undermines the trust in maintaining such systems. As a result, countries' needs for MHEWS have been addressed largely through external financing, on a project-by-project basis. This approach addresses the short-term equipment needs but has many shortcomings considering that the "ecosystems" that make use of MHEWS require long-term support and financing.

- 199 This GCF program focuses on establishing a global support architecture that intends to be long-lasting, is built on the principle of multi-partners to address a wide range of needs which to date have been addressed, if at all, in a fragmented manner, and offers a platform for regional and global south-south, north-south, and public-private information exchange. The quantum of GCF financing allows a program of global/multi-country coverage to establish a pathway towards EW4All not just for the initial 7 countries but more; and their mandate allows multiple partners not only the pillar leads but also CREWS, SOFF and other initiatives, to come together to overcome the fragmentation. The multi-partners approach, particularly the involvement of the likes of ITU, allows exploring the possibility of meaningful public-private sector collaboration around the dissemination and use of CIEWS.

Key characteristics and dynamics of the private sector

- 200 Considering that EWS also affect various economic sectors, project interactions with the private sector are expected to be diverse, and the nature and the extent of private sector engagement in this project will show significant variations depending on country contexts. However, the following commonalities can be drawn to guide private sector engagement within the proposed project.
- 201 The private sector, especially in the telecommunication industry, plays a critical role in both the collection/transmission of basic observational data and dissemination of early warnings. This project will, with the help of ITU and GSMA seek collaboration and partnerships with the telecommunications sector in each of the countries. This is critical as universal coverage of early warnings will not be met unless the existing mobile cellular networks, managed by private operators, are utilized to assist in automatic collection of observations and to complement radio, TV and other networks in disseminating alerts. However, in many developing countries, regulatory barriers exist in effectively engaging these private operators in the dissemination of early warning information.
- 202 Telecommunication infrastructure is vulnerable to extreme weather events and therefore, companies owning such infrastructure are also prospective clients for climate services. The same infrastructure can also be leveraged to host observation systems such as lightening sensors and generic weather observations, particularly in remote areas with limited power infrastructure and security challenges.
- 203 Limited public-private partnerships with providers of digital networks and services limit the reach of early warnings and weather information through multiple communication channels including radio, TV, internet, mobile phones, satellites and social media. In remote locations (particularly mountainous environments) satellite-based transmission and dissemination mechanisms may offer the only effective means of communication and therefore also need to be considered.
- 204 Private sector engagement in the project will also take place at a different scale involving multi-national corporations who are expected to supply hydrometeorological equipment to target countries. National governments will be supported to adopt requisite policy frameworks and principles for successful public-private partnerships between government entities and private businesses. This will facilitate alignment with corporate social responsibility and help leverage mechanisms such as the UN Global Compact which offer businesses the opportunity to adopt sustainability principles and support UN goals. The project will leverage the global commitments and engagement of the private sector as part of the EW4All Advisory Panel e.g. GSMA, for the telecom industry, IDF for insurance and Microsoft and Planet Labs for digital/frontier technologies.
- 205 This will also facilitate the adoption of emerging technologies such as artificial intelligence and provide access to new and innovative digital platforms, social media and technological platforms such as remote sensing systems for high resolution hydrometeorological data acquisition, big-data and cloud computing. This can facilitate data analysis and delivery to the most vulnerable, enhance warning, improve dissemination and increase awareness about disasters and early warnings.

At the other end of the spectrum, private companies and innovators can help design low cost, autonomous EWSs and alarms that are managed by communities, for example for floods¹⁰⁹ and landslides.¹¹⁰

- 206 The insurance sector plays an important role in using improved weather and climate data. This project will explore how CIEWS can be used for development of weather indexed insurance products that the population can benefit from. In many developing countries, risk transfer mechanisms using insurance are significantly underdeveloped due to the absence of weather/climate and other socioeconomic data needed for product development, lack of awareness about insurance products, underdeveloped regulations, all of which increase risks for insurance providers to enter into the market. Globally, economic losses from natural hazard-related disasters in 2022 amounted to USD 270 billion and 55% of the losses were uninsured. In emerging and developing countries, less than 10% of their losses were insured.¹¹¹ The absence of insurance as a mechanism to mitigate the impact when climate-induced extreme events strike continues to be a major impediment for resilient development in many developing countries.

B.6. Exit strategy

The global support structure

- 207 The speed and scale of interventions required to deliver on the EW4All objectives can only be achieved through coordinated global support involving all key actors across the EW4All value chain. The needs of LDCs and SIDS to establish, operate and maintain their national end-to-end EWSs, effective communication systems and disaster preparedness at the level of vulnerable communities requires multiple funding instruments. These instruments need to bring together complementary mechanisms that maximize country ownership and impact.
- 208 There is a need to facilitate national investment planning to achieve EW4All goals around which finance can coalesce to achieve shared value and results. This includes the development of national roadmaps and investment pipelines and the communication of a common vision of needs, priorities, and opportunities across all four pillars of EWS. It is necessary to recognize the respective business practices, funding cycles, and operational practices of various financial instruments to accelerate GCF co-finance. This will help address gaps across the continuum of country-specific technical and financial support needs for end-to-end EWSs, translating country vision into concrete investment.
- 209 The absence of a consistent, reliable and robust source of holistic technical and financial support to countries is the essential challenge to the effective generation and use of MHEWS. Support available to countries is fragmented and tends to be driven through project financing. Global initiatives such as EW4All and mechanisms like CREWS are therefore necessary for countries to receive consistent, continuous support that meets the global standards, allows them to exchange knowledge and sustainably develop and expand MHEWS to cover all sections of their vulnerable communities.
- 210 Recognizing that responding adequately to the SG's call for EW4All will require concerted efforts at the global level for a sustained period of time, developing a strong exit strategy for this project will become one of the critical deliverables of this GCF project. In particular, the global support architecture supported by the project will continue to finance itself through various channels including bilateral or multilateral public financing, philanthropists, private sector financing, or other innovative financial instrument. Sustainability and replicability of the project will be facilitated and supported through key partnerships and its alignment with key frameworks and strategies at the global and regional level. The project is aligned with the WMO Global Framework for Climate Services (GFCs), designed to facilitate delivery of best practice climate services.
- 211 This project is designed to address existing root causes and barriers to sustainable climate information services, MHEWS and Anticipatory Action in the most vulnerable countries among the LDCs and SIDS. The following strategy is proposed to ensure sustainability beyond the project's lifespan.

¹⁰⁹ "Community-Based Flood Early-Warning System | India | UNFCCC" (<https://unfccc.int/climate-action/un-global-climate-action-awards/winning-projects/activity-database/community-based-flood-early-warning-system-india>, n.d.), accessed April 14, 2023.

¹¹⁰ N. Dixon et al., "An Acoustic Emission Landslide EWS for Communities in Low-Income and Middle-Income Countries," *Landslides* 15, no. 8 (August 2018): 1631–44, <https://doi.org/10.1007/s10346-018-0977-1>.

¹¹¹ Munich Re, "Natural Disaster Risks - Rising Trend in Losses" (<https://www.munichre.com/en/risks/natural-disasters.html>, n.d.), accessed April 6, 2024.

Institutional strengthening

- 212 The project will strengthen the existing institutions and governance at the global, regional and national levels, rather than creating new parallel structures. Sustainability will be supported by the establishment of holistic national and regional frameworks, with associated institutional coordination mechanisms such as road-maps, to ensure effective governance, coordination and management of national climate services. The project will support integration of climate services in key policies, strategies, plans and budgets which will provide a foundation for uptake of climate information in decision-making and facilitate sustainable service provision in the long term. The identification of funding modalities for climate information products and anticipatory action including by integration in the national/regional budget/pools will also contribute to sustainability.
- 213 State of the art hydrometeorological monitoring equipment and tools for observations, monitoring, modelling and prediction will be integrated in the existing hydrological and meteorological institutions, based on available data management infrastructure. Institutional capacity will be further strengthened through upgrading relevant IT and communications equipment. Special attention will be given to building capacities for O&M of both equipment and data infrastructure. The project will also help NHMS and NDMA in developing and implementing long-term sustainable strategies for O&M through financial arrangements such as SOFF, for which it will provide support. NHMS will be supported in designing and disseminating a suite of tailored climate information products for national ministries, departments and agencies and the private sector to ensure some cost recovery. Transfer of Equipment: Equipment purchased through GCF Proceeds will be transferred to the governments per standard processes as detailed in Article III of each SBAA entered between UNDP and each respective government in a mutually agreed upon manner that places a priority of the sustainability of project results and objectives.
- 214 In order to ensure long term technical and capacity support, the project will support NMHS and NDMA to participate in networks based on regional and national centres of expertise. This will ensure long-term and sustainable engagement with governments and policy makers. It will allow network actors to apply to a wide variety of other funding sources, including utilising academic grants to support research as part of MEAL. This national and regional technical capacity will furthermore benefit DAEs, regional institutions and national government by increasing access to technically rigorous support and their capacity to submit transformative proposals in the future. By including DAEs and NDAs in the development of proposals, this programme will also expose them to new ideas and enhance their proposal development capacity.

Community level capacity strengthening

- 215 The project will adopt a broad multi-stakeholder approach and bring on board government agencies, multilateral and international agencies and NGOs, Organisations of Persons with Disabilities (OPDs), Red Cross and Red Crescent network, private sector actors and communities as users of climate information, early warnings and anticipatory actions as well as front-line agencies in climate adaptation and disaster response. When possible, this will be done through existing government-led coordination mechanisms. Institutional linkages will be strengthened by enacting enabling policies and procedures that improve effectiveness of delivery and use of CIEWS. Local, community-based institutions, in particular, will be strengthened to effectively access CIEWS, interpret and co-develop anticipatory action, adaptation and response strategies and plans. These activities will be implemented through government agencies and institutions with a mandate for decentralized disaster preparedness and response. Lessons learned will be used to fine tune and improve the strategies and plans and successes will be scaled up through the relevant agencies to other vulnerable regions in the country.

Programmatic alignment

- 216 This multi-country project will unlock opportunities for coordinated action and serve as an accelerator and amplifier for EWS support co-financed through complementary instruments. It will work closely with other important programmes such as CREWS and SOFF, and other mechanisms for funding such as from GEF and multilateral donors. This will be achieved by building on multilateral mechanisms such as CREWS and the GCF, which mobilizes a significant number of diverse national entities ranging from Ministries to national NGOs operating through a collaborative and partnership-based business models. This will ensure scaling up of projects supported by these agencies, particularly under SOFF and CREWS.

Financing

217 Establishing a financing strategy in each participating country is one of the deliverables of the project to address the critical challenge of financial sustainability. The strategy will assess the feasibility of various possible sources of finance, including both public and private sources, to expand and maintain end-end-to end MHEWS in the selected countries, based on their specific needs. This includes the national hydrometeorological monitoring network including public and private financing sources. The financing strategy will include financial mechanisms that ensure the sustenance of: i) the national hydrometeorological monitoring network; ii) development of climate products including IbfFs, models for scenario and trigger development; iii) support to anticipatory actions including ex-ante forecast-based financing and resilience bond or ex-post climate risk transfer mechanisms; and iv) support disaster preparedness and response capabilities down to the level of communities. These mechanisms will be operated at multiple levels and tailored to each countries fiscal policies and capacities for effective and efficient fund transfer for DRR. Contingency funds will thereby be released to targeted agencies and communities to take early action and avoid and/or mitigate the impact of disasters through effective action built on CIEWS. This arrangement applies to countries where contingency funds are already available for early and anticipatory action. Deployment of these funds is based on agreed approaches built by consensus between governments and donors. Where this coordination is lacking the project will help coordination mechanisms between relevant agencies and the government.

218 .

219 The financing plan will explore public-private partnership on data usage, analysis and dissemination. It will develop and implement campaigns and consultations with various development partners, seek synergy among donors and coordinate funding requests, including GCF submissions, across pillars to unlock financing at the global level. The project will support Direct Access Entities to fast-track access to GCF financing on early warnings. Countries will be supported in sourcing financing from international financial institutions, international organisations, vertical funds and corporate donors seeking to support climate adaptation.

Policy strengthening

Project activities will fill gaps in policies and governance arrangements to strengthen national level EWS coordination among government actors at the national and local level across sectors. Relevant national strategies and plans will be updated or developed for NMHS and NDMA agencies in each country contributing to Sendai Target G Indicator 4: All countries' local governments have a plan to act on early warnings. The project will strengthen legislation and policies governing coordination, exchanging of information, coordinated dissemination of warnings and establishing DRR structures for preparedness and response. This will mainstream early warning and anticipatory action and will leverage supportive framework and legislation within the respective countries. The strategy will be linked to social protection and DRM based on local priorities and will address assignment of roles and responsibilities at the national and subnational level for government agencies, communities and local institutions as well as other actors such as NGOs. Coordination platform will be established and inter-institutional arrangements will be specified for data sharing, inter-operability of systems, co-design of services.

C. FINANCING INFORMATION						
C.1. Total financing						
(a)	Total amount			Currency		
Requested GCF funding (i + ii + iii + iv + v + vi + vii)	103.25			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	Enter <u>amount</u>	Enter years	Enter years	Enter %		
(ii) Subordinated loans	Enter <u>amount</u>	Enter years	Enter years	Enter %		
(iii) Equity	Enter <u>amount</u>			Enter % equity return		
(iv) Guarantees	Enter <u>amount</u>	Enter years				
(v) Reimbursable grants	Enter <u>amount</u>					
(vi) Grants	103.25					
(vii) Results-based payments	Enter <u>amount</u>					
(b) Co-financing information	Total amount			Currency		
	11.34			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority

Governments Pillar Leads	In kind	10.39	million USD (\$)	<u>Enter</u> years	<u>Enter</u> %	Options
UNDP-CO	Grant	0.90	million USD (\$)	<u>Enter</u> years	<u>Enter</u> %	Options
UNDP-CO	In-kind	0.05	million USD (\$)	<u>Enter</u> years	<u>Enter</u> %	Options
Total financing (c) = (a)+(b)	Amount			Currency		
	<u>114.59</u>			<u>million USD (\$)</u>		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	1) Parallel financing in the amount of 40.1 mln. from System Observations Financing Facility (SOFF) will go towards supporting the countries at the early stages of the EWS value chain by investing in meteorological equipment like automated weather stations and radio sounds. 2) Parallel co-financing US\$ 563,985 in the form of a loan has been provided by the National Secretariat of Risk Management of the Republic of Ecuador to streamline ongoing initiative “Strengthening the National Early Warning System for Multiple Threats”.					

C.2. Financing by component

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount Options	Financial Instrument	Name of Institutions
1	1	13.50	6.68	Grants	6.82	In-kind	<u>IFRC, ITU, WMO</u>
2	2	92.50	88.93	Grants	3.57	Grants, In kind	<u>National Government s, institutions of Ecuador, Ethiopia, Cambodia)</u>
M&E	M&E	3.04	2.79	Grants	0.25	Grants, In- kind	<u>Participating country offices</u>
PMC	PMC	5.54	4.84	Grants	0.70	Grants, In- kind	<u>Participating country offices</u>
Total cost (USD)		114.59	103.25		11.34		

Financing per country:

Antigua & Barbuda US\$ 12,373,360

Cambodia US\$ 13,218,000

Chad US\$ 12,262,938

Ecuador US\$ 12,369,770

Ethiopia US\$ 12,975,801

Fiji US\$ 12,938,526

Somalia US\$ 12,792,367

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

- 220 The project is expected to support capacity building, technology development and technology transfer through multiple mechanisms. Under Activity 1.3 knowledge exchange between countries and across initiatives with a focus on countries seeking to improve their MHEWS and Anticipatory Action approaches and systems and through development of analytic frameworks and models for IbFs and scenarios to inform early and anticipatory actions.
- 221 Under Activity 2.1 the project will support capacity development towards meeting the needs of Sendai Framework Target G. This includes collection of risk data, information and its analysis leading to improved production and use of risk information to support IbFs, better integration of relevant agencies with the work on NDMAs and NMHYS for more efficient use of this information and improved monitoring, evaluation and learning systems.
- 222 Under Activity 2.2, the observation and forecasting capacities of NMHS will be enhanced by addressing technical gaps leading to better accuracy and timeliness of these forecasts leading to GABON compliance. This will also lead to installation of EWS and forecasting services for all priority services leading to meeting Sendai Target G Indicator 4.
- 223 Activity 2.3 will lead to improved use of technological solutions for data integration and sharing for development and broadcasting alerts, communications and dissemination of early warnings across multiple channels, but specifically focusing on mobile cellular networks and communications. This will lead to the strengthening or establishment of the Common Alerting Platform in all countries backed with multiple channels of dissemination by working with network operators. The activity will also build capacities of extension services in co-development of early warnings which integrate weather forecasts and information with indigenous knowledge and practices. Innovative technologies for autonomous community based EWSs will be transferred for relevant hazards to adapt to local EWS needs, particularly in remote, "off-grid" locations and areas with poor reach of the NMHS and most dissemination channels.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

- 224 Globally, 3.3 to 3.6 billion people are highly vulnerable to climate change¹¹². Between 2015 and 2021, 1.05 billion people were affected by disasters and \$330 billion USD in economic losses of 145 countries reporting to the Sendai Framework. Yet, only 4.1% of disaster-related development assistance was allocated to ex-ante prevention and preparedness, of which the majority was for emergency response¹¹³. As of 2022 half the countries in the world, less than half of LDCs and less than a third of SIDS had a MHEWS.
- 225 Numerous studies have assessed the costs and benefits of EWSs. EWSs save lives and assets worth at least ten times their cost. Just 24 hours warning of a coming storm or heat wave can cut the ensuing damage by 30 percent, and spending \$800 million on such systems in developing countries would avoid losses of \$3–16 billion per year. Such investments benefit from high benefit-cost ratios. For each of the 7 countries included in this project, an economic analysis will be conducted following the GCF Guidelines. The outcome of these 7 analyses will be aggregated into a single economic analysis for the entire proposed project.
- 226 The project contributes to the achievement of GCF strategic-level impacts through increased resilience and enhanced livelihoods of the most vulnerable people and communities globally through a programmatic approach which accelerates access to EWSs for the most critical climate hazards. It does so by investing in technologies and finance for generation, dissemination of climate information and in its effective use through preparedness and anticipatory action. It will enhance and increase the impact and long-term sustainability of these investments both at the national and regional levels. It responds to the update of the GCF strategic plan 2024-2027¹ which contributes to the EW4All call of the Secretary General with a dedicated target result on EWS.
- 227 Seven countries will receive holistic investment and technical assistance to establish an effective end-to-end EWS in accordance with the actions proposed in the EW4All Executive Action Plan. A particular focus will be placed on improving last-mile communication of and community-level capacities to receive, understand and act on climate information and early warnings by ensuring each country prepares and updates disaster response plans and ensures communities have access to communications, the capacities and knowledge and the preparation to react to warnings. These activities will directly contribute to improving the resilience of people potentially at risk from climate-related extreme events and hazards in LDCs and SIDS where communities are highly vulnerable to climate induced disasters on account of socio-economic vulnerabilities and physical exposure.
- 228 Actionable climate information products for agriculture, energy and water management will increase resilience across sectors, particularly agriculture, reducing the risks from climate-related hazards and thus reduce losses and damage to crops, infrastructure and other assets. By the end of the project, it is expected that these target countries are well on track to achieve the target of universal early warning coverage and hence the total population of the countries will be receiving adaptation benefits from the project. Section B (pg. 43 to 48) summarizes the proposed interventions at the country level which address barriers and gaps in governance, coordination, policies, implementation of EWS and in delivering EWS services and response. The section also describes the expected adaptation benefits of these interventions for vulnerabilities across relevant sectors and geographic areas.
- 229 The project will support existing WMO frameworks such as WIGOS, WIS, WIPPS and is aligned with existing initiatives around early warning and early action such as CREWS and the Risk-informed Early Action Partnership (REAP) which aims to make one billion people safer from disasters by 2025. The project will facilitate knowledge exchange, coordination and

¹¹² The UN Global Early Warning Initiative for the Implementation of Climate Adaptation. 'Early Warnings for All - Executive Action Plan 2023-2027'. World Meteorological Organization (WMO), 2023.

¹¹³ UNDRR and WMO. 'Global Status of Multi-Hazard Early Warning Systems: Target G'. United Nations Office for Disaster Risk Reduction, 2022.

collaboration with key initiatives such as the Alliance for Hydrometeorological Development, Anticipatory Hub, CREWs, SOFF, and with UN agencies and IFIs to ensure that they collectively strive towards meeting the EW4All target.

230 The project will contribute to the achievement of the GCF indicator of reduction in the number of people affected by climate-related disasters and will deliver climate change adaptation benefits by:

1. Improving the resilience of people potentially at risk from climate-related extreme events and hazards in LDCs and SIDS where communities are highly vulnerable to climate induced disasters on account of socio-economic vulnerabilities and physical exposure. It will provide early warnings to communities who are susceptible to cyclones, coastal flooding and sea-level rise, floods, landslides and droughts.
2. Strengthening NMHSs and NDMAs in participating countries through regional and global collaborations which enhance their access to sustainable finance and technical knowledge. This will result in up gradation and improved O&M of hydrometeorological infrastructure, access to innovative and state of the art tools and techniques and streamlined protocols which increase institutional effectiveness leading to more reliable impact-based forecasts and targeted, actionable climate information products for early and effective on-ground action.
3. Enhancing regional and national MHEWS with targeted and timely IbFs which help communities take effective measures to secure their lives, assets, and livelihoods during climate-related hazards and triggers to inform early and anticipatory actions, including forecast based financing so that resources are made available to at-risk communities ahead of disasters.
4. Increasing the safeguarding of assets and infrastructure from climate-related hazards through accurate, timely and actionable early warning services. Actionable climate information products for agriculture, energy and water management will increase resilience across sectors, particularly agriculture, reducing the risks from climate-related hazards and thus reduce losses and damage to crops, infrastructure and other assets.
5. Strengthening national DRM legal frameworks and financing mechanisms as well as coordination, collaboration, and knowledge sharing from the local to the international level leading to an effective DRM cycle and increased opportunities for regional cooperation and up-scaling of climate resilient sustainable development.

D.1. Impact potential

Country	No. Direct Beneficiaries (m/f)	% of Direct Beneficiaries of Total Population	No. Indirect Beneficiaries (m/f)	% of Indirect Beneficiaries of Total Population
Antigua and Barbuda	91,500 (43,712/ 47,788)	100%	114,000 (54,461/59,539)	N/A ¹¹⁴
Cambodia	514,098 (254,324/ 259,774)	2.91%	2,153,028 (1,065,103/ 1,087,925)	12.21%
Chad	1,200,000 (593,939 / 606,061)	6%	5,500,000 (2,722,222/ 2,777,778)	28%
Ecuador	513,900 (250,367/ 263,533)	2.6%	14,000,000 (6,820,675/ 7,179,325)	72%
Ethiopia	16,000,000 (8,035,200/ 7,964,800)	12.12%	21,600,000 (10,847,520/ 10,752,480)	13.64%
Fiji	416,282 (211,217/ 205,065)	46%	488,308 (247,762/ 240,546)	54%
Somalia	7,800,000 (3,910,920/3,889,080)	40.25	19,247,455 (9,650,674/9,596,781)	100%

231 The approach used to estimate direct and indirect beneficiaries is as follows. Direct beneficiaries are vulnerable populations (as per official census or assessments) directly involved in project activities falling largely under Pillar 3 and 4, such as recipients of EWS through enhanced dissemination systems, capacity building in the use of MHEWS, disaster

¹¹⁴ Indirect beneficiaries for Antigua and Barbuda are considered the significant number of tourists that visit the country during the hurricane season.

preparedness and response, and those residing within the service area of sub-national emergency coordination and response facilities that the project would enhance and support. Indirect beneficiaries are populations (as per latest official census or assessments) in areas targeted by the project under Pillars 1, 2 and 3, who would receive enhanced MHEWS through improved risk information, forecasting and dissemination.

D.2. Paradigm shift potential

- 232 This project will help shift the emphasis of DRM in countries from response to preparation and anticipatory action by strengthening MHEWS, IbF. It will facilitate the generation and dissemination of critical climate risk information and the design and implementation of evidence-informed transformative policies and plans which reduce the exposure and strengthen the capacities of communities to prepare for and take pre-emptive action against disasters. It will thereby reduce the vulnerability of communities and critical infrastructure and strengthen the resilience of vulnerable sectors to climate-related hazards and their impacts of climate change. It will ensure the needs of most vulnerable population groups during extreme events such as women, children, elderly and persons with disabilities will be reflected in relevant policies, procedures and processes.
- 233 The global support architecture that the project establishes a sustainable mechanism to support countries both technically and financially in rolling out a comprehensive MHEWS value chain. It will catalyse learning and impact beyond one-off, project-based investments, which characterises investments in CIEWS in many developing countries. Elements that contribute to the paradigm shift potential of this project include the emphasis on utilizing global standards for assessing the gaps and needs and fostering synergies with other initiatives with complementary objectives. These include the Alliance for Hydrometeorological Development, CREWS and SOFF, which will lead to support towards knowledge generation; knowledge synthesis and sharing among countries; and long-term financing strategy for both countries and for the global support architecture.
- 234 Finally, these investments and coordination between initiatives will be based in EW4All Roadmaps and standardized assessments (CHD) leading to an unprecedented level of cooperation and collaboration among key actors across the MHEWS value chain, from global to regional, national and even sub-national levels. This will leverage multi-institutional partnerships at the regional and global level coupled with national frameworks and road-maps which ensure multi-sectoral collaborations within government ministries and multi-level implementation mechanisms for disaster preparedness and response. The mechanism will enhance partnerships between government and non-government agencies, including societies and circles of Pillar Leads, iNGOs and national entities and mechanisms such as CREWS and SOFF.

Potential for scaling-up and replication:

- 235 Scale-up and replication to other countries is feasible given the global support structure, should further funding be made available. Substantively, the establishment of selected IbF schemes and Anticipatory Action initiatives have a high potential for replication within and across the target countries. The use of appropriate technologies and innovations will also contribute to scaling up and replications. For example: i) The implementation of the CAP and use of multiple channels for dissemination, but those specifically leveraging on the cellular networks will facilitate uptake by different telecommunication companies; ii) Successful applications of AI in for development of early warning products and vulnerability assessments; iii) Co-development of agricultural advisories using PICSA and E-PICSA; and iv) Community based autonomous early warning and alarm systems.

Potential for knowledge and learning:

- 236 The enhancement of MHEWS and production of IbF which utilise and combine traditional indigenous knowledge with modern and scientific techniques will facilitate adoption of adaptation and responses to early warnings, facilitate anticipatory actions for disaster preparedness and response by DRM institutions and grassroots communities. This will catalyse wider acceptance and lead to safeguarding of lives, livelihoods and assets. It will support regional and country level dialogues, knowledge exchange, coordination and collaboration with key initiatives such as on-going projects and programmes of the UN, the Alliance for Hydrometeorological Development, CREWS and SOFF. Through these initiatives, the project will also tap into communities of practice in place for knowledge sharing between countries and key actors

involved in effective MHEWS. For example, the community of peer advisors working with countries through SOFF investments to develop CHD and Gap Analysis used for EW4All baseline.

Contribution to the creation of an enabling environment:

- 237 The project will support enabling conditions at regional and global level, building on the significant investments of SOFF in human and institutional capacity development and thereby promoting country-level actions towards the EW4All goals. It will improve the hydrometeorological infrastructure of participating countries and strengthen capacities of NMHS and NDMA to improve observations and O&M of infrastructure as well as the generation of risk and vulnerability information. It will streamline national and local protocols to improve institutional effectiveness of coordinating with multiple actors in the dissemination of reliable, actionable IbF and climate information to trigger and support decisions for anticipatory action. It will strengthen national DRM legal frameworks as well as coordination, collaboration, and knowledge sharing from the local to the international level leading to an effective DRM cycle and increased opportunities for regional cooperation and up-scaling of climate resilient sustainable development. It will help link sectoral and technical government entities, science institutions, civil society and other stakeholders to promote anticipatory actions. All of these will ultimately contribute to an enabling environment where climate risk information is effectively produced and disseminated, in a sustainable fashion, to the most vulnerable groups to safeguard their lives, livelihoods and assets in the long-term.

Contribution to the regulatory framework, policies and financial mechanisms:

- 238 The project will catalyse a systemic change in the operational procedures and processes of participating countries in the generation and use of early warnings. It will facilitate the integration of climate information services in key decision making, planning and operational processes by: i) revising institutional frameworks to optimise coordination between hydrological and meteorological organisations with other governmental institutions by embedding early warnings, IbFs and anticipatory action into government contingency plans and DRM budgets; ii) advocating for and providing links to international financing opportunities for long-term funding arrangements for hydrological and meteorological institutes to provide tailored climate information services on climate change threats; iii) linking with initiatives such as SOFF to ensure funding for the installation and O&M of hydrometeorological infrastructure as well as human and institutional capacity of NMHS as per WMO standards; and iv) delivering fine grained and updated vulnerability assessments to assist in long-term disaster mitigation and adaptation planning.

Overall contribution to climate-resilient development pathways:

- 239 The development and dissemination of timely, tailored and targeted climate information products will be transformational in building the long-term resilience of key economic sectors such as agriculture, tourism and health, to climate risks. Delivered through a suite of outreach, learning and knowledge management activities, information products will increase efficiency and provide more detailed and better-informed approaches and response actions to climate risks.
- 240 A summary of the paradigm shift potential for each participating country is presented below.

Antigua and Barbuda

Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

- 241 The proposed approach for Antigua and Barbuda will create a paradigm shift in the end-to-end MHEWS process, enabling effective anticipatory action and planning across multiple sectors. In particular, the project will be transformative in the marine environment, significantly improving the monitoring and forecasting capacities for ocean swells, storms, strong winds and sea temperature and other parameters. This will have a significant impact in terms of building resilience of fisheries communities, as well as to enabling preparatory action to safeguard coastal infrastructure and related economic activities – including shipping and tourism – against the impacts of swells, coastal flooding and erosion. The marine monitoring will also enable tracking and forecasting of environmental impacts in coastal ecosystems – including coral reefs early warnings and related links to the fisheries and tourism sectors. Additionally, the project's focus on improved disaster risk knowledge, monitoring, forecasting and response for Barbuda will enable a paradigm shift in disaster preparedness and response on the island which is currently almost entirely reliant on information coming from Antigua.

Contribution to the regulatory framework and policies, and the creation of an enabling environment

- 242 The approach for Antigua and Barbuda is grounded in a strong cross-cutting governance and coordination component, mainstreaming disaster preparedness and response into national sectoral policies and strategies, as well as ensuring effective coordination between stakeholders in the full end-to-end process of MHEWS. The regulatory framework will be enhanced by the development of a national strategic plan for national emergency management agencies, that together with the policy updates will provide a solid foundation for the effective functioning and sustainability of MHEWS and ensuring clarity and alignment among all involved parties.
- 243 In addition to strengthening the regulatory framework for MHEWS, the project will build capacity among key agencies and response services, ensuring clear delineation of roles and responsibilities in preparedness and response processes. Together with the strategic action plan, such capacity development will create an enabling environment for collaborative actions, ensuring smooth and efficient implementation of preparedness and response strategies.

Potential for scaling up and replication

- 244 The expansion of Community Based Early Warning Systems (CBEWS) and the incorporation of localised response plans offer a replicable framework for scaling CBEWS to additional districts or even other countries facing similar risks. These scalable components, combined with robust training and capacity-building efforts, ensure that the project's successes can be replicated and adapted to enhance resilience in diverse contexts.

Potential for knowledge sharing and learning

- 245 By establishing a centralised online risk data platform and integrating it with sector-specific knowledge management tools, the project facilitates the collection, sharing, and dissemination of risk information across government agencies and the public. Antigua and Barbuda is also an active player in the Caribbean in terms of climate action and response, positioning the country well to act as a catalyst for enhancing MHEWS and associated response actions in the region. As a regional focal point for the CDEMA, Antigua and Barbuda actively participates in regional knowledge, learning and support systems in the DRM space, creating a platform for sharing lessons and promoting the replication of actions across the region. The involvement of the University of the West Indies (UWI) in the risk knowledge components further facilitates knowledge sharing across the Caribbean region.

Ecuador

Potential for Scaling Up and Replication

- 246 The MHEWS project in Ecuador is designed with a flexible framework that can be adapted to different regions and hazard types, making it highly scalable. The establishment of a NFCS and a comprehensive climate data management strategy provides a model that can be replicated across other sectors or regions within Ecuador and even in other countries facing similar climate risks. The use of innovative technologies for data collection and forecasting, the inclusion of communication strategies and the implementation of community emergency plans in selected communities provides support for replication and scaling up in other regions of Ecuador and countries in the region.

Potential for Knowledge Sharing and Learning

- 247 The project emphasises the development of a harmonized MEAL strategy, which will systematically capture lessons learned and best practices. This creates a strong foundation for knowledge sharing at both national and international levels. The integration of diverse stakeholder groups, including gender and development initiatives, ensures that a wide array of perspectives and knowledge bases are shared, enriching the overall learning process. The focus on capacity building and the creation of a robust data management framework will enable continuous learning and adaptation.

Contribution to the Creation of an Enabling Environment

- 248 The project contributes to the creation of an enabling environment by strengthening governance structures and establishing inter-ministerial and multi-stakeholder coordination platforms. By developing a National Climate Finance Framework and creating a sustainable funding mechanism, the project ensures long-term viability for climate services. The development of standardised data frameworks and the establishment of a climate data centre will also foster an environment where climate data is readily available, enhancing decision-making and planning across sectors.

Contribution to the Regulatory Framework and Policies

- 249 The MHEWS project will directly contribute to the enhancement of Ecuador's regulatory framework by creating a functional regulatory framework for MHEWSs. The project's focus on developing and strengthening governance capacities, including the establishment of clear warning procedures and communication protocols, will result in more coherent and effective climate-related policies. Additionally, the creation of binding documents for warning dissemination and the development of anticipatory action protocols will contribute to more robust and enforceable policies.

Overall Contribution to Climate-Resilient Development Pathways Consistent with Relevant National Climate Change Adaptation Strategies and Plans

- 250 The project aligns with Ecuador's national climate change adaptation strategies by enhancing the country's ability to anticipate, prepare for, and respond to climate-related hazards. By improving disaster risk knowledge, enhancing forecasting capabilities, and strengthening early warning dissemination, the project supports Ecuador's climate-resilient development pathways. The focus on integrating climate services with national development plans ensures that the project contributes meaningfully to the broader goals of sustainable development and climate adaptation in Ecuador, consistent with its commitments under international agreements like the Paris Agreement.

Chad

Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

- 251 This initiative will strengthen climate information systems by supporting the development of EWSs, the monitoring of climate variables and observation facilities, and building the capacity of the stakeholders involved. It will contribute to strengthening the capacity of relevant ministries and directorates dealing with DRM in Chad, the NDA, Red Cross volunteers acting as first responders, local authorities, traditional leaders and communities at different levels of governance (from the national to the local levels).

Contribution to the regulatory framework and policies, and the creation of an enabling environment

- 252 It will address the disconnect between climate information systems and the end-users in Chad, by investing across the entire early warning value chain, supporting an inclusive multi-channel approach to disseminate early warning alerts to the people most at risk of climate disasters, and build systems for two-way communication as a way of promoting a culture of risk and an understanding of local capacities, and local knowledge systems for disaster preparedness and disaster management. It will work closely with the range of national and international actors engaged in climate finance, and in EWSs, to ensure a coordinated approach that contributes to a longer-term vision in Chad. This will include advocacy at all levels for resourcing, regulatory evolution and ensuring that institutional mandates reflect the expected roles and capacities of each of the national departments and institutions.

Potential for scaling up and replication

- 253 Qualitative research and dialogue conducted to inform the design and delivery of inclusive and people-centered EWSs (including, language, gender and conflict sensitivity) will seek to address the global gap in community influence over climate programming. This will ensure that anticipatory action programmes are aligned with community perceptions, preferences, and priorities and foster greater community ownership and responsiveness in disaster preparedness. Learnings from this process and its results will be relevant for comparable contexts, particularly in the Sahel and sub-Saharan African countries with similar challenges.

Potential for knowledge sharing and learning

- 254 The project will invest substantially in knowledge management, M&E, accountability and learning. Research and dialogue conducted to inform design and delivery will also shape and influence global and regional policy discussions on early warning through sharing project findings, the process of developing the tools and learning. Broad dissemination of findings, process and tools through targeted regional engagement, ensuring community views and voices are included in targeted discussions with decision-makers, policymakers and donors. Data and learning will be shared at the global level through the Anticipation Hub and the REAP. Community members and local organizations will be engaged in advocacy efforts at both regional and global levels.

Cambodia

Potential for scaling up and replication

- 255 Project interventions in the four target provinces will act as a proof of concept for what a comprehensive approach to EW4All at a local level can achieve. For this reason, the project will monitor progress carefully and leveraged for good practices and lessons learned and replication as part of the overall EW4All country initiative as well as potential for scaling up in other provinces around the country. Additionally, the project will advocate and build capacity for improved monitoring and learning across the EW4All initiative.

Potential for knowledge sharing and learning

- 256 The Cambodia EW4All project will form a central part of the already ongoing EW4All initiative started in October 2023. Many activities in the project arise from the development of the country EW4All Roadmap, developed in conjunction with three government ministries and 34 other agencies engaged with the initiative.
- 257 The EW4All initiative is planning regular semi-annual meetings where facilitators will ensure a focus on cross-pillar knowledge sharing and learning.

Contribution to the creation of an enabling environment

- 258 A key focus of the project will be on capacity development for the National Committee for Disaster Management (NCDM) as well as for the Ministry of Water Resources and Meteorology (MoWRAM) at both the central as well as the subnational levels. Training interventions will emphasise making the best use of regional centres of excellence and investing in skills in newly emerging technologies such as AI and remote sensing capacities. In addition, trained local youth will play a key role in accessing existing online early warning resources as part of Cambodia's commitment to digital transformation. In the target provinces, for the first time, the project will operationalise the newly developed government Shock Resilient Social Protection mechanism.

Contribution to the regulatory framework and policies

- 259 In line with existing laws and decrees on disaster management¹¹⁵, in 2022 UNDP developed and advocated for the implementation of Standard Operating Procedures for disaster preparation and response, on behalf of the NCDM. This project will review these SOPs as well as develop protocols for shared data management collaboratively with government ministries through the EW4All initiative. In addition, as part of the planned DRR capacity development initiatives at national and sub-national levels, existing DRR regulations will be publicised and further operationalised.

Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

- 260 Universal access to EWSs for natural hazards in Cambodia could substantially reduce risks to its population's well-being. Studies suggest that implementing these systems could prevent annual well-being losses of around US\$340 million and asset losses of up to USD 170 million, potentially decreasing losses by up to 21% and 20% respectively.
- 261 This project seeks to directly address the needs and priorities outlined in the National Action Plan on DRR (NAP-DRR) 2024-2028.

Ethiopia

Potential for scaling up and replication

- 262 Project interventions in the four target provinces will act as a proof of concept for what a comprehensive approach to EW4All at a local level can achieve. For this reason, the project will monitor progress carefully and leveraged for good practices and lessons learned and replication as part of the overall EW4All country initiative as well as potential for scaling up in other provinces around the country. Additionally, the project will advocate and build capacity for improved monitoring and learning across the EW4All initiative.

¹¹⁵ See <https://opendevdevelopmentcambodia.net/topics/disaster-preparedness-and-emergency-response-policy-and-administration/> for a comprehensive source of data on existing national disaster management legislation.

Potential for knowledge sharing and learning

- 263 The EW4All project will form a central part of the already ongoing EW4All initiative started in October 2023. Many activities in the project arise from the development of the country EW4All Roadmap, developed in conjunction with government ministries and agencies engaged with the initiative.
- 264 The EW4All initiative is planning regular semi-annual meetings where facilitators will ensure a focus on cross-pillar knowledge sharing and learning.

Contribution to the creation of an enabling environment

- 265 A key focus of the project will be on capacity development for the NCDM as well as for MoWRAM at both the central as well as the subnational levels. Training interventions will emphasise making the best use of regional centres of excellence and investing in skills in newly emerging technologies such as AI and remote sensing capacities. In addition, trained local youth will play a key role in accessing existing online early warning resources as part of the country's commitment to digital transformation. In the target provinces, for the first time, the project will operationalise the newly developed national Shock Resilient Social Protection mechanism.

Contribution to the regulatory framework and policies

- 266 In line with existing laws and decrees on disaster management¹, in 2022 UNDP developed and advocated for the implementation of SOPs for disaster preparation and response, on behalf of the NCDM. This project will review these SOPs as well as develop protocols for shared data management collaboratively with government ministries through the EW4All initiative. In addition, as part of the planned DRR capacity development initiatives at national and sub-national levels, existing DRR regulations will be publicised and further operationalised.

Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

- 267 Universal access to EWSs for natural hazards could substantially reduce risks to its population's well-being. Studies suggest that implementing these systems could prevent annual well-being losses of around US\$340 million and asset losses of up to US\$170 million, potentially decreasing losses by up to 21% and 20% respectively.

This project seeks to directly address the needs and priorities outlined in the NAP-DRR 2024-2028.

Somalia

Potential for scaling up and replication

- 268 The proposed activities will initiate Somalia's EW4All initiative, fulfilling the immediate programme of activities under the Roadmap of the EW4All initiative 2024-2027, for building disaster resilient communities in Somalia. The activities under the three project outputs, which represent the four pillars of the EW4All initiative and early warning more broadly, will establish key needs and national apparatus for further future early warning activities to be implemented at local scales.

Potential for knowledge sharing and learning

- 269 The database of disaster impact data and loss and damage tracking (aligned with UNFCCC, Sendai and SDG targets and indicators) will provide potential learning from past events alongside Interpretation and communication materials developed. These materials are to include hazard maps that identify specific geographic areas, sectors and groups that may be affected by particular hazards; tailored for EWS and DRR plans/local risk management plans. They will incorporate climate risk projection summaries, tailored in clear, accessible language (as appropriate) that incorporate traditional knowledge and understandings, developed and disseminated for the national and local levels and applied/interpreted and translated for Somalia's critical sectors (health, agriculture, livestock, forestry, fisheries and aquaculture etc) and in considerations of key vulnerability factors (e.g. age, gender, disability, access to infrastructure and services, IDP, conflict affected areas, economic diversity, social inequalities and environmental sensitivities). As knowledge sharing and learning opportunities, these will complement the Early Warning Coordination and Information Sharing Platform proposed to be established as a sub-structure of the National Platform for DRR (under the NMHEWC) committed to coordinating and distributing early warning information across the country.

Contribution to the creation of an enabling environment

- 270 Despite establishing an NMHEWC as a central point for early warning coordination and information sharing, early warning and information management in Somalia still need to improve in the early warning value chain. As a result, competition and overlapping roles and responsibilities continue across the various government departments and agencies. The Early Warning Coordination and Information Sharing Platform is therefore proposed for coordinating and distributing early warning information across the country, with many of the activities (e.g. Comprehensive Risk Assessment Framework and development of database, disaster risk knowledge materials and communication developed for national and local levels) aimed to contribute to these. The activities of the EW4All Initiative, and therefore the proposed activities under Somalia's project, implement key areas of Somalia's National Disaster Risk Management Policy (NDRMP) and its accompanying national and local DRR plans, contributing to creating a better enabling environment for enhancing the availability of disaster risk knowledge in the country.

Contribution to the regulatory framework and policies

- 271 The project's proposed activities include specific harmonisation of Somalia's early warning with Somalia's commitments to the UNFCCC, Sendai Targets and SDGs. Specifically, seeking to harmonise the development of a comprehensive hazard risk assessment methodology and assessments for key risks with regional approaches and UNFCCC, and Sendai and SDG targets and indicators and the harmonisation of early warning data collection, analysis and exchange modalities among federal line ministries, sectorial offices, and regional bodies and improved data automation, modelling and forecasting of early warning harmonised among sectorial offices. The project directly contributes to Somalia's commitment to the Sendai Framework for Disaster Risk Reduction (2015–2030) Target G (substantially increase the availability of and access to multi-hazard EWS and disaster risk information and assessments to the people by 2030) and Priority 4 (enhancing disaster preparedness for effective response and to 'build back better' in recovery, rehabilitation, and reconstruction). Furthermore, the project interventions align with Somalia's commitment to the Paris Agreement (Article 7, paragraph 7c) on adaptation and the Africa Program of Action for implementing the Sendai Framework in Africa. The EW4ALL initiative is now operationalised at the national level under the NMHEWC leadership. The activities accompany the process of the Roadmap, which establishes a systematic reform initiative aimed at aligning it to Article 3, para 6 and Article 7, para 2, of the establishment Law 17/2016 of SoDMA. These Articles state the clear responsibility of SoDMA to gather information and report on disasters and the risks facing the country and send out early warnings in collaboration with organisations inside the country and abroad working on those activities. While implementing key aspects of the NDRMP and its accompanying national and local DRR plans, the proposed project activities also interlace with Somalia's NAP Framework¹¹⁶, the 2021 NDC¹¹⁷, Somalia's Biennale Update Report (BUR) to the UNFCCC¹¹⁸, and the 2021 National Climate Change Policy¹¹⁹ which provides the framework for addressing climate change impacts focused on both adaptation and DRR measures.

Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

- 272 As noted, the proposed activities kick off the implementation of Somalia's Roadmap, thereby contributing to climate-resilient pathways consistent with national climate change strategies and plans. These contribute to climate-resilient development pathways through the alignment with Somalia's commitments on the Sendai Target G (substantially increase the availability of and access to multi-hazard EWS and disaster risk information and assessments to the people by 2030) and Priority 4 (enhancing disaster preparedness for effective response and to 'build back better' in recovery, rehabilitation, and reconstruction) as well as the Africa Program of Action for implementing the Sendai Framework in

¹¹⁶ Somalia's National Adaptation Plan (NAP) Framework, 2022, <https://napglobalnetwork.org/wp-content/uploads/2022/11/napgn-en-2022-somalia-nap-framework.pdf>

¹¹⁷ The Federal Republic of Somalia updated nationally determined contribution (NDC), JULY 2021

¹¹⁸ The Federal Republic of Somalia Ministry of Environment and climate change Somalia's first biennial update report under the United Nations Framework Convention on climate change (UNFCCC).

¹¹⁹ The Federal Republic of Somalia Ministry of Environment and Climate Change Somalia National Climate Change Policy, <https://moecc.gov.so/wp-content/uploads/2023/07/Somalia-National-Climate-Change-Policy.pdf>

Africa; and Somalia's commitment to the Paris Agreement (Article 7, paragraph 7c) on adaptation. Somalia's NAP Framework establishes local State-level climate change adaptation plans for Jubaland, Hirshabele, Puntland, Somaliland, and the South West State to which the proposed project activities will bolster a number of the priority actions through improved education and awareness raising, as well as additional climate and disaster hazard mapping and interpretation and communication materials tailored for EWS and DRR plans, and local risk management plans. These plans also cross-cut with DRM, preparedness, and response capacities.

Fiji

Potential for scaling up and replication

- 273 Integration of AI-powered systems into EWS will provide more accurate forecasts, which will also enhance people's faith in early warning messages. By analysing historical weather data, AI can predict future events effectively, especially for specific regions like agriculture and emergency services. It can optimize resource allocation and assess risks of extreme weather events, improving preparedness and mitigation strategies.
- 274 Although Cell Broadcast (CB) is not a new technology for public warning systems, it has yet to be implemented in Fiji due to the lack of infrastructure requirements for mobile service providers. However, the current infrastructure can be upgraded to support cell broadcasting technology. Consequently, the project aims to establish cell broadcasting services in Fiji and will enhance these mechanisms by integrating additional systems to disseminate early warning messages utilising the Common Alert Protocol system.
- 275 The initiative to use selected sugarcane farms as reference points to monitor crop status and provide farmers with guidance on necessary actions is a simple but effective way to implement an impact-based early warning system. Although the system may not be perfect, it will be relatable to farmers. Furthermore, this system has the potential to be extended to other crops in Fiji.

Potential for knowledge sharing and learning

- 276 Most importantly, people often do not take appropriate actions in response to early warnings; therefore, delays can be critical in fast-moving disasters. This lack of action may be due to a lack of faith in the messages or insufficient knowledge of the actions they could take to reduce impacts. It can also be attributed to resource or financial constraints, in the cases of sugarcane farmers. Furthermore, false alarms can lead to people becoming desensitised to warning and are less likely to take them seriously. Therefore, to have an effective EW4All system in Fiji, there is a need to change people's mindsets and attitudes towards early warning messages, so they are willing to take responsive actions, knowing that it may not eventuate. This effort is required not only at the community level but also at the government level to adopt and implement an effective Anticipatory Action system in Fiji.
- 277 It is important to understand how people perceive early warning messages and what influences their decision-making and actions. This understanding can inform the development and implementation of innovative approaches and technological advancements that enhance community capacity and inclusivity for people with disabilities and the elderly, empowering them to respond effectively to early warning messages. It is anticipated that, over time, there will be changes in people's mindsets regarding how to perceive and respond to early warning messages.
- 278 The project will facilitate a stocktake and harmonise multi-hazard risk assessment methodologies and tools across government and non-government institutions in Fiji. It will encompass data on hazards, vulnerabilities, exposure, capacity, and impacts. This will necessitate the development of a centralised database or repository system for disaster-related data to be shared among all national and subnational EWS actors. Furthermore, the project will prioritise enhancing the capacity to assess and gather information about risks at the local level. This will include documenting data and information on community-specific risks, incorporating traditional, local, and indigenous knowledge. Once in a mature stage, this centralised database can be used to inform the development monitoring of future EWS or DRR initiatives in Fiji and other countries in the region.

Contribution to the regulatory framework and policies

- 279 The project will facilitate development of interagency protocols and SOPs for data access and sharing between relevant MHEWS agencies. It will also strengthen coordination mechanisms for implementing MHEWS plans and protocols. Furthermore, it will aim to enhance technical support for continuously collecting risk data, analysing it, and applying

information for emerging hazards. It will involve evaluating coordination among different actors in communities to identify overlaps, gaps, and inefficiencies in response efforts, clarifying roles within and between agencies with a focus on involving communities in early warning systems.

Contribution to the creation of an enabling environment

- 280 Furthermore, the project will focus on building communities' ability to understand and adopt new climate-resilient livelihood practices through the use of an anticipatory action system. Efforts will be made to advocate and gain the government's support and willingness to implement AA in Fiji.

Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

- 281 The project activities are well aligned with Fiji's Scaling Up Early Warning Systems – Implementation Roadmap (in draft) and contribute to fulfilling the country's commitments to the Nationally Determined Contribution and the implementation of the National Adaptation Plan, which includes specific actions aimed at strengthening early warning systems in Fiji. Furthermore, it is also well aligned with and contributes to the National Disaster Risk Reduction Policy 2018-2030, which highlights the importance of early warning systems, community preparedness, and building resilience at all levels; the National Climate Change Policy, which recognises the need for improved early warning systems to anticipate and respond to climate-related risks; and the National Disaster Management Plan, which outlines strategies for disaster risk reduction and response. It emphasises the role of early warning systems in providing timely information to communities and authorities for informed decision-making during emergencies. Additionally, the project supports the Fiji Meteorological Service Act, which provides the legal framework for the operation of the Fiji Meteorological Service, responsible for monitoring and issuing weather warnings, as well as local Disaster Risk Management Plans, which incorporate early warning systems and outline strategies for response and preparedness at the local level. Moreover, Fiji has signed the Sendai Framework for Disaster Risk Reduction (2015-2030) and the Sustainable Development Goals. The country is also actively involved in regional initiatives and policies, such as the Framework for Resilient Development in the Pacific (2016), which prioritise early warning systems.

D.3. Sustainable development (max. 500 words, approximately 1 page)

- 282 This project will contribute to long-term social, economic and environmental benefits from avoided human and economic losses and support the objective of climate risk-resilient economies. Poor and vulnerable communities suffer disproportionately from natural hazards due to increased exposure, fewer assets and economic resources and poor access to information and financing to promote adaptation. In these countries, resources required to recover from disasters are often covered through external borrowing, leading to significant debt and increased exposure to economic shocks that result in large reductions in GDP and exports during global crises. Climate information services and MHEWS can be vital tools in reducing these impacts and thereby promoting social development. This will enable preparation, planning and anticipatory action based on early warnings which will result in avoidance, mitigation and adaptation to climate induced hazards, directly contributing to health and safety for communities most vulnerable to such disasters. The project is fully aligned with the SDGs, Paris Agreement and Sendai Framework, SDG 13 on urgent action to combat climate change and its impacts and related target 13.1 to "Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries". The project also speaks to the SDGs 1, 2, 9, 11, 12 and 15¹²⁰.
- 283 The project is aligned with the following priorities under Article 8 of the Paris Agreement:¹²¹ i) EWSs; ii) emergency preparedness; iii) slow onset events; iv) events that may involve irreversible and permanent loss and damage; v) comprehensive risk assessment and management; and vi) resilience of communities, livelihoods and ecosystems. It is also

¹²⁰ Goal 1. End poverty in all its forms everywhere; Goal 2. End hunger, achieve food security and improved nutrition and promote sustainable agriculture; Goal 9. Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation; Goal 11. Make cities and human settlements inclusive, safe, resilient and sustainable; Goal 12. Ensure sustainable consumption and production patterns; Goal 13. Take urgent action to combat climate change and its impacts; Goal 15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

¹²¹ United Nations Framework Convention on Climate Change, "Paris Agreement," 2015.

aligned with the strategies and actions on DRR as per the Sendai Framework¹²² which lists four priorities for action to prevent new and reduce existing disaster risks: i) Understanding disaster risk; ii) Strengthening disaster risk governance to manage disaster risk; iii) Investing in disaster reduction for resilience and; iv) Enhancing disaster preparedness for effective response, and to "Build Back Better" in recovery, rehabilitation and reconstruction. It is closely aligned with Sendai Target G Indicator 4: All countries' local governments have a plan to act on early warnings and Sendai Target G Indicator 5¹²³.

- 284 The proposed project is closely aligned with and will support and collaborate with a number of WMO programmes, initiatives and frameworks such as WIGOS,¹²⁴ WIS and WIPPS.¹²⁵ This includes support to NMHS in the area of DRR, i.e.: i) preparedness through EWS; ii) providing hazard information for risk assessments, prevention, response and recovery, and risk transfer across sectors; iii) mitigating existing risks and prevent the creation of new risks; iv) responding to user requirements; and, v) cooperating and engaging in disaster risk governance structures at all levels.
- 285 Additionally, the project will leverage CREWS and collaborate with the SOFF for the provision of a global public good: basic weather and climate observations.¹²⁶ SOFF's primary objectives are to address the gaps in systematic observations¹²⁷ as described by GBON.¹²⁸

Environmental benefits

- 286 Climate information and MHEWS will inform the management of natural resources and adaptation to climate impacts such as floods and droughts, particularly in the agricultural sector. The project will scale up the co-development and use of climate advisories using participatory approaches such as PICSA which is expected to contribute significantly to the adoption of climate resilient agricultural practices which improve productivity and reduce agricultural losses. The project will also support water resources management by providing IbFs and scenarios to help in managing dam releases and irrigation scheduling.

Economic benefits

- 287 Improved accuracy and coverage of MHEWS and more effective use of early and anticipatory action will have significant economic benefits. These economic benefits will be in the form of reduced losses and damages across multiple sectors, including agriculture, transport and energy. Better design, planning and management of private and public infrastructure will yield substantive economic returns. The project will also deliver significant economic benefits reducing the projected impacts of climate change on morbidity and mortality of exposed and vulnerable populations.

Social benefits

- 288 The health sector is likely to benefit significantly from disaster preparedness and anticipatory action to prevent the spread of diseases in the wake of disasters. The project will directly support the livelihoods of agricultural communities, including those engaged in raising livestock and fisheries. Advisories and warnings will reduce the loss of live and livelihoods by supporting decisions to adopt climate resilient practices and reducing and avoiding risk from extreme weather events, particularly among fishing and coastal communities. The project will also help prepare relevant agencies and communities for possible conflicts over resources and access that emerge from climate mediated hazards. This is most relevant to the northern and eastern African context where migratory pastoral communities can come in conflict with settled farmers.

¹²² UN, "Sendai Framework for DRR," 2015.

¹²³ All countries have accessible, understandable, usable and relevant disaster risk information and assessment available to people at national and local levels.

¹²⁴ Luis Nunes, "The WMO Global Basic Observing Network (GBON) A WIGOS Approach to Securing Observational Data for Critical Global Weather and Climate Applications," 2018.

¹²⁵ Michael Peter Schwab et al., "Bridging the Gap between Operational Meteorology and Hydrology: Including Hydrological Forecasts into the WMO Integrated Processing and Prediction System (WIPPS/GDPFS)," Other (pico, May 2023).

¹²⁶ WMO, "Systematic Observations Financing Facility."

¹²⁷ World Meteorological Organization (WMO), *SOFF Series, 02. The Gaps in the Global Basic Observing Network (GBON)*.

¹²⁸ Luis Nunes, "The WMO Global Basic Observing Network (GBON) A WIGOS Approach to Securing Observational Data for Critical Global Weather and Climate Applications," 2018.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

289 Least Developed Countries, SIDS and most African countries are particularly vulnerable to a range of climate change impacts and climate-related hazards, which affect people, livelihoods, assets and economy. Affected countries have highlighted the urgent need for better climate information and early warnings to help adapt to, mitigate and recover from increasingly frequent climate induced disasters. Early warning systems are a critical need to improve the resilience of their population to natural hazards and to strengthen and scale up early and anticipatory action. It is noted that national policies on areas related to EWS and DRM often neglect the differential needs of people with disabilities and those with special needs on account of age, gender and social and cultural exclusion¹²⁹. The country projects will address such gaps through an inclusivity strategy that is mainstreamed across project activities.

290 The project will address the following needs of recipients:

- Strengthening climate information services through enhancing institutions, coordination mechanisms, policies and financial mechanisms and frameworks. This will sustainably address the needs and priorities of the NMHS and NDMA by leveraging modern technologies, automation, digitisation and establishing a MHEWS. The enhanced capacity of these agencies will meet the needs for enhanced Ibf, access to data and effective and anticipatory actions to safeguard lives, livelihoods and assets against climate risks. The NMHSs and NDMA will be key responsible parties in the respective projects, which directly addresses their needs and priorities.
- Through this project, systems building of national and local actors will be addressed. This can include strengthening the decision-making upon the early warnings received by setting up and equipping the national emergency operations centres, improving logistics and supply chains for disaster preparedness and response along with relevant procurement processes, establishing partnerships/MoUs between NMHSs, NDMA, ministries, agencies and departments responsible for critical climate vulnerable sectors, National Red Cross and Red Crescent Societies and others.
- The project will address the needs of vulnerable groups within communities who are often neglected in the design of MHEWS, their dissemination and for disaster preparedness and planning. These specifically include people with disabilities, elderly, women, youth, cultural minorities, urban and rural communities. In addition, specific attention will be given to women headed households, households with infants, elderly, children and youth.
- Needs of communities exposed to climate-related hazard for timely and accurate early warning that will enable them to take anticipatory actions. Populations living in coastal areas, along floodplains and in mountainous areas are particularly vulnerable to climate-related hazards. The specific climate information needs of women, elderly and disabled people require particular attention considering their increased socio-economic vulnerabilities and relatively weaker representation and more limited access to resources and channels of communication.
- Communities dependent on natural resources, especially those reliant on rain-fed agriculture, pastoralism, forest resources and fisheries are disproportionately affected by climate-related hazards. These communities are often socially and economically more vulnerable and spend a high proportion of their income on subsistence which limits their ability to adapt. Climate change impacts often undermine progress that has been made in poverty reduction in countries with large proportion of such populations, adversely impacting food security and economic growth, particularly in rural areas. This project will help these rural populations reduce these associated risks.
- Pastoral communities practising transhumance, particularly those countries in Sub-Saharan, Eastern and the Horn of Africa, face unique challenges associated with climate change. Pastoral communities are often forced to migrate to wetter areas which are already inhabited in order to adapt to changing conditions and overgrazing.

¹²⁹ Aleksandra Kosanic et al., "An Inclusive Future: Disabled Populations in the Context of Climate and Environmental Change," *Current Opinion in Environmental Sustainability* 55 (April 2022): 101159, <https://doi.org/10.1016/j.cosust.2022.101159>.

This often creates conflicts.¹³⁰ People moved by climate-related events are often treated differently from those displaced by conflict, mostly because climate shocks, such as floods, are seen as temporary and affected people are expected to move back to their original areas. However, the occurrence of climate-related displacements have and will continue to increase. Overall, conflicts are expected to increase over rights and access to water and natural resources.¹³¹ Early warnings about the onset of adverse climatic conditions and disasters can help avert such conflicts by providing timely information to agencies involved in humanitarian work, peace building and development.

- The project will address needs of sectors such as agriculture, tourism, health and water infrastructure for targeted and actionable climate information products to facilitate adaptation planning and action.

291 A summary of the needs of each country is presented below

Antigua and Barbuda

Vulnerability of the country and/or specific vulnerable groups, including gender aspects (for adaptation only)

292 As a SIDS, Antigua and Barbuda is particularly vulnerable to extreme climate events such as hurricanes, droughts and coastal flooding, ranking 85th on the ND-GAIN index. The country's NDC identifies five primary reasons for the country's vulnerability, including: i) inefficient planning and management of the built environment; ii) high costs of repairing damage caused by recurrent extreme climate events; iii) the composition of the country's economy; iv) high population density in the coastal zone; and v) limited availability of freshwater resources.

Economic and social development level of the country and the affected population

293 Climate change is increasing the frequency of high-intensity hurricanes, which will have severe impacts on the country, including loss of life, economic losses and damage to infrastructure. These impacts are further exacerbated by the potential to disrupt key economic activities, including tourism which makes up the largest proportion of the country's GDP (~60%) and accounts for the highest overall investment, at ~40%. Antigua and Barbuda is also heavily reliant on imports of basic supplies, including food, medicine and building materials, increasing vulnerability to disruptions to shipping routes and associated increases in insurance costs during and after extreme events. Fisherfolk are also particularly vulnerable given the limited availability of risk information and forecasts for ocean conditions or environmental impacts of climate change on reef fisheries.

Absence of alternative sources of financing (e.g. fiscal or balance of payments gap that prevents government from addressing the needs of the country; and lack of depth and history in the local capital market)

294 While Antigua and Barbuda invests heavily in building resilience, there capacity to do so is severely constrained by the current economic conditions in the country. Although classified as high-income in terms of per capita GDP, the distribution of wealth is largely skewed by a few high-income persons who have moved to the country from abroad, with much of the local population having much lower income. Moreover, the country's small tax and market base, as well as its high public debt — more than 87% its GDP in 2022 — constrain the government's ability to allocate funding from the national budget for adaptation. External investment is therefore critical to enhance MHEWS and Anticipatory Action measures needed to build climate resilience and preparedness of Antigua and Barbuda's population.

Need for strengthening institutions and implementation capacity

295 The gap assessment conducted for Antigua and Barbuda identified several areas where capacity development and institutional strengthening is needed to enable effective MHEWS and Anticipatory Action. This includes the need for: the revitalization of the linkages between national development organisations and each major stakeholder ministries responsible for key parts of the economy; updates to key policies for disaster management, emergency response, and environmental management; strengthening the institutional and human capacity of ABMS.

Ecuador

¹³⁰ BRACED, "Building Climate Resilience in Fragile Contexts: Key Findings of BRACED Research in South Sudan," 2018.

¹³¹ Christopher B. Field, ed., *Global and Sectoral Aspects*, Climate Change 2014, Impacts, Adaptation, and Vulnerability ; Part A (New York, NY: Cambridge Univ. Press, 2014).

Vulnerability of the Country and/or Specific Vulnerable Groups, Including Gender Aspects (for Adaptation Only)

- 296 Ecuador is highly vulnerable to a range of climate-related hazards, including floods, landslides, and droughts. These events disproportionately affect vulnerable populations, including Indigenous communities, low-income households, and women, who often have less access to resources and information needed for effective disaster preparedness and response. The project's focus on integrating gender-sensitive approaches into EWSs directly addresses these vulnerabilities. Through tailoring climate services and early warning messages to the needs of these groups, the project aims to reduce their exposure to climate risks and enhance their resilience.

Economic and Social Development Level of the Country and the Affected Population

- 297 Ecuador's economic and social development is uneven, with significant disparities between urban and rural areas. Many rural and Indigenous communities rely on agriculture and natural resources, making them particularly susceptible to climate hazards. The MHEWS project will contribute to enhance the resilience of these vulnerable populations by providing timely and accurate early warnings, which can prevent significant economic losses and support sustainable development.

Absence of Alternative Sources of Financing

- 298 Ecuador faces significant fiscal constraints, including a high debt burden and limited access to international capital markets, which hinder the government's ability to independently finance large-scale climate adaptation initiatives. The MHEWS project addresses this gap by establishing a National Climate Finance Framework, which will help mobilise and sustain funding for climate services and disaster preparedness efforts.

Need for Strengthening Institutions and Implementation Capacity

- 299 Ecuador's institutions, while developing, still require significant strengthening to effectively manage and implement comprehensive climate adaptation strategies. The MHEWS project addresses this by enhancing governance structures, improving inter-ministerial and multi-stakeholder coordination, and building technical capacities within key institutions such as the National Meteorological and Hydrological Institute (INAMHI). The project also emphasises the development of a harmonized MEAL strategy, which will improve institutional learning and adaptability, ensuring that Ecuador's climate adaptation efforts are both effective and sustainable over the long term.

Cambodia

Vulnerability of the Country and/or Specific Vulnerable Groups, Including Gender Aspects (for Adaptation Only)

- 300 Cambodia, a country still recovering from decades of conflict, faces significant development challenges and is particularly susceptible to compounding and cascading risks. These risks disproportionately affect vulnerable populations, including rural communities, women, children, and ethnic minorities. Climate change intensifies the frequency and severity of floods, droughts, and storms, exacerbating poverty and food insecurity. Economic downturns and health crises further amplify poverty, unemployment, and inequality, while discrimination based on gender, ethnicity, and disability exacerbates vulnerability to shocks and stresses. Specific examples in Cambodia include rural communities highly vulnerable to climate-related shocks, women and girls facing gender-based discrimination and limited economic opportunities, children affected by displacement and lack of education, and ethnic minorities experiencing discrimination and increased vulnerability to exploitation.

Economic and Social Development Level of the Country and the Affected Population

- 301 Between 2014 and 2021/22, Cambodia made significant progress in reducing poverty, dropping the poverty incidence from 36.7% to 16.6%. However, after the 2019 COVID-19 outbreak and economic shock, approximately 4.5 million people were still classified as falling under ID poor 1 or 2, and at-risk households are in danger of falling into poverty due to economic instabilities, natural hazard-related disasters, and environmental degradation. Cambodia is highly vulnerable to the impacts of climate change, which is projected to exacerbate flooding and drought risks and decrease the country's GDP by as much as 10% by 2050.

Absence of Alternative Sources of Financing

- 302 This GCF project will make a critical difference in its systemic approach to addressing the vital EW/EA value chain gaps, from risk knowledge to early response. Over the years, funding has been ad hoc and has primarily focused on forecasting

and response. Using the newly developed consultative Cambodia EW4All Roadmap, this project will emphasise the linkages and gaps between the four EWS pillars, with a particular emphasis on ‘joined-up’ data management and a new approach to early warning dissemination leading to increasingly effective detection and response.

Need for Strengthening Institutions and Implementation Capacity

- 303 The project aims to enhance Cambodia's ability to collect and utilize weather data to mitigate the impact of disasters and build resilient communities. By establishing a system for providing early warning information to at-risk communities, the project seeks to transform uninformed and unprepared populations into proactive and well-prepared communities. The focus is on addressing gaps in the existing EWS by bringing stakeholders together in a coordinated effort to create a more significant impact.

Chad

Vulnerability of the Country and/or Specific Vulnerable Groups, Including Gender Aspects (for Adaptation Only)

- 304 Chad's extreme vulnerability to the impacts of climate change is reflected in its position on the ND-GAIN index (185): a combination of high poverty, frequent conflicts, and the risk of both droughts and floods have placed Chad as the lowest ranked country overall, with a score of 27.0. In 2024, Chad had an Inform risk index of 7,8, placing it in the “very high” risk class, meaning that the country is at substantial risk of a humanitarian crisis in case of natural or man-made disaster. The gender inequality index is 0.7, indicating pervasive inequality in women's empowerment. With a literacy rate of 28.1% (compared to 57.6% for men), women face challenges in accessing information. The specific vulnerabilities of refugees, returnees, minority groups and people living with disabilities also need to be considered.

Economic and Social Development Level of the Country and the Affected Population

- 305 Chad remains among the poorest and least developed countries in the world, with a 2023/4 Human Development Index ranking of 189 out of 193. The World Bank estimates that 40% of Chadians live in poverty and face severe deprivation of a range of basic needs. The percentage of households with access to electricity is 8% (NAP). Climate change itself is also having a negative impact on the viability of rural production systems including agricultural production, rangeland production capacity and livestock watering possibilities. This has led to increased competition between users of rural areas and a deterioration in social cohesion.

Absence of Alternative Sources of Financing

- 306 Access to climate finance is insufficient to meet the acute challenges posed by climate change. There is a financing shortfall to build climate resilience in Chad, with annual climate finance are estimated at a quarter of annual needs. Recently-approved projects are investing in critical aspects of the EW/EA value chain, however additional financing is needed to ensure that effective, actionable and intelligible communication and dissemination strategies are delivered, to work specifically on institutional governance and coordination challenges, and to deliver a MHEWS that is functional, impact-based, people-centered, and end-to-end.

Need for Strengthening Institutions and Implementation Capacity

- 307 Due to fragility and conflict within the country and beyond its borders, individuals, households and communities in Chad are vulnerable to shocks and climate change. Fragile national and sub-national institutions (e.g. governmental, legal and financial) are responsible for under-investment and weak systems for essential services, disaster and climate risk management, and natural resource management.

Fiji

Vulnerability of the country and/or specific vulnerable groups, including gender aspects

- 308 Fiji faces a high risk from various natural hazards, including tropical cyclones, storm surges, floods, landslides, droughts, extreme temperatures, earthquakes, and tsunamis. The country is particularly vulnerable to the anticipated effects of climate change, such as rising sea levels, more severe tropical cyclones, and heavy rainfall. For Fiji, the probability of catastrophic damage and loss of life from meteorological and geophysical hazards is considered very high. Fiji's extreme vulnerability to natural, human-induced, and technological hazards increases associated risks, particularly due to rising population numbers, poorly regulated land use, and climate change. Although key hazards and risks have been identified and priority issues recognised, they are not integrated into national and sector plans and policies, often resulting in a lack

of preparedness. Agriculture in Fiji, mainly rain-fed and small-scale, supports over 117,000 people, primarily in rural areas. Economic impact does not fully capture its nutritional and livelihood significance. However, agricultural livelihoods are vulnerable to climatic extremes like cyclones, droughts, and floods.

Economic and social development level of the country and the affected population

- 309 Fiji's GDP is around US\$4.98 billion, with a per capita GDP of US\$5,356.2. The economy is mainly driven by tourism and agriculture with 91% of the population living near the coastline. By the end of the century, natural hazards may annually push over 4% of the population into poverty. Cyclones result in loss of life, injuries, property damage, and disease spread. A 100-year river flood in Fiji can lead to 12.5% falling into poverty, with an average annual loss of \$130 million attributed to storm surges and high winds. In Fiji, impoverished remote communities face water security challenges and rely on subsistence fishing. Climate change worsens their situation, impacting future extreme weather events. Recognized for their vulnerability, these populations will suffer more from tropical cyclones and floods. There is a rising risk of disasters and losses in Fiji due to increased vulnerability. Unplanned development rose by 24% between 2007 and 2015, with new developments in hazard-prone coastal areas. Climate impacts such as floods, droughts, coastal erosion, and coral bleaching pose significant threats to Fijian communities, the economy, and infrastructure, potentially resulting in loss of life and displacement. Key sectors of Fiji's economy, including agriculture and tourism, are especially vulnerable to changing climate conditions.

Absence of alternative sources of financing

- 310 Though Fiji has access to many global and regional donors and has been the recipient of numerous projects, either bilateral or multinational, none have focused on establishing an end-to-end Early Warning System (EWS) in Fiji. Most projects have been fragmented, concentrating on one or two pillars, mainly Pillar 2 (strengthening observation) and Pillar 4 (enhancing preparedness and response). This is the first project on EW4All in Fiji, and there are a few other multinational projects in the pipeline or in their early stages that also focus on EWS, which includes Fiji (such as Weather ready and CREWS 2.0). Directly relevant to this project, NDMO is currently leading a US\$500K initiative titled "Mapping, Assessing and Planning for Comprehensive Multi-Hazard Early Warning Capabilities for Fiji." This project aligns well with the EWA4All Fiji component, particularly in the areas of risk management, multi-hazard early warning systems, and capacity building. Consequently, it allocates the entire project budget of US\$500K as co-financing.

Need for strengthening institutions and implementation capacity

- 311 However, despite efforts to improve the quality, reach, and effectiveness of an Early Warning System (EWS), there are still gaps and challenges across four essential pillars in Fiji. These challenges stem from fragmented investments, limited technical capacity, and a lack of coordination among key stakeholders in the EWS. To address these issues, it is necessary to enhance observation equipment and ensure its proper use and maintenance. Additionally, there is a need to improve the accessibility and inclusivity of early warning messages, establish community feedback mechanisms, and integrate traditional knowledge into EWSs. The current early warning system faces challenges due to the absence of a designated WMO Registered Alerting Authority and lack of cell broadcast implementation. The regulatory framework for mobile manufacturers/importers to support cell broadcast is also missing. There is no automated systems in place to mitigate impacts in the case of events with a short time frame. Certain remote areas have poor mobile coverage, hindering receipt of warnings by fishermen at sea. Furthermore, there is no coordination stocktake among different actors working in communities, which can lead to potential overlaps, gaps, and inefficiencies in response efforts. Coordination with other stakeholders, including government agencies, NGOs, and community groups, can be challenging due to differing priorities, mandates, and approaches. It is necessary to conduct studies to understand how people access and interpret early warning messages. The findings from these studies should then be incorporated into message formats and dissemination processes. Additionally, the lack of feedback mechanisms and surveys currently negatively impacts the quality of warnings. Therefore, it would be beneficial to invest further in feedback mechanisms.

Somalia

Vulnerability of the Country and/or Specific Vulnerable Groups, Including Gender Aspects

- 312 Somalia is one of the most climate-vulnerable countries in the world, with a ranking of 178 out of 182 countries on the ND-GAIN Index 2024, and with the least capacity to respond and recover from extreme weather events.¹³² Somalia consistently undergoes prevalent violence, conflict and fragility, which in turn has resulted in (and vice versa) harsh and severe environmental conditions, affecting the depletion, use and access to natural resources and livelihoods. Moreover, Somalia has a large number of IDPs and is one of the highest refugee-producing countries.¹³³

Economic and Social Development Level of the Country and the Affected Population

- 313 Somalia is a low-income country with a fragile economy that relies largely on agriculture, livestock, remittances and international assistance. Desertification is reducing the viability of rain-fed agriculture and livestock production. Unemployment, particularly among youth, is alarmingly high while poverty rates are highest in rural areas.

Absence of Alternative Sources of Financing

- 314 Somalia has faced recurrent humanitarian crises, including droughts, famines, and displacement due to conflict. These crises have exacerbated food insecurity, malnutrition, and displacement, with millions of Somalis reliant on humanitarian assistance for survival. Somalia is considered the world's fifth-poorest country¹³⁴, and with ongoing and regular climate induced disasters, Somalia's early warning and response needs are significant, without the domestic resources to invest effectively in MHEWS and Anticipatory Action.

Need for Strengthening Institutions and Implementation Capacity

- 315 In Somalia, interagency coordination gaps exist alongside overlapping mandates, resulting in poor coordination among governmental agencies and non-governmental organisations as well as a duplication of efforts and inefficient resource utilisation. There is a need for policy development and policy harmonisation, which means creating and enforcing comprehensive climate policies that align with international standards. There is also the need for capacity building by training policymakers and stakeholders to improve implementation and compliance.

Ethiopia

Vulnerability of the country and/or specific vulnerable groups, including gender aspects (for adaptation only)

- 316 Ethiopia is highly vulnerable to a range of climate-related hazards, including recurrent droughts, floods, landslides, and wildfires, which are increasingly exacerbated by climate change. This vulnerability significantly impacts specific populations, particularly women, children, and rural communities that heavily depend on agriculture for their livelihoods. Women of reproductive age face substantial challenges due to systemic inequalities, with over 22% suffering from chronic under-nutrition. Furthermore, more than 1.2 million children under five are affected by acute malnutrition, illustrating the severe repercussions of climate-induced food insecurity on these vulnerable demographics. Addressing these vulnerabilities is crucial for enhancing preparedness through effective EWSs, ensuring that critical information reaches those most at risk, and empowering them to respond proactively to climate-related shocks.

Economic and social development level of the country and the affected population

- 317 Ethiopia, despite notable economic growth over recent years, remains a low-income country with a significant reliance on agriculture, which is particularly susceptible to climate variability. Approximately 80% of the population resides in rural areas where livelihoods are closely tied to rain-fed agriculture. Rural households are often severely impacted by climate shocks, leading to food insecurity, displacement, and heightened poverty rates. This economic vulnerability is compounded by limited access to essential services such as healthcare, education, and financial resources, further exacerbating the challenges faced by these communities. As part of the feasibility preparation for the GCF-funded Multi-Hazard EWSs (MHEWS) project, it is vital to improve resilience and support sustainable economic development by integrating climate risk data into national and regional planning processes, thereby addressing the specific needs of the most affected populations.

¹³² ND-GAIN Index Country Rankings, Somalia. <https://gain-new.crc.nd.edu/country/somalia> [accessed 7 July 2024]

¹³³ <https://www.ifad.org/en/web/operations/w/country/somalia>

¹³⁴ <https://worldpopulationreview.com/country-rankings/poorest-countries-in-the-world>

Absence of alternative sources of financing (e.g. fiscal or balance of payments gap that prevents government from addressing the needs of the country; and lack of depth and history in the local capital market)

- 318 Ethiopia faces substantial fiscal constraints that hinder its ability to effectively address the extensive needs for climate adaptation and disaster risk reduction. The country grapples with a persistent balance of payments gap, which limits the financial resources available for critical climate resilience projects. Additionally, the depth of the local capital market is insufficient to mobilize the necessary investments for large-scale climate initiatives. Consequently, the Ethiopian government heavily relies on external financial support, such as that provided by the Green Climate Fund (GCF), to bridge these funding gaps. This reliance underscores the urgent need for effective investment in EWSs, disaster preparedness, and community resilience initiatives to mitigate the impacts of climate change on vulnerable populations as part of the feasibility preparation process.

Need for strengthening institutions and implementation capacity

- 319 Ethiopia's institutional capacity for managing climate risks and implementing EWSs requires significant strengthening. While key institutions such as the Ethiopian Meteorological Institute (EMI) and the Ethiopian Disaster Risk Management Commission (EDRMC) exist, they often lack the resources, coordination, and technical expertise necessary for effectively managing multi-hazard EWSs. In the context of feasibility preparation for the GCF Early Warning for All project, there is a critical need to support initiatives such as establishing a National Risk Data Hub for climate risk information, modernizing and expanding hydro-meteorological observing stations, and integrating anticipatory action into emergency response and disaster risk reduction programs. Additionally, strengthening or establishing Community-Based EWSs (CEWS) in priority vulnerable communities is essential for ensuring that local populations are better prepared for climate-related events and can respond effectively to emerging risks.

D.5. Country ownership (max. 500 words, approximately 1 page)

- 320 This project will equip countries to establish state of the art EWS which increase the resilience of the most vulnerable communities to climate hazards. It will facilitate a paradigm shift wherein global and regional collaborations and responsive, people centred policies will enable the production and delivery timely, targeted and contextual, community-centred early warnings. These will be coupled with mechanisms for sustainable financial and technical support to NDMA and NMHSs. The project will substantively support the call of the UN Secretary General for EW4All by 2027.
- 321 By focusing on multiple delivery mechanisms, enhancing capacities to be able to deliver support and create a knowledge base through national projects, the project will lead to participating countries developing timely, credible, holistic and easily accessible climate, weather information designed to reach and equip communities with actionable and targeted early warning. It will reduce vulnerability of communities to climate risks by supporting countries to reduce their own exposure through timely alerts and protocols designed to reach the grassroots. Furthermore, the project will establish links between participating countries and international financial and development institutions for sustainable financing and support for adaptation and disaster recovery.
- 322 In each country, the NDA was approached with an idea note of the project to explain what the aim would be, and to present the initial architecture of the project. There were extensive discussions and consultations with the NDA in each country during the FP development. Furthermore, the NDA convened larger consultations with a wide range of stakeholders including CSOs and CBOs, to anchor the project design as much as possible on ground realities. National and regional DAEs were engaged with to establish their willingness to contribute to the design of country specific projects (through both output 1 & 3). Where DAEs are financial institutions, these discussions and those with other lenders were used to ascertain the potential for GCF grant funding to be used to unlock additional loans/lending through de-risking measures.

National priorities, as expressed through NAPs, NDCs, communications to the UNFCCC and the Sendai framework have been used as a basis for discussions and determining of appropriate actions. Further engagement on private sector financing and insurance, gaps in national and regional MHEWS and regional coordination mechanisms were conducted with national and regional agencies, including the Caribbean Community Climate Change Centre (CCCCC), Southern African Development Community (SADC), Secretariat of the Pacific Regional Environment Programme (SPREP), IGAD

Climate Prediction and Applications Centre (ICPAC), Pacific Community (SPC), Asian Disaster Preparedness Center (ADPC), Regional Integrated Multi-Hazard Early Warning System for Africa and Asia (RIMES), etc. Antigua and Barbuda

- 323 The project ensures country ownership by closely aligning with Antigua and Barbuda's national climate strategy and relevant frameworks. By integrating these national commitments and plans, the project not only supports the country's climate goals but also strengthens its institutional and technical capacity for DRM.

Alignment with national climate strategy and frameworks

The project is designed to complement Antigua and Barbuda's NDCs, which focus on enhancing climate resilience and reducing vulnerability to climate-induced hazards. By updating the national disaster management legislation and integrating disaster risk reduction into sectoral policies, the project directly supports the NDCs' objectives of improving adaptive capacity and resilience. The project also supports the implementation of the NAP by addressing key areas such as enhanced risk knowledge, observation systems, and community-based EWSs. These efforts align with the strategic priorities outlined in the NAP, particularly in building adaptive capacity and integrating climate resilience into national and sectoral planning.

Engagement with stakeholders

- 324 The project approach was developed through an extensive multi-stakeholder engagement process, including two stakeholder workshops to identify gaps and implementation actions, culminating in an implementation roadmap for reinforcing a people-centred MHEWS in the country. These multi-stakeholder workshops were followed by multiple bi-lateral engagements with key stakeholders in country, as well as the EW4All regional pillar leads, to drill into the specifics of the various gaps and needs identified, shaping them into an actionable project.
- 325 Civil Society Organisations: The project engages a wide range of stakeholders, including CSOs, by incorporating their input into the planning and implementation phases. Civil society organisations are involved in the development of community-based EWSs and response plans, ensuring that local knowledge and needs are considered. Local communities are also consulted to integrate their traditional knowledge and practices into the disaster management framework, enhancing cultural relevance and effectiveness.
- 326 Women and vulnerable groups: The project prioritises the inclusion of women and other vulnerable groups by specifically addressing their needs in disaster preparedness and response strategies. This includes developing inclusive communication frameworks for early warnings and ensuring that women, persons with disabilities and other vulnerable communities are actively involved in the planning and execution of community-based EWSs. Training and capacity-building activities are designed to empower these groups and enhance their participation in DRM processes.
- 327 By fostering strong partnerships with national institutions, civil society, and local communities, the project ensures that all stakeholders are actively engaged and that their perspectives are integrated into the project's design and execution. This participatory approach not only enhances the relevance and sustainability of the interventions but also strengthens national ownership and commitment to achieving the project's goals.

Chad

- 328 This initiative is fully in line with the priorities of the Government of Chad reflected in its development programs and plans, particularly in terms of disaster and risk management programs. This project supports the Government's National Development Plan 2024-2028 which is undergoing validation for adoption, and the objectives of the 2030 vision, specifically under "Pillar 4: Improving the quality of life of the Chadian population". The project is also in line with Priority #10 "Climate Risk Management" under the First National Adaptation Plan of Action and in line with the updated National Determined Contributions (October 2021).
- 329 This project will ensure that gender issues are addressed equitably across all its components by being aligned with the National Policy on Gender for 2011-2020, which aims at the following: "By 2020, Chad is a country free from all forms of gender inequalities and inequities, from all forms of violence, where men and women have equal opportunities to access and control resources and participate equitably in decision-making processes in the interests of achieving sustainable development."

Cambodia

330 Cambodia published an updated NDC in 2020¹³⁵. The project has developed various activities to address the NDAs EWS priorities. These activities are aligned with specific pillars and coordination efforts. The proposed activities include establishing a standard information management system through PRISM, implementing a cross-pillar approach with a focus on gender and inclusion in EWS/EA, improving sectoral EWS in agriculture and health, reinstating the multi-sectoral monsoon forum, and focusing on IBF in Capacity Development activities for the MoWRAM. Additionally, the project aims to involve youth in developing community-based EWS, improve public knowledge of climate change risks and EWS, and build the capacities of the NDCM and the NAP-DRR at national and sub-national levels. Overall, the activities are designed to address the prioritized NDC EWS needs and enhance coordination across different pillars.

Ecuador

- 331 The MHEWS project aligns seamlessly with Ecuador's national climate strategy, GCF Country Programme, and existing policies like NDCs, NAMAs, and NAPs. It aims to improve EWSs, enhancing climate resilience in vulnerable sectors.
- 332 The project will be implemented by accredited entities with strong track records in climate resilience and DRM, such as the INAMHI. These institutions are being further strengthened through the project's capacity-building components, ensuring they have the technical expertise and institutional capacity to deliver on the project's objectives. By enhancing the capabilities of these entities, the project contributes to the long-term sustainability and effectiveness of Ecuador's climate adaptation efforts.
- 333 The NDA plays a pivotal role in coordinating and endorsing projects that align with the country's climate priorities. For the MHEWS project, the NDA will ensure that the project is integrated into national planning processes and receives the necessary governmental support and coordination. This ensures that the project is aligned with national climate strategies and receives the political backing needed for successful implementation.
- 334 The MHEWS project emphasizes inclusive stakeholder engagement, ensuring that civil society organizations, Indigenous peoples, women, and other vulnerable groups are actively involved in the project's design and implementation. This approach ensures that the project addresses the specific needs and vulnerabilities of these groups, making the EWSs more effective and equitable. The integration of IKS and gender-sensitive approaches also ensures that the project is culturally relevant and socially inclusive, enhancing its overall impact on community resilience. in feedback mechanisms

Somalia

- 335 The proposed project activities for Somalia directly respond to the needs outlined in the Roadmap for Implementation of the EW4ALL Initiative 2024-2027 in Somalia. The National Multi-Hazard Early Warning Centre (NMHEWC) was established to provide effective and timely warnings of possible risks posed by hazards. However, the country lacks a robust EWS infrastructure and an enabling policy and institutional environment for the dissemination of warnings. The Roadmap for EW4All builds on the country's existing National MHEWS. It is also concurrent with Somalia's commitment to the Sendai Framework for Disaster Risk Reduction (2015–2030). In this context, the SoDMA has called for the national rollout of the EW4All initiative, finalising gap analysis, and agreeing on a roadmap to consolidate and grow coverage of MHEWS to empower and enable early action by individuals and communities, especially those most in need. In collaboration with the UNDP Somalia country office, the EW4ALL initiative is now operationalised under the leadership of the NMHEWC.
- 336 The activities are also represented in Somalia's NDRMP, and the Roadmap development process has been accompanied by a systematic reform initiative aimed at aligning it to Article 3, para 6 and Article 7 para 2, of the establishment Law 17/2016 of the SoDMA. These Articles state the clear responsibility of SODMA to gather information and report on disasters and the risks facing the country and send out early warnings in collaboration with organizations inside the country and abroad working on those activities. Aside from implementing the interventions also link with Somalia's 2013 NAPA which includes an adaptation strategy and risk mitigation plan.¹³⁶, Somalia's NAP Framework¹³⁷, the 2021 Initial

¹³⁵ https://ncsd.moe.gov.kh/sites/default/files/2021-01/20201231_NDC_Update_Cambodia.pdf

¹³⁶ Federal republic of Somalia National Adaptation Programme of Action on Climate Change (NAPA) April 2013

¹³⁷ Somalia's National Adaptation Plan (NAP) Framework, 2022, <https://nappglobalnetwork.org/wp-content/uploads/2022/11/napgn-en-2022-somalia-nap-framework.pdf>

Nationally Determined Contributions (INDC)¹³⁸ Report to the UNFCCC¹³⁹, draft 2021 National Climate Change Policy¹⁴⁰ which provides framework for addressing climate change impacts, and the 2018 Initial National Communication (INC)¹⁴¹ to the UNFCCC vulnerability assessments, and adaptation strategies.

Fiji

- 337 The Fiji National Climate Change Policy (NCCP) 2018-2030 is the instrument to protect Fiji's development priorities from current, future, and intergenerational climate change risks. It also aims to anchor Fiji's national climate change response and nationally determined commitment under the Paris agreement within national policy and planning processes. The policy identifies and utilises interlinked and related priorities in DRR and climate change adaptation to allow for the advancement of integrated actions that effectively address both fast and slow onset hazard events, as well as long-term climate impacts and risks.
- 338 All community service organizations involved with the EWS, including church groups, women's groups, and people with disabilities and other minorities, fall under the umbrella of the Fiji Council of Social Services (FCOSS). The FCOSS office coordinates with the government whenever there is a disaster, and the FCOSS director is a member of the NDRM Council. The importance of early warning to these CSOs is to assist in planning for response and recovery purposes. Therefore, accurate and timely early warning is crucial for these organizations.

Ethiopia

Alignment with national climate strategy and frameworks

- 339 The GCF Early Warning for All project aligns strategically with Ethiopia's national climate and disaster management strategies, reinforcing essential frameworks such as the NDRMP and the Climate Resilient Green Economy (CRGE) strategy. Specifically, the Roadmap for Multi-Hazard, Impact-Based Early Warning and Early Action System (2023–2030) provides a comprehensive plan for implementing a robust EWS that integrates IbF and anticipatory actions. The National Framework for Climate Services, alongside the Strategic Programme and Investment Framework (SPIF), outlines priority actions and investments needed to enhance Ethiopia's DRM system. Additionally, the project supports Ethiopia's updated Nationally Determined Contributions (NDC), emphasizing the integration of climate resilience measures into development initiatives. As part of feasibility preparation, the project reinforces Ethiopia's commitments to sustainable development and climate adaptation by embedding EWSs into broader national and regional strategies.

Engagement with national stakeholders

- 340 The feasibility preparation for the GCF-funded Early Warning for All project has involved extensive engagement with national stakeholders, coordinated through the Emergency Coordination Center (ECC) led by the EDRMC. The EW4All Coordination Platform serves as an integrated framework to strengthen national EWSs by coordinating actions across four key pillars: Risk Knowledge, Monitoring and Forecasting, Communication and Dissemination, and Response Capability. This platform is designed to align stakeholders at all levels—government agencies, civil society organizations, international partners, and local communities—ensuring that EWSs are effective, inclusive, and sensitive to the local socio-political context. Given Ethiopia's complex landscape of socio-economic challenges, conflict dynamics, environmental risks, and gender inequality, the EW4All platform will adopt a multi-sectoral and context-specific approach, ensuring that early warning mechanisms are not only operationally effective but also conflict-sensitive and gender-responsive.

D.6. Efficiency and effectiveness (max` . 500 words, approximately 1 page)

¹³⁸ The Federal Republic of Somalia updated Nationally Determined Contribution (NDC), July 2021

¹³⁹ The Federal Republic of Somalia Ministry of Environment and climate change Somalia's first biennial update report under the United Nations Framework Convention on climate change (UNFCCC).

¹⁴⁰ The Federal Republic of Somalia Ministry of Environment and Climate Change Somalia National Climate Change Policy, <https://moecc.gov.so/wp-content/uploads/2023/07/Somalia-National-Climate-Change-Policy.pdf>

¹⁴¹ <https://moecc.gov.so/initial-national-communication-inc/> . <https://moecc.gov.so/2023/07/24/national-communication/>

- 341 This project will ensure a cost-effective alternative to reactive approaches to climate-related hazards that focus on ad-hoc recovery and investment. It will significantly enhance the risk knowledge and early action capabilities of sectors and communities and will contribute to the capacity building of relevant stakeholders. The enhanced risk knowledge will enable stakeholders, particularly local communities and first responders, to take pro-active action based on timely and accurate information, thereby reducing the impacts of disasters on lives, livelihoods and assets. The project will ensure a larger spatial coverage of ibF for vulnerable regions and establish protocols, supported by national policies and procedures, for communities to prepare for and take pre-emptive action against disasters.
- 342 A high expected benefit-cost ratio (BCR) on investments in strengthening the hydrological and meteorological agencies and all partners in the MHEWS value cycle can only be achieved with a sizeable and multi-year investment which addresses gaps in the value cycle and ensures that actionable warnings reach the last mile and those at risk being able to respond. Improvements in MHEWS and preparedness make it possible to limit losses from hydrometeorological hazard-induced disasters, which would not be possible without the informed use of constantly improving meteorological, hydrological, social, behavioural and related information, which this programme will seek to do sustainably by leveraging additional co-finance, as well as ensuring country ownership by governments that are able to fund the ongoing costs of maintaining all elements of the MHEWS cycle.¹⁴²
- 343 MHEWS are a public good and for under-developed countries and SIDS, the financial and technical barriers that prevent the implementation of a reliable, robust, widely accessible and effective MHEWS for disaster preparedness and planning can only be removed by public investment financed by grant resources from international development institutions. The project does not entail revenue generation or cost recovery during its duration and cannot be expected to attract investments from the private sector, which is nascent in many of the targeted counties. However, the project does plan to engage the private sector by stimulating demand for climate information-based value-added products and services, by engaging with the private sector particularly in telecommunications, agri-businesses, tourism and transport – with a focus on aviation and financial services such as insurance. The level of engagement and the type of industry will be determined by the national context. These engagements may generate some revenue and cost sharing or co-financing in the future and could support the project's financial sustainability beyond its proposed duration.
- 344 On the other hand, there is overwhelming evidence of substantial benefits of investing in weather and climate information and DRR. Reported benefit to cost ratios are four and higher¹⁴³, the socioeconomic benefits are estimated to be at least US\$ 160 billion per year¹⁴⁴ and the net benefit of investing in the resilience of infrastructure in developing countries would be \$4.2 trillion over the lifetime of new infrastructure¹⁴⁵. The return on investment from DRR offer “triple dividends” from avoiding direct impact, enhancing economic potential and generating sustainable development co-benefits.¹⁴⁶

Efficiency and effectiveness

- 345 Efficiency and effectiveness will be ensured by building on existing national capacities and infrastructure across the MHEWS value chain and leveraging investments from other donors, particularly from SOFF and CREWS. Existing institutional capacities and linkages will be effectively used at all levels across government, non-government, private sector – particularly telecommunications and among community based organizations. This will minimize time and costs and provide a foundation to strengthen and enhance MHEWS to the community level.

¹⁴² WMO, “Valuing Weather and Climate: Economic Assessment of Meteorological and Hydrological Services” (World Meteorological Organization, 2015).

¹⁴³ United Nations International Strategy for Disaster Reduction (UNISDR). ‘Costs and Benefits of Disaster Risk Reduction’. High Level Dialogue. Geneva, Switzerland, 10 May 2007; World Bank. ‘World Development Report 2014: Risk and Opportunity - Managing Risk for Development’, 2013. World.; WMO. ‘Valuing Weather and Climate: Economic Assessment of Meteorological and Hydrological Services’. World Meteorological Organization, 2015.

¹⁴⁴ SOFF. ‘The Value of Surface-Based Meteorological Observation Data: Costs and Benefits of the Global Basic Observing Network’. Information Brief. Systematic Observations Financing Facility, 2020.

¹⁴⁵ Stéphane Hallegatte, Jun Rentschler, and Julie Rozenberg. ‘Lifelines: The Resilient Infrastructure Opportunity’. World Bank Group, 2019.

¹⁴⁶ Thomas, Tanner, Swenja Surminski, Emily Wilkinson, Robert Reid, Jun Rentschler, and Sumati Rajput. ‘The Triple Dividend of Resilience: Realising Development Goals Through the Multiple Benefits of Disaster Risk Management’. Overseas Development Institute (ODI), International Bank for Reconstruction and Development / International Development Association or The World Bank., 2015.

- 346 Synergies between ongoing government initiatives and projects, including those supported through vertical funds will be used to co-finance and share costs. Coordination with ongoing initiatives will prevent duplication of effort while optimizing investments and operation and maintenance of observation infrastructure. Technical support from regional facilities, multilateral agencies and the private sector will be used to minimise costs of analysis, technology transfer, training and data acquisition. Costs of hiring technical staff and their training and capacity building within NMHS, NDMA and other relevant partners will also be reduced through collaborations with regional and national technical and research agencies.
- 347 The choice of technologies for observation, networking and dissemination for each country are governed by a scientific gap analysis and consideration of operation and maintenance costs and capacities. An optimal mix of technologies will be used, including gridded and remotely sensed datasets from other agencies to complement and supplement expensive observation infrastructure. Conformance to WMO and international public standards for data and communication and the use of open-source software and IoT devices for local observations will further ensure inter-operability and reduce costs. The development of support networks comprising regional and national academic institutions and centres of expertise is expected to be more cost effective and ensure longer term sustainability. By working with NDAs, DAEs and regional coordinating bodies (SPC, SPREP, SADC, IGAD, CCCCC, CAF etc) investments will be able to target the most critical gaps and ensure the full MHEWS cycle is achieved, reaching as many people, businesses and communities as possible.

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

- ☐ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Persistent and widening gaps between climate risks and support to countries that are highly sensitive to climate change prevents the establishment and management of MHEWS for early and anticipatory action, perpetuating the vulnerability of 3.3 to 3.6 billion people living in these countries.	<u>Low</u>	Paradigm shift would imply the establishment of 1) increased community resilience to climate hazards through well-coordinated, responsive and inclusive disaster adaptation and preparedness through early and anticipatory action which is informed by an effective end-to-end MHEWS; and 2) mechanisms for long term support to the entire EWS value chain, both technical and financial, which raise and leverage resources from both national as well as regional, multilateral and international agencies, including private corporations.	This project is based on an unprecedented level of cooperation and collaboration among key actors across the EWS value chain, from global to regional, national and sub-national levels. The project establishes a global support architecture and sustainable mechanism to support countries both technically and financially in rolling out a comprehensive EW4All value chain through: i) robust scientific information and analytics on weather, water and climate hazards and vulnerabilities within NMHS; ii) optimised internal coordination among various government entities that are responsible for different elements of managing a national EWS; and iii) a long-term financing plan to maintain the necessary hydrometeorological infrastructure, and community level capacities to receive, understand and respond to EWS.

Replicability	Investments in developing countries in the CIEWS sector are typically one-off and project-based. Technological innovations and gains and lessons in institutional collaboration and successes at the community level remain in isolated silos, with little opportunity for cross learning. There are limited mechanisms or opportunities for replicating lessons learned or scaling up these projects to other countries or even across sectors and regions in the same country.	<u>Low</u>	The establishment of selected IbF schemes and Anticipatory Action initiatives have a high potential for replication within and across the target countries. There is a substantial potential for replicating innovations. This includes i) the use of emerging technologies such as cloud computing, AI and remotely sensed acquisition of observations; ii) IoT and mobile ICT devices for community based EWS, and communication protocols and technologies for targeted dissemination; but perhaps more importantly iii) in terms of multi-institutional coordination and collaboration for technical and financial support, and support for on-ground implementation of interventions, such as in fragile and conflict affected states.	The project will make comprehensive country-level investments in select countries to ensure the roll-out of EW4All. Regional and international mechanisms that are being put in place provide the framework for continuous cross learning and collaboration as well as for rapid scale-up and replication to other countries.
Sustainability	Many developing countries struggle with establishing and operating a broad based, end-to-end MHEWS to inform adaptation to climate hazards. There is a gap in institutional structures and resources required for a broad based, community centered system for disaster preparedness and planning which makes effective use of the MHEWS through early and anticipatory action. MHEWS is fundamentally a public service and resources for its operation, maintenance and expansion to vulnerable communities need to be raised from scarce public funds for which political will is often lacking in poor and developing countries.	<u>Low</u>	The targeted scenario would involve long term regional and global support for country level actions towards the EW4All goals, including countries with fiscal and technical constraints and those facing challenges due to insecurity. Long term and continuous investments will be ensured through mechanisms such as SOFF and will result in improved hydrometeorological infrastructure, its O&M and strengthened human and institutional capacities of NMHS and NDMA for generation and international sharing of weather and climate data. Strengthened institutional coordination and collaboration will result in effective use of climate information to trigger and support decisions for anticipatory action. Deliberate and detailed impact assessments will lead to increased awareness and knowledge targeted at increased political commitment coupled with institutional mechanisms that ensure community involvement in disaster preparedness and planning.	The project will create increased opportunities for regional cooperation and up-scaling of climate resilient sustainable development. It will link sectoral and technical government entities, science institutions, civil society and other stakeholders to promote anticipatory actions. This will contribute to an enabling environment where climate risk information is effectively produced and disseminated, in a sustainable fashion, to the most vulnerable groups to safeguard their lives, livelihoods and assets in the long-term. Legal frameworks for DRM and protocols for inter-agency coordination and involvement of local institutions will be strengthened along with regional and international level collaborations and partnerships.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ¹⁴⁷	
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Direct: Surveys and feedback forms, including USSD and electronic means, statistics from mobile network operator and radio call-in statistics. Indirect: Radio, cell phone network coverage and population statistics from national statistical authorities or other recognized data sources.	Ecuador: 0/0 Cambodia: 0/0 Antigua and Barbuda: 0/0 Fiji: 0/0 Ethiopia: 0/0 Chad: 0/0 Somalia: 0/0	Ecuador : 128475 (62592, 65883)/ 3500000 (1705169, 1794831) Antigua and Barbuda : 22875 (10928, 11947)/ 28500 (13615, 14885) Cambodia : 128524.5 (63581, 64944)/ 538257 (266276, 271981) Fiji : 208,141 (106,152/101,989)/ 244,154 (124,519/119,635) Ethiopia : 4000000 (2008800, 1991200)/ 5400000 (2711880, 2688120) Chad : 425000 (210354,	Ecuador : 513900 (250367, 263533)/ 14000000 (6820675, 7179325) Antigua and Barbuda : 91500 (43712, 47788)/ 114000 (54461, 59539) Cambodia : 514098 (254324, 259774)/ 2153028 (1065103, 1087925) Fiji : 416282 (211217, 205065)/ 488308 (247762, 240546) Ethiopia : 16000000 (8035200, 7964800)/ 21600000 (10847520, 10752480) Chad : 1700000	Population estimates may be revised for some countries. Agencies, including private mobile network operators, may be unwilling to share statistics. Field based assessments and surveys may not be possible in some inaccessible or insecure areas.

¹⁴⁷ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

				214647)/ 1275000 (631061, 643940) Somalia : 292500 (145070, 147430)/ 1950000 (967137, 982863)	(841414, 858586)/ 5100000 (2524242, 2575758) Somalia : 1170000 (580282, 589718)/ 7800000 (3868548, 3931452	
ARA1 Most vulnerable people and communities	Supplementary 2.4: Beneficiaries (female/male) covered by new or improved early warning systems	Direct: Surveys and feedback forms, including USSD and electronic means, statistics from mobile network operator and radio call-in statistics. Indirect: Radio, cell phone network coverage and population statistics from national statistical authorities or other recognized data sources.	To be provided in first APR	Ecuador : 256950 (125184, 131767) Antigua and Barbuda : 45750 (21856, 23894) Cambodia : 257049 (127162, 129887) Fiji : 208,141 (106,152/101,989) Ethiopia : 8000000 (4017600, 3982400) Chad : 850000 (420707, 429293) Somalia : 146250 (72535, 73715)	Ecuador : 436815 (212812,224003) Antigua and Barbuda : 77775 (37155,40620) Cambodia : 436983 (216175,220808) Fiji : 353840 (179534,174305) Ethiopia : 13600000 (6829920,6770080) Chad : 1445000 (715202,729798) Somalia : 994500 (493240,501260)	Population estimates may be revised. Agencies, including private mobile network operators, may be unwilling to share statistics. Field based assessments and surveys may not be possible in some inaccessible or insecure areas. Assumption: 50% of the direct beneficiaries for mid-term and 85% of direct beneficiaries in long term
ARA1 Most vulnerable people and communities	Supplementary 2.7: Change in expected losses of lives due to the impact of extreme climate-related disasters in the geographic area of the GCF intervention	Registers maintained by relevant authorities and agencies involved in disaster response, field based sample surveys	Missing baselines will be provided in first APR Ecuador: 220:	Ecuador: 60% reduction Cambodia: 60% Antigua and Barbuda: 10% Fiji:80% Ethiopia: 20% Chad: 20%	Ecuador: 90% reduction Cambodia:90% Antigua and Barbuda:30% reduction Fiji: 95% Ethiopia: 50%	Baseline figures will be updated by participating countries in the first APR. Agencies may be unwilling to share data Data may be inaccurate Field based assessments and surveys may not be possible in some inaccessible or insecure areas.

				Somalia: 20%	Chad: 50% Somalia: 50%	

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Most participating countries have significant challenges in governance, institutional collaboration and coordination as well as policy frameworks for the delivery of comprehensive and effective MHEWS. These challenges are particularly acute in fragile and conflict affected states such Somalia, Chad and Ethiopia.	Low	Strengthened governance and coordination amongst relevant actors, including government agencies, private sector, community and faith based organisations for efficient exchange of information and effective dissemination of warnings for early and anticipatory action .	Country specific policies, road-maps and strategies will be strengthened and developed through a process of consultation with relevant stakeholders. Roles, procedures and protocols for information sharing and dissemination will be established and mechanisms instituted for coordination between agencies involved in the generation of EWS and disaster information, its dissemination and its use by vulnerable communities.	Multi-countries
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	Most countries have rudimentary or dysfunctional observation and communication systems, inadequate analysis of climate information, uncoordinated dissemination and weak institutional mechanisms and capacities for on ground disaster preparedness and planning. They usually lack formal collaborations which	low	Observation and dissemination systems optimized to local conditions that leverage technologies and technical collaborations for long term investments and sustained technical support . Coordinated early and anticipatory actions down to the level of communities, adopting best practices in terms of institutional	Formal linkages of the NMHS and NDMA with multiple sources for long term technical and financial support including local, regional and international research agencies, multilateral agencies and international corporations including those involved in technical innovations in AI/machine learning, cloud computing, remote sensing	Multi-countries

	would facilitate technology transfer and build local capacities in these diverse areas where there is rapid innovation and development at the global level.		mechanism and protocols for disaster preparedness and the use of IbFs and triggers .	and low cost citizen based EWS.	
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	Investments in CIEWS have typically been one-off with limited effort on knowledge generation and learning from earlier experiences. External consultants and firms are typically engaged for analysis and development of models with no mechanisms to develop in-house expertise or sharing and replicating the learnings.	low	Systematic and deliberate documentation, sharing of knowledge and lessons and building of technical capacities in-house within NMHS and NDMA . Establishment of formal partnerships and collaborations with academia and facilities at the national, regional and global level to ensure long term technical support, development and innovation .	Increased capability for the production, access to and application of quality, timely and contextually relevant risk information at the national, sub-national and local levels. Improved capacities of relevant agencies in data access, metrics and analysis to support decision-making, and learning and incorporation of risk knowledge into EWS. Creating enabling conditions for better integration and use of existing capacities of relevant ministries and entities with the work of NDMA's and NMHS. Raising awareness on crucial intersections of MHEWS with gender inequality and social exclusion. Strengthened collaboration and risk knowledge capability between key ministries, academia and the private sector, including at the community level. Combining local, traditional, Indigenous, generational and scientific knowledge to enable resilience under a range of future risk scenarios. Strengthen locally led, people-centred and citizen science-based and innovative approaches to leverage new and existing technologies to	Multi-countries

				expand risk knowledge capability at all scales, including continuous monitoring / crowdsourcing of important vulnerability metrics as a basis for responding to and anticipating emerging crises and disasters. Improved monitoring, evaluation and learning systems integrated within project and national structures to provide up to date status of activities and emerging risks and measuring effectiveness of EWS to promote adaptive approaches to the implementation of MHEWS.	
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E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Sustained support for global coordination and support architecture for EW4All	Each region has at least one technical and one financial mechanism in place for sustained support to countries implementing the EW4All MCP	Agreements, conventions, letters of support.	0	0	1 Technical and 1 Financial mechanism in place	Assumptions: Participating countries agree to collaborations, joint programmes and road-maps with the regional agencies/ consortia. Data sources: Global and regional offices of respective coordination agencies/consortia
Output 1.1. Enhancing enabling conditions globally	Global (1) and regional (4) structures for	Agreements and resolutions affirming the formation and	0	1, 0	1, 4	Assumption: Participating countries agree to

and regionally to promote country-level actions towards the EW4All goals	coordination of technical and financial support for EW4All, built on existing mechanisms in the region.	formalization of the global and regional support structures				collaborations, joint programmes and road-maps with the regional agencies/ consortia. Data sources: Global and regional offices of respective coordination agencies/consortia; EW4All Financial Observatory.
Outcome 2: End-to-end EWS that is inclusive and builds community resilience to climate hazards	Improved resilience of communities in seven countries to climate hazards. Population (m, f) in targeted sites supported by end-to-end EWS.	Impact surveys and assessments including disaster losses and damages reports	0	Ecuador : 128475 (62592, 65883) Antigua and Barbuda : 22875 (10928, 11947) Cambodia : 128524.5 (63581, 64944) Fiji : 208,141 (106,152/101,989) Ethiopia : 4000000 (2008800, 1991200) Chad : 300000 (148485, 151515) Somalia : Somalia : 2,600,000 (1296369/1303631)	Ecuador : 513900 (250367, 263533) Antigua and Barbuda : 91500 (43712, 47788) Cambodia : 514098 (254324, 259774) Fiji : 416282 (211217, 205065) Ethiopia : 16000000 (8035200, 7964800) 1200000 (593939, 606061) Somalia: 7,800,000 (3889080/3910920)	Assumption: Full participation of relevant countries in project implementation Data sources: Govt. agencies (NDMA, NMHS, statistical bureaus), sub-national line and extension agencies, telecommunication companies, local DRM committees, communities
Output 2.1: Accelerating actions in select countries towards EW4All	Number of countries with operational end-to-end EWS	CHDs Country disaster risk assessments and common	0	3	7	Assumption: Full participation of relevant countries in project implementation Data sources: Govt. agencies

		country analysis reports				(NDMA, NMHS, statistical bureaus), sub-national line and extension agencies, telecommunication companies, local DRM committees, communities
Project/programme co-benefit indicators						
Co-benefit 1: Improved livelihoods and food security of communities dependent on climate-sensitive natural resources.	Number of people dependent on natural resources (including agriculture) provided with climate information to support adaptation and resilience building.	Baseline surveys at start, mid point and end of project	0	50% of NR dependent communities in project areas	85% of NR dependent communities in project areas	Assumptions: Security environment and logistics conducive to field work for impact assessments and surveys Data sources: field surveys using a mix of methods Government records
Co-benefit 2: Improved management and conservation of the environment and critical ecosystems.	Area under critical ecosystems conserved and managed using climate information services for decision support.	Management plans utilizing climate information products based on project interventions	# management plans to be provided by country projects	# management plans to be provided by country projects	# management plans to be provided by country projects	Assumptions: The concerned country has resources dedicated to management and conservation of critical ecosystems Data sources: Relevant ministries and departments
Co-benefit 3: Better design and planning of settlements and infrastructure.	Population living in settlements benefiting from design and planning of infrastructure informed by climate information generated by the project.	Infrastructure design and town planning documents utilizing climate information products generated by the project	# settlements to be provided by country projects	# settlements to be provided by country projects	# settlements to be provided by country projects	Assumptions: The concerned country has operational processes and procedures in place for design and planning of settlements Data sources: Relevant ministries and departments

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables

Activity 1.1: Enhanced Coordination and Technical Support for EW4All Implementation	Enhancing coordination and providing comprehensive joint technical support to ensure the effective implementation of the Early Warnings for All (EW4All) initiative	Sub- Activity 1.1.1: Develop normative guidance and tools across all four pillars & at the inter-pillar level Sub-Activity 1.1.2: Accelerate Technical Support to Countries for EWS implementation	1.1.1: Guidance and tools developed for four pillars and at the inter-pillar level. 1.1.2: Technical support to 7 EW4All countries
Activity 1.2: Strengthened M&E to support EW4All Implementation	Enhanced and expanded global M&E framework for EW4All, including the EW4All Maturity index, a regularly updated dashboard to track progress of the initiative globally and at country levels across all four pillars	Sub-Activity 1.2.1: Develop and operationalise the EW4All maturity index Sub-Activity 1.2.2: Strengthen global EW4All M&E and reporting to enable regular progress updates Sub-Activity 1.2.3: Regular update of EW4All Dashboard	1.2.1: EW4All maturity index developed and operationalized 1.2.2: EW4All M&E global report and reporting for 7 countries 1.2.3: EW4All dashboard updated annually at global/regional level and periodically for 7 countries
Activity 1.3: Strengthened knowledge management and lessons learned	Support knowledge exchange among countries at different stages of implementation of the EW4All EAP across all Pillars and implementing partners	Sub-Activity 2.3.1: Development of knowledge management framework that guides learning and exchange Sub-Activity 1.3.2: Documentation and dissemination of good practices and lessons learned Sub-Activity 1.3.3: Development of global and regional technical expert working groups	1.3.1: Knowledge management framework (one report) 1.3.2: Reports on good practices and lessons from seven EW4All countries (one report) 1.3.3: One global and four regional technical working groups formed
Activity 2.1: Strengthened Governance and Coordination	Facilitate coordinated dissemination of warnings between different actors, including strengthening of policies	Sub-Activity 2.1.1: Strengthened coordination mechanisms Sub-Activity 2.1.2: Enhanced governance capacities Sub-Activity 2.1.3: Monitoring, Evaluation and Learning	2.1.1: Formal coordination mechanisms (SOPs, protocols) developed for seven countries. 2.1.2: Governance arrangements for seven countries enhanced through capacity development of relevant institutions 2.1.3: MEAL systems established in seven countries and at the global PMU level
Activity 2.2: Disaster risk knowledge and observations	Systematic collection of risk data, information and analysis leading to improved awareness about hazards, exposure populations, services, infrastructure, vulnerabilities of communities, and improved loss and damage accounting	Sub-Activity 2.2.1: Develop disaster risk knowledge systems Sub-Activity 2.2.2: Broaden observation and monitoring networks Sub-Activity 2.2.3: Strengthen MHEWS forecasting capacity	2.2.1: Seven countries establish disaster risk knowledge system (dashboards, data-bases) 2.2.2: Gaps in climate/weather observation and monitoring networks addressed in seven countries 2.2.3: Capacities of seven countries enhanced in IbF and trigger development for early/anticipatory action
Activity 2.3: Warning dissemination and preparedness	Strengthening and/or establishment of meteorological, hydrological and climate forecast and monitoring systems for each participating country	Sub-Activity 2.3.1: Strengthen early warning dissemination systems Sub-Activity 2.3.2: Develop preparedness and anticipatory action protocols	2.3.1: Seven countries with enhanced dissemination systems geared towards reaching communities and first responders 2.3.2: Contextual SoPs and protocols for

		Sub-Activity 2.3.3: Strengthen disaster response capacity	early/anticipatory action developed for seven countries 2.3.3: Institutions and communities in seven countries trained for effective disaster preparedness and response
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E.7. Monitoring, reporting and evaluation arrangements

- 348 Monitoring and evaluation will be at two levels, at the global project level and at the country level in each of the seven countries. The global MEAL system will feed into overall project management and oversight, reporting as well as knowledge management. It will prompt the global teams of experts to support national projects and to ensure a continuous, two-way flow of information. The national MEL system will support adaptive management, reporting and knowledge sharing at the national level.
- 349 Project-level monitoring and evaluation at both levels, will be undertaken in compliance with the UNDP POPP and the UNDP Evaluation Policy. The primary responsibility for day-to-day project monitoring and implementation rests with the respective Project Managers. The Project Managers will develop annual work plans to ensure the efficient implementation of the project. The Project Managers will inform their respective Project Boards and the UNDP Country Offices of any delays or difficulties during implementation, including the implementation of the M&E plan, so that the appropriate support and corrective measures can be adopted. The Project Managers will also ensure that all project staff maintain a high level of transparency, responsibility and accountability in monitoring and reporting project results.
- 350 The UNDP Bureau for Policy and Programme Support (BPPS) will support the global Project Manager through the Technical Specialist on Climate Change Adaptation through regular meetings and oversight. The UNDP Country Office will support their respective Project Managers as needed, including through annual supervision missions.
- 351 The BPPS and the respective UNDP Country Offices are responsible for complying with UNDP project-level M&E requirements as outlined in the UNDP Programme and Operations Policies and Procedures (POPP). Additional M&E, implementation quality assurance, and troubleshooting support will be provided by the UNDP Regional Technical Advisor as needed. The project target groups and stakeholders including the NDA Focal Point will be involved as much as possible in national project-level M&E.
- 352 A project inception workshop will be held globally and at each country after the UNDP project document has been signed by all relevant parties to: a) re-orient project stakeholders to the project strategy and discuss any changes in the overall context that influence project implementation; b) discuss the roles and responsibilities of the project team, including reporting and communication lines and conflict resolution mechanisms; c) review the results framework, re-assess baselines as needed, and discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E plan; d) review financial reporting procedures and mandatory requirements, and agree on the arrangements for the annual audit; e) plan and schedule Project Board meetings and finalize the first year annual work plan. The Project Manager will prepare the inception report no later than one month after the inception workshop. The final inception report will be cleared by the BPPS and the Technical Specialist on Climate Change Adaptation at the global level and the respective UNDP Country Offices and the UNDP Regional Technical Advisor at the national level. These will be approved by the respective Project Boards.
- 353 A project implementation report will be prepared for each year of project implementation at both the global and national levels. The respective Project Managers, the BPPS and respective UNDP Country Office, and the UNDP Technical Specialist Climate Change Adaptation and Regional Technical Advisor will provide objective input to the Annual Performance Report (APR).
- 354 The global and national Project Managers will ensure that the indicators included in the project results framework are monitored annually well in advance of the APR submission deadline and will objectively report progress in the Development Objective tab of the APR. The annual APR will be shared with the respective Project Board and other stakeholders. The UNDP Country Offices will coordinate the input of the NDA Focal Point and other stakeholders to the APR. The quality rating of the previous year's APR will be used to inform the preparation of the next APR. The final project APR, along with the terminal evaluation report and corresponding management response, will serve as the final project report package. The national APRs will be collated and consolidated into a global APR which will be reviewed and shared with the global Project Board and other stakeholders for comments.

- 355 An independent Interim Evaluation (IE) will be undertaken at the mid-term of the project, and the findings and responses outlined in the management response will be incorporated as recommendations for enhanced implementation during the final half of the project's duration. The report will be based on an evaluation of both the global project as well as the country level projects. The terms of reference, the review process and the final IE report will follow the standard templates and guidance available on the UNDP Evaluation Resource Center. The national level IE report will be cleared by the UNDP Country Office and the UNDP Regional Technical Advisor and will be approved by each of the national level the Project Boards. These approved reports will then be collated and consolidated into a project level IE. The final IE report will be available in English.
- 356 An independent terminal evaluation (TE) will take place no later than three months prior to operational closure of the project. The terms of reference, the review process and the final TE report will follow the standard templates and guidance available on the UNDP Evaluation Resource Centre. The national TE report will be cleared by the UNDP Country Office and the UNDP Regional Technical Adviser and will be approved by the respective national Project Board after which they will be consolidated into the project level TE report. The TE report will be available in English. Each UNDP Country Office will include the planned project terminal evaluation in the UNDP Country Office evaluation plan and will upload the final terminal evaluation report in English and the management response to the public UNDP Evaluation Resource Centre (ERC) (www.erc.undp.org). The IE and TE will be carried out by an independent evaluator. The evaluation report prepared by the independent evaluator is then quality assessed and rated by the UNDP Independent Evaluation Office.
- 357 The BPPS and each UNDP Country Office will retain all M&E records for this project for up to seven years after project financial closure in order to support ex-post evaluations as part of the data management and M&E continuous learning and evaluation as project management cycle and beyond, UNDP stores all records up to seven years. A detailed M&E budget, monitoring plan and evaluation plan will be included in the UNDP project document.
- 358 UNDP will perform monitoring and reporting throughout the Reporting Period, including semi-annual reporting, in accordance with the AMA and Funded Activity Agreement (FAA). UNDP has country presence and capacity to perform such functions. In the event of any additional post-implementation obligations over and above the AMA, UNDP will discuss and agree these with the GCF Secretariat in in the final year of the project and will prepare a post-implementation monitoring plan and budget for approval by the GCF Board as necessary.

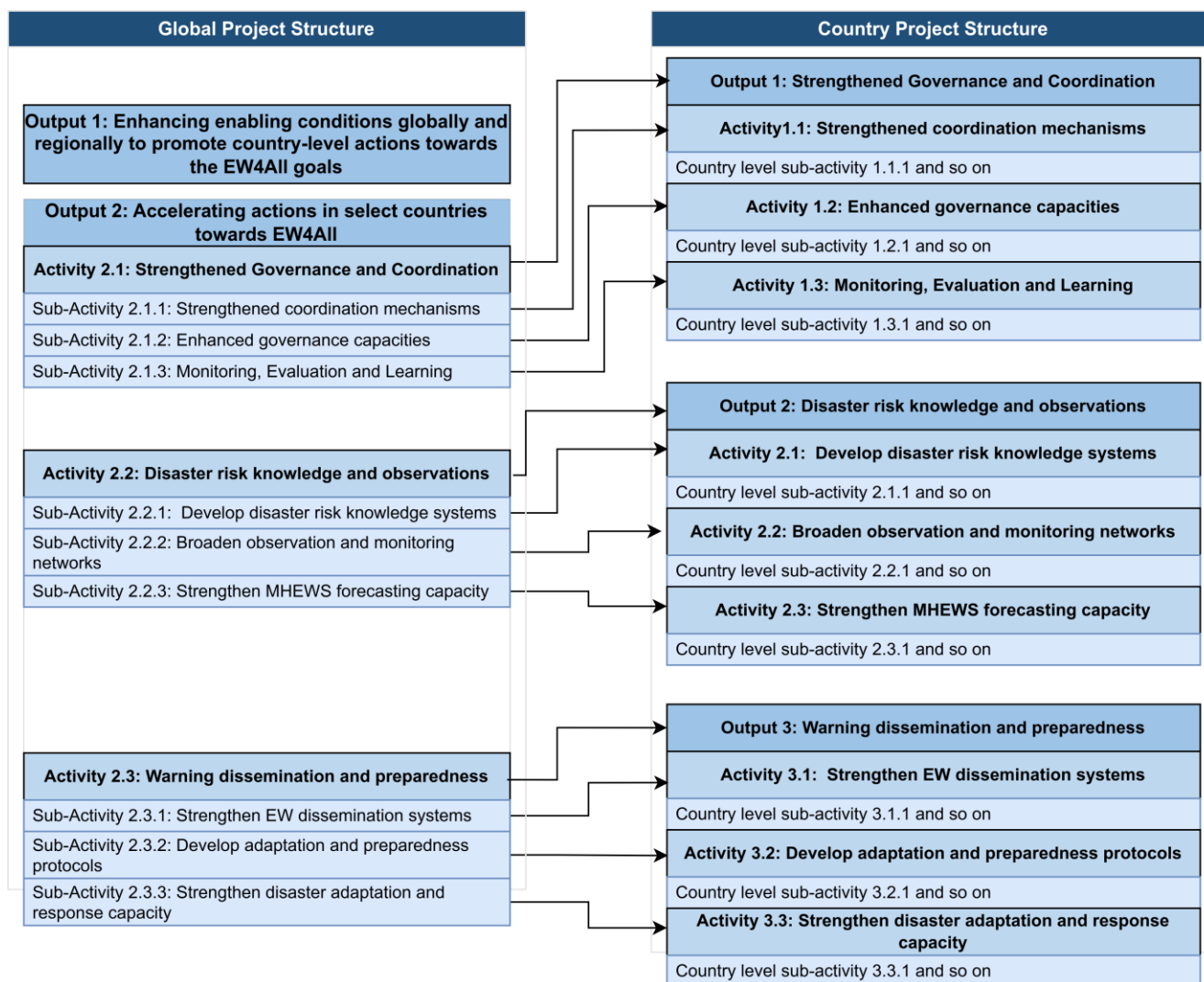


Figure 4: Conceptual relationship between the Global Output and Activities and the Country-level Outputs and Activities Note: specific wording of titles used for Outputs and Activities in the country projects may differ based on each country context

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1 Gaps in governance and institutional mechanism at the national level

Category	Probability	Impact
Governance	Medium	High

Description

Some countries may lack or have gaps in their NMHS and NDMA and in governance and institutional mechanisms necessary for coordinated action required for an operational EWS, disaster preparedness and response. This is often on account of local or regional conflict and insecurity, which is a risk to implementing the proposed activities, particularly under Output 2 of the project.

Mitigation Measure(s)

The project will work towards strengthening government structures for delivery of EWS to communities. It will simultaneously engage with national and local agencies, including offices of pillar leads such as local societies of the IFRC, multilateral agencies and NGOs to implement activities that are aligned with their core areas of expertise and comparative strengths. This implementation will, as far as possible, involve government agencies and seek to build their capacities through partnerships and hand holding.

Selected Risk Factor 2 Operational delays due to cumbersome procedures

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

Cumbersome procurement and administrative procedures may result in delays in procurement of critical equipment necessary to set up the hydrometeorological observation networks and communications as well as hiring of technical expertise from outside the country.

Mitigation Measure(s)

The UNDP Country Offices will extend support to executing entities and responsible parties where necessary, including support for procuring and importing equipment and technical expertise during project implementation. This will reduce the risk to low.

Selected Risk Factor 3 Risk of damage to equipment due to vandalism

Category	Probability	Impact
Governance	Medium	High

Description

Levels of insecurity in some participating countries may constrain the number and types of investments in observation and communications equipment and its regular operation and maintenance. This may lead limit the gathering of observational data and dissemination of early warnings affecting both the quality of climate information products as well as their effective communications to those in need.

Mitigation Measure(s)

The selection of equipment and its siting were done based on a comprehensive analysis of risks of vandalism and damage, keeping in mind constraints on O&M operations. The project will site equipment in secure government owned and operated sites or on telecommunications towers and installations where possible. The project will also utilise in-expensive IoT based community managed and operated EWS with an emphasis on their ownership of the systems through mobilization and sensitization of local groups and leaders. The equipment will be maintained and replaced as necessary, timing the field missions ahead of the relevant season, so that it is operational at the time when climate hazards are most likely to occur.

Selected Risk Factor 4 Poor inter-agency coordination and collaboration		
Category	Probability	Impact
Governance	Medium	High
Description		
Poor coordination and cooperation both within and between government, multilateral and private agencies can reduce the collection and sharing of observational data, the sectoral analysis of climate information, its dissemination and effective use for disaster preparedness and response. Private telecommunication companies can be reluctant to adopt standards and share data on customers which may be required to operationalize technologies such as cellular broadcast. Coordination and collaboration can be even more difficult for countries where climate hazards are of a trans-boundary nature, such as floods that originate in a neighbouring country, or where communications networks in one country overlap with another.		
Mitigation Measure(s)		
The project will strengthen institutional arrangements and governance mechanisms between agencies both at the national and regional level. This will be done by leveraging the national and regional initiatives, programmes and facilities being led by the EW4All pillar leads and other partner agencies. Where possible, policies, procedures and road-maps will be strengthened and formulated to ensure governments play a coordinating role and ensure compliance and cooperation between agencies, including the private sector such as telecommunication companies. This will ensure key dissemination mechanisms such as the CAP and technologies such as cellular broadcast can be deployed and leveraged.		
Selected Risk Factor 5 Poor community outreach and engagement in disaster preparedness		
Category	Probability	Impact
Technical and operational	Medium	High
Description		
The reach of government and non-governmental agencies to communities can constrains their ability to support communities to prepare and plan for disasters or to build their capacities to adapt. Community level preparedness and planning is critical for effective early and anticipatory action, particularly for marginalized and more vulnerable groups such as women, people with disabilities, elderly and households with infants and children. This poses a significant risk to the project's effectiveness in fragile and conflict affected countries as well as countries with limited access to remote areas.		
Mitigation Measure(s)		
The analysis of hazards, exposure and vulnerabilities done in the feasibility studies identifies the regions where communities are most at risk. This has informed the design of community engagement strategies, leveraging local actors from both within government and non-government agencies and innovative technologies where available. The project will strengthen the capacities of relevant agencies such as extension and health services and line departments to better equip them to support community level disaster planning and preparedness. It will strengthen institutional linkages between sub-national agencies and community level institutions and invest in preparedness activities and infrastructure at the village level with specific attention to vulnerable and marginalized groups.		
Selected Risk Factor 6 Limited technical and financial capacities to sustain the EWS value-chain		
Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
There is a risk that the gains made by the project are not backed up by sufficient resources, both technical and financial, to ensure the O&M of equipment and infrastructure or to sustain the production of MHEWS, specifically IbFs and decision support systems to inform early and anticipatory action. This is a risk to projects sustainability, particularly in poor and developing countries which have severe fiscal and technical constraints.		
Mitigation Measure(s)		
The project will address this risk through a multipronged approach. The knowledge management strategy will specifically target policy makers, presenting them with compelling evidence of financial returns from investments into the early warnings value chain, demonstrating the savings through avoiding and mitigating the impacts of climate hazards through adaptation and early and anticipatory action. Institutional linkages and collaborations will be established with national and regional agencies and academia to ensure ongoing and continuous support to the NMHS and NDMA with the latest technologies and techniques. This will help steer these away from any dependence on consultancies and one-off assignments for technical inputs. Finally, the project will strengthen existing regional financial facilities and funding mechanisms for long term support to national governments in sustaining their MHEWS. As part of this effort, the project will also foster collaborations with large technical firms to offset costs of data and computational facilities and technologies.		
Selected Risk Factor 7: Funds from the project are used for money laundering and terrorist financing		
Category	Probability	Impact

ML/FT	Low	Low
Description		
There is a small risk of the funds from the project are used for money laundering and terrorist financing in some of the participating countries.		
Mitigation Measure(s)		
This project will be implemented under UNDP's Direct Implementation Modality (DIM) and as such the entire execution of the project will fall under the responsibility of UNDP. UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is beholden to the United Nations Security Council Sanctions List and the UN General Assembly Resolutions. As such, the sanctions regime is embedded in all aspects of UNDP's operations and UNDP screens against the list for any entities being considered to be contracted by or partner with UNDP. Pursuant to the terms and requirements set out in its own framework of policies and procedures, UNDP will, as an integral part of its due diligence process for this Project, carry out the requisite partner capacity assessments and AML/CFT screening of any proposed partner that may be engaged for the delivery of some of the intended activities or any vendor procured for the purpose of delivery of goods or services to the project.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

- 359 The UNDP Social and Environmental Screening Procedure (SESP) is provided in Annex 6 for the global project. The project is assigned a risk category of "C" for GFC (and low risk based on the UNDP SESP). The project's activities are mainly focused on cooperation, dialogue, capacity building and training. The main risk of these activities is that they are not inclusive of all stakeholders, including Indigenous peoples, women, people with disabilities and other marginalized groups which is well-managed through project design as well as Stakeholder Engagement Plans (SEPs) and Gender Analysis and Action Plans (GAAPs) prepared. In order to avoid these risks, Output 2 will involve extensive engagement with communities, including vulnerable and marginalized groups. These consultations will ensure representation of and equitable benefits and consideration to the needs of women, Indigenous groups, people with disabilities, elderly and groups who may otherwise be culturally and economically excluded. The project recognizes the differential access to resources, information and social roles and will proactively work towards overcoming cultural biases and other limitations which may prevent women and marginalized groups from equitable participation and sharing of project benefits. The project therefore embeds consultations and stakeholder engagement in its activities from project design through implementation. Gender Analysis and Actions Plans (GAAP), as well as Stakeholder Engagement Plans (SEP) have been prepared, which will integrate considerations relevant for Indigenous peoples. These plans will help to ensure that the project is inclusive of stakeholder's needs, and that vulnerable groups access constraints and opportunities are taken into account to ensure they can effectively benefit from the project's activities.
- 360 Output 2 also has minor EWS small works activities that are being defined. These will be small-scale, such as small hydrometeorological observation stations that carry small footprints which will need to be sited, small boreholes, or repairs or small upgrades to buildings, such as waterproofing and installing toilets. The repairs/upgrade/installation of these would be site-specific and contained, with limited, predictable, site-specific impacts that can be readily addressed through standard mitigation measures. These impacts would mainly include limited pollution, waste and risks of Occupational Health and Safety (OHS). The project excludes activities from project financing which require an ESIA/ESMP or which would have a GCF Category A or B rating. The project has developed an Environment and Social Management Framework (ESMF) which includes an assessment of all these risks and proposes mitigation measures, including through an Environment and Social Code of Practice (ESCP) which would guide the implementation of small works. There will be no displacement of people or their livelihoods as a result of project interventions. No installation or works will be done in sites of cultural, historical or environmental importance and as such no biodiversity impacts are expected. New installations will be sited on government property or public areas free of encumbrances, with existing access so that O&M operations do not require crossing private lands, biodiversity areas or other zones that would pose any impact to the environment, workers or communities (including zones where there are ongoing conflicts). As a result of limited numbers of staff, consultants and workers implementing project activities, there could also be some potential low risks of Gender Based Violence (GBV)/Sexual Harassment (SH). Preventing SEAH guidance has been provided in the ESMF, as well as guidance to prepare country-specific SEAH plans during implementation if SEAH risks (moderate or higher) are identified.
- 361 In some countries, the population includes indigenous peoples. The project will not negatively impact Indigenous Peoples. The project does not envision any small works on IP lands, and in the unlikely event this was identified during project implementation, or any other impact to IPs identified, Indigenous Peoples Plans would need to be prepared following the Indigenous Peoples Planning Framework (IPPF) annex in the ESMF. Nevertheless, it is important that meaningful consultations are conducted with IP representatives during project implementation, to ensure their needs are considered in EWs and that they are able to receive EW in a way that is meaningful to them. Guidance for consultations and general stakeholder engagement, including specifically with IP groups, is included in the global SEP.

The project complies with UNDP's Social and Environmental Standards which were reviewed by the GCF accreditation panel and deemed sufficient to accredit UNDP to submit low and medium risk projects.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

363 A global Gender Analysis and Gender Action plan (GAAP) is provided in Annex 8. The project proposal emphasises gender inclusiveness and gender responsive design of outputs and activities. It recognises the differential impacts of climate change on women on account of the disadvantages they face in terms of representation in decision making, ownership and access to resources, finances and information. The project also acknowledges the impact of gender roles and cultural biases on the ability of women to prepare and plan for climate hazards and to receive timely and actionable early warnings. The project design factors in the dependence of women on climate sensitive natural resources and their roles in livelihoods, particularly those related to agriculture. Activities pertaining to climate change adaptation and preparedness and response to climate mediated disasters, incorporate critical elements of gender mainstreaming and equity.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

- 364 The financial management and procurement of this project will be guided by [UNDP financial rules and regulations](#). Further guidance is outlined in the financial resources management section of the UNDP [Programme and Operations Policies and Procedures \(POPP\)](#), which also includes comprehensive procurement policies as outlined in the 'Contracts and Procurement' section. These policies outline formal procurement standards and guidelines across each phase of the procurement process and apply to all procurements in UNDP.
- 365 National projects will be implemented following the Direct Implementation Modality (DIM) by UNDP country offices. The DIM project management and implementation approach takes full responsibility for executing a project, with no further delegation of the implementation to a government entity or other partner.
- 366 If the NGO or private sector to be engaged as a Responsible Parties they will be engaged through Responsible Party Agreements. Each Responsible party who will be engaged to support the implementation of project, following the Framework for Cash Transfers to Implementing Partners (part of the Harmonized Approach to Cash Transfers - HACT). All projects will be audited following the UNDP financial rules and regulations noted above and applicable audit guidelines and policies. The HACT framework consists of four processes: (1) macro assessments; (2) micro assessments; (3) cash transfers and disbursements; and (4) assurance activities. Assurance activities include planning, periodic on-site reviews (spot checks), programmatic monitoring, scheduled audits and special audits. Weaknesses identified during micro-assessments will require actions to addresses the gaps. When a spot check finds that the gaps are not addressed it will mean that the level of assurance activities will have to remain higher and modalities of engaging with that implementing partner will have to be reviewed if necessary, [as described here](#).
- 367 The project will be audited in accordance with UNDP policies and procedures on audits, informed by and together with any specific requirements agreed in the AMA established between GCF and UNDP, signed on 16 of March 2023 and made effective in May 3, 2023. According to the current audit policies, UNDP will be appointing the auditors. Scheduled audits are performed during the programme cycle as per UNDP assurance/audit plans, based on the implementing partner's risk rating and UNDP's guidelines. A scheduled audit is used to determine whether the funds transferred to the implementing partner were used for the appropriate purpose and in accordance with the work plan. A scheduled audit can consist of a financial audit or an internal control audit.
- 368 All GCF resources will be provided to the responsible party, less any agreed cost recovery amount. Under UNDP's direct implementation modality, UNDP advances cash funds on a quarterly basis to the responsible party for the implementation of agreed and approved programme activities, in accordance with UNDP standard policies-. The responsible party reports back expenditure via a financial report on quarterly basis to UNDP. Any additional requirements will be in accordance with the AMA as and when it is agreed. A draft procurement plan (which will be further discussed and revised prior to UNDP Project Document signature) is provided in Annex XIII.

G.4. Disclosure of funding proposal

- ☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be

disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☐ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☐ Annex 2 Feasibility study - and a market study, if applicable
- ☐ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☐ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☐ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
 - Annex 6a SESP
 - Annex 6b ESMF
- ☐ Annex 7 Summary of consultations and stakeholder engagement plan
- ☐ Annex 8 Gender assessment and project/programme-level action plan
- ☐ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☐ Annex 10 Procurement plan [\(template provided\)](#)
- ☐ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☐ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)¹⁴⁸

¹⁴⁸ Annex 22 is mandatory for mitigation and cross-cutting projects.

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.

Appendices

Appendix 1: Climate hazards and associated risks (impacts, exposures and vulnerabilities) in participating countries

Antigua and Barbuda

- 369 Antigua and Barbuda is on the frontline of climate change, facing a range of intensifying climatic hazards that threaten their very existence. Ranked as the 5th highest in the World Risk Index in 2023, the islands are exposed to an array of baseline hazards, including hurricanes, extreme rainfall, droughts, and rising sea levels. These risks are exacerbated by the islands' geographical position and limited land area, making them particularly vulnerable to the cascading and compounding impacts of climate change.
- 370 Hurricanes pose the most significant threat in Antigua and Barbuda, with the country having faced 15 hurricanes and 14 tropical storms between 1995 and 2021. The devastation from Hurricanes Irma and Maria in 2017 serving as stark reminders of the destructive power of these storms. These hurricanes not only caused widespread destruction of infrastructure but also disrupted essential services such as healthcare, electricity, water, and waste management. In Barbuda, nearly 90% of the island's infrastructure was destroyed, rendering the island temporarily uninhabitable. The increasing intensity of these storms, fueled by climate change, is expected to lead to even more catastrophic events in the future. Moreover, hurricane intensity is anticipated to increase by 18% for category 4 and 5 hurricanes over the next 30 years, based on General Circulation Models (GCMs)¹⁴⁹. By 2030, the general building stock in Antigua and Barbuda, using RCP 8.5, will experience direct losses from storm surges and wind in the amount of USD 451 million for a historic event, USD 4.03 billion for a 1:100-year event, and USD 9.4 billion for a maximum credible event (MCE).¹⁵⁰
- 371 Droughts, while less dramatic than hurricanes, pose a persistent and growing threat to the islands. Historically, droughts have led to severe water shortages, impacting utilities, agriculture, and even leading to economic crises, such as the need to import water during the drought of 1983-1985. With projections indicating a 30%-50% reduction in annual rainfall by 2090, droughts are expected to become more severe, further straining the already limited water resources.¹⁵¹ Under the RCP 4.5 scenario, models predict 9 drought years for the period 2040–2069, representing an 8% increase in drought severity¹⁵².
- 372 The low-lying geography of Antigua and Barbuda makes it particularly susceptible to land loss due to sea-level rise, which presents a long-term existential threat to Antigua and Barbuda. Projections indicate that by 2060, Antigua could lose approximately 20 km² of land, while Barbuda could lose 15 km². This loss of land, combined with rising sea surface temperatures and increased ocean acidification, threatens not only the islands' ecosystems but also the livelihoods of those who depend on the sea, particularly in the tourism and fishing sectors.¹⁵³
- 373 The interplay of these climatic hazards creates a highly vulnerable environment for Antigua and Barbuda, where the effects of one event can exacerbate the impacts of another. For example, a severe hurricane can lead to flooding, which disrupts essential services and exacerbates water shortages caused by droughts. The cascading effects of these hazards, coupled with the islands' low-lying topography and concentration of economic activities along the coast, underscore the urgent need for comprehensive climate adaptation strategies.
- 374 Local stakeholders have identified several key areas that require urgent attention to strengthen the islands' resilience. The first is the need for improved observation and forecasting systems, particularly for marine hazards. Given the islands' exposure to ocean-related hazards such as storm surges and coastal erosion, there is a critical need for accurate and locally specific advisories to protect vulnerable groups. The built environment is another area of concern, as many buildings are not resilient to high-intensity storms due to the cost of building to code and the existence of outdated

¹⁴⁹ Caribsave. 2012. Caribsave Climate Change Risk Profile for Antigua & Barbuda.

¹⁵⁰ GoAB 2022. Adaptation Communication for Antigua and Barbuda

¹⁵¹ GoAB 2022. Food Security Sectoral Action Plan

¹⁵² Johnston, Janeil. 2021. Sectoral Adaptation Planning for Climate Resilience in Antigua and Barbuda: Risk Modelling Data and Adaptation. Presentation at Workshop on Sectoral Adaptation Planning for Climate Resilience in Antigua & Barbuda, July 29, 2021.

¹⁵³ GoAB 2022. Adaptation Communication for Antigua and Barbuda.

infrastructure. An improved understanding of building resilience is crucial for planning emergency responses, including evacuations and sheltering.

- 375 The tourism sector, which employs a large percentage of the population and is heavily concentrated in exposed coastal areas, also requires targeted preparatory actions. International visitors, who may lack local knowledge of climate hazards, are particularly vulnerable, highlighting the need for better integration of the tourism sector into the islands' Multi-Hazard EWSs (MHEWS) and Anticipatory Action systems. Finally, the island of Barbuda, still recovering from the devastating impacts of Hurricane Irma, has been flagged as a key focus area for improving preparedness and response capabilities.

Ecuador

- 376 Due to a combination of political, geographic, and social factors, Ecuador is recognized as vulnerable to climate change impacts, ranking 115th out of 185 countries in the 2021 ND-GAIN Index¹⁵⁴. When coupled with a low readiness score, this makes Ecuador highly susceptible to climate-related impacts. These impacts, such as droughts, floods, and landslides, undermine development and reduce the availability of natural resources, affecting the majority of the population. This vulnerability was evident during the droughts of 2023¹⁵⁵ and 2024¹⁵⁶, as well as the floods of 2016¹⁵⁷, 2023¹⁵⁸, and 2024¹⁵⁹.
- 377 Ecuador's climate is highly diverse due to its location on the equator and its varied topography, which includes the Amazon rainforest, the Andes mountains, and the Pacific coast.
- 378 From 1960 to 2010, Ecuador experienced increases in temperature as well as spatial and seasonal variations in rainfall across its territory. The observed average temperature increase was 1.4°C, with maximum temperatures rising by 1.0°C and minimum temperatures increasing by 1.1°C in the Galapagos Islands¹⁶⁰. These changes varied across the country's regions, leading to impacts such as: (i) approximately 40% loss of glacier cover on the Antisana, Carihuayrazo, Cotopaxi, and Chimborazo volcanoes; (ii) severe droughts affecting 66.7% of agricultural areas and 53.7% of livestock areas (pastures); and (iii) extreme rainfall increasing the floodable surface area by 15.9%, affecting 49.5% of the population.
- 379 Projections foresee changes in the frequency and magnitude of both average and extreme precipitation and temperature values. By the 2090s, average temperatures in Ecuador are projected to rise significantly, potentially increasing by 3.5°C to 4.0°C under the highest emissions pathway (RCP8.5), compared to the baseline of 1986–2005¹⁶¹. Annual precipitation is expected to decrease in the Amazon (by 2–6%) but increase along the coast and in the mountain areas under the highest emissions pathway by the 2090s¹⁶². Intense precipitation is also expected to rise across the country, especially in the central mountain region, with increases of up to 50 mm for the average largest 1-day precipitation¹⁶³.
- 380 Floods are the most frequent hydro-meteorological disaster in Ecuador and rank second in terms of casualties, with landslides, storms, and droughts also contributing significantly. According to the Desinventar database¹⁶⁴, there has been an increase in both the number of events and deaths over the past 40 years, although with significant fluctuations. The rise in the intensity and frequency of these extreme climatic events has caused economic damages and losses to goods

¹⁵⁴ University of Notre Dame (2023). Notre Dame Global Adaptation Initiative. URL: <https://gain.nd.edu/our-work/country-index/>

¹⁵⁵ <https://www.wwf.org.ec/?uNewsID=385722>

¹⁵⁶ <https://www.bbc.com/news/world-latin-america-68835127>

¹⁵⁷ <https://reliefweb.int/report/ecuador/informe-de-situaci-n-emergencia-inundaciones-etapa-invernal-en-esmeraldas-fecha-29-de>

¹⁵⁸ <https://elpais.com/internacional/2023-06-08/las-lluvias-dejan-15077-damnificados-en-cuatro-ciudades-de-ecuador.html>

¹⁵⁹ <https://www.primicias.ec/noticias/sociedad/fuertes-lluvias-balance-inundaciones-27febrero/>

¹⁶⁰ <https://unfccc.int/documents/77568>

¹⁶¹ Ministry of Environment, Water and Ecological Transition of Ecuador (MAATE). (2022). Fourth National Communication to the United Nations Framework Convention on Climate Change. MAATE, Quito, Ecuador

¹⁶² <https://climateknowledgeportal.worldbank.org/>

¹⁶³ Ministry of Environment, Water and Ecological Transition of Ecuador (MAATE). (2022). Fourth National Communication to the United Nations Framework Convention on Climate Change. MAATE, Quito, Ecuador

and services, with annual losses reaching between \$927 million and \$3.3 billion¹⁶⁵. Associated non-economic losses include loss of life, health impacts, human mobility, biodiversity, cultural heritage, and indigenous knowledge. In this context, 18 indigenous peoples and 13 nationalities are at risk from the threats and impacts of climate change¹⁶⁶.

- 381 Climate change projections indicate significant impacts, including the continued melting of glacier cover on volcanoes, variations in sea surface temperatures, changes in spatial and temporal precipitation patterns, expansion of flood-prone areas, intensification of droughts, reduced water quantity and quality, the spread of disease-transmitting vectors (such as dengue and malaria), and loss of biodiversity. The increased intensity and frequency of both extreme and slow-onset climatic events will continue to cause economic and non-economic losses to goods, services, and human well-being¹⁶⁷.

Fiji

- 382 Historical Climate Trends: According to the NextGen Projections report¹⁶⁸, Fiji has warmed up by 0.7 degrees relative to the pre-industrial baseline of 1850-1900, up to the period of 2011-2020. In Fiji, observed and projected warming shows that 2°C of global warming corresponds to 1.2 to 1.9°C in Fiji. The temperatures in Fiji have increased at a faster rate in recent decades, as shown by commonly used baselines (WMO – 1981 to 1991, AR5 – 1986 to 2005, AR6 – 1995 to 2014, and 2011 to 2020). The strongest increase has been in daily maximum temperatures (around 0.2°C per decade) and a similar increase in daily minimum temperatures (around 0.1°C per decade). There are no significant trends in gridded rainfall datasets for the entire region of Fiji including surrounding oceans. Historical observations show large year-to-year variability in rainfall with no significant trends since 1960. The average of all climate models considered shows no significant change in annual rainfall for Fiji between the pre-industrial baseline of 1850-1900 to the more recent baselines.¹⁶⁹
- 383 Projected changes in climate: Projections for Fiji are presented for a range of ‘global warming levels’ (defined as the warming of the global average temperature since the pre-industrial era), including the 1.5°C and 2°C limits from the Paris Agreement. The range of possible change is larger under higher global warming levels compared to lower levels.¹⁷⁰
- 384 Temperatures: Historical and projected changes in temperatures (relative to 1850 – 1900) are:
- There is a warming by 0.7°C up to 2011-2020.
 - By 2030, the warming is 1.2°C regardless of the emission pathway.
 - By 2050, the warming is 1.3°C (RCP2.6) to 1.9°C (RCP8.5).
 - By 2070, the warming is 1.3°C (RCP2.6) to 2.5°C (RCP8.5).
- 385 According to the World Bank report, based on the Coupled Model Intercomparison Project Phase 5 (CMIP5) models, under the medium warming scenario (RCP4.5), Fiji is projected to experience a warming of 0.9°C by 2040-2059 and 1.3°C by 2080-2099 and under the high warming scenario (RCP8.5), the projected warming is 1.3°C and 2.7°C, respectively.
- 386 The Fifth Assessment Report (AR5) of the Intergovernmental Panel on Climate Change (IPCC) projections in rainfall are less certain with a high degree of variability which indicates that average annual rainfall may remain the same, but there will be increase in the intensity for extreme rainfall events.¹⁷¹ The NextGen Report states that future changes in rainfall are uncertain but could range from wetter to drier, depending on the South Pacific Convergence Zone (SPCZ). The SPCZ is one of the main climate drivers in the south Pacific and is influenced by the presence of El Niño and La Niña.

¹⁶⁵ UNDRR. <https://www.desinventar.net/>

¹⁶⁶ Secretaría de Gestión de Riesgos. <https://www.gestionderiesgos.gob.ec/>

¹⁶⁷ Ministry of Environment, Water and Ecological Transition of Ecuador (MAATE). (2022). Fourth National Communication to the United Nations Framework Convention on Climate Change. MAATE, Quito, Ecuador

¹⁶⁸ *ibid*

¹⁶⁹ NextGen’ Projections for the Western Tropical Pacific: Current and Future Climate for Fiji.

¹⁷⁰ *ibid*

¹⁷¹ Fifth Assessment Report (AR5) of the Intergovernmental Panel on Climate Change (IPCC), 2018

- 387 Southwest Pacific tropical cyclone (TC) activity projection for a 2°C global warming indicates a likely decrease in the total number of TCs, and a possible increase in the frequency of severe (category 4-5) TCs. The projected increase in average TC intensity, combined with sea level rise and increases rainfall rate would increase TC impacts.
- 388 Resulting Impacts: Fiji is considered a high-risk country due to its vulnerability to natural hazards, ranking 75th out of 193 countries in the World Risk Report 2023. The country faces significant impacts from TCs and floods, with annual asset losses exceeding F\$500 million, reaching over 5% of the GDP. In extreme cases like a 100-year fluvial flood, losses could surpass F\$2 billion, affecting the transport and building sectors the most. Droughts, particularly during El Niño events, have led to reduced crop production and the need for emergency water and food supplies in Fiji. Climate change poses a threat to food security, especially in coastal areas impacted by sea level rise and salinity intrusion. Traditional crops exhibit more resilience to climate change than non-traditional ones, but staples like rice, taro, and yams remain vulnerable. Fiji's limited capacity to manage and adapt to climate risks contributes to health risks and increased poverty rates. Natural hazards like TCs and floods annually push around 25,700 people into poverty, disproportionately affecting communities reliant on subsistence fishing and facing water security challenges. Changing weather patterns also threaten sectors like agriculture and tourism, critical to Fiji's economy, as evidenced by the increasing impact of 124 natural hazards over the past 37 years, with TCs, floods, and earthquakes being the most frequent occurrences.

Chad

- 389 Between 1971 and 2022, 62 natural disasters were recorded in Chad, of which 68 percent (42 events) occurred since 2000 (EM-DAT, 2024). Successive droughts have affected the food security and livelihoods of households across large swathes of the country, with 1.5 million people affected in 1981-85, and 2.4 million people affected in 2009-2010. The most recent drought in 2022 affected 2.1 million people while floods in the same year rendered 121,088 people homeless. Floods and droughts have also been associated with outbreaks of disease, and with undernutrition. Together, these disasters led to a total of 10,556 recorded deaths, of which two thirds were a result of viral or bacterial disease epidemics such as cholera, meningitis and measles.
- 390 Over the past 10 years, IFRC has launched 18 Disaster Relief Emergency Fund (DREF) operations supporting the national Red Cross society in Chad; and raised a further 30 emergency appeals to mobilise resources to respond to 21 emergencies, targeting 5.8 million people affected by emergencies.
- 391 Studies have shown a continuous increase in temperatures, as well as increased variability in rainfall in the subregion since the 1990s. After a sequence of wet years covering the 1950s and 1960s, followed by a sequence of dry years in the 1970s and especially dry years in the 1980s, inter-annual variations have become more irregular since the 1990s, though with relatively better rainfall. Overall precipitation in Chad has been above average over several years since the 1990s, particularly in the Sahelian zone, but with variability in its distribution over the year. The spatial distribution of precipitation has changed, with the retreat of the 300 mm and the 1,200 mm isohyets.
- 392 Air temperature is expected to increase by 2.1°C to 4.3°C (very likely range) by 2080. Relative to pre-industrial levels, the median temperature increase from climate models in Chad is 2.1°C in 2030, 2.6°C in 2050, and 3.5°C in 2080 under RCP6.0. The number of very hot days per year is also expected to increase across the country with a high degree of certainty. For the RCP6.0 scenario, the median of the multiple models (averaged over the entire country) predicts 17 additional very hot days per year in 2030, 31 additional days in 2050 and 49 additional days in 2080, compared to 2000. In some parts of the country, particularly in the center, this equates to over 300 very hot days per year by 2080.
- 393 Projections of precipitation and variations in rainfall for 2030, 2050 and 2100 vary greatly, according to the country's zones owing to the high natural variability from year to year and high level of uncertainty and differences between models. The more conservative models would lead to a reduction in precipitation while the models with higher emissions driving higher global temperature increases suggest higher precipitation, with higher variability. Critically, all models project more variability in precipitation than the pre-1980 situation.
- 394 Chad remains among the poorest and least developed countries in the world and is ranked 119th of 125 countries of the global hunger index 2023, with a level of hunger classified as serious. A combination of high poverty, frequent conflicts, and the risk of both droughts and floods have placed Chad as the lowest ranked country overall on the ND-GAIN index, with a score of 27.0.

- 395 With rain-fed agriculture accounting for almost 46% of GDP in 2019, and 77% of the labour force working in agriculture, the food production and livelihood systems in Chad are highly dependent on climatic factors, particularly rainfall patterns and distribution. The effects of high temperatures, and of hunger, are more acute for children, the elderly, and those with chronic diseases.
- 396 Chad also faces severe environmental degradation, exemplified by the drying up of Lake Chad, increased desertification, declining fish stocks, the disappearance of certain animal and plant species, and soil degradation (National Adaptation Plan, 2022). Increased competition for access to pasture and water, soil depletion and lower agricultural productivity have the potential to exacerbate existing conflict factors or contribute to new forms of dispute.
- 397 The proposed project aims to provide technical support and capacity building at national, institutional level to ensure essential climate services can be delivered on a sustainable basis. Improved data on vulnerability and risk, knowledge management, governance of multi-hazard EWSs and disaster preparedness/disaster management will be improved at national and sub-national levels. The intervention addresses the fact that people's information needs are not being met, in the face of changing climate patterns with increased exposure to the impacts of droughts and floods, where they can no longer rely solely on traditional knowledge and experience. The project will equip communities with relevant and timely risk education, training and advice on adaptation and preparedness measures and early warning information to reduce both the loss of life and the economic impact of disasters. It will support the piloting, and integration of early action protocols by WFP and the Red Cross, into national systems and will engage with communities on their needs in terms of information, preparedness and disaster management to ensure these processes are inclusive, needs-based and people-centered.

Ethiopia

- 398 Ethiopia's climate varies significantly across its regions, classified into three primary zones based on seasonal rainfall patterns. The northern and central western areas experience a single rainy season, with peak precipitation occurring in July and August. The southern areas are characterized by two distinct rainy seasons: a short rainy period from September to November and a long rainy period from March to May. The eastern and central parts experience two rainy periods: a spring season from February to May and a summer rainy season from June to September (Paleoclimate Research Institute, 2021). The variability in Ethiopia's rainfall is influenced by large-scale climatic phenomena such as the El Niño-Southern Oscillation (ENSO) and sea surface temperatures (SSTs) in the Indian and Pacific oceans. SSTs, with their gradual variations, affect long-term rainfall patterns and are crucial for improving the accuracy of regional-scale climate predictions (World Meteorological Organization, 2022)
- 399 Climate change poses a significant threat to Ethiopia's development, exacerbating existing vulnerabilities. The country faces high impacts from natural hazards and increasing climate variability, which threaten its development goals. Ethiopia's ambitious reform agenda aims to address these challenges amidst conflict and inflation. Balancing development needs with climate adaptation and mitigation efforts is crucial for ensuring sustainable growth (World Bank Group, 2023).
- 400 Ethiopia is committed to a green development path, adopting various policies, strategies, and plans to guide its climate interventions. The legal framework for climate action includes specific sectoral responsibilities and coordination mechanisms. However, gaps and overlaps in institutional mandates require further refinement. Ethiopia's climate change targets and commitments are integrated into public expenditure management, with efforts to mainstream climate action across sectors and levels of government (World Bank Group, 2023).
- 401 Climate change impacts various sectors in Ethiopia, including water, agriculture, energy, transport, and urban development. The country faces significant water challenges, with variable and unpredictable rainfall affecting millions of people. Essential adaptation measures include improving storage efficiency and integrated watershed management. The agricultural sector must become more productive and climate-resilient, while energy, transport, and urban infrastructure need to be adapted to withstand climate impacts. Additionally, diversifying the economy through mining and tourism can enhance resilience (World Bank Group, 2023).

- 402 Climate change has profound implications for Ethiopia's macroeconomic stability and poverty levels. Rising temperatures and increased rainfall variability threaten economic growth and exacerbate poverty. Structural reforms are crucial to expanding fiscal space and enabling private sector participation in climate adaptation. Managing residual risks and mobilizing external finance are essential for addressing climate impacts. Supporting the most vulnerable populations through targeted interventions is critical to building resilience and ensuring equitable development (World Bank Group, 2023).
- 403 Ethiopia faces significant economic losses from various climate hazards. Droughts cause substantial economic damage, with a drought event having a 1-in-5 year return period leading to losses exceeding US\$1 billion, and a 1-in-100 year drought event resulting in losses of over US\$3 billion. Flooding also has a severe impact on GDP, with a flood event with a 25-year return period causing a 3.3% deviation in GDP baselines, and a 100-year flood leading to a 7% deviation. Additionally, heat stress results in significant losses due to its impact on labor productivity and livestock yields. Infrastructure damage, particularly to roads and bridges, can incur annual costs of up to US\$755 million under certain scenarios. These economic impacts highlight the urgent need for effective climate adaptation and mitigation strategies in Ethiopia (World Bank Group, 2023). The country faces critical challenges, including ethnic conflicts and climate-related risks such as droughts and floods, which impact agriculture and water resources. Effective climate services and EWSs are essential for enhancing resilience (Food and Agriculture Organization, 2021).

Somalia

- 404 Somalia is one of the most climate-vulnerable countries in the world. Somalia is ranked 178 out of 182 countries on the ND-GAIN Index 2024, with the least capacity to respond and recover from extreme weather events.¹⁷² Located in the Horn of Africa, Somalia extends south of the Equator northward to the Gulf of Aden, between sub-Saharan Africa and the Middle East. As a predominately arid and semi-arid country with a hot desert climate¹⁷³, the country is susceptible to both droughts and flash floods. Prolonged dry seasons result in water scarcity, reduced grazing land, reduced livestock production, livestock perishes, reduced agricultural productivity, food insecurity, and ultimately, famine and the loss of human life.¹⁷⁴ Somalia's chronic climate-associated shocks, including cycles of floods, droughts, and locust infestation, are compounded by increased conflict and insecurity, and have aggravated the nation's growth and slowed the transition from fragility. Climate change has exacerbated this vulnerability by increasing the frequency, persistence and intensity of droughts in Somalia.¹⁷⁵
- 405 Historical and projected climate change impacts in Somalia are drought and flood, although Somalia is also affected by cyclones and coastal flooding is also classified as being high.¹⁷⁶ Daily temperature maximums have increased over the last two decades¹⁷⁷ and are projected to continue over the next decade.¹⁷⁸ The very southern areas of Somalia are projected to experience extreme heat, and the southern and some parts of the northern areas are projected to experience very high heat.¹⁷⁹ Climate change projections suggest that the duration, frequency and intensity of droughts will increase by the end of the 21st Century under all scenarios and that drought areas will increase by 16 per cent, 36 per cent and 54 per cent under increasing Representative Concentration Pathways (RCPs) (2.6, 4.6, 8.5 respectively).¹⁸⁰ At the same time,

¹⁷² ND-GAIN Index Country Rankings, Somalia. <https://gain-new.crc.nd.edu/country/somalia> [accessed 7 July 2024]

¹⁷³ classified according to the Köppen-Geiger Classification 1991-2020, World Bank 2021, Climate Change Knowledge Portal: Somalia, <https://climateknowledgeportal.worldbank.org/country/somalia>

¹⁷⁴ National Drought Plan For Somalia, 2020.

https://www.unccd.int/sites/default/files/country_profile_documents/FINAL%20NATIONAL%20DROUGHT%20PLAN%20FOR%20SOMALIA%28final%29%2016%20Dec%202020%28%20PDF%20version%29.pdf

¹⁷⁵ Somalia: Drought Situation Report No.2 (As of 21 December 2021), <https://www.unocha.org/publications/report/somalia/somalia-drought-situation-report-no2-21-december-2021>

¹⁷⁶ GFDRL 2024, Somalia Coastal Flood: <https://thinkhazard.org/en/report/226-somalia/CF>

¹⁷⁷ World Bank 2021, Climate Change Knowledge Portal: Somalia, II – Variability and Change in Variability

<https://climateknowledgeportal.worldbank.org/country/somalia/trends-variability-historical>

¹⁷⁸ World Bank 2021, Climate Change Knowledge Portal: Somalia, I – Trends within Variability

<https://climateknowledgeportal.worldbank.org/country/somalia/trends-variability-projections>

¹⁷⁹ World Bank 2021, Climate Change Knowledge Portal: Somalia, High-level Summary: Compound Heat Risk

<https://climateknowledgeportal.worldbank.org/country/somalia/heat-risk>

¹⁸⁰ Haile, G.G., Tang, Q., Hosseini-Mogharri, S., Liu, X., Gebremicael, T.G., Leng, G., Kebede, A., Xu, X., & Yun, X. (2020). 'Projected impacts of climate change on drought patterns over East Africa', *Earth's Future*, 8(7). <https://doi.org/10.1029/2020EF001502>

higher rainfall and runoff are expected across the various climate scenarios, leading to increased flash flooding.¹⁸¹ The frequency and intensity of heavy precipitation events are projected to increase with potential impacts of flooding and soil erosion.¹⁸²

406 Drought is a critical issue for Somalia, which disparately affects nomadic and agro-pastoral communities who are more vulnerable to such droughts. During 2015 – 2022, 6.1 million people were affected by the drought, and 760,000 - 1.2 million people were displaced by drought.¹⁸³ However, the following year of 2023 saw a further 468,000 people impacted by flood, including 247,000 displaced people.¹⁸⁴ Somalia experiences regular river floods and flash flooding that result in major economic impacts and damage, as well as many human casualties, with riverine flooding typically occurring along the Juba and Shabelle Rivers in southern Somalia and flash flooding tending to occur in the north along intermittent watercourses.¹⁸⁵ Yet, intense rainfall events and subsequent floods have become more frequent, resulting in water runoff on the earth's surface, washing away fertile soil and causing widespread devastation. Most recently, the 2023 Deyr flood has been reportedly the worst flood Somalia has faced, resulting in total damages of US\$126.6 million and total losses of \$49.5 million across sixteen districts, mostly affecting agriculture, livestock, and WATSAN and transport.¹⁸⁶ Around 2.48 million people were affected, with over 1.2 million displaced from their homes.¹⁸⁷ The floods also resulted in the loss of 188 lives.¹⁸⁸ Cyclones, such as Cyclone Sagar (2018) and coastal inundation, tend to affect northern Somalia's Somaliland and Puntland regions¹⁸⁹ rather than the southern areas.

Cambodia

407 This table lists major disasters in the past 25 years with over 200 deaths, mainly due to riverine or flash floods. However, while droughts and lightning strikes caused fewer deaths, they still had significant impacts.

Year	Deaths	Houses Destroyed	Houses Damaged	Directly affected	Evacuated	Damages in crops Ha.
2000	388	1,305	7,920	3,305,648	80,599	574,236
2011	458	963	1,838	1,890,178	155,102	359,515
2013	375	2,684	7,509	1,921,502	191,587	209,013
2018	206	1,385	3,112	754,000	65,125	167,264

408 Historically, Cambodia has experienced consistent temperatures that peak above 32°C in early summer and range from 25 to 27°C annually. The summer monsoon from May to November brings the heaviest rainfall to the southeast and northwest regions. Annual rainfall varies from 1,400–2,000 millimetres, with coastal and highland areas receiving higher

¹⁸¹ Government of Somalia. (2021). National Water Resource Strategy 2021–2025. Ministry of Energy and Water Resources. <https://www.climate-laws.org/geographies/somalia/policies/national-water-resource-strategy-2021-2025>

¹⁸² Seneviratne, S. I. & Zhang, X. (2021) Chapter 11: Weather and Climate Extreme Events in a Changing Climate [MassonDelmotte, V., P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J. B. R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekçi, R. Yu & B. Zhou. Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change In Press, Cambridge University Press. https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Chapter_11.pdf.

¹⁸³ IOM, OCHA, UNHCR 2022 Somalia: Drought and Famine Displacement Monitoring Dashboard (September 2022), <https://reliefweb.int/report/somalia/somalia-drought-and-famine-displacement-monitoring-dashboard-september-2022>

¹⁸⁴ OCHA 2023, Somalia: Gu rainy season 2023 Flash Floods Update No. 3 (13 April 2023), UN Office for the Coordination of Humanitarian Affairs, <https://reliefweb.int/report/somalia/somalia-gu-rainy-season-2023-flash-floods-update-no-3-13-april-2023>

¹⁸⁵ FAO 2022. Somalia water and land information management. Food and Agriculture Organization. <http://www.faoswalim.org>

¹⁸⁶ SODMA 2024. Deyr Floods Somalia 2023 Rapid Post-Disaster Needs Assessment: United Nations (UN), United Nations Development Programme (UNDP), the World Bank (WB), the European Union (EU) and Government of Somalia, https://www.undp.org/sites/g/files/zskgke326/files/2024-06/somalia_deyrfloods_r-pdna_executive_summary_07062024.pdf

¹⁸⁷ SODMA 2024. Deyr Floods Somalia 2023 Rapid Post-Disaster Needs Assessment: United Nations (UN), United Nations Development Programme (UNDP), the World Bank (WB), the European Union (EU) and Government of Somalia, https://www.undp.org/sites/g/files/zskgke326/files/2024-06/somalia_deyrfloods_r-pdna_executive_summary_07062024.pdf

¹⁸⁸ SODMA 2024. Deyr Floods Somalia 2023 Rapid Post-Disaster Needs Assessment: United Nations (UN), United Nations Development Programme (UNDP), the World Bank (WB), the European Union (EU) and Government of Somalia, https://www.undp.org/sites/g/files/zskgke326/files/2024-06/somalia_deyrfloods_r-pdna_executive_summary_07062024.pdf

¹⁸⁹ FOA(2019) Drought Action Plan <https://openknowledge.fao.org/server/api/core/bitstreams/98faacae-9ab5-4d18-bfbb-57c5cbbbe5b7/content>

amounts. El Niño Southern Oscillation impacts monsoons, leading to warmer and drier winters during El Niño and cooler conditions during La Niña.

- 409 Under SSP3-7.0, projected precipitation patterns show increased regional variability by mid-century and drier dry seasons nationally. Annual precipitation is expected to rise by 14.48 mm from Cambodia's 1995–2014 reference period over 2020–2039. Conversely, a decrease of 11.27 mm is projected over 2040–2059, with southern and eastern provinces seeing the most significant increases and northern areas experiencing the greatest decreases in annual precipitation.
- 410 Climate change will likely exacerbate Cambodia's vulnerability to extreme weather events such as floods and droughts, which can damage infrastructure, disrupt agriculture, and lead to food insecurity. Rising temperatures and changing precipitation patterns may affect water resources and biodiversity, threatening ecosystems and livelihoods—additionally, sea-level rise risks coastal areas, potentially displacing communities and impacting the nation's economic stability.

Appendix 2: Summary of climate hazards across critical sectors and proposed interventions for the participating seven countries

Country	Sector	Climate Hazard	Climate Impact	Proposed Intervention and its expected impact
Cambodia	Water resources	Increasingly intense precipitation events both within Cambodia, as well as upstream on the Mekong. Increasingly drier dry seasons with higher numbers of extreme heat events.	The likely impacts of increasingly intense precipitation events within Cambodia and upstream on the Mekong include severe flooding, which can destroy homes, infrastructure, and crops, leading to economic losses and displacement of communities. Upstream, intense rainfall can exacerbate river flooding and affect water management and hydroelectric operations, causing disruptions downstream. Increasingly drier dry seasons with higher numbers of extreme heat events can lead to water scarcity, negatively impacting agriculture and reducing crop yields. This can result in food insecurity and strain on water resources, affecting both rural and urban populations. Additionally, extreme heat events can increase the risk of heat-related illnesses and stress on public health systems	Improved data management and sharing, both before and after disasters. Enhanced forecasting particularly in terms of timeliness of dissemination and relevance. Country wide improvements to the speed and utility of early warning messaging. Heightened awareness around preparedness activities and upgraded response options. Systemic coordination, legislation and monitoring across pillars.
Fiji	Agriculture, water, health	Drought, TCs and extreme weather events (high intensity rainfall and winds), flooding, and sea level rise	Water scarcity, coastal erosion, food security (reduction in subsistence crop), outbreak of climate-sensitive diseases (usually occurs soon after floods and TCs)	Standardizing risk assessment methods, establishing a centralized database, and enhancing data collection and analysis capabilities for climate projections, evaluating hazard impacts and future risks. Efforts to improve coverage in certain areas include expanding or upgrading observation networks with AWS and enhancing EWS with historical climate data. AI and a centralized database will be utilized for forecasting and early warning. Develop National Emergency Telecom Plan (NETP) to strengthen emergency structure, and review Emergency Telecommunications Cluster (ETC) and create SOPs for stakeholders. Automate geo-targeted messaging, implement cell broadcasting for mass messaging, and improve translation of early warning messages. Developing communication strategies based on risk perception studies and integrating anticipatory actions into national plans. Expanding the Community Based DRM Program to improve community level preparedness and capacity.
Chad	Meteorology and hydrology Agriculture Disaster preparedness	Flash floods, River flooding Drought, dry spells	Loss of life. Lost livestock. Damages to houses, damages to community and household assets (wells, latrines), infrastructure, crops, seed and food stores. Displacement. Vector borne and epidemic disease. Loss of crops, reduced crop yields. Water scarcity. Loss of life through malnutrition and increased disease burden. Increased conflict over natural resources including pasture	Sustainability of climate services, multihazard warning systems and disaster preparedness / management at national and subnational levels. Improved data on vulnerability and risk, improved data transfer, analysis and sharing, and knowledge management

			and water. Proliferation of crop pests.	Enhanced forecasting in terms of timeliness as well as relevance, including IbF. Developing capability to disseminate risk information, preparedness and early warning messaging to end users ensuring messaging meets the needs of users. Heightened awareness around preparedness activities and upgraded response options.
Ecuador	Agriculture	Droughts	Reduced crop yields due to insufficient water; increased risk of crop failure.	Intervention: Improve drought monitoring. Expected Impact: Timely alerts will allow farmers to adjust planting schedules and use water-efficient practices, reducing crop losses and enhancing resilience to drought conditions.
		Extreme Weather Events (Floods)	Damage to crops and soil erosion from heavy rainfall; potential loss of harvest.	Intervention: Develop flood EWSs and integrate them with agricultural risk management practices. Expected Impact: Early warnings will help farmers take protective actions to safeguard crops, minimising damage and losses from flooding.
	Water Resources	Changes in Water Availability (Droughts)	Reduced river flows and water availability for irrigation and consumption.	Intervention: Implement hydrological forecasting. Expected Impact: Improved water resource management and planning will mitigate the impacts of drought on water supply, ensuring more effective use of available resources.
		Extreme Weather Events (Floods)	Flooding of water infrastructure, affecting water supply and quality.	Intervention: Enhance EWSs for flooding and integrate them with water management practices to protect infrastructure. Expected Impact: Reduced damage to water infrastructure and better preparedness for flood events will ensure continued water supply and quality.
	Energy	Hydrological Changes (Droughts and Reduced Precipitation)	Decreased water availability for hydropower generation; potential for reduced energy production and reliability.	Intervention: Implement hydrological forecasting activities for hydropower. Expected Impact: Improved anticipation of water availability issues will enable adjustments in energy production, reducing the risk of power shortages and maintaining energy reliability.
		Extreme Weather Events (Floods)	Damage to hydropower infrastructure (e.g., turbines, dams) from flooding; potential for energy production disruptions.	Intervention: Develop and implement flood EWSs specifically for hydropower infrastructure and integrate these with maintenance and response plans. Expected Impact: Enhanced infrastructure resilience and preparedness will minimise damage and operational disruptions from floods, ensuring stable energy production.
Antigua and	All	Increasingly intense	Hydrometeorological hazards are the primary threats in the country. A significant recent disaster was the combined	Intervention: The project will improve the marine observation and forecasting capacity of ABMS to generate more accurate

Barbuda		hurricanes	impact of Hurricanes Irma and Maria in 2017. Given the islands' small size, a single storm can impact the entire population, leading to severe flooding, storm surges, powerful winds, and widespread destruction. In September 2017, Hurricane Irma struck as a Category 5 storm, causing extensive damage across many Caribbean islands, including Antigua and Barbuda. Essential services like healthcare, electricity, water, sewage, and waste management were heavily disrupted, and livelihoods were severely affected. On Barbuda, the hurricane devastated about 90% of the island's infrastructure, rendering it uninhabitable for its population of over 1,600 people. The country's First Biennial Update Report to the United Nations Framework Convention on Climate Change (UNFCCC) highlighted the risks posed by hurricanes and the exacerbating effects of climate change on these tropical systems. The report noted that between 1851 and 2011, the country was impacted by 93 tropical storms, 44 of which were hurricanes. It also emphasised that hurricanes frequently result in the total loss of GDP.	and locally specific warnings for ocean conditions, wave action and storm surge. Impact: Warnings will enable fisherfolk to take better precautions when conditions are bad, as well as enable preparedness actions for coastal infrastructure (including clearing vulnerable areas, securing movable assets, evacuating flood zones etc.). Preparedness of fisherfolk will be further enhanced through the development of landing zone disaster management plans. Such actions will significantly reduce the potential impact of storms and hurricanes on coastal infrastructure as well as the reduction in loss of lives. Intervention: Update the crisis response manual for tourism and capacity development of the tourism sector to improve enforcement of DRR regulation for tourism establishments. Impact: Improved emergency response protocols for the tourism sector will reduce impact on tourism sector during extreme events, including enabling more efficient evacuations.
	Water resources Agriculture	Drought	Drought is a recognized hazard within the national risk planning framework. The 2015 Nationally Determined Contribution underscores the country's vulnerability to water shortages due to droughts, which typically occur every 5-10 years. With climate change projected to reduce annual rainfall by 30%-50% by 2090, in a country that is already water-scarce, alongside rising temperatures and increased sea levels leading to saltwater intrusion and contamination of freshwater supplies, the risk of drought on the islands is a serious concern. From 1928 to 2007, 24 drought events were documented on the islands. Between 2014 and 2015, the country experienced a severe drought that disrupted utilities nationwide. A drought from 1983 to 1985 significantly impacted the tourism sector, leading to economic repercussions as the country had to import water from Dominica. Historically, droughts have also led to loss of life. These far-reaching impacts highlight the critical need for comprehensive planning for drought, especially in the context of climate change.	Intervention: Preparation of a disaster readiness and response manual for the agricultural extension service. Impact: Farmers will be better informed on how to prepare for and recover from extreme events.
	All	Extreme rainfall and floods	Flood risk is highly prevalent in the country, leading the 2015 Nationally Determined Contribution to emphasise the importance of adapting all buildings to withstand extreme climate events such as droughts, floods, and hurricanes. While flooding is commonly associated with tropical storms and hurricanes, it can also occur outside these events, with the country experiencing riverine, coastal, and flash flooding. Antigua's natural topography causes rapid runoff after heavy	Intervention: The project will improve the hydrological observation and forecasting capacity of ABMS to generate more accurate and locally specific flash flood warnings. Impact: Improved flash flood warnings will enable people to avoid flash floods, limiting the potential loss of life as well as reducing disruptions to other economic activities (for example, by enabling diversion of traffic).

			rains, often resulting in flash floods. Development practices that increase impermeable surfaces have exacerbated this problem by increasing runoff and flash flooding. Intense rainfall further contributes to flooding as waterways overflow due to limited retention capacity. Coastal flooding poses another significant risk due to the country's relatively low-lying topography and the concentration of tourism-related activities along the coasts. Additionally, other hydrometeorological hazards contribute to the national risk profile. Although rare, tsunamis are a recognised threat due to the country's location in a seismically active region, with historical earthquakes triggering tele-tsunamis that have impacted several Caribbean islands. On a much smaller scale, the country also faces hydrometeorological threats such as heat and cold waves, hailstorms, downbursts, and tornadoes.	H
Ethiopia	Meteorology and hydrology	Increasingly drier dry seasons. Increasingly intense precipitation events.	Drought Riverine and flash floods	Improved multi-hazard and multi-sector data collection, management and sharing (national risk data hub accessible to all). Improved impact-based risk analysis and forecast, especially for floods; in high-risk woredas at local levels. Improvement and continuation of upgrade and coverage of hydrometeorological observing systems. Assessment of relevant early warningscommunication channels in view of most vulnerable. Development of media training and messaging on disaster risk, Anticipatory Action and mitigation. Development of an emergency call short code. Improved community-level early warnings systems in high-risk locations; with gender focus; linking to Anticipatory Action recommendations. Agreement on national Anticipatory Action and Emergency Thresholds. Capacity development and integration of Anticipatory Action into preparedness planning at local levels, linked to financing mechanism, and integrating social protection mechanisms. Stockpiling and pre-positioning essential items. Inter-pilar coordination and monitoring.
Somalia	Agriculture and food security, livestock, livelihoods, water resources	Water scarcity, drought	As a predominately arid and semi-arid country with a hot desert climate , prolonged dry seasons result in water scarcity, reduced grazing land, reduced livestock production, livestock perishes, reduced agricultural productivity, food insecurity, and ultimately, loss of livelihoods, displacement, famine and the loss of human life. Drought is a critical issue for Somalia, which disparately affects nomadic and agropastoral communities who are more vulnerable to such droughts.	TBC/under discussion-development Intervention: Integrated Actions for Detection, Monitoring, Analysis and Forecasting Impact: harmonise early warning data collection, analysis and exchange between federal line ministries, sectorial offices, humanitarian organisations and regional bodies, as well as improving data automation, modelling and forecasting of early

	Agriculture, livestock, transport, WATSAN, meteorology and hydrology	Flash flooding, floods and inundation	Somalia experiences regular river floods and flash flooding that result in major economic impacts and damage, as well as many human casualties, with riverine flooding typically occurring along the Juba and Shabelle Rivers in southern Somalia and flash flooding tending to occur in the north along intermittent watercourses.¹⁹⁰ Yet, intense rainfall events and subsequent floods have become more frequent, resulting in water runoff on the earth's surface, washing away fertile soil and causing widespread devastation. Floods have resulted in huge losses and damage that has been widespread across districts mostly affecting agriculture, livestock, and WATSAN and transport but also result in displacement and losses of life.¹⁹¹	warning. Intervention: Activities to improve the interoperability of early warning systems Impact: Improved early warning information exchange between sectors. Intervention: early warning communication strategies and SOPs for early warning information, communication and dissemination governance, functions, roles and responsibilities. Impact: Accessible nationwide warning dissemination and communication Intervention: activities covering simulation exercises, CAP adoption, telecommunication service provider agreements, automated system setups and protocols for AA and communication strategies and feedback mechanisms through local networks. Impact: Comprehensive functional infrastructure networks and early warning dissemination services and equipment Intervention: Hazard risk assessment methodology harmonised with regional approaches, and accompanying hazard risk assessments Impact: Risk at national and local levels translated for critical considerations of key vulnerability factors. Intervention: Disaster risk knowledge materials and communication for national and local levels that include hazard maps that identify specific geographic areas, sectors and groups that may be affected by particular hazards, tailored for EWS and DRR plans/local risk management plans. Impact: available climate risk summaries accessible for national and local levels and considering key vulnerability factors Intervention: A database of disaster impact data and loss and
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¹⁹⁰ FAO 2022. Somalia water and land information management. Food and Agriculture Organization. <http://www.faoswalim.org>

¹⁹¹ SODMA 2024. Deyr Floods Somalia 2023 Rapid Post-Disaster Needs Assessment: United Nations (UN), United Nations Development Programme (UNDP), the World Bank (WB), the European Union (EU) and Government of Somalia, https://www.undp.org/sites/g/files/zskgke326/files/2024-06/somalia_deyrfloods_r-pdna_executive_summary_07062024.pdf

				damage tracking. Impact: Improved reporting and analysis of climate disaster impacts over time. Intervention: Training and enhancing public awareness and education Impact: Capacity and coordination at national and local levels through empower communities to plan and respond. Intervention: Developing participatory, gender-responsive disaster preparedness measures and plans for district levels, integrating Anticipatory Action and preparedness, forecast based plans and protocols/EAPs and SOPs for IBFWS, CAP Impact: building local preparedness capacities.
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Table 2: A summary of the relevant projects and programmes in each country. S/N	Name of Project/Programme	Funding Source	Amount	Period	Key Achievements	Gaps to be addressed/ complementarity
Cambodia						
CREWS/RProj/08/South-East Asia	CREWS Cambodia and Laos (Note - CREWS Cambodia and Lao People's Democratic Republic (PDR) phase 2 has been approved to start from July 2025)	Implemented by WMO, WB - GFDRR, UNDRR	USD 5,540,000	June 2022 – June 2025	Specific technical inputs on flood and drought monitoring and projections being provided including human resource development especially on IbF .	Due to a lack of funding, more costly infrastructure inputs and upgrades remain to be met, however questions remain around ongoing O & M capacities within MoWRAM. Human resource development needs for MoWRAM are considerable and this project will work on a common HR strategy with WMO MoWRAM have requested inputs on improving their online presence.
In the pipeline	SOFF (Systematic Observations Financing Facility)	United Nations Multi-Partner Trust Fund (MPTF).	Estimated USD 3.5 million	In the pipeline	In the pipeline	The SOFF project will install 5 AWS and one upper air station, upgrade data

						transmission to regional and global offices and invest in HR development.
FP-199	Public-Social-Private Partnerships for Ecologically-Sound Agriculture and Resilient Livelihood in Northern Tonle Sap Basin (PEARL)	GCF	USD 42,820,231 of which USD 36,231,981 grant is provided by the GCF	April 2022 – April 2028	As far as early warnings, PEARL has a primary focus on adapting local agricultural systems to the impact of drought and floods.	FAO as part of the Cambodia EW4All initiative have committed to coordinating closely on how to ensure linkages and support for MoWRAM, especially of IbF for agriculture with an emphasis on drought.
GEF Project ID: 5235 UNDP/GEF ID: 5318	Strengthening Climate Information and EWSs in Cambodia to Support Climate Resilient Development and Adaptation to Climate Change Project	GEF	USD 4.9 million	Oct 2016 – Oct 2020	This project substantially increased the capacity of MoWRAM, both in terms of HR development and hydro-met infrastructure.	MoWRAM was unable to provide a sufficient O & M budget to properly maintain the equipment installed and the production of weather data remains limited. In addition, substantial HR development needs still remain.
	Applying Seasonal Climate Forecasting Southeast Asia	BMU	EUR 7977045	April 2018 – June 2024	The project assisted Cambodia in developing national/regional adaptation and risk management strategies.	Funding for the strategies and plans remains a challenge.
	Multi-Hazard Hydrometeorological Early Warning & Capacity Building (EWS-F) & Flash Flood Guidance System (FFGS) Phase III	USAID	USD 25 million	September 2016 – September 2021	This project provided support to the Mekong River Commission (MRC) which has been assisting Cambodia with Multi-Hazard EWSs and Advancing Flood Risk Reduction.	MRC data is critical to the work of MoWRAM ¹⁹² . Although the Mekong is tracked relatively closely, more remains to be done in other river basins in Cambodia. The project will assist with the installation of automated river observation sensors in the target provinces.
	A number of World Bank projects are currently active, including the Cambodia Southeast Asia DRM Project I	World Bank	USD 44.2 million	I from Jun 2017 -Mar 2022 and II in the pipeline.	- Improved rural resilient road connectivity. - Financial planning for disaster resilience strengthened the Ministry of Economy and Finance (MEF's)	Improved financial planning for disaster in the MEF has been a welcome addition. However, contingency funding remains largely

¹⁹² DHRW uploads the water level data from manual stations to MRC's HYDMET software, and subsequently refers to the water level and flood forecasts through MRC's flood forecasting portal at <http://ffw.mrcmekong.org/bulletin.php>.

					capacity for financial planning for disaster resilience. - Contingent emergency response component.	unfunded and instead budgets are redirected from running costs in case of disaster.
Fiji						
1	CREWS Pacific SIDS 2.0 Pre-Approved: CREWS Pacific Drought Project (USD 5,000,000) & CREWS Pacific SIDS 2.0 Additional Financing (USD 5,500,000)	CREWS Implemented by WMO, WB, UNDRR	USD 3.8 million	Jan 2021-Dec 2024		The objective is to further enhance the effectiveness of Pacific Island National and Regional EWSs by improving the integration of national and regional hydrometeorological centres, their production of meaningful severe hazard information and its communication to benefit local population in times of need.
2	Multi-Hazard Hydrometeorological Early Warning & Capacity Building (EWS-F) & Flash Flood Guidance System (FFGS) Phase III	USAID	USD 25 million	September 2021 – September 2016		The goal of this initiative is to collaborate on advancing hydrometeorological end-to-end EWSs and capacity to enable countries to take early action and reduce the risk of climate, weather and water related hazards such as floods and flash floods, cyclones, severe weather, storm surges, extreme temperatures, droughts climate variability and change and other hydrometeorological disaster. It will help countries to reduce impact of weather, climate and water related hazards through advancing EWSs, developing tools, guidance and plans, forecasting events and impact, building capacity and other actions to enable people them to access early warning of these events to take early action. The initiative has two major areas

						of intervention: Multi-Hazard EWSs and Advancing Flood Risk Reduction.
3	WMO HydroHub Phase II	SDC	CHF 2400000	Jan 2021 – Aug 2026		
4	Intra-ACP Climate-Services Programme (ClimSA)	EU	EUR 85 million	Jan 2020 – June 2024		The project aims to improve the production, access to and use of climate information, services, and applications for decision makers. The project will foster sustainable development through the prevention of desertification, preservation of ecological biodiversity and the sustainable use of water management in ACP countries.
Ecuador						
1	ENANDES+	Swiss Agency for Development and Cooperation (SDC)	6,359,676USD (CHF 5,800,000)	November 2022 - October 2026	To reduce human and socioeconomic impact caused by hydrometeorological hazards such as floods, droughts, landslides and severe weather, through increased access to early warnings and risk information.	Gaps: National-Level Coordination: ENANDES+ focuses on regional and local levels, particularly in the upper Pastaza River basin. However, there are gaps in national- level coordination and integration of climate services across different regions and sectors in Ecuador. Broader Hazard Coverage: ENANDES+ is centered around hydroclimatic hazards, like frost, which are critical but do not cover the full range of hazards (floods, landslides, droughts) faced by Ecuador. Comprehensive EWSs: While ENANDES+ includes local adaptation practices and forecasts, there is a need for more comprehensive EWSs that include real-time data

						management and user interfaces for broader community engagement. Complementarity: The proposed MHEWS project will complement ENANDES+ by establishing a national-level coordination platform that integrates various climate services, ensuring broader hazard coverage beyond hydroclimatic risks. It will also enhance the forecasting and real-time data management capabilities initiated under ENANDES+, thereby improving the overall resilience of Ecuador's communities and productive sectors.
2	Euroclima+ Project	European Union	144 million euros	2019-2022	Energy Component of Ecuador's NDC (Nationally Determined Contribution) Regional Amazon Platform of Indigenous Peoples and Climate Change of OTCA (Amazon Cooperation Treaty Organization) Support for the institutional restructuring process of the Meteorology and Hydrology Institute of Ecuador (INAMHI) Support for the implementation of a platform and geovisor	Gaps to be Addressed: Local Hazard and Risk Assessments: Euroclima+ has a broad regional focus on floods and droughts, but there may be gaps in localized hazard and risk assessments, particularly for hazards that are specific to Ecuador. Communication and Inclusivity: The project has worked on capacity building and information sharing but may not fully address the communication needs of remote and Indigenous communities in Ecuador. This includes the development of multi-language alert systems and inclusive communication strategies that will be developed by this project. Complementarity: The proposed project will

						build on the achievements of Euroclima+ by implementing hazard and risk assessment nationwide. It will also enhance communication strategies to ensure that early warnings are inclusive and accessible. Additionally, the development of a National Climate Finance Framework will ensure the sustainability of climate services initiated under Euroclima+.
3	Strengthening the Multi-Hazard National EWS project	IADB	15,300,000 USD	2020-2022	The general objective of the project is reducing the number of people potentially affected by floods (including river overflows), landslides, tsunamis and volcanic eruptions in Ecuador, through the universal provision of accurate early warning information and the implementation of activities to strengthen community resilience capacity	Gaps to be Addressed: Integration of Climate Services: While the project focuses on specific hazards like floods, landslides, tsunamis, and volcanic eruptions, there is a gap in integrating broader climate services, such as financial frameworks, and comprehensive data management strategies. Governance and Coordination: The project primarily concentrates on strengthening community resilience and improving alert communication but lacks a broader governance framework that coordinates multiple stakeholders at the national level, including gender and development initiatives. Complementarity: The proposed MHEWS project will complement this by establishing a National Framework for Climate Services (NFCS) and creating a comprehensive climate data management strategy.

						These efforts will support the existing project by providing a more robust institutional framework and enhancing the sustainability of EWSs.
4	Multi-Hazard Hydrometeorological Early Warning & Capacity Building (EWS-F) & Flash Flood Guidance System (FFGS) Phase III	USAID	USD 25 million	Sept- 2021-2026		Gaps to be addressed: The implemented FFGS was undertaken with input from the NHMS in Ecuador. However, there is a lack of use of the results, and the capacities for using the system are limited. Complementarity: The existing FFGS will be used in the implementation of the flood forecasting system within the NHMS in Ecuador. While the forecasting platform will be complemented with other forecasting approaches, the outputs from this will be used.
Chad						
1	SOFF Investment Phase	SOFF	USD 6,945,084	2024	It is expected that 33 GBON Surface Stations and 6 GBON upper air sites operating and transmitting stations will be functional by the end of the project (2029). This is expected to lead to a substantial increase in availability and reliability of observational data and improvement of the quality of global NWP models outputs including products and forecasts at national and regional level.	SOFF will cover a large part of Pillar 2 activities, through co-financing.
2	CREWS Chad (Chad Hydrometeorological and Early Warning Services)	CREWS (World Bank/World Meteorological Organisation)	USD 3,150,000	2021-2026	Supports the development of the capacities of the ANAM and the DRE to respond to early warning needs. Piloting community-based early warning and dissemination of agro-meteorological bulletins to farmers Install additional hydrological stations	Covers Pillar 2, and pilot work on Pillar 3 which will inform the lessons learned and project design. Institutional support to civil protection services will support Pillar 4 activities.
3	CREWS Central Africa	CREWS (World	USD 4,900,000	2022-2026	Seamless approach to forecasting and	Collaboration between

		Bank/World Meteorological Organisation)			warning for meteorological, hydrological and climate extremes Regional support to capacity building of hydrometeorological institutions, provision of services	regional centres and NMHSs: forecasting, capacity building Support risk information and use of common alert protocols
4	West Africa regional Food Systems Resilience Program (FSRP-2) / PRSA – Chad component	World Bank	USD 105,000,000 (for Chad)	2023-2027	Improving the production and timely delivery of climate, hydrometeorological, agromet and impact-based information for use by policy makers, farmers, pastoralists and other actors in the food system.	Provides approximately USD 8 million in support of weather services, including training of engineers and technicians. Runs PROPAD hotline.
5	The Africa Integrated Climate Risk Management Programme: Building the resilience of smallholder farmers to climate change impacts in 7 Sahelian Countries of the Great Green Wall (GGW)	GCF	USD 143,300,000 (7 countries)	2021-29	Strengthening climate weather information services. Use of forecast-based approach combining climate preparedness with risk transfer.	Execution by WFP and ANAM, complementarity with SOFF and EW4ALL.
6	Flood Crisis Support Project (Projet d'Appui aux Crises d'Inondations - PACIN)	AFD	TBC		Set up a municipal flood risk management service. Strengthen devolved state institutions to prevent and respond to fluvial and pluvial flooding. The resilience of urban and peri-urban communities is strengthened to cope with rain and river flooding.	Pillar 3 activities will be complementary to PACIN. Pillar 4 activities will be designed in a way that is consistent with lessons from the project in building capacity and strengthening institutions for preparedness and crisis management at subnational level.
7	Community-based management of climate risks in Chad (PGCRCT)	UNDP-GEF	USD 17,750,000	2021-2024	Covers seven of the provinces most affected by flooding in Chad. Supports 44 communities in the two Logones areas to embrace community-based climate risk management processes and support resilience to climate change risks.	The new intervention will build both on the achievements and lessons of this project. It introduce new approaches for warning dissemination (pillar 3) to communities which have already benefited from support. It will further strengthen sub-national authorities for preparedness and response (pillar 4) and will introduce IbF to meet the expressed needs of communities.

8	Integrated water resources management and EWS for climate change resilience in the Lake Chad Basin (project under development)	Adaptation Fund	USD 10,755,000	2024-2029	Improve hydrometeorological observation networks, information systems, products and services. Training and awareness raising. Response planning and medium term adaptation measures in pilot areas.	Sustainable funding mechanism for hydrometeorological monitoring; interoperable database and data sharing in place.
9	Anticipatory Action: CERF	CERF pilot funding	USD10,000,000	2022-24	New CERF windows expected.	Integration and complementarity between the players involved in early warning and anticipatory action
10	Anticipatory Action: Red Cross, anticipatory action protocol	Red Cross Climate Centre / DREF	TBC	TBC	TBC	TBC
11	Anticipatory Action: WFP : 2m\$	Austrian Government	USD2,000,000	2024-28	Strengthening and maintaining national forecasting and communication capacities; Reducing the impact of foreseeable climate hazards through anticipatory action. Funds for response are additional.	Integration and complementarity between the players involved in early warning and anticipatory action
12	Pilot Programmatic Partnership Project	EU/ECHO	TBC	TBC	Community based early warning, disaster management component and training local authorities (CPA, CLA and CDA).	Lessons learned, consolidation of community based early warning approaches.
13	Project to strengthen the productivity and resilience of family-run agro-pastoral farms (RePER)	IFAD	USD 62,900,000	2018 - 2025	integration of climate information and observation stations in support of agriculture. The weather stations will also support the agricultural insurance program. ANAM must collect, analyze and disseminate data for agricultural risk management.	Covers Lac, Hadjer Lamis and Kanem Nord regions. Complementarity with pillar 3 activities and IbF.
14	N'Djamena Urban Resilience Project (PILIER)	World Bank	USD 150,000,000	2023-29	Reduce flood risk and strengthen climate resilient urban planning and selected flood-related services in N'Djamena Capacity building of hydrometeorological services on addressing flood issues faced by the capital N'Djamena. Establishing an operational center for	Strategic vision to institutionalise DRR, preparedness and DRM approaches and to build national and sub-national capacity to reduce the impact of disasters.

					disaster management in N'Djamena.	H
Antigua and Barbuda						
1	Systematic Observations Financing Facility (SOFF) Readiness Funding Request	Systematic Observations Financing Facility (SOFF)	USD 182,970	2023-2024	Under this project, UK Met Office, provided SOFF peer advisory services to Antigua and Barbuda to develop the outputs of the SOFF Readiness Phase. Outputs included: National GBON Gap Analysis GBON National Contribution Plan CHD	The EW4ALL project will make use of the improved hydrometeorological systems while linking these to EWS, operational procedures and anticipatory actions.
2	GBON National Contribution Plan of Antigua & Barbuda	Systematic Observations Financing Facility (SOFF)	USD 2.5million	2024	N/A – still in inception phase.	The EW4ALL project aims to address some of the recommendations not targeted for investment during this SOFF investment phase. In particular, EW4ALL will strengthen marine monitoring capacity through the installation/upgrades to existing marine observation infrastructure. EW4ALL will also provide updates to the hydrological data management system since SOFF is planning updates for the CDMS but not the HDMS. In addition to strengthening these Pillar 2 processes, EW4ALL will target interventions across the other pillars to strengthen policies, institutions and processes throughout the MHEWS value chain.
3	Resilience to hurricanes in the building sector in Antigua and Barbuda	GCF	USD46.2 milliion	2021-2027	Under Output 2, climate services will be strengthened to facilitate early action during extreme weather events through two activities: establish a centralised online server within the ABMS to enhance real-time processing of climate data; and establish a formalised communication protocol to facilitate rapid information sharing and early action preceding an extreme	The ongoing GCF project focusing on strengthening the resilience of the country's built environment to the impact of hurricanes. This includes actions to strengthen climate information systems and post-disaster responses, with a focus on coordination and

					weather event.	preparedness. The proposed project will complement and build on the GCF Build projects efforts in strengthening EWS in Antigua and Barbuda. Coordination activities under this project will draw on the assessments and tools developed under component 3 of GCF Build. Moreover, the proposed project will use the guidelines being prepared for CEWS to scale up the approach, drawing lessons from the planned pilot that will be conducted in one district.
4	CREWS Caribbean 1.0 Strengthening Hydro-Meteorological and Multi-Hazard Early Warning Services in the Caribbean	World Bank (Lead) WMO UNDRR	USD 6.5 million	2018–2023	Outcomes were achieved largely at the regional level through the development of a regional roadmap to strengthen and streamline early warning and hydrometeorological services, beginning with a Situation Analysis to identify foundational strengths, weaknesses, opportunities and emerging threats. Institutional strengthening and operational capacity building of regional partners —CDEMA, CIMH, and CMO HQ. The model was adapted and endorsed by eight countries, including Antigua and Barbuda. Further, ten NSPs including frameworks for weather, water and climate services (FWWCS) and detailed action plans for a period of five years have been developed and accepted in multiple countries, including Antigua and Barbuda. Additionally, model SOPs between NMHS and NDRMOs were developed and are ready to be adapted on a national level. In May 2023, the national adaptation of these model SOPs were discussed with four pilot states, including Antigua and Barbuda.	The proposed EW4ALL project aims to implement several of the outputs as recommended in the Situational Analysis Priority Areas for Action in Antigua and Barbuda. The project will also draw on the cost-benefit data generated by CREWS to strengthen advocacy support in the country.

					Assessment of radar capacity in the region resulted in the development of an action plan for strengthening the regional network to ensure its sustainable operation. Four pilot priority activities on end-to-end EWS including IbF tested at the regional and national levels.	H
5	CREWS Caribbean 2.0 Strengthening Multi-Hazard EWSs in the Caribbean Region	CREWS Implemented by WMO, UNDRR	USD 7 million	2024-2027	Project just initiated – no key achievements yet.	Effective Multi-Hazard EWSs (MHEWS) in the Caribbean region are possible only through the strong leadership of strengthened National Meteorological and Hydrological Services (NMHSs) and National DRM Offices (NDRMOs) which integrate co-design and delivery approaches in their services. This will enable improved disaster risk knowledge, better monitoring and forecasting, stronger warning and dissemination capabilities, and enhanced response capabilities of individuals, communities, business and institutions and organizations, with a special focus on reaching and involving the last mile and the most vulnerable groups. The main objective of the project is to strengthen the operational capacities of NDRMOs and NMHSs, in the four+one pillars of MHEWS, through regional cooperation and improved governance mechanisms that are conducive for users to participate and be engaged in the design and delivery of early warnings. The project will closely collaborate with the CREWS initiative, drawing

						on the work being conducted regionally to strengthen MHEWS approaches in Antigua and Barbuda. This will include sharing of disaster risk knowledge, and ensuring alignment and complementarity of capacity development support.
6	National Adaptation Planning in Antigua and Barbuda (NAP)	GCF	USD 3,000,000	2017-2020	The NAP provided valuable assessments of climate risk and prioritisation of key hazards.	The project will draw on the outputs of the NAP to ensure that EWS meet the needs of vulnerable communities.
7	EWSs for Floods in Central America and the Caribbean (EWS-F CA&C) focusing on four priority countries, namely Antigua and Barbuda, Barbados, Guatemala, and Haiti. Phase III	USAID	USD 25 million	Sept- 2021-2026		The EW4All project will collaborate with WMO and USAID in the development of flood EWS, ensuring alignment of efforts to maximise the coverage and bridge gaps in the observation and modelling network.
Ethiopia						
	GBON	SOFF	USD 9.9 million	2023 to X?	GBON Gap Analysis; GBON National Contribution Plan Report CHD Report	tbc
	Integrated DRM Project (IDRMP) Floods Management Project (FMP) Enhancing resilience and DRM including early warnings for flood risks. develop and operationalize impact-based flood EWSs targeting 34 million people living in poor communities in the Awash Basin, the Omo Basin, and the Rift Valley Lakes Basin. The project will support efforts for forecast based financing, to allow the rapid deployment of resources in advance of an impending disaster.	WB_ IDA	USD 300 million			tbc
	Water at the Heart of	Dutch Ministry of	USD 6.8 million	2023-2027		tbc

	Climate Action (W@H) programme Integrated Water Management	Foreign Affairs with a total budget of	in Ethiopia, South Sudan, Uganda			H
	Improving climate and water information services (project name?) Improving early warning information and action chain, linking observations, forecasts, warnings, disaster communication and action	Swedish Meteorological and Hydrological Institute (SMHI)		2023-2025		tbc
	Building Resilience in Ethiopia (BRE) Developing an effective, people-centred, integrated, coordinated, accountable, and decentralized DRM system.	Oxford Policy Management.				Tbc
	CREWS Strengthening Impact Based hydro-meteorological EWSs at community levels	WB	USD 745,000	2022- 2026		Tbc
	ACACIA					Tbc
	ACREI (ended 2020?)					Tbc
Somalia						
	Systematic Observations Financing Facility - Somalia	SOFF	USD 150,000	2024-		TBC
	CREWS Greater Horn of Africa: Strengthening early warning and early action systems for meteorological, hydrological, and climate extreme	Climate Risk and EWSs (CREWS) initiative - United Nations Office for DRR (UNDRR), World Meteorological Organization (WMO), and the World Bank	USD 5,240,998	2022-2026	- Enhancing capacity to produce, disseminate and communicate information services through National Climate Outlook Forums (NCOFs) - Developing Climate Services Toolkits (CSTs) tailored to the operational needs of NMHSs - Enhancing regional and national capacities of WIGOS and WIS (observations and data exchange)	CREWS in Somalia will improve the capacity development for staff in priority technical areas (observation, data management and analysis, meteorological and hydrological forecasting), strengthen the development and delivery of services to the priority farming and pastoralist communities and support technical design of minimum basic observing, data management,

						forecasting and service delivery systems.
	Denmark's support to Early Warning for All (EW4All) through WMO	DANIDA	DKK 40 million	2023-2026	Still in early stage – aims to - Improving the observational and forecasting capacity of NMHSs to generate, customize and disseminate weather and early warning services - Common Alert Protocol (CAP) implementation and training, and introduction to IBF - Improving the quality and availability of observational data regionally and internationally - Support Data Rescue at NMHSs - Provision of ICT equipment such as workstations and laptops etc. for operational forecasting - Education and training of staff from each NMHS for better operational and service delivery capacity	Production, analysis, interpretation and use of climate and risk information to strengthen MHEWS; proposes the installation of 16 stations for GBON compliance and 3 upper air station; National EW4All Roadmap in Place (four pillars and enabling environment meet minimum requirements of the maturity index)
	EW4All Sweden	Swedish International Development Cooperation Agency (SIDA)		2024	- DRM Plans at the FMS level - DRR plans target disaster-vulnerable districts - Gender Mainstreaming in the DRR efforts at the FMs level - The establishment of community-based EWSs and a network of disaster volunteers aimed at enhancing the community's capacity to prevent disasters and respond effectively to emergencies	Expanding and strengthen early warning and early response processes to enhance community and state actors' ability to protect citizens. Coordinates different ministries and hydrometeorological actors, including WMO, FAO, UNDP, UNICEF, the World Bank and ICPAC for mapping the situational context regarding the current status of infrastructure and services related to hydrometeorological and seeks to develop a plan to proceed to support capacity building and improve service

						delivery in Somalia Organize and host a workshop to review and endorse the national roadmap to achieve EW4All objectives by 2027 Technical assessment of NMHS to enhance technical know-how in critical matters related to pillar 2 of the national roadmap Development of National Strategic Plan (for NMHSs)
	FAO SWALIM	EU	Phase 6 – completed (2018-2022) USD 19 million		SWALIM has continued to maintain the hydro meteorological network of over 130 stations in collaboration with MoAI and MoEWR. SWALIM Continues monitoring and issuing Drought and Flood early warning information Mapping of open and weak river embankment breakages bi-annually. The breakages are mapped using very high-resolution images with minimum field verification Capacity development initiatives targeting the government partners as part of the SWALIM transition SWIMS Platform Information dissemination	
	UNDP – Training and Community Based Services/ Enhancing Climate Resilience of the Vulnerable Communities and Ecosystems in Somalia	GEF-LDCF.	USD 8 million	Completed	Over 700 officials and experts trained in Early Warning services nationwide. 42% of trainees are women - EWSs established at Federal and Member State levels, including Somaliland. - Comprehensive assessment of Somalia's EWSs	

					conducted by the Ministry of Energy and Water Resources.	H
	Somalia Crisis Recovery Project (SCRCP)	World Bank	USD 187.5 million	2020-2025	Immediate basic services and livelihood support to the flood-affected states and locust-affected areas nationally through: (a) establishing a cash-for-work scheme for vulnerable households; (b) controlling the desert locust population through ground and aerial spraying operations; (c) restoring and protecting farmers' capacity for agricultural production; (d) revitalizing basic health provisions strengthening response services; and (e) supporting household hygiene promotion and treatment. Medium-term flood recovery to rehabilitate critical public and community infrastructure. The component will rehabilitate water and sanitation systems, health facilities, bridges, and small feeder roads. It will also support local mitigation efforts for risk reduction, such as slope protection and environmental rehabilitation. Longer-term disaster risk preparedness will strengthen the institutional capacity and preparedness of governmental ministries agencies and departments to respond to flood and drought-related emergencies. Project Management will strengthen the institutional capacity PIU (Project Implementation Unit) and State Project Teams for the implementation of the Project. Contingency Emergency Response Component (CERC) will serve as a first-line financing option for an emergency response.	Support the recovery of livelihoods and infrastructure in flood and drought affected areas and strengthen capacity for disaster preparedness nationwide and includes physical investments in flood risk management, and; supporting hydrometeorological and EWSs for the generation and dissemination of hydrometeorological data flood hazard mapping and the establishment of a National Emergency Operations Center.
	Gender, Climate, and Conflict Analysis	UN Women, EU	USD 3 million	Government of Japan	Gender, Climate and Conflict Analysis in Somalia and Assessment of	Integrating gender-sensitive approaches

					Opportunities for Climate Smart Agriculture and Livelihood Opportunities for Crisis-affected and At-risk Women in Somalia	H
	National Adaptation Plans	UNDP	USD 2,957,213	GCF	Somalia's National Adaptation Plan (NAP) Framework (NAP Framework) (2022)	Analysis of climate change risks and vulnerabilities, a synthesis of state-level plans and priority adaptation options.

Appendix 3: Eligibility criteria for project Outputs and Activities

The proposals should meet the requirements of the EW4All initiative (Sendai framework Target G). They should have a strong rationale describing the circumstances that justify using this programmatic approach. The proposals should also be closely aligned with national strategies and policies based on the national climate investment priorities.

The proposals should be designed to satisfy the following criteria:

1. They should have a clearly defined governance structure for decision-making and ensuring country ownership. This structure should be compatible with the global project structure and organization to ensure smooth coordination and collaboration at the global level and with other participating countries throughout the duration of the programme.
2. The outputs of the proposal must be closely aligned with the activities listed under Output 2 of the global funding proposal. This will directly influence the design of the log-frame of the proposal which must be compatible with the logical framework of the global funding proposal.
3. The proposal design process must include comprehensive engagement with government agencies and national, regional as well as global focal points of the four EW4All pillar leads. All these partners must be facilitated to contribute to proposal design and given an opportunity to weigh in on the proposed activities. This consultative process must ensure the respective governments retain full ownership of the design process.
4. The support of multilateral agencies and non-governmental agencies should be ensured for countries with challenges emanating from fragility and conflict. Proposals designed for FCAS states should demonstrate appropriate governance arrangements and mechanisms to implement proposed activities.
5. The proposal should meet the reporting requirements for both physical and financial reporting.
6. The proposal needs to be fully compatible with the monitoring, evaluation and learning system and knowledge management framework provided in the global funding project.
7. The projects should demonstrate sustained impact of investments on a continued basis and should specifically provide for the development of an Operations and Maintenance plan in the initial period of implementation.
8. The proposal should adhere to the legal arrangements and frameworks of the global project and all contractual structures should assure continued operations of the sub-project.
9. The proposal should demonstrate ownership of the respective country government to take onboard and absorb the investments proposed. It should also demonstrate financial and economic sustainability and cost efficiency.
10. The proposal should be compliant with environmental and social safeguards and ensure environmental and social risks are within the GCF risk category “C” or low risk.

11. The proposal must be designed in accordance with the GCF template for funding proposals including all relevant annexes.

The criteria for eligibility of country proposals are based on the following principles. These are closely aligned with the GCF investment criteria.

1. Impact potential: Following criteria will be used to determine the project's impact potential
 - a. The project must clearly set out the specific risks and vulnerabilities, under current and future conditions of climate change, that the project aims to address, articulating a clear causal narrative between specific activities and their objective of reducing vulnerability or climate risk. Projects that include arguments that are well evidenced and quantified using reliable data sources and information will be prioritized.
 - b. The project must clearly describe the adaptation benefit that is expected to be generated by the implementation of the project activities.
 - c. The project must align to GCF adaptation result areas.
2. Paradigm shift potential: The proposal should demonstrate its potential to catalyze impact beyond a one-off project investment by:
 - a. Clearly describing how the project outcome will be sustainable in the long run through securing additional public and/or private financing and building the capacity of national institutions to implement similar activities without GCF funding.
 - b. Demonstrating the potential of being replicated elsewhere in the same country or other countries in region.
 - c. Promoting innovation such as, through the development and/or adoption of new technology or approaches which can contribute to more accurate forecasts, improve dissemination of and access to early warning information or allow communities to set up autonomous early warning and alarm systems.
 - d. Promoting sharing of knowledge and learning within and with other countries and contribute to the global project knowledge sharing.
3. Sustainable development potential: The proposals should demonstrate sustainable development potential by identifying at least one positive co-benefit from the following with associated indicators.
 - a. Economic co-benefits –explicitly aim at poverty reduction.
 - b. Social co-benefits – human health and social development through improved food and nutrition security and reduced loss of life resulting from the impact of extreme weather events.
 - c. Environmental co-benefits –improved ecosystem services and biodiversity.
4. Gender: The project must meet the gender criterion independently of the environmental, social and economic co-benefits. The project must fall under the UN Gender Equality Marker 2 or 3 .

5. Needs of the Recipient: The proposal must meet the need of the recipient by :
 - a. Overcoming barriers related to technical, institutional and/or financial barriers to meeting the objectives of the four pillars of Multi Hazard Early Warning Systems as described in the Early Warnings for All Executive Action Plan.
6. The project must strengthen the capacity of National Meteorological and Hydrological Services and National Disaster Management Agencies institutions to deliver effective MHEWS, enhance coordination and improve governance.
7. Country Ownership: The project must demonstrate alignment with national priorities climate priorities as reflected in nationally determined contributions (NDCs), national adaptation plans (NAPs) and/or national disaster risk reduction plans. It must also demonstrate stakeholder engagement in developing the project, including consultation with relevant stakeholders among government and civil society organizations and with climate vulnerable communities across all relevant climate related hazards. Finally, the project must demonstrate engagement with national designated authorities (NDAs) in developing the project.
8. Efficiency and Effectiveness: To demonstrate efficiency and effectiveness the project should demonstrate that grants from GCF are being used to offset the incremental cost of climate resilient investments, provide targeted climate resilient technical assistance, or for non-revenue generating outputs and non-market elements. By blending grants from GCF with additional grants and/or loans in order to augment the volume of financing available for strengthening MHEWS, disaster response and adaptation and better tailor concessionally to achieve transformative investments which would otherwise not proceed.

In addition to these criteria, the project must have a framework to manage environmental and social risks and impacts of the activities through the design of and implementation of an Environmental and Social Management Framework (ESMF). The ESMF must adhere to compliance benchmarks UNDP as its AE, as per the terms set forth in the AMA.

No-objection letter issued by the national designated authority(ies) or focal point(s)



GOVERNMENT OF ANTIGUA AND BARBUDA, MINISTRY OF HEALTH, WELLNESS, SOCIAL
TRANSFORMATION AND THE ENVIRONMENT

Lwr High and Long Streets
St. John's, Antigua, W.I.
Telephone: (268) 462-5522
Facsimile: (268) 562-5174
E-Mail Address: gheslin2015@gmail.com

H 31/95

To: The Green Climate Fund ("GCF")

6 December 2024

Re: Funding proposal for the GCF by the United Nations Development Programme regarding Early Warning for ALL (EW4ALL): Antigua and Barbuda

Dear Madam, Sir,

We refer to the programme titled **Early Warning for ALL (EW4ALL): Antigua and Barbuda in Antigua and Barbuda** as included in the funding proposal submitted by the United Nations Development Programme to us on **29 November 2024**.

The undersigned is the duly authorized representatives of the Ministry with responsibility for the Environment and the Ministry of Finance and Corporate Governance, the Co-National Designated Authority of Antigua and Barbuda.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of **Antigua and Barbuda** has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of **Antigua and Barbuda**;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Ena Dalso-Henry
Permanent Secretary
Ministry with responsibility for the Environment
Co-National Designated Authority

Peter Abraham
Debt Manager
Ministry of Finance and Corporate Governance
Co-National Designated Authority



KINGDOM OF CAMBODIA
Nation Religion King

Ministry of Environment

N° : 6538 / 1026 MoE

Phnom Penh, 11 October 2024

To: The Green Climate Fund ("GCF")

Subject: Funding proposal for the GCF by United Nations Development Programme regarding Early Warning for All Multi-Country Project

Dear Sir / Madam,

We refer to the project titled " Early Warning for All Multi-Country Project in Cambodia (EW4ALL) " as included in the funding proposal for the Global Programme titled Early Warning for All Multi-Country Project.

The undersigned is the duly authorized representative of the Ministry of Environment, the National Designated Authority of Cambodia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Cambodia has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with the national priorities, strategies, and plans of Cambodia; and
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.



Your Sincerely,
Minister of Environment
National Designated Authority

Dr. Eang Sophalleth



MINISTRY OF ENVIRONMENT, FISHERIES AND SUSTAINABLE DEVELOPMENT

Re: Funding Proposal for the GCF by the United Nations Development Programme regarding the Early Warnings for All (EW4ALL) Project.

Dear Madam, Sir,

We refer to the Project titled Early Warning for all (EW4ALL) as included in the funding proposal submitted by the United Nations Development Programme to us on 30 October 2024.

The undersigned is the duly authorized representative of the Ministry of Environment, Fisheries and Sustainable Development, ABAKAR MORNON, the focal point of Chad.

Pursuant to GCF decisions B.08/10 and B.13/21, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Project as included in the funding Proposal.

By communicating our no-objection, it is implied that:

(a) The government of Chad has no-objection to the Project as included in the funding Proposal.

(b) The Project as included is in conformity with Chad's national priorities, strategies and plans; and

(c) In accordance with the GCF's environmental and social safeguards, the Project as included in the PPF Proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the Project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

ABAKAR MORNON



To: The Green Climate Fund ("GCF")

Quito, 8 November 2024

**Re: Funding proposal for the GCF by United Nations Development Programme regarding
EW4ALL Ecuador Country Proposal.**

Dear Madam, Sir,

We refer to the programme titled *EW4ALL Ecuador Country Proposal* in Ecuador as included in the funding proposal submitted by United Nations Development Programme to us on 6 November 2024.

The undersigned is the duly authorized representative of Ministry of Environment, Water and Ecological Transition/Angel Sandoval, the National Designated Authority of Ecuador.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Ecuador has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Ecuador;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Firmado electrónicamente por:
ANGEL JAVIER
SANDOVAL TORRES

Angel Sandoval
NDA for GCF project in Ecuador
Ministry of Environment, Water and Ecological Transition
Ecuador



FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA
MINISTRY PLANNING AND DEVELOPMENT

To: The Green Climate Fund (“GCF”)

Addis Ababa, November 28, 2024

Re: : Funding Proposal for the GCF by the **United Nations Development Programme** regarding the **Early Warnings for All (EW4ALL) Project**

We refer to the Project titled Early Warning for all (EW4ALL) as included in the funding proposal submitted by the United Nations Development Programme to us on 01 November 2024.

The undersigned is the duly authorized representative of the Ministry of Planning and Development, **Abas Mohammed Ali, the National Designated Authority/focal point of Ethiopia.**

Pursuant to GCF decisions B.08/10 and B.13/21, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Project as included in the funding Proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Ethiopia has no-objection to the Project Preparation Facility request as included in the PPF Proposal.
- (b) The PPF Proposal is in conformity with Ethiopia’s national priorities, strategies and plans; and
- (c) In accordance with the GCF’s environmental and social safeguards, the PPF activities as included in the PPF Proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the PPF Proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Abas Mohammed Ali, CEO Climate change Policy and Actions
NDA for GCF projects in Ethiopia
UNFCCC National focal point
Ministry of Planning and Development
Addis Ababa, Ethiopia
Cell: +251 91 154 45 26
P. O. Box 4472 (Office)



MINISTRY OF ENVIRONMENT AND CLIMATE CHANGE

Levels 1 & 2, Bali Tower, 318 Toorak Road
P. O. Box 2109
Government Buildings, Suva, Fiji

TELEPHONE NO: (679) 8921476

Email: info6@govnet.gov.fj

28 November 2024

By E-mail: countries@gcfund.org

Mafalda Duarte
Executive Director
Secretariat of the Green Climate Fund
175 Art Center-daero
Yeonsu-gu, Incheon 406-840
South Korea

Dear Sir/Madam

Re: GCF Funding Proposal by the United Nations Development Programme (UNDP) on Early Warnings for All (EW4ALL)

We refer to the programme Early Warnings for All (EW4ALL) in Fiji as included in the funding proposal submitted by United Nations Development Programme (UNDP) to us on 5 November 2024.

The undersigned is the duly authorized representative of the Ministry of Environment and Climate Change, the National Designated Authority of Fiji.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Early Warnings for All (EW4ALL) programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Fiji has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Fiji; and

(c) In accordance with GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the Early Warnings for All (EW4ALL) programme, as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Thank you.

A handwritten signature in blue ink, appearing to read "Sm Michael".

Dr. Sivendra Michael
Permanent Secretary / NDA Focal Point
Ministry of Environment and Climate Change



FEDERAL REPUBLIC OF Somalia
MINISTRY OF ENVIRONMENT & CLIMATE CHANGE
Office of the Minister

Ref:MOECC/0365/2024
To: The Green Climate Fund ("GCF")

Date:2024-11-28

SUBJECT: Funding proposal for the GCF by the United Nations Development Programme Regarding Early Warnings for All (EW4All) Multi-Country Project

Dear Madam, Sir,

We refer to the project titled "Early Warning for All Multi-Country Project in Somalia (EW4All)" as included in the funding proposal for the Global Programme titled Early Warning For All Multi-Country Project.

The undersigned is the duly authorized representative of the Ministry of Environment and Climate Change, the National Designated Authority of Somalia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- The government of Somalia has no-objection to the project as included in the funding proposal;
- The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Somalia; and
- In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


H.E Ambassador Khadija Mohamed Amakhzoum
Minister, Ministry of Environment and Climate Change
Federal Government of Somalia



Secretariat's assessment of FP258

Proposal name:	Multi-country Project Advancing Early Warnings for All (EW4All)
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Antigua and Barbuda, Cambodia, Chad, Ecuador, Ethiopia, Fiji, Somalia
Project/programme size:	Medium

I. Overall assessment of the Secretariat

- The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The programme, Multi-country Project Advancing Early Warnings for All (EW4All), presents innovative solutions for improving, at scale, the adaptive capacity of communities in the target countries through improved production, dissemination and use of climate information and multi-hazard early warning systems (MHEWS) among the most vulnerable communities. This will contribute towards ensuring that vulnerable communities receive early warnings in a timely fashion and develop the capacity to undertake anticipatory action to mitigate, prepare for and respond to the climate hazards and impacts that they face.	As three of the seven target countries, namely Chad, Ethiopia and Somalia are faced with conflict and social and institutional fragility status, there are additional risks for a timely and synchronized implementation of the programme. A range of mitigation measures have been built into the programme's design, including disaggregated disbursement per country and reallocation clauses and provision of a global platform to enable the participating countries to share experiences and learn from each other.
It focuses on promoting the design of pragmatic climate information and early warning actions and activities that build on implementation strategies and approaches based on lessons learned from earlier work. This will help the participating countries to establish, operate, and maintain and expand the coverage of end-to-end MHEWS that are tested, sustainable and effective in reducing the exposure of the vulnerable communities to climate hazards and impacts.	

The Multi-country Project Advancing EW4All focuses specifically on addressing barriers and challenges that prevent National Disaster Management Agencies and National Meteorological and Hydrological Services in the least developed countries and small island developing States, including those in fragile and conflict-affected states, from operating. In addition, the programme's implementation through both national and global institutions is commendable, because it will facilitate the improvement of the existing capacities, structures and governance mechanisms across the global early warning support system and mechanism, while continuing to build the capacity of the participating countries at all levels.	
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2. The Board may wish to consider approving this funding proposal in accordance with the terms listed in the term sheet agreed between the Secretariat and the accredited entity (AE), and, if considered appropriate, subject to the conditions set out in annex II to document GCF/B.41/02.

II. Summary of the Secretariat's assessment

2.1 Programme background

3. The Multi-country Project Advancing Early Warnings for All is an intervention aimed at promoting climate-resilience among vulnerable communities in Antigua and Barbuda, Cambodia, Chad, Ecuador, Ethiopia, Fiji, and Somalia, by helping to institute relevant policies and governance mechanisms for end-to-end MHEWS, establishing early warning system (EWS) infrastructure and setting up systems for its long-term operation, and supporting local capacities in the generation and dissemination of early-warning products. The objective of the project is to strengthen the resilience of vulnerable communities in the participating countries through enhancing production of actionable multi-hazard early warnings and impact-based forecasts, strengthening of communication and last-mile dissemination channels, and building capacity for the effective use of early warnings and advisories for early and anticipatory action, with a view to reaching 80 per cent of the most vulnerable communities in the targeted seven countries.

4. The project's seven target countries face growing disaster risk resulting from increasing prevalence and severity of different types of hazardous events, which pose adverse impacts for both the social and natural systems in the countries, including exacerbated food, water, health and livelihood insecurity, and infrastructure and economic losses. The countries are faced with extremely high levels of vulnerability to climate change, with limited readiness to access knowledge, or technical and financial capacity to build the resilience of communities and climate-proof their social and economic infrastructure.

5. The project will overcome these challenges by accelerating the development of EW4All projects in the participating countries, enhancing the enabling conditions for the implementation of EW4All actions that meet each participating country's needs, and rolling out EW4All investment activities in the seven countries based on the Climate Information and Early Warning System (CIEWS) paradigm shift and EW4All target G (Sendai Target G) as selection criteria. It will support the countries to establish and sustain EWS infrastructure, set up systems for its long-term operation, expand the coverage of end-to-end MHEWS and support local capacities in the generation and dissemination of early-warning products, and help institute relevant policies and governance mechanisms for end-to-end MHEWS.
6. The project is requesting USD 103.25 million of GCF financing. The total cost of the project, including co-financing, is USD 114.59 million, with co-financing primarily in the form of cash and in-kind grants from the governments of the participating countries, EW4All pillar leads, and the United Nations Development Programme (UNDP). The programme will also receive parallel financing of USD 40.10 million from the Systematic Observations Financing Facility (SOFF).
7. In terms of environmental and social safeguards (ESS), this proposal has been categorized as environmental and social (E&S) risk category C. The review by the Secretariat confirms the E&S risk category assigned by the AE.

2.2 Component-by-component analysis

Component 1: Enhancing enabling conditions globally and regionally to promote country-level actions towards the EW4All goals (total cost: USD 13.50 million; GCF cost: USD 6.68 million)

8. Component 1 will accelerate and scale up national access to global and regional technical and operational support for the countries to develop comprehensive MHEWS and respective capabilities. This component will focus on enhancing enabling conditions for the countries to access the global coordination and support architecture for EW4All and for them to undertake country-level EW4All actions in a way that will be sustained beyond the project implementation period. This will entail establishing enhanced coordination and technical support for EW4All implementation, strengthening monitoring and evaluation mechanisms and systems to support EW4All implementation, and strengthening knowledge management and learning.

Component 2: Accelerating actions in select countries towards EW4All (total cost: USD 92.50 million; GCF cost: USD 88.93 million)

9. Component 2 will focus on addressing gaps in policies, inter-agency coordination and strategies; building technical and scientific capacities; improving capacity for observation; and promoting effective local actions. The activities will be based on the four pillars of the EW4All executive action plan and will be implemented through the country's national governments with support from UNDP, the EW4All pillar leads (World Meteorological Organization, United Nations Office for Disaster Risk Reduction (UNDRR), International Telecommunication Union and International Federation of Red Cross and Red Crescent Societies) and their in-country national offices and other relevant partner agencies. The component will entail strengthening governance and coordination mechanisms and capacities; bolstering disaster risk knowledge and observation and forecasting systems, network and capacities; and fortifying warning dissemination, preparedness and response.

Project management and coordination and monitoring and evaluation (M&E) (total cost: USD 8.58 million; GCF cost: USD 7.63 million)

10. Project implementation and M&E costs have been adequately provided for, and specific activities have been planned and integrated within the design of the project to ensure that

project implementation and monitoring and evaluation processes will be effectively coordinated.

Gender and environmental and social management framework aspects

11. A costed environmental and social management framework (ESMF) was developed for the project, outlining specific activities that will sufficiently ensure that ESS safeguards are put in place and observed throughout the project's lifespan. A well-designed gender action plan was also developed, laying out costed plans of actions that will be undertaken in the seven project countries to ensure that gender equality, equity and safeguards will be upheld throughout the lifespan of the project.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: High

12. Currently, one-third of the world's population, communities in Africa, least developed countries and small island developing states being particularly affected, lacks comprehensive coverage by early warning systems. At the launch of the EW4All Initiative in March 2022, United Nations Secretary General Antonio Guterres called on all governments, financial institutions, and civil society to support this crucial effort to ensure that every person on Earth is protected by early warning systems by 2027.

13. The proposal presents a robust climate change context and demonstrates how the ambitious EW4All targets can be achieved through an integrated and comprehensive global project that supports the establishment of bespoke country-led enhancements to MHEW in the target countries. All seven countries that are targeted by the project are faced with significant climate variability and change, affecting livelihoods, food security, and economic stability. Since the 1970s, extreme weather events have led to agricultural losses, food crises, water scarcity, flooding, and environmental degradation. The countries face a wide range of climate risks and hazards, including:

- (a) Hurricanes, extreme rainfall, droughts, and rising sea levels in Antigua and Barbuda;
- (b) Floods, extreme heat and longer droughts in Chad;
- (c) Increased intensity and frequency of extreme and slow-onset climatic events such as melting of glacier cover, increased variations in sea-surface temperature, increased floods and intensified droughts in Ecuador;
- (d) More frequent intense temperatures and droughts, increased frequency and intensity of extreme rainfall events, more severe cyclones and increased floods, flash floods and landslides, sea level rise, and increased storm surges and coastal erosion in Fiji;
- (e) More erratic rainfall, prolonged dry spells and worsening drought conditions, wild land and forest fires, and intensified rainfall events leading to severe flooding, land and mudslides in Ethiopia;
- (f) Exacerbated flooding and drought risks resulting from more intense rains, a delayed rainy season and longer dry seasons, unexpected dry periods during the rainy season, and untimely rains in Cambodia; and
- (g) Drought, floods, cyclones and extensive coastal flooding in Somalia.

14. The project seeks to support the countries to address these climate challenges by harnessing the crucial role that climate information and early warning play in enabling individuals, communities, and organizations to take appropriate measures to minimize the

potential impact of the disasters that the climate hazards would create. The project aims to achieve this by supporting concrete investment in infrastructure and systems needed to observe weather events, forecast climate hazards and provide vulnerable communities and public and private sector decision makers with advance notice of the potential hazards in a way that would be timely enough to enable them to prepare and respond before the hazards occur, and thereby save lives and livelihoods and reduce damage and losses.

15. The project will support at least 26.34 million direct beneficiaries and 51.66 million indirect beneficiaries. It would effectively increase climate-change resilience among this large number of people (at least 80 per cent of the population) across the seven countries by providing technical assistance, implementation support for installation and operation of early warning infrastructure and systems (component 1), and support for strengthening of climate information and early warning governance and coordination frameworks and processes in the countries (component 2). These interventions are all vital in ensuring that the countries attain the kind of climate information and early warning systems that will be effective and sustainable in supporting communities and businesses to undertake adequate anticipatory action in response to the threats that climate hazards pose to their lives, livelihoods and social and economic systems and infrastructure.

16. The project will specifically support adaptation and resilience-building across seven core areas for local climate resilience-building in the seven countries: climate change-adapted agriculture, forestry, agroforestry, water and health, energy efficiency, renewable energies, and climate-resilient infrastructure and coastal areas (Antigua and Barbuda, Fiji and Somalia). The grant finance will cover additional costs of public early warning investments.

17. Within the context of majority of these countries, where a significant percentage of the population (over 80 percent in Ethiopia, Chad and Somalia) derive their livelihoods from climate-sensitive sectors such as rain-fed, subsistence agriculture, access to climate information and early warnings remains a major challenge.

18. The universal access target of the EW4All programme aims to ensure that it will ultimately benefit 100 per cent of the target countries' population, as everyone can use climate information for dynamic decision-making, planning, and reducing the risks associated with climate variability and change over a range of decision timelines. The proposal is designed to align with this target.

19. The benefits and impact potential of this intervention is therefore assessed to be very significant.

3.2 Paradigm shift potential

Scale: High

20. The updated Strategic Plan for the GCF 2024–2027 (USP 2) notes that GCF can substantially contribute to the EW4All Initiative. This is evidenced by strong support from the Board with an MHEWS target of supporting up to 60 countries. Realizing this target requires aligning the GCF pipeline to the EW4All priorities with the aims of prioritizing support for the most vulnerable, optimizing the impact of investments, coordinating various investments sources and improving innovation to deliver quality at scale in a timely manner.

21. GCF has a range of tools and approaches that could be harnessed to deliver the USP 2 and EW4All targets. These include the projects and programmatic approaches and ongoing work under the simplified approval process through the Climate Risk and Early Warning Systems Initiative (CREWS)–simplified approval process scaling up framework and fast-tracking approaches. The proposal has been co-designed by the EW4All pillar leads (World Meteorological Organization, UNDRR, International Telecommunication Union and International Federation of Red Cross and Red Crescent Societies, International

Telecommunication Union) and a broad range of public- and private-sector partners. The ambitious targets required a coordinated approach by all the pillar leads and partners. A meeting with all pillars leads, SOFF and CREWS was held in Bangkok, 3–5 May 2023 to develop a conceptual framework (EW4All Global support programme) for the timely delivery of the EW4All targets.

22. The key issues identified to be addressed to deliver the targets include:
 - (a) The need to develop a **financing plan** to accompany the EW4All implementation strategy. (What is the role of GCF in the financing plan? How does it crowd in other investors?)
 - (b) The need to support **pipeline generation**, and proposal maturation, in particular, via direct access entities;
 - (c) The need to provide funding for **coordination**, fit-for-purpose prioritization, monitoring, lesson learning, knowledge management, and regular meetings; and
 - (d) The need to **invest in/pilot early movers and examples** to serve as inspiration.
23. The conceptual framework arising from this exercise has led to the development of a GCF project proposal as an “**incubator**” with a project preparation grant to create the financing plan and prepare the project with the following elements:
 - (a) **Pipeline generation** – Prioritize support for pipeline development particularly by direct access entities, leveraging from frameworks like CREWS and SOFF;
 - (b) **Coordination** – Use existing mechanisms via the EW4All pillars to promote a coordinated approach nationally, regionally and globally. Finance regular meetings to discuss opportunities, pipeline development, and share good practices and lessons learned. Attract co-investors, including the private sector, and develop standards, templates, and indicators; and
 - (c) **Pilot projects** – Invest in pilot projects to serve as inspiration of what can be achieved.
24. The EW4All Facility will be established that supports the preparation of four to five projects (including multi-country programmes) for GCF Board approval annually. This should result in systematic and technically sound, predictable project maturation, increased quality of projects, local partner-owned and implemented, co-invested by donors. Other projects will be developed following this initial project. The approach has the potential to meet the USP 2 target of covering 60 countries by 2027. This proposal is the first attempt to operationalize the EW4All conceptual framework at scale through a programmatic approach.
25. The project demonstrates a high paradigm-shift potential in that it seeks to deliver a shift of disaster risk management processes in the target countries from reactive approaches that are focused on response to impacts of climate hazards and disasters, towards an approach that also prioritizes preparedness and anticipatory action, where vulnerable communities and decision makers in climate-sensitive sectors have easy access to widely available reliable science-based climate information and early warnings.
26. The project promotes an approach that fosters the development of an integrated and comprehensive MHEWS value chain, which integrates long-term financing strategies to enable the sustainability of the operation and maintenance of the MHEWS infrastructure and systems. The project uses lessons learned from previous work, such as the Afghanistan Community Resilience Programme, to inform the design of climate information and early warning systems and processes in a way that has the potential for scaling up and replication across the globe, particularly in fragile and conflicted affected situations.
27. The project demonstrates a paradigm-shift potential across all the components of the project and ensures the strategic focus is maintained over the long term.

28. The project presents an innovative and strategic approach to promoting and increasing the participation of the national and local governments, communities and local-level private-sector actors in the development and implementation of CIEWS.

3.3 Sustainable development potential

Scale: High

29. According to UNDRR, advances in early warning systems and preparedness have saved many lives and significant financial resources. People-centred, MHEWS can effectively minimize harm by triggering timely and well-prepared actions. These systems are among the most proven and cost-effective methods for reducing disaster deaths and losses. However, as at 2022, only half of the world's countries have such systems, with even lower coverage in developing countries, where less than half of the least developed countries, and only one-third of the small island developing States are protected.

30. The project will significantly reduce the exposure of poor and vulnerable communities to natural hazards and will thereby be instrumental in helping these communities to safeguard their lives, assets and economic resources from potential damage from such hazards. These avoided human and economic losses and the resultant development of climate risk-resilient local and national economies will in the long term yield vast social, economic and environmental benefits for the vulnerable communities and for the sustainable development aspirations and goals that the global community seeks to achieve.

31. The project contributes directly towards efforts to achieve the Sustainable Development Goals (SDGs), especially SDG 1 (No poverty); SDG2 (Zero hunger); SDG 9 (Industry, innovation and infrastructure); SDG 11 (Sustainable cities and communities); SDG 13 (Climate action); and SDG 15 (Life on land).

32. The project also significantly aligns/contributes to the GCF targeted results for 2024–2027 (USP 2), specifically:

- (a) Target 1. Developing countries directly supported to advance nationally determined contribution/national adaptation plan/long-term strategy implementation (four developing countries);
- (b) Target 3. CIEWS (project areas in the four countries would be protected by robust early warning systems);
- (c) Target 4. Food (promote investment in transformative agricultural practices);
- (d) Target 5. Ecosystems (promote investment in ecosystem-based adaptation measures);
- (e) Target 6. Infrastructure (social and community infrastructure made resilient through investment in climate-proofing);
- (f) Target 7. Clean energy (renewable energy sources to drive the interventions); and
- (g) Target 9. Adaptation and locally-led action (the proposal is an adaptation proposal).

33. The economic, social and environmental co-benefits that the project offers would be achieved and sustained through capacity-strengthening that the project will provide to the participating countries to enhance their ability to develop, operate, maintain and sustainably manage climate information and early warning systems and dissemination channels. This will contribute directly to building awareness of climate hazards, and capacity to mitigate the risks that they pose and adequately respond to their impacts.

34. The sustainable development potential of this investment is therefore estimated to be high.

3.4 Needs of the recipient

Scale: High

35. The project targets some of the world's most vulnerable countries, where water, food and livelihood insecurity conditions arise from the various hazards posed by extreme climate variability, which are compounded by the political fragility in some of the target countries.

36. The target communities are facing severe challenges from climate change threatening their current livelihood strategies, and yet, they lack reliable access to usable climate information and early warning systems to support their need, and effort to secure their livelihoods and natural capital from the impacts of the climate hazards that afflict their localities. The current local livelihood, food, water and health systems in the participating countries are not able to withstand the pressure from climate hazards. Through the development of MHEWS, the proposed interventions will help communities to use advance information to develop strategies for safeguarding and transforming their livelihood options, local food, water and health systems as well as ecosystems to become resilient to climate hazards, thereby building the resilience and adaptive capacity of the vulnerable communities.

37. The target countries are heavily indebted and do not have sufficient resources to invest in the actions to safeguard their populations from climate-induced hazards and the associated livelihoods, water and food insecurity. The countries and beneficiaries are currently struggling to access usable climate information and early warnings that they need to prepare and secure their well-being, livelihood and food, water and health systems. Without access to climate information and early warning, the local communities and economies in the target countries will not be able to prepare and develop, in a timely fashion, the relevant the safeguards and safety nets that they need. The project will fill an important gap through working with key global, national and local institutions in both the public and private sectors to foster long-term solutions for enhanced access and use of climate information and early warnings for building resilient infrastructure and health, water, food and livelihood systems.

3.5 Country ownership

Scale: High

38. Country ownership is well embedded in the project, which is fully aligned with the climate change and broader development aspirations, strategies and policies of all seven target countries. The project will promote the development of capacity among local governmental and non-governmental actors in the target countries to disseminate and use climate information and early warnings to identify their own solutions and to prepare for and undertake anticipatory action. The project also places an important focus on ensuring better synergy with ongoing regional and national strategies and actions being implemented in the target countries.

39. The project is aligned with the nationally determined contributions, national adaptation plans, climate change policies and programmes, as well as the GCF country programmes of the target countries. Furthermore, the project has been developed with the strong participation and commitment of the governments concerned, as well as the participation and commitment of other national and local-level stakeholders, involving all levels of stakeholder groups, including grass roots, women and youth.

40. The AE is the UNDP, which, along with the four early warning pillar leads that are participating in the project as implementing partners, has a long-standing presence in all the target countries, and a wealth of experience working with stakeholders and dealing with the development and environmental challenges in the countries. UNDP offers a unique structure as well as experience and partnerships that will facilitate meaningful engagement and involvement of all relevant stakeholders in the target countries in the successful implementation and scaling up of the programme activities and outputs.

41. The architecture of the project consists of a Project Board and Project Management Unit at the country level where the country governments, acting through relevant national institutions, will participate directly in the governance and management of the project and its activities. This will bolster the integration of local contexts and ownership in the project's activities and outputs.

42. The project is built on well-designed stakeholder consultations that were conducted throughout the project development process, ensuring local knowledge, innovation and sustainability, and participation is fully embedded in the design of the project.

3.6 Efficiency and effectiveness

Scale: Medium to high

43. The overall EW4All action plan proposes an investment of USD 3.1 billion over five years, which equates to just USD 0.50 per person per year. This funding aims to enhance disaster risk knowledge and management, improve observation and forecasting, and strengthen the dissemination and communication of warnings, as well as preparedness and response capabilities. To achieve these goals, this proposal plans to leverage existing pooled funding mechanisms like CREWS and SOFF as well as global multilateral funds, including development banks.

44. This programme will directly support poor and vulnerable communities in the participating countries to strengthen their ability to secure, diversify and transform their own adaptive capacity and livelihood systems. It will target 26.84 million direct beneficiaries and 51.26 million indirect beneficiaries and will drive innovation and transformation by providing access to climate information and early warning for communities and businesses to use in undertaking anticipatory action to protect themselves from potential climate hazards and disasters.

45. The funding amount is well justified to provide the support needed to safeguard lives, livelihoods and local health, water and food systems and to provide innovative approaches and lessons for transforming climate governance in the participating countries and around the globe in general. The project is fostering an enabling environment for private- and public-sector investments in climate information and early warning systems. It aims to achieve this through the structuring and deployment of a global support mechanism for EW4All and of a value-chain approach to addressing the barriers to EW4All; and through developing robust and sustainable CIEWS and thereby foster an innovative approach to helping countries to access long-term financial innovation for EWS and maintain their sustainability.

46. The project also has an innovative exit strategy that will enable scaling up its approach across the globe and will build access and partnership between global actors and governments, communities and businesses to tackle local climate-change challenges with innovative and locally appropriate and developed solutions based on reliable and science-based climate information.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

47. The programme is categorized as C for environmental and social impacts and risks in accordance with the GCF Revised Environmental and Social Policy and the accreditation level of the AE. While most of the activities are focused on enhancing risk knowledge, upgrading information and forecasting systems, capacity-building, technical assistance, disaster preparedness and planning and information dissemination in seven countries, small

infrastructure works will be undertaken under output 2. Output 2 will involve the procurement of goods and equipment to upgrade and/or strengthen forecasting services in the seven selected countries, which includes installation or upgrading of hydrometeorological stations, installation of water storage tanks, installation of toilets and minor improvements to existing access paths. The upgrading and installation works will have minimal physical footprints. An ESMF has been prepared to include an assessment of all the associated E&S risks and impacts, proposed mitigation measures, including through an environment and social code of practice, which would guide the implementation of minor works. There will be no displacement of people or their livelihoods as a result of programme interventions. No work will be undertaken on sites of cultural, historical or environmental importance. New installations will be sited on government property or public areas free of encumbrances, with existing access so that operation and maintenance (O&M) tasks will not require the crossing of private lands, biodiversity areas or other zones that would pose any impact to the environment, workers or communities (including zones where there are ongoing conflicts). The ESMF will guide safeguard implementation as the specific activities and locations are not fully known at project preparation stage. The ESMF will be the basis for the screening of subprojects during implementation, and includes exclusion criteria where activities considered as category A or B will not be considered for programme financing. Nevertheless, all activities will be required to apply any national regulatory permits that would apply to minor works.

48. The project is multi-country in nature and will be implemented in countries that have been classified by the World Bank as fragile and conflict-affected states (i.e. Chad, Ethiopia and Somalia). Any subproject located in these countries will develop a corresponding conflict sensitivity analysis informed by consultations to assess institutional needs and capacities so that interventions will be designed in a way that ensures they will not create or exacerbate conflicts and ideally will contribute to peace. In such countries, no minor works will take place in areas where there is ongoing conflict or where works could exacerbate existing tensions.

49. At the programme level, a global Project Management Unit will be established within UNDP which will include a Project Manager), Technical Advisor, Monitoring and Evaluation and Learning Officer, Communications Expert, and a Safeguards, Gender and Inclusivity Expert (SGIE). The Project Manager will work together with the SGIE to oversee overall compliance with the ESMF. At the country level, a national Project Management Unit will be established as well. It will include a national Project Manager who will work with the national SGIE to ensuring compliance with E&S measures laid out in the ESMF. The national SGIE will provide capacity-building for central and field staff, contractors, and community members who will be involved in ESMF and environmental and social code of practice implementation. Periodic monitoring of ESMF implementation will be undertaken by the national SGIEs with oversight from the global SGIE.

50. A global stakeholder engagement plan (SEP) has been developed through extensive stakeholder consultations at the national and local levels, including some vulnerable groups in the seven countries from 2023 to 2024. The global SEP provides a framework for the overall programme, outlining the stakeholders, methodologies, grievance redress mechanism (GRM), and timelines for engagement during the programme duration. Country-specific SEPs have been prepared to more accurately capture country and local contexts and will be updated as needed during programme implementation once the specific locations are known. During project design and implementation, special considerations for vulnerable groups will be taken into account. The programme will ensure that they are meaningfully consulted and that their needs and constraints are considered. The global Project Manager, global SGIE, national Project Manager, and national SGIE will be working together to implement the SEPs. A programme-level GRM and a country-level GRM will be established and communicated through stakeholder engagement sessions. The GRMs will facilitate complaints related to the E&S performance of the programme. The UNDP Accountability Mechanism and GCF Independent Redress Mechanism will also be

made available as additional channels for stakeholders' reporting of complaints related to E&S performance of the programme.

51. **GCF Indigenous Peoples Policy & ESS 7 (Indigenous Peoples).** Indigenous Peoples are potentially present in most of the countries covered by the funding proposal. The overarching initiative has an Indigenous Peoples Plan Framework that will be systematically applied to any activity implemented by the national-level projects involving Indigenous Peoples. Project exclusions ensure that no activity goes beyond minimal adverse impacts on Indigenous Peoples, including those that would result in impacts on lands, natural resources subject to traditional ownership or under customary use or occupation; that no relocation happens of Indigenous Peoples from their lands and natural resources; and that no activity impacts the cultural heritage – tangibly or intangibly – of Indigenous Peoples, or results in the commercial use of such heritage. No activity requiring that free, prior and informed consent be secured will be undertaken, and no impact will be made on Indigenous Peoples in isolation or through initial contact. The Secretariat's Office of Sustainability and Inclusion (OSI) can support the AE with recommendations on building in Indigenous knowledge to ensure co-benefits and also that it will be in line with USP 2 commitments and guidance from the Conference of the Parties to the United Nations Framework Convention on Climate Change on this matter. The AE is recommended to have a sound participative approach at the national and local levels, and to engage with the GCF Indigenous Peoples Advisory Group on relevant Indigenous Peoples' issues that may be encountered.

52. **Sexual exploitation, abuse and harassment (SEAH) safeguarding.** SEAH risks are included in the social and environmental screening report and have been assessed as low risk. Potential SEAH issues may arise from interaction of project workers with communities during outreach activities and small works such as hydrometeorological equipment installations and repairs. Some project workers are expected to work in various locations where communities are located when implementing the programme in the beneficiary countries, however this is expected to be temporary. A SEAH action plan is included in the programme-level ESMF with activities to address SEAH risks that include training on SEAH and the code of conduct as part of capacity building for various programme actors, such as local contractors; timelines; responsibilities for implementation and monitoring. The grievance redress mechanism (GRM) is described in the ESMF and includes information disclosure and stakeholder engagement. Specific procedures for SEAH cases are outlined to make the GRM accessible, inclusive, survivor-centred, and gender-responsive, including facilitating confidential reporting and ensuring safe and ethical documentation of incidents, and protocols for reporting incidents and the follow-up actions that will be taken. The Fund's independent redress mechanism will also be available to stakeholders.

4.2 Gender policy

53. The AE has provided a gender analysis and action plan and therefore complies with the GCF Gender Policy. The gender assessment identifies barriers to equitable participation and benefit-sharing in MHEWS. Women face restricted access to early warning information and communication technologies, particularly in rural and marginalized areas, due to lower digital literacy, socioeconomic disadvantages, and cultural norms. Gender roles often limit women's involvement in decision-making and disaster preparedness, as their primary responsibilities for unpaid care work reduce opportunities for active participation. Additionally, systemic issues such as the underrepresentation of women in leadership positions, lack of gender-disaggregated data, and minimal integration of gender-sensitive policies in disaster risk management exacerbate these disparities. Vulnerable groups, including women with disabilities, Indigenous women, and those in rural areas, face compounded challenges in accessing resources, safe evacuation facilities, and targeted interventions during emergencies.

54. The gender action plan proposes a comprehensive set of actions to address these challenges by mainstreaming gender considerations across MHEWS. The gender action plan defines activities, indicators, and targets for women's participation and benefit with corresponding timelines, budget and gender expertise. Key actions include increasing women's representation in decision-making and leadership roles, enhancing gender-responsive data collection and analysis, and promoting community-based training programmes tailored to women's needs. It intends to develop accessible, multichannel early warning dissemination systems, prioritizing formats that reach women and marginalized groups effectively. The plan emphasizes capacity-building initiatives to empower women as agents of change in disaster preparedness and response. Furthermore, it underscores the need for gender-sensitive infrastructure in evacuation sites, integrating separate facilities and addressing safety concerns. Institutional capacity will be strengthened to ensure gender equity in disaster risk governance, with mechanisms to monitor progress through gender-focused indicators. Each country will have its own GAP that its gender staff will report on to the global PMU which will in turn consolidate inputs for annual reporting to the Fund.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

55. The programme aims to implement EWS in seven countries, including small island developing States, least developed countries and fragile and conflict-affected states. The GCF is requested to provide a full grant of USD 103.25 million. Governments, Pillar Leads and AE are contributing co-financing of USD 11.34 million in the form of grants and in-kind contributions.

4.3.2. Accredited entity/executing entity capability to execute the current programme (low-medium risk)

56. UNDP will serve as both AE and executing entity (EE) for the project. The AE has an extensive track record and global experience in implementing climate adaptation and climate information projects. This funding proposal represents the first multi-country project submitted by UNDP and the largest UNDP project to date in the GCF portfolio.

4.3.3. Programme-specific execution risks (medium risk)

57. **Political risks and coordination challenges:** The programme spans multiple countries with diverse political and social dynamics, different coordination capacities and governance structures, which can pose risks to the implementation of EWS. The AE identifies these challenges and expects to address them by focusing on strengthening governance structures in collaboration with national and local entities, non-governmental organizations and other multilateral entities. Also, the AE will centralise the disbursement plans to maintain the overall pace of the project implementation ensuring progress continues even if delays occur in a single country.

58. **O&M plan:** The government commitment for O&M beyond the project implementation is critical for the project's success. However, the O&M expenses for each participating country are going to be submitted at the later stage. This uncertainty poses an implementation risk that remains unaddressed at the time of drafting this assessment.

59. The economic analysis was submitted only for Ecuador at the time of drafting the assessment. For the other countries, the economic and financial analysis expected to be submitted by the end of January 2025.

4.3.4. Compliance risk (low risk)

60. As a multi-country programme that is part of the global EW4All Initiative, which will improve MHEWS use in the seven targeted countries, the programme's proposed interventions include the scaling up of global and regional support for country-level MHEWS initiatives and national-level investment activities in the seven target countries to foster the achievement of the EW4All targets. Programme activities are generally lower risk from a money-laundering/the financing of terrorism (ML/TF) perspective.

61. UNDP will serve as both AE and EE through its Direct Implementation Modality, wherein it will be responsible for the entire programme execution. All programme implementation counterparties will undergo the UNDP Harmonized Approach to Cash Transfers microassessment, which takes a risk-based approach to assessing partners' financial management capacity. In addition to partner capacity assessments, the UNDP due diligence process to be applied to the programme also includes anti-money laundering/countering the financing of terrorism screening of any proposed partners or vendors. As such, UNDP has assessed the ML/TF risk as low.

62. Based on the proposed activities, the direct implementation and oversight of the AE, as well as the ML/TF controls to be applied during programme implementation, the overall ML/TF risk is determined to be low.

4.3.5. GCF concentration risk (medium risk)

63. In case of approval, the impact of this proposal on the GCF concentration risk remains within the monitoring thresholds of the Risk Appetite Statement in terms of result areas, single proposal or AE concentration.

4.3.6. Recommendation

64. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Medium
Accredited entity/executing entity capability	Low-Medium
Project-specific execution	Medium
GCF portfolio concentration	Medium
Compliance	Low

4.4 Fiduciary

65. UNDP will be the AE as well as the EE for this project.

66. The financial management and procurement of this project will be guided by UNDP financial rules and regulations. National projects will be implemented following the modality applicable to each country based on the national capacities of the implementing partner by undertaking an evaluation of capacity following the Framework for Cash Transfers to Implementing Partners (part of the Harmonized Approach to Cash Transfers). All projects will be audited in accordance with the UNDP financial rules and regulations noted above and applicable audit guidelines and policies. Assurance activities include planning, periodic on-site reviews (spot checks), programmatic monitoring, scheduled audits and special audits.

67. During implementation, UNDP will provide oversight and quality assurance in accordance with its policies and procedures, and any specific requirements in the accreditation master agreement (AMA) and project confirmation to be agreed with GCF. This may include, but

is not limited to, monitoring missions, spot checks, facilitation and participation in project board meetings, quarterly progress and annual implementation reviews, and audits at project level or at implementing partner level on the resources received from UNDP.

68. According to the current audit policies, UNDP will be appointing the auditors. Scheduled audits will be performed during the programme cycle as per UNDP assurance/audit plans, based on the implementing partner's risk rating and UNDP guidelines. A scheduled audit is used to determine whether the funds transferred to the implementing partner were used for the appropriate purpose and in accordance with the workplan. A scheduled audit can consist of a financial audit or an internal control audit.

4.5 Results monitoring and reporting

69. The funding proposal contains a programme-level theory of change demonstrating three outcomes that will focus on distinct dimensions of the global EW4All initiative. The programme's overall goal is to contribute to enhanced avoidance of the adverse effects of climate change and adaptation of lives, livelihoods, and assets to climate hazards through effective early warning systems. While the implementation of EWS in each country needs to consider the respective country's context and specificities, the AE is planning to submit individual theories of change for each country investment.

70. The adaptation impact of the programme stems from increasing resilience to climate-change hazards in the population of the seven targeted countries. The approach to estimating direct and indirect beneficiaries of these country-level investments is provided under the impact potential investment criteria. The AE will submit further information on applying this approach in the country context and outputs within the individual country-level document. The impact estimates will be reviewed and finalized once this information is available, and hence, the Secretariat expects a revision in the present beneficiary figures.

71. The programme's logical framework is predominantly in line with the requirements of the Integrated Results Management Framework, with a few identified areas for improvement. The potential for a paradigm shift and the selection of enabling environment indicators are appropriately addressed. Given the programme's focus on adaptation, the logical framework has applied the core 2 indicator and related supplementary indicators. While country and sex-disaggregated targets have been provided for the indicators, targets for supplementary indicators are only partially consistent with the indicator definitions. The AE has proposed conducting baseline assessments in the first year of project implementation to inform the targets and baseline for the indicators. The programme-specific logical framework is high-level and requires the inclusion of indicators specifically on reporting results from output 3 on the implementation of activities related to the four pillars of EW4All in the targeted countries. To ensure the required granularity and robustness for measuring GCF and programme-specific indicators, the AE will be submitting country-level logical frameworks that will be mapped to programme logical frameworks before project approval by the Board.

72. The programme proposal contains an M&E plan, laying out the role and responsibilities of AE for various M&E activities and reporting to GCF as per various policy requirements. The programme will conduct an interim and final independent evaluation as per the GCF Evaluation Policy.

4.6 Legal assessment

73. The Accreditation Master Agreement was signed with the Accredited Entity on 5 August 2016, and became effective on 23 November 2016, which was amended and restated pursuant

to a first amendment and restatement agreement dated 16 March 2023, and which became effective on 3 May 2023 (the “AMA”).

74. The Accredited Entity has provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

75. The proposed multi-country Programme will be implemented in Antigua and Barbuda, Chad, Cambodia, Ecuador, Ethiopia, Fiji, and Somalia. The GCF has signed bilateral agreements on privileges and immunities with one of these seven host countries, namely Antigua and Barbuda. In the remaining six countries, namely Chad, Cambodia, Ecuador, Ethiopia, Fiji, and Somalia, GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in these countries, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in such countries may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

76. Therefore, it is recommended that the Board considers whether disbursements of GCF proceeds should only be made after the GCF has obtained satisfactory protection against litigation and expropriation in the countries, or has been provided with appropriate privileges and immunities for GCF and its personnel.

77. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP258

Proposal name:	Multi-country Project Advancing Early Warnings for All (EW4All)
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Antigua and Barbuda, Cambodia, Chad, Ecuador, Ethiopia, Fiji, Somalia
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. This is the first submission of the proposal for the Multi-country Project Advancing Early Warnings for All (EW4All) programme under the GCF standard proposal approval process.¹ The funding proposal is part of a global initiative to advance EW4All by 2027, as announced by the United Nations Secretary General in March 2022 and articulated in the *Early Warnings for All Executive Action Plan*.² Presented as an umbrella programme, the funding proposal includes a global component as well as prioritized support to seven participating countries across four geographies: Antigua and Barbuda, Cambodia, Chad, Ecuador, Ethiopia, Fiji, and Somalia.
2. The objective of the programme is to accelerate the design and implementation of EW4All through coordinated global support involving all the key actors across the early warnings value chain. Additionally, the project will directly support the financial and technical needs of seven participating countries to enhance the production of actionable multi-hazard early warning systems (MHEWS) and impact-based forecasts. This information will be disseminated to effectively reach the most vulnerable communities in each of the seven countries and to trigger preparedness, early and anticipatory actions. To illustrate how this will work in practice, one country funding proposal for Ecuador has been submitted alongside this funding proposal. The accredited entity has clarified to the independent Technical Advisory Panel (iTAP) that the remaining six country proposals are expected to become available by the end of January 2025.
3. The funding proposal responds to several challenges related to MHEWS: (i) insufficient financial resources and limited capacities to access financing at country and community levels; (ii) inadequate policies and strategies for inter-agency coordination, data-sharing and governance for preparedness and response; (iii) limited technical, scientific and institutional capacities for a comprehensive MHEWS; (iv) ineffective dissemination of early warning information to climate-vulnerable communities; (v) insufficient national- and local-level disaster preparedness and anticipatory action planning; and (vi) limited opportunities for women and other marginalized groups to engage with and benefit from early warning systems.

¹ The assessment of the independent Technical Advisory Panel (iTAP) is based on the version of the funding proposal received by the iTAP on 12 December 2024 and is informed by a set of written questions and answers further discussed in an online meeting between the iTAP and the accredited entity on 6 January 2025.

² Available at <https://digitallibrary.un.org/record/4001006?v=pdf>

4. The Multi-country Project Advancing EW4All is a medium-sized public-sector adaptation programme in environmental and social safeguards risk category C. The total budget of USD 175.49 million comprises a request for GCF grant funds of USD 107.49 million, with expected co-contributions of USD 68 million from the Systematic Observations Financing Facility (SOFF),³ the seven participating governments, and country offices of the United Nations Development Programme (UNDP). The implementation period is 5 years with an overall benefit stream of 30 years.
5. The project will be delivered through two outputs:
 - (a) Output 1: Enhancing enabling conditions, both globally and regionally, to promote country-level actions towards the EW4All goals; and
 - (b) Output 2: Accelerating actions in the seven participating countries towards EW4All goals, through context-specific activities that address the four key pillars of MHEWS namely: (i) disaster risk knowledge and management; (ii) detection, observations, monitoring, analysis and forecasting; (iii) warning dissemination and communication; and (iv) preparedness and response capabilities.⁴
6. The UNDP Bureau for Policy Programme Support is the accredited entity (AE). UNDP has been accredited to GCF since March 2015, with a cumulative GCF funding of approximately USD 1.2 billion.⁵
7. There are two distinct governance and management structures for the global- and national-level components respectively.
 - (a) Output 1 will be executed by UNDP, supported by four pillar leads – World Meteorological Organization (WMO), United Nations Office for Disaster Risk Reduction, International Telecommunication Union, and International Federation of Red Cross and Red Crescent Societies (IFRC) – in collaboration with global and regional organizations; and
 - (b) For output 2, UNDP country offices will implement the project as executing entities under a Direct Implementation Modality, in collaboration with the relevant ministries housing the national disaster management authorities or the national meteorological and hydrological services (NMHS), and with other key partners in each country. UNDP will also play the role of senior supplier.

1.2 Impact potential

Scale: Medium

8. Of the 30 focal countries identified under the global EW4All Initiative,⁶ seven countries from across four geographies were selected for participation in this project. All seven countries are highly vulnerable to the impacts of climate change, including increases in temperature (leading to heatwaves, damages to agriculture and human health), increases in intensity of rainfall, typhoons, storms and cyclones (causing floods, landslides and coastal inundation) and in some cases decreases in rainfall (leading to agricultural and hydrological droughts), and increased disaster risks from hazardous climate events.
9. In response to questions raised by the iTAP, the AE has clarified that country selection was based on exposure to climate variability and change, including from multiple

³ SOFF is a United Nations multi-partner Trust Fund (created in 2021) which provides grant funding to support countries in collecting basic weather and climate observation data. It acts as a delivery mechanism for the United Nations Secretary General's EW4All Initiative.

⁴ From action plan referred to in footnote 2 above.

⁵ See www.greenclimate.fund/ae/undp

⁶ UNDRR. 2023. *Global Status of Multi-Hazard Early Warning Systems 2023*. Available at <https://doi.org/10.18356/9789213587058>

hydrometeorological climate-related impacts. Other factors that informed country selection include varying levels of economic development, diversity of livelihoods, and varying levels of fragility and insecurity. The selection process also considered partner priorities, regional representation, and countries with prior or planned investments from Climate Risk and Early Warning Systems Initiative (CREWS)⁷ and SOFF programmes.

10. The climate hazards and associated risks vary across countries and are clearly explained in the funding proposal (appendices 1 and 2) and summarized in table 1 below.

Table 1: Country climate hazards

No	Country	Geographical region	Predominant climate hazards
1	Antigua and Barbuda	Latin America and the Caribbean	Hurricanes, drought, extreme rainfall and floods, rising sea levels, and other threats such as heat and cold waves, hailstorms, downbursts, and tornadoes
2	Cambodia	South-West Pacific	Intense precipitation events, dry seasons, extreme heat events
	Chad	Sahel/Central Africa	Flash floods, river flooding, droughts, dry spells
3	Ecuador	Latin America and the Caribbean	Droughts and reduced precipitation, floods
4	Ethiopia	Horn of Africa/East Africa	Droughts, riverine and flash floods
5	Fiji	South-West Pacific	Droughts, tropical cyclones, high intensity rainfall and winds, flooding, sea-level rise
7	Somalia	Horn of Africa/East Africa	Water scarcity, droughts, flash flooding, floods, inundation

Source: Adapted from the multi-country funding proposal and appendices.

11. The proposal outlines how the impacts of climate and climate-related hazards can wreak havoc on the lives and livelihoods of millions of people; impact human health, food and water security; disrupt ecosystems and ecosystem services; and cause losses and damages to communities. Within communities, the poor and vulnerable, including women, children, the elderly, and persons with disabilities, often lack the means to anticipate and adapt to climate-related disasters.

12. The funding proposal clearly sets out the rationale for how well-functioning early warning systems can help countries to prepare for and respond to climate risks and climate-induced hazards.⁸ The Multi-country Project Advancing EW4All programme therefore proposes to strengthen the role of MHEWS in informing timely decisions by governmental and non-governmental users across sectors, as a foundation for risk-informed development processes.

13. The project presents two outcome areas of: sustained support for global coordination and support architecture for EW4All (outcome 1); and end-to-end early warning systems that are inclusive and build community resilience to climate hazards (outcome 2). The two associated outputs could potentially bring impacts at different levels and scales. By catalysing global resources and collaboration, and by enhancing the enabling environment, output 1 could help to accelerate efforts towards universal coverage of early warning systems by 2027. By helping to improve the effectiveness and reach of end-to-end MHEWS in seven participating

⁷ CREWS funds projects in the least developed countries and small island developing States to establish risk-informed early warning services, supported by three implementing partners: UNDRR, the World Bank Global Facility for Disaster Reduction and Recovery and the World Meteorological Organization.

⁸ As set out in table 2 of the funding proposal (p. 18).

countries, output 2 could help benefit vulnerable communities in being better prepared for multiple hazards.

14. Programme impacts are listed under the four GCF adaptation results areas, as follows:

- (a) Livelihoods of people and communities (85 per cent of GCF contribution);
- (b) Health, food and water security (5 per cent of GCF contribution);
- (c) Infrastructure and built environment (5 per cent of GCF contribution); and
- (d) Ecosystems and ecosystem services (5 per cent of GCF contribution).

15. The project anticipates reaching 26.84 million direct beneficiaries and 51.26 million indirect beneficiaries (see table 2 below for details), delivered through output 2. The target beneficiary numbers are provisional figures, and the funding proposal notes that details on the approach are included in individual country proposals and stakeholder plans (not available to the iTAP as part of this funding proposal package).

Table 2: Expected beneficiaries of the Multi-country Project Advancing Early Warnings for All by country

Country	Estimated number of direct beneficiaries	% of total population(direct)	Estimated number of indirect beneficiaries	% of total population(indirect)
Antigua and Barbuda	91,500	100	114,000	NA ⁹
Cambodia	514,098	2.91	2,153,028	12.21
Chad	1,700,000	10	5,100,000	30
Ecuador	513,900	2.6	14,000,000	72
Ethiopia	16,000,000	12.12	21,600,000	15.9
Fiji	416,282	46	488,308	54
Somalia	7,600,000	40.9	7,800,000	42
Total	26,835,780		51,255,336	

Source: multi-country funding proposal

16. Direct beneficiaries are considered to include vulnerable populations that are direct recipients of project activities notably early warning dissemination systems, capacity-building in the use of MHEWS, and disaster preparedness and response; whereas indirect beneficiaries are wider populations located in areas covered by the project and likely to receive measurable benefits from the enhanced MHEWS through improved risk information, forecasting and dissemination.

17. In the view of the iTAP, the scope of both outputs is ambitious, and the extent of impact to be realized is dependent on the ability of multiple stakeholders at global, regional, national and local levels to deliver a coherent project.

18. The iTAP also notes that country-based activities (under output 2) are likely to vary depending on country contexts, prioritization of sectors based on national climate strategies and plans, selection of locations for strengthening monitoring networks, and criteria applied for selecting country-specific investments. The country proposal for Ecuador articulates the country-specific approach. While the overall funding proposal summarizes the proposed interventions for each of the countries, the iTAP notes that the remaining six country proposals are yet to be finalized. Therefore, in the view of the iTAP, it will be important for the AE to ensure that the full country proposals are targeted at the most vulnerable sectors, remain consistent with the overall funding proposal, and are able to implement actions at pace and

⁹ The funding proposal states that indirect beneficiaries for Antigua and Barbuda are considered the significant number of tourists that visit the country during the hurricane season.

scale to achieve the intended reach to project beneficiaries and to meet overall intended project targets.

19. The feasibility study for Ecuador prioritizes six focal areas and sectors, and the iTAP recommends that the approach of prioritizing sectors vulnerable to climate impacts is also applied to all country proposals. The iTAP underscores the importance of the sector focus that could allow for more accurate historical baselines on losses and damages, enabling robust analysis and targeted interventions to address unique vulnerabilities. Success in one sector could also serve as a model for replication in others; whereas a broader national-based approach could enable broader coverage of early-warning information and coordinated action across government.

20. Overall, the iTAP assesses the impact potential as medium.

1.3 Paradigm shift potential

Scale: Medium

21. The framing of the theory of change for the overall funding proposal is aligned with the EW4All Initiative. The two outputs are focused on tackling the barriers to the effective generation, dissemination, communication and use of early warnings for climate change adaptation at two levels as follows:

- (a) At the global level, output 1 seeks to achieve a paradigm shift by establishing a sustainable architecture, enhancing global and regional coordination, developing a long-term financing strategy for an end-to-end MHEWS, strengthening the global monitoring and evaluation framework across all four pillars, augmenting knowledge management, aligning guidance with international standards, and providing operational and technical assistance to countries and for early warning actions. This could help to build momentum and support for more country-level actions towards EW4All; and
- (b) By extending technical and operational support to the seven countries, output 2 will address barriers related to policy coherence and coordination, technical and scientific capacity gaps; missing observations particularly in small island developing States and the least developed countries; lack of forecast-based planning and observation at national and grass-roots levels; ineffective dissemination to the last mile; and lack of preparedness and response. The activities under output 2 are closely aligned to the four pillars of EW4All, and support will be tailored to the unique needs, capacities, and contexts of countries as presented in their respective feasibility studies that are yet to be finalized. Activities will support concrete investments in EWS hardware and software; improve inter-agency coordination at national and local levels; bolster risk knowledge; and strengthen systems to disseminate and communicate early warnings and preparedness so that communities can take early action. The use of innovative digital technologies can also support a paradigm shift in countries by employing a more efficient means of generating observational data, transmitting early warnings using mobile networks, and ensuring that this information reaches communities in a timely manner.

22. The potential for creating an enabling environment is high as the project could achieve synergies across global partners supporting the EW4All target. The project's intent to strengthen coordination across partners providing similar support towards the EW4All target is sound. The funding proposal includes a useful appendix that illustrates synergies with the CREWS and SOFF initiatives respectively.

23. The planned development of a longer-term financing strategy will be essential to help unlock much needed finance for sustaining the EW4All architecture to 2027 and beyond. This is a critical element of the project's sustainability and the iTAP recommends that this activity is

undertaken in consultation with a wide range of partners, including government, non-governmental organizations, the private sector, and donor partners.

24. The potential for knowledge-sharing and learning is high, and the funding proposal broadly indicates how the project will develop, disseminate and support the application of detailed guidelines and other artificial intelligence-based tools to strengthen national EWS capabilities. There is also provision for knowledge exchange between countries, including through the establishment of global technical expert groups and committees, a cell broadcast expert group, and building shared knowledge and awareness through international events and workshops, although the details have yet to be fully developed.

25. The project could also offer critical learning and insights to scale up or replicate this approach more widely across other countries. In the view of the iTAP, the project also has the potential to facilitate cross-learning between countries through the governance structure. Output 1 could set a solid foundation for replication and scaling up of EW4All across other countries. National-level activities delivered through output 2 could also help to set the basis for the approach to be replicated and scaled up more widely, subject to existing barriers being overcome, but this is unlikely to emerge in the short term. The proportion of direct beneficiaries varies significantly by country and represents a small proportion of the total population. Without having seen the country packages, this raises some questions about the scale of impact to be achieved through this programme alone.

26. The programme has the potential to bring about a paradigm shift in each of the seven countries. The iTAP finds that the programme is ambitious in scope as it spans a global component plus in-depth activities in seven countries. While the funding proposal sets out the broad framework, there are likely to be differences in approaches and investments envisaged at the country level as encapsulated in individual country proposals. In particular, the readiness, capacities and varying country contexts will determine the project activities, the pace of implementation, and the extent to which these are effective and sustainable. The assumptions in the theory of change are significant and, if not managed well, could bring risks for the project (e.g. operation and maintenance (O&M) of hydromet infrastructure, retaining trained staff, ability to attract investments).

27. The funding proposal highlights three (of seven) countries as fragile and conflict affected states (FCAS) which could affect the extent to which project interventions could achieve a paradigm shift particularly if occurrences of instability or conflict derail project progress.

28. The results framework for the overall programme is presented at a high level, and the detailed indicators for the outputs are yet to be developed. The specificity of how each of the seven countries will take forward activities under output 2 are not yet presented and the iTAP understands that these are currently under finalization in the respective country proposals.

29. Overall, the iTAP assesses the paradigm shift potential as medium.

1.3 Sustainable development potential

Scale: Medium to high

30. The funding proposal is designed to support strategic frameworks and actions at national and global levels to reduce disaster risk and the impacts of climate shocks and hazards. In particular, the project will contribute specifically to target G of the Sendai Framework for Disaster Risk Reduction 2015–2030 to “substantially increase the availability of and access to multi-hazard early warning systems and disaster risk information and assessments to the

people by 2030”,¹⁰ the four priorities of the Sendai Framework for Disaster Risk Reduction,¹¹ and the priorities identified in Article 8 of the Paris Agreement.

31. The programme could also contribute to the Sustainable Development Goals (SDGs), specifically SDG 13 (Climate action) and SDG 1 (No poverty), SDG 2 (Zero hunger), SDG 9 (Industry, innovation and infrastructure), SDG 11 (Sustainable cities and communities), SDG 12 (Sustainable production and consumption), and SDG 15 (Life on land).

32. **Economic co-benefits:** Improved coordination and use of effective end-to-end MHEWS and impact-based forecasts in design, planning and management of private and public infrastructure could enable early and anticipatory action. This enhanced use of MHEWS could help to reduce losses and damages across multiple sectors, including agriculture, transport and energy. The project could also benefit communities whose livelihoods depend on climate-sensitive sectors and reduce the negative impacts of climate change on morbidity and mortality of vulnerable populations. In the longer term, the involvement of the private sector will be crucial in sustaining the overall support architecture for MHEWS – for instance in the use of private communications infrastructure for disseminating early warnings and advisories.

33. **Environmental co-benefits:** Activities undertaken through country-specific actions, as encapsulated in the country funding proposals, could bring positive environmental co-benefits. The timely uptake and use of climate and disaster information could inform the management of natural resources, and the impact on the environment from climate events such as floods and droughts. It could also inform how regions are managed, particularly those that are more exposed to climate risks, including small islands, coastlines and in flood and landslide prone regions. Timely and accurate climate information and impact-based forecasting could also be used to guide sectoral policies, planning and implementation – for instance, in supporting the adoption of climate-resilient agricultural practices, managing ecosystems, and improving water resource management. Capacity-building and awareness-raising will also help to sensitize communities and local stakeholders to the impacts of climate change.

34. The funding proposal (annex 6B) provides a global-level environmental and social management framework, with overarching guidelines and principles applicable to all participating countries. Additionally, each country-level funding proposal will contain environmental and social management frameworks to meet the unique needs of the participating country. The UNDP social and environmental screening template is also part of the package (annex 6a). The funding proposal confirms that only low risk (category C) activities will be supported by the project.

35. **Social and gender co-benefits:** The funding proposal adopts a people-centred approach to MHEWS. As such it recognizes the value of participatory stakeholder engagement in project design and implementation to ensure that the specific needs of vulnerable groups are addressed. Through country-level actions, the project could contribute to enhancing the resilience of vulnerable populations by ensuring that EWS are timely, accessible and responsive to the needs of women, persons with disabilities, and Indigenous Peoples. Much will depend on the country context and the specifics are to be included in the country-level proposals. The funding proposal also integrates fragility and conflict in its approach, primarily focused on anticipating and preparing relevant agencies and communities for possible conflicts over resources due to climate-related hazards. It also anticipates benefits to the health sector from disaster preparedness and anticipatory action to prevent the spread of diseases. Output 1 also lays the foundation for expanding end-to-end MHEWS to other countries that are not covered by this project.

¹⁰ See www.undrr.org/quick/74257

¹¹ <https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030>

36. A global-level gender analysis and action plan (GAAP) sets the overarching framework and guiding actions that recognizes the constructive and productive role that women can play in early warning and disaster risk management (annex 8 to the funding proposal). The funding proposal highlights that EWS should consider the unique needs and capacities of different groups, including those who are marginalised or living in vulnerable situations, and aims to tackle gender gaps in access to and use of MHEWS technology, communications, and services.

37. Overall, in the view of the iTAP, while the sustainable development potential is strong, the impacts can vary substantially across countries based on several factors, including the country and local contexts. Each country has unique opportunities and barriers to implementing effective EWS, and much will depend on the activities and sub activities that are taken forward in the forthcoming country project proposals. On this basis, the sustainable development potential of the project is assessed as medium to high.

1.4 Needs of the recipient

Scale: High

38. The funding proposal estimates that, globally, between 3.3 billion and 3.6 billion people live in conditions that are highly sensitive to climate change, with the majority living in countries that have gaps in early warning systems. Most of these countries are among the least developed countries, small island developing States and African countries.

39. The funding proposal has summarized the vulnerability of all seven participating countries. The iTAP notes that all seven countries, spanning a range of geographies, are highly vulnerable to climate change and climate-related hazards, which affect people, livelihoods, assets and economy.

40. The seven participating countries have limited ability to access financing, face institutional and capacity gaps in tackling climate risks, and have varying degrees of socioeconomic vulnerabilities and systemic inequalities. Some countries also face compounded risks from fragility and conflict.¹² Vulnerable populations, including Indigenous People, ethnic minorities, persons with disabilities, women and children, rural communities, and low-income households are disproportionately affected by the climate crises as they lack the means to receive the timely information needed to respond to climate-related risks and hazards.

41. The programme is designed to be inclusive, so that early warning systems can reach vulnerable communities, especially those whose livelihoods depend on climate-sensitive sectors; and those with greater physical exposure to climate hazards, including women and other marginalized groups within these communities. The programme aims to use information dissemination channels that are context-specific to each country and can reach citizens regardless of their gender, status, ethnicity or location. Annex 8 to the funding proposal presents the GAAP for the overall multi-country proposal, while noting the availability of country-specific GAAPs that analyse gender, including in the context of early warnings and disasters, with specific actions, targets and budget for the gender action plan.

42. Overall the iTAP assess the needs of the recipient as high.

1.5 Country ownership

Scale: Medium

43. All seven no-objection letters have been received from national designated authorities of participating countries (annex 1 to the funding proposal). Of the total of USD 68 million expected as co-finance, approximately USD 35 million is envisaged as co-finance contributions from countries, but specifics on individual country contributions are not yet available.

¹² The funding proposal classifies Chad, Ethiopia and Somalia as either “in conflict” or “fragile” (annex 6b, p. 16),

44. The iTAP recognizes that the funding proposal is aligned with national priorities, as expressed through national adaptation plans, nationally determined contributions, communications to the United Nations Framework Convention on Climate Change secretariat and the Sendai Framework. The AE has also clarified to the iTAP that the project seeks to align with work across updated country nationally determined contributions, particularly as the lack of climate information and EWS have been cited in each of these countries as necessary investments for enhanced action and delivery for adaptation. However, the extent of national- and local-level stakeholder buy-in to the project is unclear as the detail will become available in country-level project proposals. The iTAP has extracted available data from the main funding proposal and summarized it in table 3 below.

Table 3: Alignment with country climate change priorities

Country	Country climate change plans and national policies
Antigua and Barbuda	• Nationally determined contributions (NDCs) (2021) • National adaptation plan (NAP)
Cambodia	• NDCs (updated 2020)
Chad	• NDCs (updated October 2021) • Priority No. 10 (Climate risk management) under the First national adaptation plan of action • National development plan (2024–2028) • 2030 vision, specifically under Pillar 4: Improving the quality of life of the Chadian population • National policy on gender (2011–2020)
Ecuador	• National climate strategy • NDCs • Nationally appropriate mitigation actions • NAP
Ethiopia	• NDCs (updated July 2021) • National disaster risk management policy (2013) • Climate resilient green economy strategy (2011) • Roadmap for multi-hazard, impact-based early warning and early action system (2023–2030) • National framework for climate services (2021–2030) • The strategic programme and investment framework outlining priority actions and investments to enhance Ethiopia’s disaster risk management system
Fiji	• National climate change policy (2018–2030)
Somalia	• Roadmap for implementation of the Early Warnings for All Initiative 2024–2027 • NAP framework (2022) • NDCs (updated July 2021) • National climate change policy (draft 2021) • Initial national communication (2018) to the United Nations Framework Convention on Climate Change • National disaster risk management policy (December 2020) • First biennial update report under the United Nations Framework Convention on Climate Change • National adaptation programme of action (2013)

Source: Summarized from the multi-country funding proposal, with years reflected in brackets where information has been made available in the funding proposal.

45. The iTAP has observed from the funding proposal (table on p. 44) that some countries have governmental agencies/ministries as counterparts for activity 1 (e.g. Antigua and Barbuda), whereas others lack a national government partner agency (e.g. Chad). In its response to the iTAP, the AE has clarified that governance arrangements vary across countries. The AE

has noted that FCAS often lack government mandated agencies, or that existing agencies lack the capacity to implement specific project activities. Consequently, country projects are designed to be implemented by in-country agencies, including non-governmental and multilateral agencies, with the best comparative advantage in terms of capacities. The AE response to the iTAP also notes that in some cases, “a great deal of the actual implementation will be managed by UNDP or government approved Responsible Parties”.¹³

46. Annex 7 to the funding proposal summarizes the overall approach to the stakeholder engagement plan. The document spells out that preliminary stakeholder consultations were undertaken during project preparation in 2023 and 2024. The consultations for outputs 1 and 2 seem to be conducted separately.

47. The consultations for global and regional activities (output 1) appear as comprehensive and include representatives from across all four pillar leads as well as from SOFF, IFRC, Microsoft Artificial Intelligence, GCF, and UNDP. Regional agencies were also consulted, including the Caribbean Community Climate Change Centre, Southern African Development Community, Secretariat of the Pacific Regional Environment Programme, Intergovernmental Authority on Development Climate Prediction and Applications Centre, Pacific Community, Asian Disaster Preparedness Centre, and the Regional Integrated Multi-hazard Early Warning System for Africa and Asia.

48. For output 2, national-level consultations were conducted in all seven countries and are summarized in appendix 4 to annex 7 to the funding proposal, with full details to be made available in the country-level proposals. The consultations covered topics such as country-specific stakeholder needs, gaps and vulnerabilities, potential roles for various stakeholders, ways to maximize different technologies, including artificial intelligence, partnerships, potential environmental and social impacts from project activities, and management plans developed, as well as the grievance redress mechanism, and identification of critical actors and implementing partners.

49. The consultations mainly with national-level stakeholders, appear to be one-off, with some limited engagement of local-level stakeholders. The funding proposal clarifies that detailed consultation with local-level stakeholders will follow during project implementation. This will be important in identifying project sites and in informing the implementation of specific activities. However, in the view of the iTAP, later consultations could also bring some risks of limited stakeholder engagement or responsiveness to project activities which could stall implementation processes.

50. The stakeholder engagement plan also makes provision for consultations with local-level authorities and community people, including men, women, people with disabilities, the elderly, youth and Indigenous Peoples’ groups as relevant. In some countries, key groups may also include refugees and internally displaced persons. However, the details of how the programme will go about undertaking these consultations is not as well articulated in the overall funding proposal. The extent to which the private sector was involved in consultations is less clear.

51. Overall, country ownership of the project is assessed as medium.

1.6 Efficiency and effectiveness

Scale: Medium

52. The estimated benefit–cost ratio and socioeconomic benefits in the funding proposal make a compelling case for investing in weather and climate information and disaster risk reduction. For example, the funding proposal notes that 24 hours warning of a coming storm or

¹³ The AE’s written answers in response to iTAP’s questions (Q10 (iii)) on 18 December 2024.

heatwave can cut the ensuing damage by 30 per cent, while spending USD 800 million on such systems in developing countries could avoid losses of USD 3–16 billion per year.¹⁴

53. The programme will be implemented over a period of 5 years, with a project lifespan of 30 years. The funding proposal notes that total project funding of USD 175,490,730 comprises a request for GCF grant funds of USD 107,490,730 alongside co-finance of USD 68 million suggesting a co-finance ratio of 1:0.63.

54. However, the iTAP has observed some variances in co-financing figures presented in the funding proposal as compared to the more recent AE's written responses and understands that these are yet to be firmed up. In response to queries from the iTAP, the AE has clarified in its written response that a project co-finance of USD 54,984,863 is expected from three sources, based on letters of commitment received so far. These include:

- (a) SOFF investments of USD 40.1 million (USD 18 million has already been approved and an additional USD 22 million is yet to be approved to support the project in seven countries) – whereas the funding proposal states a figure of USD 30 million. (The AE has clarified to the iTAP that country proposals have been designed in close coordination with the SOFF and CREWS secretariat to ensure complementarity with this project.)
- (b) Co-financing commitments from governments – letters have so far been received from Ethiopia (USD 2,315,000) and Ecuador (USD 3,720,072), while the funding proposal envisaged a total contribution of USD 35 million and stated that the remaining letters of commitment would be received by December 2024; and
- (c) Pillar leads comprising the United Nations Office for Disaster Risk Reduction (indicative USD 3,124,650), IFRC (USD 3,751,681), and International Telecommunication Union (USD 1,973,460); whereas the funding proposal envisaged contributions of USD 3 million.

55. The allocation of funds across components seems to be aligned with the project's ambition. The majority of requested GCF resources are allocated to output 2 (87 per cent), with the balance allocated to output 1 (6 per cent), project management costs (5 per cent), and monitoring and evaluation (2 per cent).

Table 4: Overview of the global Early Warnings for All project outputs and financing

Project component	GCF grant contribution (USD million)	GCF funds as a proportion of the total GCF budget (%)	Total component budget (USD million)
Outcome 1: Sustained support for global coordination and support architecture for EW4All			
Output 1: Enhancing enabling conditions globally and regionally to promote country-level actions towards the EW4All goals	6.41	6	6.41
Outcome 2: End-to-end EWS that is inclusive and builds community resilience to climate hazards			
Output 2: Accelerating actions in select countries towards EW4All	93.17	87	161.17
Monitoring and evaluation	2.79	2	2.79
Project management costs	5.12	5	5.12
Total GCF	107.49	100	175.49

Source: Adapted from the multi-country funding proposal and its annex 4.

Abbreviation: EW4All = Early Warnings for All

56. The allocation of resources by country is set out in annex 4, but the funding proposal notes that these are tentative. While there are significant differences in infrastructure gaps,

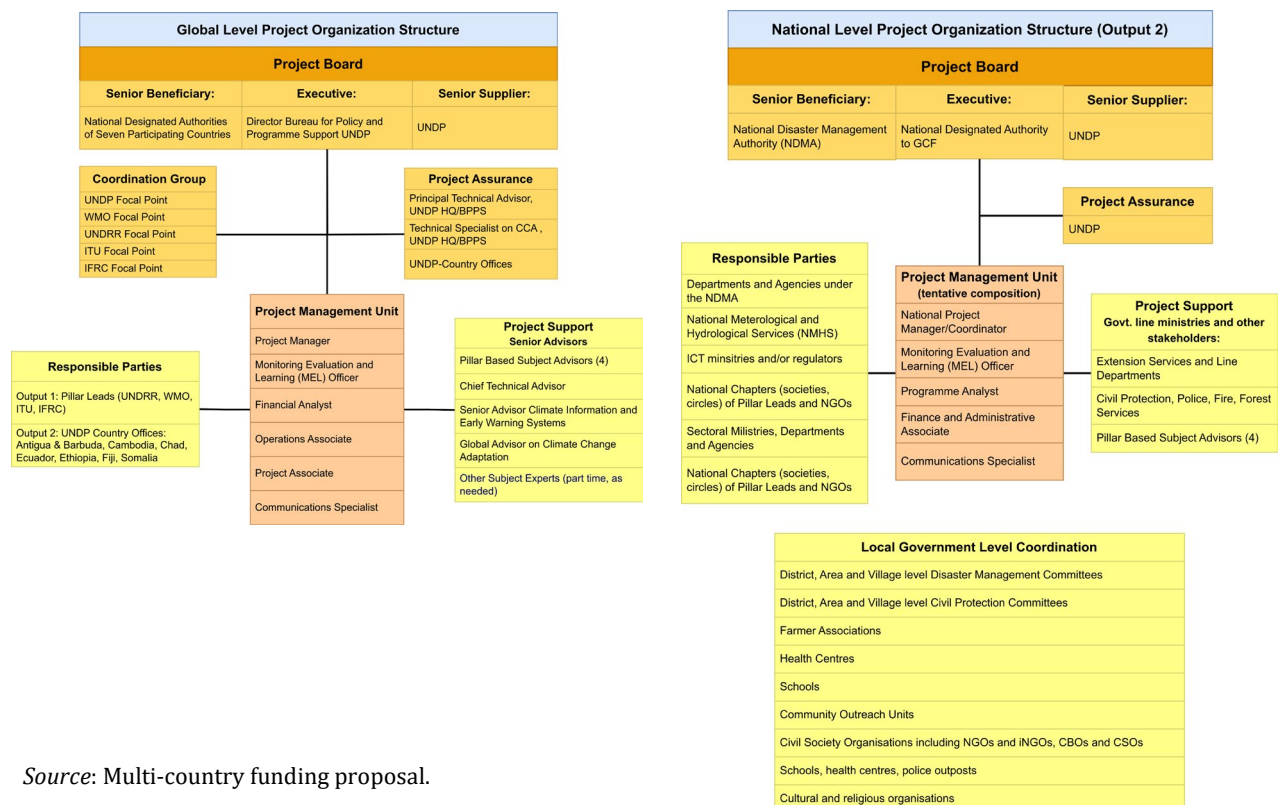
¹⁴ See <https://gca.org/reports/adapt-now-a-global-call-for-leadership-on-climate-resilience/>

technical capacities and frequency/intensity of hazards and disasters between countries, each country has a notional funding envelope of between USD 12 million and 13 million. The AE has clarified to the iTAP that these notional country-level allocations are under revision as, so far, budgets for only three (of seven) participating countries have been finalized, with the remainder expected to be finalized by the end of January 2025.

57. Given the public nature of MHEWS and the nascent contexts across countries, the funding proposal acknowledges the limitations in engaging with the private sector. The project does not expect to attract private-sector investments but instead plans to engage the private sector through stimulating demand for climate information-based value-added products and services – for instance, in telecommunications, agribusinesses, tourism and transport, including a focus on aviation and financial services such as insurance. The level and type of engagement is likely to vary according to country context, and the iTAP recommends that these approaches are further defined as part of the project’s financial sustainability planning.

58. The project has a two-layered governance structure (see fig. 1 below) for each of the outputs. The global-level arrangement, clearly outlined in the funding proposal, will involve UNDP leading coordination efforts across all four pillar leads with collective decision-making authority. However, there will be seven separate governance and implementation arrangements in each of the participating countries. It will be important to ensure that decision-making is coordinated across the two levels and for the programme as a whole to leverage synergies and maximize coordination.

Figure 1: Governance arrangements for outputs 1 and 2 respectively



Source: Multi-country funding proposal.

Abbreviations: BPPS = Bureau for Policy and Programme Support, CBO = community-based organization, CCA = climate change adaptation, CSO = civil society organization, HQ = headquarters, ICT = Information and Communications Technology, IFRC = International Federation of Red Cross and Red Crescent Societies, INGO = international non-governmental organization, ITU = International Telecommunication Union, NDMA = National Disaster Management Authority, NGO = non-governmental organization, UNDP = United Nations Development Programme, UNDRR = United Nations Office for Disaster Risk Reduction, WMO = World Meteorological Organization.

59. A long-term strategy to unlock finances and to manage the global architecture will be an important aspect of sustainability and a basis for countries to make progress at pace and scale towards developing national MHEWS. The exit strategy aspires to forge partnerships, strengthen alignment and coordination, and leverage financing through multiple sources for the global support architecture to be sustainable, but the details are as yet undefined. At the country level, institutional strengthening, coordinated action, financing strategies, and access to global and regional expertise are intended to strengthen national capacities and enhance proposal development capacities.

60. For output 2, each country proposal will contain an O&M plan, and the AE has clarified to the iTAP that a multipronged strategy for O&M may be required for some countries, especially FCAS and those with limited fiscal space. Additionally, O&M costs for observation equipment will be borne by country NMHS's and, while the task of technically demanding tasks will fall to staff of NMHS's, communities will also be trained to help with basic O&M (e.g. cleaning and testing of sensors).

61. The funding proposal contains a high-level monitoring and evaluation framework encompassing two levels: the global level for all outputs; and the country level for all the seven participating countries under output 2. Separate arrangements will be put in place to track and report on logframe indicators at two levels; while the Global Project Monitoring Unit will be responsible for monitoring and evaluation of the project, Country Programme Management Units will track and report on country-level indicators for output 2.

62. The iTAP has noted that country-level monitoring frameworks have yet to be designed and the logframe currently does not include detailed indicators to cover the depth of the programme, including across countries. The AE has clarified to the iTAP that this logframe is intended to simplify reporting arrangements as multiple agencies are involved in the delivery of each of the project outputs (four pillar leads for component 1 and seven countries for component 2). The iTAP also understands that the forthcoming country-level logframes, which will be aggregated for output 2, are intended to be more detailed.

63. The iTAP recommends that, once country logframes become available, the overarching logframe be reviewed and revised during the project inception period to ensure that overall targets are achievable, inclusive, and disaggregated. The iTAP also recommends that the monitoring arrangements are revisited and strengthened during the inception phase to ensure coherence between global and country-level efforts. The preparation of baseline assessments under output 2, planned in the initial phases of the project, will be critical to reliably assess the effectiveness of the project.

64. Overall, the iTAP assesses the efficiency and effectiveness potential as medium.

II. Overall remarks from the independent Technical Advisory Panel

65. The fundamental premise of the overall Multi-country Project Advancing EW4All funding proposal is sound and presents an opportunity for GCF resources to be deployed in response to the urgent call from the United Nations Secretary General to advance EW4All by 2027 in the face of the climate crisis.

66. The funding proposal presents a programmatic umbrella approach, and the AE has clarified that details will emerge in separate country proposals for the remaining six participating countries, by the end of January 2025. While the Ecuador country proposal serves as a useful model, the iTAP review has been somewhat limited due to the lack of detail on the remaining six country proposals, particularly on relating country vulnerabilities and needs to country-level activities and targeted priority sectors. Information on allocation of financial

resources and country-level results is limited, as well as on how the AE will engage with stakeholders in different country contexts.

67. The scale of ambition of the programme, and its reach across diverse country contexts spread over geographies means that there could be some associated risks in implementation. The development of a long-term financing strategy will be critical to project success. Additionally, country readiness to take forward MHEWS, and factors such as political, conflict, and security risks remain key considerations that could undermine programme efforts.

68. The role of UNDP is spread across multiple functions including as the AE, executing entity, and senior supplier which could bring some risks to coordination and country ownership. UNDP plans to implement the country programmes under the Direct Implementation Modality, with UNDP country offices effectively responsible for the day-to-day operations and collaborative engagement with government agencies and other stakeholders. This carries some risk that primary ownership may be with the United Nations agency rather than with the respective government.

69. The iTAP recommends that the AE undertake the following actions:

- (a) Prior to the AE approving implementation for each individual participating country under output 2, make available to GCF, the full country proposal, including the feasibility study, detailed budget and results framework, highlighting any variances with the multi-country funding proposal on which the iTAP assessment is based.
- (b) Ensure that future post-disaster needs assessments in all the selected countries are targeted and EWS-oriented by focusing on assessing the functionality, gaps, and outcomes of early warning systems. This includes integrating lessons learned from post-disaster forensics into post-disaster needs assessments and ensuring that recovery frameworks are informed by actionable insights to enhance EWS performance and resilience.
- (c) During the inception phase, update the MEL plan with further details on the steps and responsibilities for knowledge management and exchange.
- (d) Include in each annual progress report an analysis of how each country project contributes to the overall programme's objectives and results.

70. The iTAP recommends that the Board approve this multi-country funding proposal with the following conditions:

Condition to be met prior to the effectiveness of the Funded Activity Agreement:

- (a) The AE shall deliver, in a form and substance satisfactory to the GCF Secretariat, an updated logframe that is consistent with the theory of change and accurately captures the milestones and final targets (with beneficiary calculations) across all seven participating countries of the Programme, disaggregated by sex. The updated logframe should include an explanatory note setting out how the country-level disaster risk knowledge systems and Monitoring, Evaluation and Learning (MEL) dashboards (activity 2.2.1) for each participating country will link to the overall EW4All MEL system and to national reporting.

Condition to be met prior to the first disbursement of funds by GCF under the Funded Activity Agreement:

- (a) The Accredited Entity shall deliver, in a form and substance satisfactory to the GCF Secretariat, a short summary that clearly explains how eligibility criteria will be applied in the selection of investments, project sites, and beneficiaries to be supported by GCF proceeds for each of the seven participating countries. This note should also clarify any

changes in approach across particular country contexts and with the overall multi-country funding proposal; and

- (b) The Accredited Entity shall deliver, in a form and substance satisfactory to the GCF Secretariat, a short note confirming the institutional set-up for output 2 in each participating country, as well as the levels of representation of country participants in the multilevel governance mechanism under output 1.

Condition to be met prior to the second disbursement of funds by GCF under the Funded Activity Agreement:

- (a) The Accredited Entity shall deliver, in a form and substance satisfactory to the GCF Secretariat, a short note outlining the overall financial sustainability plan for the Programme, including how this will be approached in each of the seven participating countries.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP258)

Proposal name:	Multi-country Project Advancing Early Warnings for All (EW4All)
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Antigua and Barbuda, Cambodia, Chad, Ecuador, Ethiopia, Fiji, Somalia
Project/programme size:	Medium

Impact potential

The detailed analysis and observations of the iTAP on the Impact Potential rated as Medium are noted and appreciated.

We confirm that the estimated target beneficiary numbers for each of the seven participating countries have been provided to the GCF. All seven country proposals have followed Ecuador's approach prioritizing climate vulnerable sectors in line with the overall funding proposal.

We recognize the ambitious scope of the outputs and acknowledge that the extent of impact will depend on successfully engaging and working with multiple stakeholders at all levels, from global to local. We assure the iTAP that the Pillar Leads and UNDP are well positioned to do this at the global and country levels.

We confirm that the required pace and scale of implementation will be ensured to reach the intended beneficiaries and meet stated targets.

We fully accept the iTAP's recommendation and confirm that the approach of prioritising sectors vulnerable to climate impact has been applied to all country projects. The project will follow the recommended sector focus to enable analysis and targeting of interventions that address the unique vulnerabilities in each country. We also fully agree that successes in one sector will be replicated and scaled up to national levels through coordinated action.

Paradigm shift potential

We note the Medium rating given to the paradigm shift potential and appreciate the careful analysis that has been provided.

We note the critical need for a longer-term financial strategy to sustain the EW4All architecture beyond 2027 and accept the recommendation that this will be undertaken through wide consultations with a range of partners. The project will develop a financial sustainability strategy in each participating country to codify sustainability pathways relevant to each country's context.

We acknowledge and confirm that mechanisms for knowledge exchange between countries, the establishment of technical expert groups and sharing of lessons through events and workshops will be put in place. We fully accept the recommendation to leverage Output 1 to replicate EW4All across other countries, provided there is additional financing, and Output 2 to define approaches that can be scaled up over the long term.

The concern that the number of beneficiaries is a small proportion of the population is noted. The figures that are provided in the country proposals are deliberately made conservative, particularly for FCAS. However, all country projects are designed to maximise the impact through deliberate integration and coordination with other projects and programmes of the government. The use of technologies such as cell broadcasting will facilitate the national reach of early warning communications, hence increasing the potential for project impact. It is important to note that the project has also looked to ensure that early warnings are able to reach those most vulnerable, hence ensuring an important focus of additionality that is consistent with UNDP's leave no one behind approach.

We appreciate the acknowledgement by iTAP of the different approaches varying by country capacities and contexts and appreciate their recognition that conditions in FCAS could affect project delivery. We also acknowledge that the assumptions in the theory of change are significant and take note they need to be managed well to avoid the risks listed.

The detailed indicators for outputs for the seven countries, including how activities under Output 2 will be implemented are provided in the country proposals. We confirm that these will be provided to the GCF by end of January to mid-February 2025.

Sustainable development potential

The "Medium to High" rating given to the sustainable development potential for the project is noted. We thank the iTAP and concur with the detailed assessment of how the project supports strategic frameworks and the national and global levels and the economic and environmental co-benefits.

We confirm that each country level proposal contains environmental and social management frameworks addressing their unique needs. We also confirm that only low risk (category C) activities will be supported by the project. We appreciate the acknowledgement that the project design adopted a people centred and participatory approach with specific attention to needs of vulnerable groups including women, persons with disabilities and indigenous peoples. Country proposals have all included country specific stakeholder engagement plans that will be monitored throughout project implementation to ensure that the project's participatory and people centred focus remains a priority.

We confirm that the country proposals provide the context and specificity needed for timely, accessible and responsive EWS. We appreciate the recognition that the proposal integrates fragility and conflict in its approach and confirm that the project will benefit the health sector and that Output 1 will lay the foundation for expanding MHEWS to other countries.

We appreciate the acknowledgement made to the substantial differences in factors and contexts that contribute to sustainable development in these countries. We confirm that the activities and sub-activities presented in the country proposals address their unique barriers and leverage available opportunities to maximise the sustainable development impacts of these interventions.

Needs of the recipient

We agree with iTAP's assessment that the needs of the project recipients are "High". We appreciate that iTAP recognises the programme is designed to be inclusive so it reaches vulnerable populations, low-income households and those whose livelihoods are in climate vulnerable sectors. We also appreciate that the iTAP has noted that there are significant

challenges faced by the seven participating countries in accessing finance and in institutional and capacity gaps.

We confirm that each country proposal includes a country specific gender analysis and action plan as well as a stakeholder engagement plan that will ensure project activities are gender responsive and inclusive.

Country ownership

The “Medium” rating given to country ownership is noted.

As mentioned in the assessment the project has engaged with NDAs in project development and no objection letters from the 7 NDAs have been received. We confirm that specific individual country co-finance contributions have been provided to the GCF, the Global FP has been updated accordingly. We further confirm that each country-level proposal provides details of national and local level buy-in as reflected within the country-level stakeholder engagement plan that includes a description of the consultations at country level while providing a detailed analysis and strategy for continued stakeholder engagement during the projects’ life-cycle.

We thank the iTAP for summarising the alignment of the proposals with the country's climate change priorities and confirm that the respective country proposals and feasibility studies provide a significantly higher level of detail.

We concur with the observations of the iTAP that some countries, particularly FCAS, lack governmental agencies and ministries as counterparts for activities and in these cases, in-country agencies, including NGOs, local civil society organizations and multilateral agencies with the best comparative advantage will support in the implementation of these activities. Investment in these countries look to enhance community and local governance systems while promoting opportunities for enhanced social cohesion and trust.

We would like to emphasise that consultations with national-level stakeholders were systematically completed in all countries, except in Somalia, where security considerations limited such work. Details of these engagements are captured in the country-specific stakeholder engagement plans. In the case of Ethiopia, Somalia and Chad a conflict sensitivity analysis has been included within the Stakeholder Engagement Plan that provides a framework for continued consultations and stakeholder engagement.

We confirm that the country-level stakeholder engagement plans provide the details on how consultations will be undertaken and apologise that this could not be captured sufficiently in the global FP as these reports were still under preparation.

Efficiency and effectiveness

The rating of “Medium” for efficiency and effectiveness is noted

Thank you for the detailed analysis.

The co-financing figures in the funding proposal are finalised and have been shared with GCF in the latest version of FP pp. 52-54. including 7 countries’ total budgets.

The letters of co-finance have been provided for all countries and have been submitted to GCF.

The allocation of resources by country as set out in Annex 4 is now finalised. We are pleased to inform the iTAP that the budgets for all seven participating countries are now final and the global FP has been updated accordingly.

We take note of the iTAP recommendation on further defining the country-specific approaches to engage with the private sector. It is important to note that national proposals include activities to support the development of long-term financial sustainability strategies which will include, in some countries, potential engagement of the private sector.

We also take note of the need to ensure that decision-making is coordinated across both national and global levels for the project as a whole for coordination and to leverage synergies. All seven countries are represented in the global coordination arrangements, and we have allocated resources for cross-regional knowledge sharing, lessons learned and effective coordination between members on the different country and global coordination structures at policy and programme levels. The project has foreseen a project management structure that will allow for coordination at both global and national levels through project management units while providing a space for lessons on the ground to guide knowledge management and learning.

We take note and accept the recommendation that the overarching log frame will be revised and adjusted during project inception period once the country log-frames become available. We also accept the recommendation and confirm that the monitoring arrangements will be revisited and strengthened during the inception phase to ensure coherence between the global and country level efforts. We acknowledge the critical nature of the baseline assessments under output 2 that are planned for the initial phases of the project.

Overall remarks from the independent Technical Advisory Panel:

We greatly appreciate iTAP's endorsement of the project to the GCF Board.

We thank the iTAP for their flexibility and support and confirm that further country-level details will be submitted to GCF by the end of January (and mid-January for one or two countries).

We accept the recommendations proposed by iTAP; however, we would like to provide one clarification and seek consideration on a revised timeline for one of the conditions proposed.

On iTAP recommendation in Paragraph 69 (a), we would like to clarify that, based on GCF Secretariat guidance, we confirm that country-level information ("country project proposals") will be made available to the GCF prior to the AE undertaking implementation. We also confirm there are no variances with the multi-country funding proposal on which this assessment is based.

On the condition for the second disbursement, given the complexities associated with this project, we would like to seek iTAP's and GCF Secretariat's consideration to set this condition for the third, instead of the second, disbursement. Two justifications can be provided to support this proposal: First, the financial sustainability plan would be better informed if it reflects early lessons from the project implementation, and two full years of implementation would permit such lessons to be accumulated. Second, by the end of 2027, we expect to have in place various assessments and stocktakes from the SGs EW4All initiative as it runs till the end of 2027. This will allow the sustainability plan to fully incorporate lessons from the SG's EW4All initiative.

Gender Analysis and Action Plan
Multi Country Proposal
Early Warnings for All

January 2025

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1 Introduction

This Gender Analysis and Action Plan (GAAP) has been prepared in support of a project proposal for Multi-Country Project (MCP) for Early Warning Systems for All (EW4All) to the Green Climate Fund (GCF). This project is supported by the United Nations Development Programme (UNDP) in its role as a GCF Accredited Entity. The project has two main outputs:

Output 1 will promote regional and global coordination among various stakeholders, including government agencies, civil society organizations, academia, and the private sector. This Output aims to scale up and accelerate global and regional support for national efforts to increase MHEWS' capacity and coverage. This will be done by enhancing coordination and providing technical support for the implementation of EW4All; strengthening monitoring and evaluation (M&E) processes to support EW4All implementation; and reinforcing knowledge management and the dissemination of lessons learned. Activities under this Output will focus on coordination activities; developing guidance, tools and training materials; technical assistance; capacity building; knowledge management and dissemination, and peer-to-peer learning. This component will be led by the EW4All project lead agencies in each of the seven countries.

Output 2 will specifically focus on seven countries – Antigua and Barbuda, Cambodia, Chad, Ecuador, Ethiopia, Fiji, and Somalia – to roll out national level investment activities to address country and local level needs to achieve the EW4All target. This Output addresses barriers that prevent the implementation of a comprehensive, and end-to-end MHEWS in the seven initial target countries, including:

- 1) Gaps in policies, inter-agency coordination and lack of coherent strategies;
- 2) Inadequate technical and scientific capacities;
- 3) Missing observations; and
- 4) Barriers that prevent effective local actions.

The activities in Output 2 countries for all seven countries will mirror the four pillars of the EW4All Executive Action Plan (EAP), and will be implemented through the national governments with support from UNDP, the EW4All pillar leads and their in-country national offices and other partner agencies where relevant. The EW4All pillars and leads are:

- Pillar 1: Disaster risk knowledge – led by the United Nations Office for Disaster Risk Reduction (UNDRR);
- Pillar 2: Detection, observations, monitoring, analysis and forecasting of hazards, led by the World Meteorological Organization (WMO);
- Pillar 3: Warning dissemination and communication, led by the International Telecommunication Union (ITU);
- Pillar 4: Preparedness to respond, led by the International Federation of Red Cross and Red Crescent Societies (IFRC).

Activities aligned with Pillar 1 will focus on strengthening governance and coordination, establishing clear functions, roles and responsibilities of agencies in the national warning dissemination process. Considerations of differential access to information and resources by women and vulnerable groups will be included in the dissemination strategy. This activity will also establish or strengthen existing local early warning committees or equivalent grassroots organisations with representatives from elected/traditional leaders, including women and vulnerable groups. Capacities of these local institutions, from district down to village level, will be enhanced through regular engagements and training programmes that strengthen institutional structures and procedures.

Activities aligned with Pillar 2 will focus on disaster risk knowledge and observations in order to improve awareness about hazards, exposure populations, services, infrastructure, vulnerabilities of communities, and

loss and damage accounting. There will be a focus on the strengthening and/or establishment of meteorological, hydrological and climate forecast and monitoring systems for each participating country.

Activities aligned with Pillar 3 and 4 will focus on Early Warning dissemination and preparedness. Early warning messages will be developed in a multitude of accessible formats and languages, including for persons with disabilities. Information will be disseminated in a manner which ensures that risks and warnings are understood, are clear and actionable by their target audience. The activity will support a spectrum of actions, including integrating clear protocols for early warning and early action in disaster management/preparedness plans. Best practices for data privacy will be put in place. Assessments will be carried out to understand how best to reach communities at risk. There will also be training of local governments and first responders. Communities will be engaged in developing holistic plans for early warning and anticipatory action to respond to and to better manage local disasters reducing risks and to better adapt to climate change based on feasibility studies, climate information and local knowledge. Specific attention will be given to the needs of women and vulnerable groups. Communities will be trained on Operations and Maintenance (O&M) of early warning devices such as water level recorders, rain-gauges and anemometers. A full description of activities can be found in the MCP. A Stakeholder Engagement Plan has also been prepared.

Gender considerations have been mainstreamed into all project activities. A specific gender analysis for each country has been conducted and can be found in sections 3-9 of this report. Section 10 of this report consists of a consolidated global Gender Action Plan (GAP) which covers the whole project¹.

1.1 Project's Institutional Arrangements at the Global Level

UNDP will be the Executing Entity of the EW4All-MCP and will be responsible for the overall project.

The global implementation arrangement will comprise of a Project Board or Project Steering Committee (PSC) comprised of representatives of the Pillar Leads (WMO, UNDRR, ITU and IFRC) who have co-developed this project. PSC meetings will be held virtually, two times a year. In addition, the members of the committee will meet physically every year for a knowledge sharing meeting. Three such meetings will be held during the implementation of the first seven country projects. These meetings will be attended by the EW4All Pillar Leads, GCF, UNDP and two representatives from each of the country projects, including at least one from the relevant national governments.

The Project Board is responsible for making, by consensus, management decisions when guidance is required by the Project Manager. Project Board decisions will be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition. In case a consensus cannot be reached within the Board, final decision shall rest with the UNDP Executive Coordinator.

A global Project Management Unit (PMU) will be constituted within UNDP for the overall project coordination, oversight and monitoring, including implementation of the consolidated global GAP that is found in Section 10 of this report. The PMU will also provide technical support help the country PMUs in accessing regional and international resources through networking. The PMU comprises of three full time staff: a Project Manager, an Administrative and Finance Assistant and a Procurement Specialist. Part-time staff on the PMU will include a Technical Advisor, a Monitoring Evaluation and Learning Officer, a Communications Expert, a Safeguards, Gender and Inclusivity Expert (SGIE) and other subject experts brought in on an as needed basis.

The Project Manager will run the project on a day-to-day basis on behalf of the Project Board within the constraints laid down by the Project Board. The Project Manager function will end when the final project terminal evaluation report and other documentation required by the GCF and UNDP has been completed and submitted to UNDP. The Project Manager is responsible for day-to-day management and decision-making for

¹ In addition, each country has prepared their own country-level GAP with more detail, which is part of the country-specific funding proposals.

the project. The Project Manager's prime responsibility is to ensure that the project produces the results specified in the project document, to the required standard of quality and within the specified constraints of time and cost. Safeguards, Gender and Inclusivity Expert (SGIE) will work under the supervision of the PM to implement the global GAP and oversee the country-level GAPs. The SGIE will be in regular contact with country teams and consolidate the result of country-level GAP implementation into the global GAP.

1.2 Project's Institutional Arrangements at the Country Level

Under Output 2, UNDP will implement the Project under the Direct Implementation Modality (DIM) in all seven countries. At the country level, each country will have a Project Board or Steering Committee made of the relevant country agency with overall responsibility for the project and the National Designed Authority (NDA) for GCF. As the Senior Beneficiary, the implementing partner will be part of the board. Furthermore, as the Senior Supplier, UNDP will provide quality assurance for the project and ensure adherence to the guidelines for the implementation modality. The Project Board will be responsible for making, by consensus, management decisions when the National Project Manager (NPM) requires guidance. Project Board decisions will be made per standards to ensure management for development results, best value money, fairness, integrity, transparency and effective international competition. This commitment to fairness and transparency in decision-making will help build trust in the project. If a consensus cannot be reached within the Board, the final decision shall rest with the UNDP Programme Manager.

The NPM will run the project on a day-to-day basis. The NPM is responsible for day-to-day management and decision-making for the project, including compliance with country-level GAPs developed. In this regard, the NPM will be supported by a National Safeguards, Gender and Inclusion Expert (NSGI) expert. The NSGI, through the NPM, will report to the global PMU on status of the GAP activities at the country level so that they can be consolidated into the global GAP monitoring.

2 General Context for Gender in Early Warnings

Around the world gendered inequalities and norms can be found including around household roles, access to resources, power, workloads, livelihoods and education. In most countries there are structural and normative barriers to gender inclusion, and different expectations and access to resources and decision making between men and boys, women and girls, and people of other marginalized gender groups. Gender differences can make a significant impact when it comes to receiving Early Warnings and in preparing for disasters.

Currently, there is a significant lack of equitable and meaningful decision-making opportunities and roles for women in the design and management of Multi-Hazard Early Warning Systems (MHEWS) and related areas. There are significant gender gaps in access to information, technology and digital services. For example, the divide in mobile internet use is most stark in South Asia and Sub-Saharan Africa where, more than half a million women are excluded from the mobile connection and the gender gap scores are 41% and 37%, respectively.

The impacts of hazard events are not gender-neutral. Women are often placed at greater risk through a lack of timely and relevant information about imminent hazards. Women may not have equal access to technology, communication or services, and thus miss out on critical information. This is particularly true for women and marginalized groups living in rural or isolated areas. On the other hand, women are often the drivers, strong networkers, managers, organizers and caretakers in a community, playing a constructive and productive role in EWs and Disaster Risk Management. The value added of women in MHEWS is significant and multifaceted, contributing to the effectiveness, inclusivity, and sustainability of these systems. Women can considerably improve the efficiency of MHEWS through their unique perspectives, knowledge, and roles within communities. Moreover, integrating women's voices and experiences not only improves the outcomes of MHEWS but also promotes equity and empowerment within disaster-prone communities.

To be effective, early warning systems should not only have a sound scientific basis and reliable technology, but also should be people-centred, end-to-end, multi-hazard, trusted and accessible to all. EWS should consider the unique needs and capacities of different groups, including people who are marginalized or living in vulnerable situations. This includes women. Additionally, EWS must recognize and adapt to diverse factors, including the socioeconomic situation, accessibility of basic services and resources, literacy levels, access to communication technologies, cultural values, power dynamics, freedom of movement, and physical location. Not considering gender differences when developing EWS could mean women miss out on benefits but also that opportunities are missed, such as not utilizing women-led community support groups and/or informal dissemination networks.

Barriers and Challenges that can be found in Early Warning Systems



Source: PreventionWeb²

² Budimin, M. et al. (2023) Gender in early warning and early action: why do we still need to talk about it?

Thus, Multi-Hazard Early Warning Systems can be enhanced by taking into considerations key gender aspects. The importance of gender considerations are embedded in the MCP for EW4All. Some general aspects that will guide the project's gender approach include:

- Women often have unique insights into community dynamics, including vulnerabilities and coping strategies, which are crucial for developing targeted early warning messages and response plans.
- Women's involvement helps identify and utilize communication channels that are more accessible to women, such as community networks, women's groups, and mobile technologies suited to their preferences.
- Women often play key roles in disseminating early warnings in culturally appropriate ways, ensuring messages are understood and acted upon effectively across diverse communities.
- Improving community awareness, and in particular women's understanding of climate change and disaster risks, using both traditional and modern scientific knowledge, helps to inform community level disaster risk management and empower women to participate.
- Women's active participation in MHEWS activities can enhance trust and credibility within communities, increasing the likelihood that early warnings will be taken seriously and acted upon promptly.
- Women often have extensive local knowledge and networks, facilitating rapid dissemination of early warnings and mobilization of community resources during emergencies.
- Women's involvement contributes to building community resilience by integrating gender-sensitive approaches into disaster preparedness and response strategies.
- Women often contribute innovative solutions to enhance MHEWS effectiveness, leveraging their practical experiences and knowledge of community dynamics.
- Women often advocate for the needs of vulnerable groups, including children, elderly, and persons with disabilities, ensuring their inclusion in early warning and response efforts.
- There are widespread positive ripple effects from well-supported community- and women-led MHEWS, including greater gender equality and status of women, and broader community-level engagement and empowerment.

3 Gender Assessment for Ecuador

Ecuador is a middle-income country. Its economy is highly dependent on oil and agricultural exports. In the last decade, Ecuador has increased social spending and investments in infrastructure, hospitals and schools. Nevertheless, income inequality remains a problem, with almost half the rural population living in poverty compared to a third of people nationwide. The rates of extreme poverty in rural areas are double that of the total population³.

The Human Development Index (HDI) is developed annually by the United Nations Development Programme (UNDP). This indicator not only measures a country's economic development but also analyzes the health, education, and income of the population. The HDI is a tool used to assess the progress of countries in terms of human development and to compare well-being across different nations.

In 2023, Ecuador's Human Development Index was 0.759, ranking 86th worldwide. This represents an improvement compared to 2020 (the COVID year), when it stood at 0.731⁴.

The Gender Inequality Index (GII) is an indicator developed by the United Nations Development Programme (UNDP) to reflect the disadvantages women face in three key dimensions of human development:

- Reproductive Health: measured through the maternal mortality rate and the adolescent fertility rate.
- Empowerment: measured by the proportion of parliamentary seats occupied by women and the proportion of women and men aged 25 and older with at least a secondary education.
- Labor Market Participation: measured by the labor force participation rate of women and men aged 15 and older.

This index reflects the potential loss in human development due to inequalities between men and women in the aforementioned dimensions. In 2023, Ecuador's Gender Inequality Index was 0.385, ranking 91st worldwide. This represents an improvement compared to 2022, when it stood at 0.388⁵.

According to Ecuador government data from June 2023⁶, income poverty stood at 27%, with urban poverty at 18%, and 46.4% in rural areas. Extreme poverty stood at 10.8%, reaching 5.2% in urban areas and 22.6% in rural areas⁷. Therefore, poverty remains a critical issue, with that 20.0% of women are in a situation of income poverty compared to 17.9% of men, highlighting the vulnerability of women to economic shocks. As for extreme income poverty, women reach 6.2%, while men register 6.4%.

The youth are especially affected: in 2022, 34.44% of the youth (between 15 and 24 years old) lived in poverty⁸. Although income poverty decreased for both men and women in 2022 (from 28.21% to 24.66% for men and from 29.46% to 26.22% for women), the gender gap stood at 1.56 percentage points, a widening that suggests a negative impact of other economic variables, such as inequality in the labour market, wage gaps, unpaid jobs and women spending more time on care. The rate of women who are not in education or in remunerated employment is 15.5% higher than the rate for men. The major causes relate to lack of opportunities, unpaid care work that women do and patriarchal gender norms. The multidimensional poverty rates of indigenous women, montubias⁹ and girls and adolescents are 78.1%, 57.3% and 45.8%, respectively¹⁰. Between 83% and 85% of indigenous boys and girls are in a situation of multidimensional poverty, being the most impoverished population group in the country¹¹. In the case of older adults, the percentage of women

³ [Ecuador \(ifad.org\)](https://www.ifad.org/)

⁴ Source: UNDP.

⁵ GII is a composite metric of gender inequality using three dimensions: reproductive health, empowerment and the labour market. A low GII value indicates low inequality between women and men, and vice-versa.

⁶ [202306 Boletin_pobreza_ENEMDU.pdf \(ecuadorencifras.gob.ec\)](https://www.ecuadorencifras.gob.ec/publicaciones/files/2023/06/Boletin_pobreza_ENEMDU.pdf)

⁷ [20230908-eom-ecuadore-sr-poverty.pdf \(ohchr.org\)](https://www.inec.gob.ec/publicaciones/files/2023/09/20230908-eom-ecuadore-sr-poverty.pdf)

⁸ Ibid.

⁹ Montubio is the term used to describe the mestizo people of the countryside of coastal Ecuador. The Montubio make up 7.4% of the country's population and they are recognized as having their own cultural identity.

¹⁰ Source: INEC - ENEMDU, December rounds.

¹¹ Pontificia Catholic University of Ecuador - Institute of Economic Research [PUCE-IIE], 2022

living in multidimensional poverty is 58.4%, 52.5% and 20.6% in the case of Montubias, Indigenous and Afro-Ecuadorian women, while percentages are 50.2%, 42.2% and 18% for men in the same groups¹².

Literacy, Education and Access to Information

From 2010 to 2022, the illiteracy rate in Ecuador decreased by 3.1 percentage points, from 6.8% in 2010 to 3.7% in 2022. According to the results of the 2022 Population and Housing Census, out of the total population aged 15 and older (12,625,997), 3.7% (472,228) are illiterate, of which 59.7% are women (281,834).

By age groups, people aged 65 and older have the highest percentage of illiteracy, with 51.5%, of whom 63.1% are women. The population aged 30 to 64 represents 43.9% of the total illiteracy in the country, with 57.4% of this group being women. The remaining 4.7% of the illiterate population is concentrated in the age group of 15 to 29 years, with women representing 43.4% of this group¹³.

Ecuador has an estimated population of 16,938,986¹⁴ with 48.7% of men and 51.3% of women. Country-wide 36% of the population have completed only primary level education, 29% have completed only high school and 25% have also completed tertiary education; 4.5% have no education. Literacy rates are 96.3%. However, there are significant differences between the genders, areas in the country and ethnic groups. Illiteracy, for example, is 6.8% in rural areas and nearly 12% for Indigenous peoples. Women are also more likely to be illiterate (4.3% compared to men at 3.1%). There is more parity in the genders around general school and high school attendance. For example, 76.5% of women and 75.5% of men attended high school and 70% of Indigenous groups. Primary attendance is much higher, at 90% or more for all these groups¹⁵.

Meanwhile, access to education and vocational training for women is significantly influenced by their ability to utilize digital technologies. According to the 2019 Multipurpose Survey on Information and Communication Technologies (ICT) conducted by INEC, 58.2% of women aged 5 and older own a cell phone, compared to 61.7% of men. Of these figures, 77.3% of women and 76.3% of men use a smartphone¹⁶.

In Ecuador, women's access to information varies based on several factors such as age, ethnicity, geographic location, and socioeconomic status. Below are the main ways women access information and the differences observed between men and various groups of women.

Ways of Accessing Information

Traditional Media:

- **Radio:** It is an important source of information, especially in rural areas. Women often listen to radio programs that cover health, education, and government programs; while farmers listen to the radio between 5 and 6 a.m. as they head to the fields (WFP, 2023).
- **Television:** It is accessible in many urban and rural areas. News broadcasts and general interest programs are followed by many women to stay informed (WFP, 2023).
- **Print Media:** Although less common, some groups of women, especially in urban areas, read free newspapers to obtain information (WFP, 2023).

¹² End of Mission statement by Mr Olivier de Shutter. Special Rapporteur on Poverty and Human Rights. Visit to Ecuador 28 August 8 September 2023. [20230908-eom-ecuadore-sr-poverty.pdf \(ohchr.org\)](https://www.ohchr.org/en/documents/other-documents/20230908-eom-ecuadore-sr-poverty.pdf)

¹³ Source: Instituto Nacional de Estadística y Censos (INEC), Censo de Población y Vivienda 2022. <https://www.censoecuador.gob.ec/resultados-censo/>. See table in the Annexes, page 15.

¹⁴ Ecuador Census data:

<https://app.powerbi.com/view?e=eyJrIjoiaNWUzMjQwOWMtZjFhOS00Njc2LTk0YTItNjcwZmRmY2YxMjkyliwidCI6ImYxNTNhMmU4LWZhZWMtNDQwNj1iMGFiLWY1ZTI1OWJkYTEwMjI9>

¹⁵ Ibid.

¹⁶ Source: Instituto Nacional de Estadística y Censos (INEC), Encuesta Multipropósito - Tecnologías de la Información y Comunicación-TIC. 2019. <https://www.ecuadorencifras.gob.ec/tecnologias-de-la-informacion-y-comunicacion-tic-2019/>

Digital Media and Social Networks:

- **Internet:** Internet access is growing, but significant gaps still exist. Young and urban women are the most likely to use the internet to obtain information. In the lowest income quintile, internet access is present in 43.7% of households, while in the highest income quintile, it is present in 79.4% of households (INEC ENEMDU, 2023).
- **Social Networks:** Platforms like Facebook, Instagram, and WhatsApp are popular for sharing and receiving information, especially among young women.
- **Text Messages (SMS):** Used by government programs and health organizations to send information directly to mobile phones.

Community and Personal Sources:

- **Other Women:** Many women obtain information through conversations with friends, family, and neighbors.
- **Community Leaders:** In rural communities, community leaders and local organizations play a crucial role in disseminating information. Community leaders mainly use WhatsApp to coordinate with public and local institutions.
- **Word of mouth:** this is a form of communication that primarily occurs in rural villages.
- **Schools and Educational Institutions:** Young women and mothers often receive information through their children's schools.

Women and men in Ecuador can access information in similar ways; however, there are some notable differences:

- **Digital Divide:** Men tend to have greater access to digital technologies and the internet compared to women, especially in rural areas. 63.2% of male heads of household have internet access, while 60.3% of female heads of household have internet access (INEC ENEMDU, 2023).
- **Gender Roles:** Women, due to their traditional roles in the household, may have less time and fewer opportunities to access information through certain media, such as the internet and television.

There are also differences among groups of women depending on their age, ethnicity, location and socio-economic status.

Age:

- **Young Women:** Prefer social networks and the internet.
- **Older Women:** Prefer traditional media like radio and television.

Ethnicity:

- **Indigenous Women:** In rural areas, radio and community leaders are important sources. Language barriers can also influence how they access information. Indigenous people are the ethnic group with the lowest number of active cell phones (30.2%) (INEC ENEMDU, 2023).
- **Afro-Ecuadorian Women:** Similar to indigenous women, but with a greater tendency to use digital media in urban areas.

Geographic Location:

- **Urban Women:** Greater access to the internet and social networks. At the urban level, 81.1% use the internet, while at the rural level, only 54.5% use the internet (INEC ENEMDU, 2023).
- **Rural Women:** Greater reliance on radio and community leaders. Digital illiteracy is present in 19.4% of the rural population (persons between 15 and 49 years old who meet the criteria of not having an active cell phone, not using a computer, and not using the internet) (INEC ENEMDU, 2023).

Socioeconomic Level:

- **High and Middle:** Greater access to a variety of media, including the internet, cable television, and print media.
- **Low:** More dependent on radio and free television

As far as phones are concerned, based on the results of the 2022 Population and Housing Census, a total of 12,485,705 people use cell phones, which represents nearly 100% of people aged 15 and over in the country. Of these, 48.7% are men and 51.3% are women, indicating a slight majority of female users. Based on data, 55.6% of people specifically own a smartphone¹⁷.

Broken down by age groups:

- Among those aged 5 to 29 years, 5,654,184 people use cell phones, of which 46.2% are men and 44.4% are women. The slight majority of men in this group may be influenced by usage patterns and access to technology at early and young ages.
- Among those aged 30 to 64 years, 5,953,529 people use cell phones, with 46.7% being men and 48.4% being women. Women slightly outnumber men in this age group, which may reflect a higher adoption and use of mobile technology among women in their productive years.
- In the 65 and older age group, 877,992 people use cell phones, of which 7.1% are men and 6.9% are women. The difference here is minimal, indicating an almost equal adoption of cell phone use among older men and women¹⁸.

According to the 2022 Population and Housing Census, conducted by INEC, the following results are detailed regarding the access of the population aged 5 years and older to ICT use in the last 3 months:

- The total number of women accessing mobile phones is 6,409,731, which corresponds to 51.3% of the national total. Of these, 67.9% (4,351,301 women) are in urban areas, and 32.1% (2,058,430 women) are in rural areas. This denotes a notable disparity between urban and rural areas, suggesting the need for specific interventions to improve connectivity and access to technology in rural areas to foster more equitable digital inclusion.
- Regarding women's access to the internet, a total of 5,580,284 women use this service, representing 51.4% of the national total. Of these, 70.9% (3,955,179 women) are in urban areas, while 29.1% (1,625,105 women) reside in rural areas. The disparity in internet access between urban and rural areas can widen the digital divide, contributing to inequalities in socioeconomic, educational, and employment opportunities.
- Regarding women's access to the use of computers or laptops, 2,636,736 women use these devices, representing 51.1% of the total population. Of this group, 77.2% are in urban areas (2,034,796 women) and 22.8% in rural areas (601,940 women). This indicates a significant gap compared to urban areas, which can be attributed to various factors such as lower availability of technology and internet access in rural areas, fewer opportunities for training in digital technology use, or lower purchasing power to buy devices.
- A total of 903,876 women use tablets, representing 51.0% of the total population. Of this group, 80.2% are in urban areas (725,009 women) and 19.8% in rural areas (178,867 women). Tablet use in

¹⁷ <https://datahub.itu.int/data/?e=ECU&i=28318>

¹⁸ Source: Instituto Nacional de Estadística y Censos (INEC), Censo de Población y Vivienda 2022. <https://www.censoecuator.gob.ec/resultados-censo/>

women can be broken down by: 17.2% are women aged 5 to 14 years, 31.9% are women aged 15 to 29 years, 46.1% are women aged 30 to 64 years, and 4.7% are women over 65 years. Tablet use among women is concentrated in ages 15 to 29, used for both studies and recreational and social activities, and in ages 30 to 64, indicating intensive tablet use for professional activities, personal management, and continuous learning. The gap between rural and urban areas has similar factors to the gap in access to computers and laptops.

- Regarding women's access to mobile phones by age group, of the total 6,409,731 women, 12.8% are women aged 5 to 14 years, 31.6% are women aged 15 to 29 years, 48.6% are women aged 30 to 64 years, and 6.9% are women over 65 years. This shows a diverse distribution by age group, with high penetration in productive ages and a significant presence even at the extremes of the age distribution.
- Regarding women's internet use by age group, of the total 5,580,284 women, 15.4% are women aged 5 to 14 years, 32.6% are women aged 15 to 29 years, 47.1% are women aged 30 to 64 years, and 4.9% are women over 65 years. Internet use among women varies significantly by age group, with higher adoption in productive ages and notable penetration among the youngest.
- The national total of women using computers or laptops is 2,636,736. Of this total, 16.2% are women aged 5 to 14 years, 36.2% are women aged 15 to 29 years, 44.7% are women aged 30 to 64 years, and 3.0% are women over 65 years. The use of computers or laptops by women presents higher adoption in productive ages and significant presence even among the youngest¹⁹.

Employment, Labour Participation and Typical Social Expectations on Gender

Gender inequality in employment is evident in both quantitative and qualitative terms. Women have a higher unemployment rate (4.7% vs. 3.2% for men) and are overrepresented in informal jobs, with 67.9% of women in such positions compared to 51.9% of men. Women's average income is significantly lower, highlighting a substantial gender pay gap. Additionally, women represent a smaller portion of the economically active population. The informal employment rate is higher among women (53.2%) than men (51.9%), with young women (ages 18 to 29) facing particularly high unemployment rates, reflecting barriers to labor market entry.

In 2023, the average labor income for men at the national level was USD 496.8, while for women it was USD 420.3. More starkly, the median labor income for men was USD 416.4, in contrast to USD 297.4 for women. This difference of USD 76.5 in average income indicates a gender wage gap, where men earn approximately 18.2% more than women on average. The difference in the median, which is USD 119, shows an even larger gap when considering the midpoint of incomes. Men earn 40% more than women in the median income²⁰. In 2023, the participation rate in unpaid employment for women was 75.8%, which is three times higher than the rate for men at 24.2%. This reveals significant gender disparities in unpaid work. Regarding age groups, it is mainly concentrated in the population aged 15 to 17, with a rate of 82.1% for women and 63.8% for men.²¹ This burden of unpaid work at an early age can interfere with education and development, perpetuating a cycle of gender inequality²².

¹⁹ Source: Instituto Nacional de Estadística y Censos (INEC), Censo de Población y Vivienda 2022. <https://www.censoecuador.gob.ec/resultados-censo/>

²⁰ Source: Instituto Nacional de Estadística y Censos (INEC), Encuesta Nacional de Empleo, Desempleo y Subempleo - ENEMDU Acumulada Anual 2023. <https://www.ecuadorencifras.gob.ec/enemdu-anual/>

²¹ See table in the Annexes, page 17.

²² Source: Instituto Nacional de Estadística y Censos (INEC), Encuesta Nacional de Empleo, Desempleo y Subempleo - ENEMDU Acumulada Anual 2023. <https://www.ecuadorencifras.gob.ec/enemdu-anual/>

In Ecuador, social expectations and cultural beliefs about gender roles are deeply rooted and vary according to region and age groups. These expectations are influenced by historical, economic, social, and cultural factors.

Traditionally, men are expected to be the primary economic providers for the family. They are assigned the role of protectors, and strength, authority, and independence are valued as desirable masculine traits. Meanwhile, women are assigned the role of caregivers and household managers. They are expected to take care of child-rearing and domestic chores. Valued characteristics in women include submission, patience, and dedication to the family.

The activities dominated by men are five times those dominated by women (332 versus 63), and the latter are linked to care and teaching services. According to data from the Central Bank of Ecuador (2021), the activities with the highest proportion of female employment include animal husbandry, publishing directories and mailing lists, hairdressing and beauty services, primary and preschool education, care in institutions, social assistance, clothing manufacturing, jewelry making, among others. Meanwhile, the activities with the highest proportion of male employment include central banking, tobacco product manufacturing, sugar production, wood extraction, wood sheet manufacturing, private security activities, motor vehicle manufacturing, precious mineral product manufacturing, other mineral extraction, machinery manufacturing, among others.²³

Based on the concept of Unpaid Work, which is the time dedicated to domestic activities, caregiving, and community support without receiving payment or remuneration, women's participation is 75.8% and men's is 24.2%. This reflects the social and cultural norms and expectations that assign women the primary responsibility for domestic and caregiving tasks²⁴.

In many households, domestic chores and childcare primarily fall on women, regardless of whether they also participate in the labor market. This unequal distribution of domestic responsibilities reflects a cultural expectation that women should balance work outside the home with household tasks. On average, women spend 31 hours a week on caregiving and unpaid household work, while men spend an average of 9 hours a week on caregiving and unpaid household work. This gap is particularly high for rural, indigenous, montubio women, and those aged 30 to 44 (INEC, EUT, 2012).

There are regional and generational variations to these gender expectations. In cities, there is a growing trend toward gender equity, with more women participating in the labor market and occupying leadership positions. However, traditional expectations still persist in certain sectors and there are challenges particularly around caregiving responsibilities. In rural areas, traditional gender expectations are more pronounced. Women have less access to education and job opportunities, reinforcing their traditional role as household caregivers. In addition, 36% of rural women have access to land compared to 43% of men, resulting in a 4.3% gap in land ownership. In the agricultural sector, 13% of women are remunerated, compared to 35% of men; this figure is explained by the fact that women's work is often unpaid as it is considered an extension of domestic work. Rural women perform 23 more hours of unpaid caregiving and domestic work per week compared to men (INEC, EUT, 2012). This results in lower educational levels, fewer hours available for paid work, lower income, and less access to health care, relying almost exclusively on environmental factors such as soil productivity, water quantity, and quality.

Meanwhile, younger generations (15-34 years) are increasingly challenging traditional gender expectations. Young women are seeking higher levels of education and labor market participation, more willing to question

²³ See table on Ecuador: Occupation by Economic Activity (CIU4) in the Annexes, page 18

²⁴ Source: Instituto Nacional de Estadística y Censos (INEC), Encuesta Nacional de Empleo, Desempleo y Subempleo - ENEMDU Acumulada Anual 2023. <https://www.ecuadorencifras.gob.ec/enemdu-anual/>

and challenge traditional gender roles, with greater professional aspirations, seeking more equity in the distribution of domestic tasks, and with high use of technology and social networks, facilitating access to information, there is an increase in the acceptance of more egalitarian gender roles. On the other hand, older individuals (around 50 years) tend to adhere more to traditional gender norms. Older women may have had less access to education and employment and likely have performed domestic roles for most of their lives. The cultural and social expectations that these generations grew up with strongly influence their perceptions and behaviors. Lower use of technology and access to information, compared to younger generations, may limit their exposure to new ideas about gender equality.

Representation and Participation in Decision Making

In Ecuador, women's political participation has shown significant progress in recent years. Following the legal reforms of 2008 and 2009, measures have been implemented to promote gender parity in politics. As of 2024, women hold 43.8% of the seats in the National Assembly²⁵.

At the local level, women's representation increased in 2023 compared to the 2019 elections. Of the country's 221 cantons, 34 have female mayors, 16 more than in 2019. Regarding provincial prefectures, 7 of the 24 provinces are led by women, representing an increase of 3 compared to the previous elections²⁶.

Regarding the representation of women heading State ministries, in November 2023, the current government appointed 11 women as ministers and secretaries of State, which represents 52% of the total positions in the National Cabinet²⁷.

According to the results of the 2022 Population and Housing Census²⁸, out of a total of 5,188,827 households nationwide, 38.5% of the persons recognized as household representatives are women, while 61.5% are men. In urban areas, 40.6% of household representatives are women and 59.4% are men. In rural areas, the percentages are lower, with 34.7% of women and 65.3% of men as household representatives.

According to ethnic self-identification, the gender distribution of household representatives is as follows:

- In indigenous communities, 35.9% are women and 64.1% are men.
- In Afro-Ecuadorian communities, 46.7% are women and 53.3% are men.
- Among Montubios, 34.2% are women and 65.8% are men.
- Among Mestizos, 38.7% are women and 61.3% are men.
- Among whites, 38.3% are women and 61.7% are men²⁹.

Despite progress, women's participation in public consultations still faces significant challenges in Ecuador. According to a study by CEPAL³⁰, although women attend public consultations, their decisive participation remains limited. This is due to the socioeconomic context, age, region, and the structural and cultural barriers that restrict their real influence in decision-making.

In urban areas, women have more opportunities and resources to participate in public consultations due to a greater presence of feminist organizations and inclusion policies, which have increased women's visibility

²⁵ Source: Asamblea Nacional del Ecuador. <https://www.asambleanacional.gob.ec/es/pleno-asambleistas>. See table in the Annexes, page 18

²⁶ Source: Consejo Nacional Electoral. <https://www.cne.gob.ec/resultados-electorales/>

²⁷ Source: Presidencia de la República del Ecuador. <https://www.presidencia.gob.ec/>

²⁸ See table in the Annexes, page 19.

²⁹ Source: Instituto Nacional de Estadística y Censos (INEC), Censo de Población y Vivienda 2022. <https://www.censoecuador.gob.ec/resultados-censo/>

³⁰ Annual Progress Report on Women in Latin America, 2020

and active roles. According to a UN Women report³¹, urban women tend to be more empowered and occupy more leadership spaces in social movements. In rural areas, participation is more limited due to factors such as lack of access to information, lower educational levels, and traditional gender roles. A study by FLACSO Ecuador³² indicates that in rural areas, women are primarily seen as caregivers, and their public participation is limited. Similarly, women from higher socioeconomic classes have more opportunities and resources, while those from lower strata face additional barriers such as poverty, lack of time due to work and domestic responsibilities, and the scarcity of resources for mobility.

From a cultural and social perspective, women face significant challenges in speaking openly and freely about the issues that affect them due to deeply rooted gender norms and stereotypes. In Indigenous communities, for example, women may face greater restrictions due to patriarchal traditions and social structures and may even be subject to reprisals or discrimination from community leaders or other community members.

Women with greater access to higher education and technology tend to be more empowered and have more influence in their communities. A UNICEF study³³ shows that young people participate more actively in social movements and the defense of their rights, whereas older women show less involvement in gender empowerment movements. Furthermore, there are examples of female leadership in indigenous organizations, showing a gradual change in this dynamic³⁴.

In conclusion, from a sociological point of view, women's participation in public consultations in Ecuador is influenced by a combination of geographical, socioeconomic, cultural, and generational factors. Although there are significant advances, especially in urban areas and among younger generations, structural barriers persist that limit the participation and empowerment of many women in the country. Training initiatives and inclusion policies are crucial to continue advancing gender equality and women's participation in public life.

Gender-Based Violence

Gender-based violence remains a critical issue in Ecuador. According to the National Survey on Family Relations and Gender Violence Against Women (ENVIGMU), conducted by the INEC in 2019:

- 65 out of every 100 women in Ecuador have experienced at least one incident of some type of violence in various contexts throughout their lives including (i) psychological violence 56.9%, (ii) physical violence 35.4%, (iii) sexual violence 32.7%, and (iv) patrimonial violence 16.4%.
- Data from 2019 show that 32 out of every 100 women in Ecuador had experienced at least one incident of some type of violence in various contexts in the last 12 months. The percentages by type of violence are as follows: (i) psychological violence 25.2%, (ii) physical violence 9.2%, (iii) sexual violence 12.0%, and (iv) patrimonial violence 6.1%.
- By ethnic self-identification, 72% of Afro-Ecuadorian women, 64% of Indigenous women, 58% of Montubio women, and 65% of Mestizo women have experienced at least one incident of some type of violence throughout their lives.
- By age groups, 45% of women between 15 and 17 years, 65% between 18 and 29 years, 69% between 30 and 44 years, 66% between 45 and 64 years, and 65% over 65 years have experienced at least one incident of some type of violence throughout their lives.

³¹ Women's Political Participation in Latin America and the Caribbean, 2019

³² Gender and Political Participation in Ecuador, 2018

³³ Youth and Participation in Latin America, 2021

³⁴ Radcliffe, S. and Laurie N., 2006

- By marital status, 77% of separated women, 65% of married women, and 55% of single women have experienced at least one incident of some type of violence throughout their lives.
- Nationally, throughout their lives, 19% of women experienced some type of violence in the educational sphere, 20% in the workplace, 33% in the social sphere, 20% in the family sphere, and 43% in the partner sphere³⁵.

Gender Policies and Legal Framework

Ecuador is a signatory to several international conventions and agreements aimed at promoting gender equality, including the Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) and the Beijing Declaration and Platform for Action. Ecuador has specific legislation on gender equality, gender violence, and national programs and committees dedicated to these issues. These frameworks aim to address gender disparities in various sectors, including education, health, and employment, but gaps remain in their effective implementation, particularly in rural and marginalized communities.

Gender Equality Legislation:

- Constitution of the Republic of Ecuador (2008): Establishes equal rights and prohibits discrimination based on gender.

Gender Violence Legislation:

- Comprehensive Organic Law to Prevent and Eradicate Violence against Women (2018): This law establishes measures to prevent, punish, and eradicate violence against women in its various forms.
- Three recent laws aimed at addressing gender inequalities are: the Organic Law on Voluntary Termination of Pregnancy in Cases of Rape (2022), the Organic Law to Promote the Violet Economy (2023), and the Organic Law on the Right to Human Care (2023).
- Other complementary legislation includes the Labor Harassment Prevention Law (2017), the Human Mobility Law (2017), and Executive Decree 696 (2019).

National Programs:

- National Plan for the Eradication of Gender Violence against Children, Adolescents, and Women (2022-2025): A plan that coordinates efforts to prevent and eradicate gender violence.
- National Agenda for Gender Equality (2021-2025): A territorial planning instrument that reflects collective work with the materiality of human rights.
- National Agricultural Strategy for Rural Women (ENAMR): A public policy instrument aimed at recognizing and making visible the participation and work of women and their importance to Family Farming and contributing from the competencies of the Ministry of Agriculture and Livestock to guarantee their rights.

Gender-Related Institutional Arrangements

Ecuador has established various institutional arrangements to support gender equality, including;

- The National Council for Gender Equality (CNIG): An agency responsible for ensuring the implementation of public policies for gender equality.
- The Ministry of Economic and Social Inclusion (MIES): Has specific programs aimed at protecting and promoting women's rights.

³⁵ Source: Instituto Nacional de Estadística y Censos (INEC), la Encuesta Nacional sobre Relaciones Familiares y Violencia de Género contra las Mujeres (ENVIGMU), 2019. https://www.ecuadorencifras.gob.ec/documentos/web-inec/Estadisticas_Sociales/Violencia_de_genero_2019/Principales%20resultados%20ENVIGMU%202019.pdf

- The Ministry of Women and Human Rights (MMDDHH): Ensures and promotes Human Rights through public policies of prevention, care, and reparation to promote gender equality.

These bodies are responsible for developing and enforcing policies that promote women's rights and gender equality.

Women's Representation and Participation in DRM Sectors

Women's participation in disaster risk management (DRM) sectors is essential for building resilient communities. Efforts to include women in DRM decision-making processes and leadership roles are ongoing, but greater representation and targeted support are needed. Programs that train and empower women in DRM roles can enhance community preparedness and response capabilities, making disaster management more inclusive and effective.

The Organic Law on Risk Management, in its articles 35, 36, and 39, establishes the need for the creation of Community Risk Management Committees as humanitarian response bodies. At the level of certain municipalities, such as Guayaquil, the Community Committee has already been created, which included gender organizations, vulnerable groups, and rural communities. Within this framework, training related to early warnings have been developed; however, they are neither continuous nor planned. UNDP carries out ongoing training projects on disaster preparedness and EWS at community level with the financial support of the European Commission Humanitarian Office (ECHO) and the Bank of Latin American and the Caribbean, this process has special measures to include women and men in an equitable approach. Also, the Ecuadorian Red Cross carries out ongoing training processes for communities on Early Warning mechanisms, as well as preparation for response to emergencies caused by natural phenomena.

Women's participation in activities related to climate change and disaster risk management has been increasing in recent years, although it varies between institutions. In the National Institute of Meteorology and Hydrology (INAMHI), 70 people work, of which only 21% are women (INAMHI, 2024), while in the case of the National Secretariat for Risk Management (SNGR), there are approximately 473 employees, of which 50% are women (SNGR, 2024).

Gender in Early Warnings and Disaster Risk Response

Emergencies and their impacts increase the risks to human life, endangering livelihoods, food security, and health. Women and girls, especially those living in poverty and marginalization, face higher levels of risk and an increased workload due to socially and culturally imposed gender roles. Additionally, they have limited opportunities to respond effectively to emergencies and emerging threats, such as (i) natural hazards: floods, droughts, fires, earthquakes, and volcanic eruptions; (ii) social issues: security crises and other forms of violence.

The impacts of crises on women, both rural and urban, are multidimensional, differentiated, and, in some cases, disproportionate, depending on local contexts as well as the networks and structures of communities, the State, and other public and non-governmental entities. Among the differentiated risks are:

- Depending on the location (rural or urban) of women, the loss of livelihoods varies significantly. In rural areas, a larger percentage of livelihoods depends on agricultural and livestock activities, while in urban areas, formal and informal employment is more predominant. Depending on the source of work, different needs can be identified for the recovery of these livelihoods after an emergency, as well as the implementation of preventive actions to minimize the impacts in the event of a disaster or emergency.
- Another differentiated impact relates to the state network for guaranteeing access to rights (education, health, among others) in different territories. The existence and strength of these networks during emergencies influence the degree of weakening of public systems. While everyone may experience limitations on their rights during a crisis, the weaker the state structure, the greater

the difficulty in restoring it and ensuring those rights. In rural areas, the recovery of these structures that guarantee rights can be more complex.

Additionally, women are particularly vulnerable to climate change, emergencies, and disasters, as social, economic, and political inequalities place them in a position of reduced resilience to adverse effects. A clear example of this is rural women, who, due to limitations in access to and ownership of land, as well as deeply rooted cultural values, face difficulties in accessing credit, financing, tools, or machinery that would enable them to develop enterprises or implement necessary improvements to address challenges.

Among the challenges faced by women are:

- Access to prior information (preparation and warning systems) and during the emergency, regarding actions they can take to safeguard their lives.
- Caring for others during the emergency, as women mostly take on the role of caregivers. This role, in addition to influencing the specific situation during the emergency, impacts both the loss and recovery of livelihoods due to the care of these individuals.
- During emergencies, there has been an increase in pre-existing forms of violence, such as domestic violence, gender-based violence, human trafficking, and other forms of violence that, according to available information, primarily affect women (girls, adolescents, and adults).

Climate change and resource scarcity also have a differentiated impact on women and girls. For example, the decrease in water sources forces them to travel greater distances to access this resource, increasing the daily hours dedicated to unpaid work, which, according to INEC (ENEMDU, 2023), is three times greater for women than men, both in urban and rural areas. During emergencies, such as the pandemic, women's unpaid responsibilities increased significantly, as they took on the care of the home, the elderly, and the sick, in addition to becoming teachers to support their children's education, which created an overload of responsibilities.

The impacts of natural disasters and emergencies are unequal from a gender perspective. Before the COVID-19 health crisis, women were already overrepresented in poverty situations, a situation that worsened after the pandemic. In 2023, for every 100 men in poor households, there were 111.7 women, reflecting their lack of economic autonomy. The health crisis and its economic consequences have deepened poverty and inequality, severely affecting women due to the decline in economic activity in key sectors for female employment, such as tourism, manufacturing, agriculture, commerce, and paid domestic work. This increased vulnerability during emergencies may further limit their access to land ownership, housing, education, health, and participation in political and decision-making spheres, reducing their adaptive capacities (INEC, 2023).

In rural areas, barriers to accessing productive assets (land, technical assistance, machinery) and financial instruments (credit and insurance), coupled with their limited mobility and participation in decision-making, restrict women's ability to develop their productive skills and place them at a disadvantage in producing, marketing, and expanding their businesses, which is reflected in a higher concentration of women in sectors such as agriculture, livestock, forestry, and fishing.

During and after disasters, not all evacuation sites identified by the relevant authorities incorporate a gender or disability perspective, which largely depends on local capacities and infrastructure. The risk of gender-based violence, including domestic violence, can increase, especially in temporary shelters where a lack of privacy and security is common. In rural areas, gender-based violence may be normalized and less frequently reported, making it difficult to provide adequate response and support to victims in emergency situations. Additionally, displaced women may lose their livelihoods and face discrimination when attempting to access reconstruction and economic recovery programs, particularly in rural areas where the loss of land, homes, and crops has a devastating impact, forcing them to migrate in search of resources for survival.

In summary, natural disasters tend to exacerbate pre-existing gender inequalities, and women often face greater risks and challenges compared to men. This unequal impact highlights the need for a gender-sensitive risk management approach that ensures equitable participation of women in all aspects of disaster planning and response.

Some of the differentiating factors between men and women regarding exposure to hazards and response capacity are as follows:

- **Access to preventive information:** Fewer women have access to this information compared to men. Even when they do have general access to information, due to their participation in work-related activities focused on early warnings, they are not always part of their community's alert systems.
- **Access to response information:** Women have less access to this information compared to men.
- **Legal vulnerability:** Women, especially in rural areas, face greater legal vulnerability. The levels of legal ownership of assets such as land, houses, vehicles, and others are usually lower compared to urban areas, and existing legalizations do not always include women.

However, women also have great potential as agents of change in natural resource management, although these strengths are often underestimated by society. The creation of networks and distribution services in rural areas, the development of enterprises, and the adoption of innovative decisions to address current problems arising from disasters and emergencies highlight this potential.

In the country, the main hazards affecting the population are landslides, earthquakes, floods, and droughts. Response capacity in rural areas is particularly affected by a lack of infrastructure, access to production factors, basic services, connectivity, and cultural barriers. In urban areas, the development of response capacities is weak and not adequately socialized at the community level, especially affecting vulnerable groups such as girls, boys, and women.

There are global databases, such as DESINVENTAR, that record multi-hazard disasters with a gender perspective. This information can be used to design multi-hazard early warning systems (MHEWS) and establish specific alert thresholds for communication targeted at women. In the event of disasters, organizations such as the Red Cross, NGOs, state entities, international cooperation, the Church, and volunteer groups, among others, provide support to women. These institutions participate as beneficiaries in pillar 4 of the project, which focuses on preparedness and response capacity. It is anticipated that, starting in 2025, the Ecuadorian Red Cross will establish mechanisms to consolidate and analyze information, ensuring that actions align with the Movement's approach to protection, gender, and inclusion.

Public policy, through government entities at the central and local levels, supports the management of Community Risk Management Committees, providing financing, technical assistance, training, monitoring, follow-up, distribution of goods and services, information systems, response kits, shelters, and humanitarian centers, among others. However, in many cases, specific gender policies for risk management are not considered.

New proposals for regulations and public policies that include a gender perspective represent an opportunity to implement programs and projects aimed at MHEWS that consider differentiated responses according to the needs and realities of territories and vulnerable groups. Currently, there is no significant difference in the level of awareness of early warning information between men, women, boys, and girls, as alerts are delivered through official communication channels and social networks without any specific directionality.

Recommendations

In order to enhance inclusiveness and participation it is crucial to consult women about their needs and preferences through:

Guided interviews with key women in the target communities, which will allow for the collection of detailed testimonies and experiences. Interviewers should preferably be women.

Focus group discussions with women of various ages and backgrounds to gain a deeper understanding of their perspectives and needs.

Develop and administer specific surveys for women that include questions about the project's requirements.

WhatsApp groups for women, where a very specific exchange of information can take place.

Continuous online consultation platforms, where women can provide ongoing feedback on the project.

Proposed activities for the project

1. It is necessary to start with an assessment of needs and preferred communication channels.
2. Both the needs assessment process and the communication of alerts must be in clear and accessible language, including local languages and adapted to local culture. In short, communication strategies must be inclusive, accessible, and culturally relevant.
3. For the planning of activities related to the design and dissemination of alert messages, it is important to involve local governments, leaders, civil society organizations and women's organizations. Support networks could be built, or forms of local cooperation could be fostered. In the indigenous areas, the organizational schemes, which are more developed than in the rest of the population, can be leveraged and also, ancestral knowledge for early warning systems.
4. Capacities should be developed (cascade methodology) and groups of women who have active roles within their communities, such as women leaders, women entrepreneurship and heads of households, or those actors who want to participate actively in the development of project-related activities should be empowered. They should be categorized as information disseminators and response leaders. This should include raising awareness among the families of the women participating in the project, considering the topic of positive masculinities with young people and husbands, and shared responsibility in household activities.
5. Any measure to disseminate information must include facilitating access to the technology that will be used to receive alerts and real-time updates.
6. According to the planning of activities, it is necessary to work on risk management education programs in schools, colleges, and communities, specifically aimed at women and girls.
7. Additionally, training and awareness workshops can be organized on interpreting early warnings and response measures, adapted to the needs and available time of women for their participation.
8. A monitoring and evaluation methodology must be implemented that includes gender indicators to measure the impact of the implemented measures and make adjustments during the development of the project

4 Gender Assessment for Cambodia

Cambodia is ranked 'medium' equality on the UNDP Gender Development Index 2023-24³⁶, comparing Human Development Index achievements for women and men³⁷. Women consist of 50.5% of the total population in Cambodia³⁸ and 31.9% of Cambodian households are female-headed³⁹. Cambodia has a large rural population with 74% of the population living in rural areas⁴⁰ and the majority of them work in the agriculture sector. About 24.4% of the population or 3.4 million people are living with disabilities in Cambodia, including 20.3% with mild or moderate degrees of disability and 4.1% with a severe disability⁴¹. The percentage of women living with disabilities (57%) is higher than that of men (43 %) ⁴². Cambodia's gender equality-related indexes are ranked the lowest among the neighboring countries in the Association of South-East Asian Nations (ASEAN) region. Cambodia is ranked 122th out of 193 countries in the Gender Inequality Index (GII)⁴³ and 102nd of 146 countries in the Global Gender Gap Report 2023.

Cambodia's rapid economic growth from 2009 to 2019, combined with structural change, helped to increase labor earnings and reduce poverty. Living standards have improved, helping Cambodia to narrow urban-rural gaps. However, low-income and rural households still lag in access to basic services and earning opportunities.⁴⁴

Education, Vocational Training and Literacy

Cambodia has made significant progress in achieving gender parity in education. The net enrolment in primary education is around 96% for female students and 95.5% for male students for the academic year 2022-2023⁴⁵. The enrolment rate at lower secondary schools is about 86% for female students and 81.4% for male students while the enrolment rate for both sexes is lower for upper secondary schools, 46% for women and 36.5% for men⁴⁶. However, the proportion of female students in tertiary education and technical vocational training is lower than that of male students, particularly for non-traditional areas of study, such as science, technology, engineering, and math. The Ministry of Education (MoEYS) estimates that only 17% of female students, compared to 40% of male students, studied a Bachelor's Degree in STEAM-related subjects⁴⁷. Literacy has improved steadily for both women and men but remains higher for men at 89.5% compared with 80% for women in 2021⁴⁸. There is also a literacy gap for both genders between rural and the capital of Phnom Penh, of 80.7% and 92.4 % respectively.

Gender gaps in access to education are particularly evident among ethnic minorities and people with disabilities. Indigenous women in Cambodia have a higher illiteracy rate and fewer educational opportunities than their male counterparts⁴⁹. The drop-out rate for both Indigenous female and male adolescents aged 14-18 is of significant concern. Girls are more affected than boys due to a higher demand for girls to do extra household chores during the COVID-19 pandemic, resulting in a widening educational disparity for Indigenous groups, people with disabilities and people living in poverty⁵⁰. This gender gap in education has hampered the ability of rural women to access informed decision-making, hold a job outside of the home, understand laws and regulations that relate to them, and access technologies, extension services, and finance as well as climate change and disaster risks, including early warning information⁵¹.

³⁶ Ranked Group 3 out of 5 categories.

³⁷ UNDP (2024). Human Development Report 2023-24. Group 3: Countries with absolute deviation from gender parity of 5-7.5 percent are considered countries with medium equality in HDI achievements between women and men

³⁸ [Population, female \(% of total population\) - Cambodia | Data \(worldbank.org\)](#)

³⁹ [Female headed households \(% of households with a female head\) - Cambodia | Data \(worldbank.org\)](#)

⁴⁰ [Rural population \(% of total population\) - Cambodia | Data \(worldbank.org\)](#)

⁴¹ MoP/NIS (2022). Cambodia Demographic and Health Survey (CDHS), 2021-2022.

⁴² [NIS, MoSAVY and DAC \(2023\). Persons with disabilities in Cambodia: findings from the Cambodia Demographic and Health Survey, 2014, 2021-2022.](#)

⁴³ UNDP (2023) Human Development Report 2023.

⁴⁴ World Bank Group (2022) Cambodia Poverty Assessment 2022. Toward a more inclusive and resilient Cambodia

⁴⁵ Ministry of Education Youth and Sports (2022). Education Statistics 2022-2023.

⁴⁶ Ministry of Education Youth and Sports (2022). Education Statistics 2021-2022.

⁴⁷ Ibid

⁴⁸ MoP & NIS (2022). National socio-economic survey report..

⁴⁹ IWGIA indigenous global report 2022

⁵⁰ Cambodia COVID-19 Joint Education Needs Assessment Working Group, 2021 cited in United Nations (2022).

⁵¹ USAID (2019). Green Future Activity: Gender analysis, strategy and action plan.

Employment and Labor Participation

In 2023, the women's labor participation rate in Cambodia was 79% compared to men's labor participation rate of 89%⁵². The Cambodia Social and Economic Survey (CSES) showed that 38.3% of women aged 15 to 64 years were employed in the agriculture, forestry, and fishing sectors, 24.2% in the industry sector, and 37.5% services sector (including hospitality and tourism, finance, healthcare and education)⁵³. However, only 40.6% of women are wage and salaried workers, while 59.3% are in 'vulnerable employment', usually working in the informal sector, with insecure working conditions, low pay, and no access to social protection⁵⁴.

Overall, gender gaps remain large between women and men in Cambodia, across industries and occupations. A United Nations study⁵⁵ found that Cambodian women earn about 80% of men's wages while owning fewer assets and that they face more barriers that prevent their equitable access to better-paying work opportunities. The average gender wage gap in Cambodia is around 19%. This is tied to gender disparities in education (a higher share of no primary education, and lower tertiary enrolment), as well as labor market discrimination, gender-segregated occupations, and social norms associated with women's primary domestic role.⁵⁶ Unequal caregiving and domestic loads are major structural barriers for women to take and stay in formal paid employment, undertake skills training, and pursue opportunities for career development.

Representation and Participation in Decision Making

Although there is a robust gender equality perspective reflected in national gender equality policy initiatives and legal frameworks, challenges remain for equal representation and participation of women and men in decision-making positions. Women are still under-represented at all levels of legislative and executive bodies. Women account for about 14% of total parliament members in the current mandate⁵⁷, a decrease from 21% in 2018. More than 10% of senior ministerial positions and around 16% of under-secretary and secretary of state positions are held by women in the executive body. As of 2022, women hold around 27% of leadership positions in civil service, which include roles such as deputy chiefs of offices to director generals. Overall, women make up 42% of the total civil service workforce⁵⁸.

At the sub-national level, the proportion of women remains low in the positions of governor at the capital city/province (8%) and municipality/khan/district (3.43%). The proportion is also low for the positions of chief of Commune/Sangkat (10.65%) and deputy governors at the capital city/provincial (13%), with slightly higher numbers for the position of deputy governor at the municipality/khan levels (20.3%) and members of councilors of Commune/Sangkat with 22%⁵⁹.

Gender-Based Violence and Social Norms

Gender-based violence is still widespread and remains a concern in Cambodia. An estimated 21% of women aged 15-49 years in Cambodia have experienced physical, sexual or emotional violence from intimate partners in their lifetime, and 43% of them report experiencing severe violence with sustained injuries⁶⁰. Emotional abuse by a partner is less visible but no less traumatizing⁶¹ and affects almost one-fifth of Cambodian women (19%). Rural women face higher rates of intimate partner violence than women in urban areas, as do women in their late twenties to forties.⁶² The UN in Cambodia notes that *"harmful gender norms, such as the idea that men are entitled to sex regardless of consent, directly contribute to gender-based violence"*⁶³. Women and girls from Indigenous groups who are survivors of gender-based violence face more

⁵² MoP & NIS (2022). National socio-economic survey report for 2021.

⁵³ Ibid

⁵⁴ Ibid

⁵⁵ United Nations Cambodia. (2022). Gender equality deep dive for Cambodia – common country analysis.

⁵⁶ Ibid.

⁵⁷ Neary Ratanak VI – Promotion of Gender Equality and Empowerment of Women and Children (2024-2028), Ministry of Women's Affairs (MoWA).

⁵⁸ Ibid

⁵⁹ MoWA (2023). Key gender statistics in Cambodia in December 2023.

⁶⁰ MoP/NIS (2023). Cambodia demographic and health survey (2021-22).

⁶¹ Fulu, E., Warner, X., & Moussavi, S. (2013). Men, gender and violence against women in Cambodia: Findings from a household study with men on perpetration of violence. UN Women Cambodia

⁶² MoP/NIS (2022). Cambodia demographic and health survey (2021-2022).

⁶³ United Nations Cambodia (2022).

challenges in seeking legal advice, safe shelter and mental support than non-Indigenous women⁶⁴. The Cambodian cultural practices and social norms continue to contribute to the unequal division of labor in the household and the community as a whole. Besides being involved in productive activities such as farming and generating income, women and girls are also primarily responsible for household and childcare duties. Much of their time is devoted to household chores, childcare, cooking, and cleaning, and Indigenous women in particular, carry a heavy burden in terms of household chores, including housework, farming, raising animals, and collecting forest foods, especially in times of food shortage and disasters. Women are still expected to follow social norms and beliefs that endorse men's privilege while women are undervalued for their capacity and potential. This impacts the roles of women and men in decision-making in the household and the society as a whole. A Ministry of Women's Affairs (MoWA) report⁶⁵ shows that women are involved in managing the household budget, but men are the primary decision-makers for major expenditures. Moreover, a gender analysis report by the ADB⁶⁶ also points out that social norms and low level of education and literacy, are the major barriers preventing women and girls from accessing information, technical skills, and training, having their voices heard, and enjoying preference in their families and society.

Gender Policies and Legal Framework

The Royal Government of Cambodia is committed to promoting gender equality and social inclusion in disaster risk reduction and management, aligning with the 2030 Agenda for Sustainable Development's goal of "leaving no one behind." Key national legislation and policies support this commitment:

- The Constitution of 1993 guarantees equal rights for women and men, including human rights, equality before the law, and active participation in all aspects of national life.
- The Law on the Prevention of Domestic Violence and Protection of Victims addresses domestic violence and promotes gender equality.
- The 2015 Law on Disaster Management focuses on the needs of women, children, the elderly, and people with disabilities during disasters.
- The Government's Pentagonal Strategy Phase I promotes gender equality and women's leadership in key sectors, aiming to reduce vulnerability to gender-based violence and improve social protection for the most vulnerable.
- Neary Ratanak VI (2024-2028), a five-year plan by the Ministry of Women's Affairs, focuses on gender equality and women's empowerment across various sectors, emphasizing care economy, well-being, legal protection, climate change, and governance.
- Cambodia's Updated Nationally Determined Contribution includes gender-focused measures for climate change adaptation, incorporating gender impact assessments and evaluations.
- The National Action Plan for Disaster Risk Reduction (NAP-DRR) 2024-2028 emphasizes gender and vulnerable group-responsive approaches in disaster risk reduction and climate resilience.
- The Climate Change Strategic Plan for Gender and Climate Change (2013-2023) focuses on addressing the needs of vulnerable women and other groups through policy-making, leadership development, and green growth initiatives.
- The Guideline for Gender Mainstreaming in Disaster Management provides practical guidance for integrating gender perspectives in disaster management policies.

Cambodia is also committed to international and regional frameworks promoting gender equality and social inclusion in disaster risk management, including Convention on All forms of Discrimination against Women (CEDAW), United Nations Declaration on the Rights of indigenous People (UNDRIP), United Nations Convention on the Rights of Persons with Disabilities (UNCPRD), the Sendai Framework, ASEAN Agreement on Disaster Management and Emergency Response (AADMER) Work Programme 2021-2025, and the Beijing Agenda for Global Action on Gender-Sensitive Disaster Risk Reduction.

⁶⁴ Ministry of Women's Affairs (2019). National action plan to prevent violence against women 2019-2023 (NAPVAWIII).

⁶⁵ Ministry of Women's Affairs (2014). Cambodia gender assessment.

⁶⁶ Asian Development Bank (2015), Promoting Women's Economic Empowerment in Cambodia.

<https://www.adb.org/sites/default/files/publication/156499/promoting-womens-economic-empowerment.pdf>

Despite these efforts, challenges remain, such as limited knowledge and skills within ministries for gender analysis, and insufficient resources and funding for gender-focused interventions

Gender-Related Institutional Arrangements

Cambodia has established mechanisms at both national and sub-national levels to advance gender equality and women's empowerment, coordinated by the Ministry of Women's Affairs (MoWA) in collaboration with sectoral ministries, local governments, and development partners.

MoWA plays a crucial role in promoting gender equality by coordinating efforts across public institutions, civil society, and the private sector through various programs and the Technical Working Group on Gender (TWG-G). MoWA also supports the Cambodian National Council for Women (CNCW) in implementing CEDAW and ensuring legal protection for women. MoWA has line departments that work at the provincial level.

Sectoral ministries have Gender Mainstreaming Action Groups (GMAGs) responsible for developing, implementing, and monitoring gender mainstreaming action plans (GMAPs). The National Committee on Disaster Management (NCDM) is forming a gender working group within its secretariat and the Ministry of Women's Affairs is one of the key members of NCDM.

At the sub-national level, Women and Children Consultative Committees (WCCCs) operate at provincial and district levels, and Commune Committees for Women and Children (CCWCs) at the commune level. These committees advise on gender equality and women's and children's issues.

The sub-committees for disaster management were also established at the sub-national level to handle various disaster risks, with a primary focus on emergency responses. The Provincial Department of Women's Affairs and a member of WCCC and CCWC are members of the sub-committees respectively at the province, district, and commune levels, and mostly in the health and hygiene team. Besides this, no specific provision ensures equal representation and participation of diverse groups, including women in the planning, response, and recovery phases.

Women's Representation and Participation in DRM Sectors

Women's representation in government, especially in water resource management, meteorology, hydrology, and disaster management, remains low. According to the latest data from 2023⁶⁷, the Ministry of Water Resources and Meteorology (MoWRAM) has a total of 1,401 officials, with 316 being women (22.5%). Nonetheless, only 16% of these women hold leadership and decision-making positions, including two women as secretaries of state (16%), one woman as under-secretary of state (6%), and two women as deputy general directors (15%). There are no women in the roles of department director or provincial department director. Moreover, a few women are working in technical areas of meteorology and hydrology at both the ministry and provincial levels. For example, in Kratie province, no women are working as technical staff in meteorology and hydrology, and there is only one woman in this field in Kampong Thom province⁶⁸.

There is no comprehensive national data on women in Disaster Management Committees, but in provinces like Kratie, Stung Treng, Ratanakiri, and Banteay Meanchey, women represent less than 13% of officials in existing provincial disaster management committees, which is low compared to civil servants at the provincial level and in other sectors. The majority of them are engaged in the finance and administration work. The mindset and belief that water and disaster management work is more appropriate for men than women persist in Cambodia as men are perceived to be good with science-related technical skills. This contributes to gender gaps in the education sector in the field related to STEM, and gender disparity in government institutions and other sectors.

⁶⁷ Ministry of Water Resource Management and Meteorology (MoWRAM) (2023). Annual congress on 2023 achievements and direction for 2024.

⁶⁸ PDoWRAM in Kratie – there are 5 women among a total of 35 staff but no woman working in meteorology and hydrology; in Kampong Thom – only one woman in meteorology and hydrology offices among 4 women out of 22 staff.

Table 1: Female officials in Disaster Management in selected provinces⁶⁹

Provinces/Committee's Roles	Total	Women	Percentage
Kratie			
Provincial Committee for Disaster Management	48	5	10%
Explore, rescue, and security team	13	0	0%
Response and Recovery Team	33	5	15%
Secretariat	18	4	22%
Total	112	14	12.5%
Stung Treng			
Provincial Committee for Disaster Management	38	4	10%
Explore, rescue, and security team	6	0	0%
Response and recovery team	24	2	8%
Secretariat	12	4	33%
Total	80	10	12.5%
Ratanakiri			
Provincial Committee for Disaster Management	60	3	5%
Explore, Rescue, and Security Team	12	1	8%
Response and recovery team	31	4	13%
Secretariat	18	2	11%
Total	121	10	8 %
Banteay Meanchey			
Provincial Committee for Disaster Management	55	4	7%
Explore, Rescue, and Security Team	23	1	4%
Response and recovery team	29	3	10%
Secretariat	22	4	18%
Total	129	12	9 %

(source: List of members of the Provincial Committee for Disaster Management in the Preparedness and Contingency Plan in Ratanakiri (2024), Kratie (2024), Stung Treng (2023), and Banteay Meanchey (2022).

Gender in Early Warnings and Disaster Risk Response

Efforts to improve disaster information and risk knowledge, such as the CamDi database and PRISM⁷⁰ dashboard, lack of gender-disaggregated data, which hampers understanding the specific vulnerabilities of different groups. Better data collection, analysis, and sharing are crucial for targeted interventions.

NCDM has worked to strengthen disaster information to increase risk knowledge and understanding with the support of development partners, national and international organizations, and the private sector. CamDi serves as a platform database for providing historical damage and loss information. NCDM has produced flood risk maps as well as ID Poor household maps for the 25 municipalities and provinces that were used to integrate into the national and sub-national flood contingency plans and publicly used to raise awareness, particularly in communities prone to hazards and disaster risks. However, none of these initiatives include disaggregation of data such as sex, age, disabilities, and other diversities. The lack of this data disaggregation is one of the biggest barriers to comprehensively understanding of vulnerabilities and impacts of hazards and disaster risks on different groups of people.

In partnership with WFP, NCDM also works to enhance the interactive web-based dashboard called Platform for Real-time Impact Situation Monitoring (PRISM). PRISM assesses the potential risk and impact of climate

⁶⁹ List of members of The Provincial Committee for Disaster Management in the Preparedness and Contingency Plan in Ratanakiri (2024), Kratie (2024), Stung Treng (2023), and Banteay Meanchey (2022).

⁷⁰ CamDi database is a platform database for providing historical damage and loss information. "Platform for Real-time Impact Situation Monitoring (PRISM), "assesses the potential risk and impact of climate hazards on the most vulnerable communities" by combining "information from satellites and other remote sensing sources, field assessment information, early warning systems with data on socioeconomic vulnerability to create actionable climate information, allowing prioritize assistance to those most in need".

hazards on the most vulnerable communities by combining information from satellites and other remote sensing sources, field assessment information, and early warning systems with data on socioeconomic vulnerability to create actionable climate information, allowing priority assistance to those most in need.

Gender and Access to Early Warning Information and Technologies

Cambodia's Early Warning System 1294 (EWS) has been implemented by NCDM nationwide in partnership with PIN for flood and Action Aids for drought to provide alerts and information about disaster risks for better preparedness and mitigation. The EWS 1294 requires users to have a registered mobile phone to receive warnings about natural hazards. Early warning systems (EWS) like EWS 1294 are effective but not inclusive, particularly for women, the elderly, and those without mobile phones. A recent study of PIN⁷¹ shows that most community people, including people living with disabilities, the elderly, women, and Indigenous people were not aware of the existence of the system and did not subscribe due to a lack of knowledge and/or understanding of how to utilize it. Awareness and usage of EWS are lower among women, who often receive information indirectly from local authorities.

Generally, awareness of EWS systems and related information remains limited for both women and men. However, men overall tend to have more information than women, as they regularly follow local news through TV and radio and have better access to smartphones and digital communication such as social media and mobile apps. A national survey (2021-2022) shows that women are less likely than men to own a mobile phone (85% versus 91%)⁷². The access to mass media in Cambodia, particularly in rural areas for women is also lower than that of men, with 22% of women and 30% of men watching TV and fewer women than men listening to the radio at least once a week⁷³. In Kratie, 9 out of 10 people who come to check the water level measurements are men. During the focus group discussion meeting with community men, most of them⁷⁴ also reported that they accessed information on the river water level and weather forecasting through mobile apps (weather apps) and social media, particularly Facebook. Some of them learned about weather apps and searched the Facebook page of MoWRAM from their children. Based on the national survey, none of the interviewed community women either used weather apps or Facebook (only very few of them use smartphones) and rarely listened to radio. All of them said they received flooding alert information from local authorities, particularly group/village leaders through home visits or loudspeakers in some communes, while people living with disabilities mostly received EWS information from their family members. Therefore, multi-channels of communication and dissemination are critically important to ensure everyone and all groups of people have access to EWS information.

Preparedness, Early Actions, and Responses

Contingency and emergency plans include no distinction between the needs of women and more vulnerable groups such as persons with disabilities residing in the affected areas. Moreover, despite community engagement being very important for the contingency and emergency planning process, women and the most vulnerable, including persons with disabilities, are not aware of the process and less likely to be engaged in the planning. Most of the women and men groups in the community meetings in Kratie and Stung Treng provinces reported that they have not been invited or engaged in the consultation about the preparation of their commune contingency and emergency plans or commune development plans. They are not aware of the needs and interventions included in the planning.

During the focus group discussion with groups of persons with disabilities, representatives of organizations for people with disabilities (OPD) in Kratie and Kampong Thom provinces pointed out that despite local authorities and the Committee on Disaster Management team being supportive during the emergency response, the interventions have not been appropriately responding to the specific needs of persons with disabilities, considering the different types of disabilities, including individuals with functional limitations and

⁷¹ People In Need (PIN) 2022. Impact assessment on EWS1294 read on vulnerable groups – people living with disabilities, elderly, women, and minority groups.

⁷² MoP/NIS (2022). Cambodia Demographic and Health Survey (CDHS) 2021-2022.

⁷³ Ibid

⁷⁴ Findings from focus group discussion during the project design mission from April to May 2024.

their limited access to the other services. The stakeholders in the consultation workshop also emphasized the need to enhance the capacity and skills of all involved in disaster management on gender equality and disability inclusion to ensure the interventions and responses appropriately address the needs and vulnerabilities of the affected groups of people.

Access to Safe Evacuation Places

Although efforts have been made to improve safe evacuation sites, concerns remain about the adequacy of facilities and ensuring safety. For instance, in Kratie, only 5 out of 150 listed safe evacuation sites meet the national minimum requirements adopted from WFP, which include at least separate toilets for men and women, clean water storage, and sanitation facilities. Similarly, in Stung Treng, only 2 out of 120 sites meet these basic needs. Public buildings such as schools, pagodas, high roads, and health centers are used as evacuation sites. However, people generally move to these sites only in situations when it is impossible to remain in their homes, as staying at home or on higher ground is often more convenient. Concerns include overcrowding, shared sleeping and cooking spaces, and lack of secure storage for valuables. Additionally, community members, particularly women in Kratie province who participated in focus group discussion meetings and have used these evacuation sites reported that there are inadequate facilities, including shelters, clean water, and toilets. Many had to use flood water for washing and bathing, sometimes filtering it for cooking, or collecting clean water from other places for cooking and drinking. The lack of proper facilities makes it especially challenging for the elderly and persons with disabilities to stay in these safe evacuation places.

Besides the physical facilities, a concern about being safe from other abuses was also pointed out by women who participated in the focus group discussion meetings. Women have reported incidents of physical and emotional violence against women and children in their communities which can worsen during their stay in the evacuation places. The prevalence rate of violence against women is higher in the provinces of Banteay Meanchey (35%), Stung Teng (32%), Ratanakiri (26%), and Kratie (16%)⁷⁵. Ensuring inclusive, well-equipped evacuation sites and addressing gender-specific needs and safety concerns are critical for effective disaster response.

Limited Access to Resources and Services

Vulnerable groups in Cambodia, including the poor, rural women, the elderly, persons with disabilities, and Indigenous communities, have limited capacity to prepare for disasters due to restricted access to resources and services. According to focus group discussion meetings for this project with community members in disaster-prone areas, people holding ID poor cards, and from at-risk households, reported not having extra savings to buy extra food, boats, or household supplies to prepare for emergencies, as they barely earn enough to survive day by day. Both women and men in drought-prone communities noted that during the dry season, especially with increasing high temperatures and heat in early 2024, their crops, agricultural and livelihood activities were severely affected. This impacted their income with inadequate to support their families and no savings for emergencies. They also experienced a lack of water for household use (cooking, washing clothes, and bathing) and safe drinking water, as there was insufficient water from canals, water pumps, and rainwater storage. Some families had to buy water for drinking and household use, exacerbating their vulnerability and exacerbating poverty in these already poor and near-poor families. Thus, limited access to resources for the poor, women, elderly, and persons with disabilities limited these groups' capacities to better prepare and take early anticipatory actions to mitigate the disaster risks and impacts on themselves and their households.

The Cambodian government has expanded its social protection policy to include 'shock-responsive social protection' for at-risk households, aiming to build resilience and mitigate disaster impacts. For better preparation of a more inclusive shock-responsive social protection policy and program, there needs to be a

⁷⁵ MoP/NIS (2022). Cambodia Demographic and Health Survey 2021-2022.

comprehensive understanding of different needs, vulnerabilities, and effects of a crisis on women and men, and most vulnerable groups, particularly in the context of climate disaster risk reduction.

Community-Led Early Warning and Disaster Risk Reduction Initiatives

There are two key initiatives to empower women, and people with disabilities, to engage and lead community disaster response, described below.

Women's Champion Networks: Founded by ActionAid in four provinces, these networks promote women's leadership in disaster management and support communities in understanding and preparing for hazard risks.

Disability Inclusive Climate Action Project: Implemented by Save the Children and partners, this project enhances the inclusion of persons with disabilities in climate actions and disaster risk reduction, ensuring they have access to early warning information and are more resilient.

Areas for Improvement in Early Warning and Gender Inclusiveness

The following are areas where improvement in gender inclusiveness should be considered, which have been taken into account when preparing the Gender Action Plan (GAP).

Enhancing GEDSI Analysis:

- Integrate Gender Equality, Disability, and Social Inclusion (GEDSI) analysis into disaster risk assessments to capture the needs of diverse groups.
- Improve platforms like PRISM and CamDi to include disaggregated data and make them publicly available.

Improving Early Warning Dissemination:

- A comprehensive review and action plan for Emergency Warning (EW) systems should provide analysis and strategies to include inclusivity and to address the digital gender gaps.
- Use multiple communication channels, including cell broadcasting for wider reach, community-based intervention, including outreach activities, targeting women or persons with disabilities who have no access to mobile phones.
- Provide clear, actionable information tailored to different groups' needs.
- Training and awareness campaigns should also specifically target women or ensure gender balance and youth participation.

Preparedness and Early Action:

- Ensure evacuation sites meet minimum standards and are safe for all, particularly women and girls, and are designed inclusively.
- Train and equip community members, including women and persons with disabilities, for early warning actions.
- Building evidence for shock-responsive social protection schemes for preparedness and anticipatory action in the target provinces, especially to include the poor, women, and persons with disabilities.

Cross-Pillar Coordination:

- Create GEDSI mainstreaming guidelines and training for emergency warning systems.
- Increase the representation and participation of vulnerable groups in preparedness and response activities.

Incorporate GEDSI aspects into standard operating procedures and monitoring frameworks, collecting and analyzing disaggregated data.

5 Gender Assessment for Fiji

In 2023 the estimated population of Fiji was 936,375⁷⁶, of which 469,490 are male (50.14 %) and 466,885 are female (49.86%). As of 2023, approximately 41% of Fiji's population lives in rural areas.⁷⁷

Fiji has made strides toward gender equality, yet it faces persistent obstacles rooted in traditional social norms and patriarchal values, especially impacting women's roles in rural areas. While government and non-governmental organizations have promoted gender equality in education and employment, challenges remain, including limited leadership representation, pay inequality, and gender-based violence. These structural issues have direct implications for women's health, education, and economic engagement⁷⁸.

For instance, maternal health has improved, with the maternal mortality ratio decreasing from 49 per 100,000 live births in 2000 to 38 in 2020, a rate that falls below the regional average. Adolescent birth rates have also declined, though 26 births per 1,000 females aged 15-19 were still recorded in 2022. Educational access is comparatively high, with 99% of girls and 92% of boys completing lower secondary school, highlighting progress in youth education.

Despite these advances, economic participation gaps remain substantial. In 2023, labour force participation was 39.1% for women compared to 77.3% for men, showing that female labour engagement has stagnated despite government efforts. Additionally, women remain more likely to work in vulnerable employment, though this rate is lower than the regional average. Unpaid domestic work still disproportionately affects women, who spend nearly three times as much time as men on household tasks. Gender-based violence (GBV) also remains a serious issue, with 52% of Fijian women ages 15-49 reporting intimate partner violence, far above the global average (27%). In terms of political representation, women held only 10.9% of parliamentary seats in 2023, reflecting slow progress in leadership inclusion. These indicators underscore both Fiji's achievements and ongoing barriers to achieving true gender equality.⁷⁹ Progress will require sustained initiatives that address cultural, economic, and social factors comprehensively.

Fiji is ranked 104th out of 193 countries in the UNDP Human Development Index (HDI)⁸⁰. Life expectancy at birth for female is 70.2 and for males is 66.5. Expected years of schooling are 14.4 for females and 13.3 for males in 2022. Mean years of schooling are 10.4 for females and 19.3 for males in 2022. Estimated Gross National Income per capita is 6,282 US Dollars for females and 16,158 US Dollars for males (2017).⁸¹ Fiji's Gender Development Index (GDI) value is 0.940, placing the country in group 3 countries⁸².

Literacy, Education and Vocational Training

The literacy rate in Fiji has progressively increased in recent years with an adult literacy rate of 99.08% -- 99.09% for women aged 15 years and older and 99.08% for men 15 years and older. Fiji has achieved 100% net enrolment in primary education with 100% completion rate, and 88% net enrolment in secondary school. This ratio reflects broader patterns in Oceania, where secondary retention and completion rates are often higher for girls than boys due to factors such as dropout rates influenced by economic and social pressures on young males to enter the workforce earlier.

In Fiji, tertiary education enrolment demonstrates a significant gender difference, with a higher proportion of women attending compared to men. Data from 2019 shows a female-to-male enrolment ratio of approximately 1.31, meaning that for every male student, about 1.31 female students were enrolled at this level. This aligns with broader trends across the Pacific region where female enrolment tends to exceed male

⁷⁶ [Population, total - Fiji | Data](#)

⁷⁷ [Rural population \(% of total population\) - Fiji | Data](#)

⁷⁸ [Country Fact Sheet | UN Women Data Hub](#)

⁷⁹ Indicators source: [Fiji | World Bank Gender Data Portal](#)

⁸⁰ UNDP (2024). Human Development Report 2023-24, pg.289

⁸¹ UNDP (2024). Human Development Report 2023-24, pg.289

⁸² Countries are divided into five groups by absolute deviation from gender parity in HDI values.

UNDP (2024). Human Development Report 2023-24, pg.289

enrolment in higher education. Specifically, gross enrolment rates show 59.9% for women compared to 41.4% for men, suggesting increased female participation and retention in tertiary education⁸³.

Vocational training programs in Fiji also cater to young people, especially those from rural or disadvantaged backgrounds, aiming to provide practical skills for employment. The government's ongoing investment in rural education infrastructure further seeks to equalize educational opportunities across urban and rural areas. Vocational training for Fijian girls is expanding through programs like the Makoi Women's Vocational Training Centre, which offers courses in fields such as aged care and culinary arts, helping over 1,600 women gain marketable skills⁸⁴. Additionally, initiatives like the Girls in STEM camp by Fiji National University and Graduate Women Fiji encourage young women to enter STEM fields through hands-on STEM activities and career talks⁸⁵. These efforts aim to boost women's economic opportunities and address gender gaps in vocational and technical fields.

The table below on *Student Roll by School Type, Gender, and Ethnicity* offers valuable insights into gender distribution across different school types, highlighting areas where gender imbalances persist, alongside ethnicity.

Student Roll by School Type, Gender and Ethnicity; table adapted from Ministry of Education Strategic Development Plan 2023-2026

School Type	I-Taukei	I-Taukei	Indian with Fijian Descent	Indian with Fijian Descent	Others	Others	Rotuman	Rotuman	Grand Total
	Female	Male	Female	Male	Female	Male	Female	Male	
ECE	7156	7818	1647	1737	269	272	68	67	19034
Primary	55127	60452	16566	17289	2461	2550	558	545	155548
Secondary	27348	24874	9704	9091	1509	1419	391	379	74715
Grand Total	89631	93144	27917	28117	4239	4241	1017	991	249297

Source: FEMIS 2023

Employment and Labor Participation

Traditionally women's role is to care for family members and carry out domestic work. This includes caring for children, the elderly and family members with disabilities in the household. According to the World Bank, women in Fiji spend 2.9 times as much time on unpaid domestic and care work than men⁸⁶.

Fiji has a labour force participation rate of 57.6 per cent. Labour force participation and employment for Fijian women show significant disparities compared to men. As of 2023, the female labour force participation rate was about 39.1%, while male participation was around 77.3%. This difference reflects not only lower participation rates among women but also an overrepresentation of women in informal, low-paid, or unpaid work. In the formal sector, about 35.2% of women hold paid jobs. Women tend to earn about one-third less than men, with much of their employment concentrated in lower-paying sectors, adding to financial instability and inequality.⁸⁷

In its 2018 review, the Committee on the Elimination of Discrimination against Women highlighted ongoing concerns about the low participation rate of women in Fiji's workforce. These concerns were reiterated in Fiji's 2019 Universal Periodic Review (UPR) and in a direct request from the Committee of Experts on the Application of Conventions and Recommendations (CEACR). Key issues included Fiji's wide gender pay gap—

⁸³ [Fiji Female to male ratio, students at tertiary level education - data, chart | TheGlobalEconomy.com](#) ; [Ratio of Female to Male Tertiary School Enrollment for Fiji \(SEENRTERFMZSFJI\) | FRED | St. Louis Fed](#) ; [Fiji - Ratio of female to male tertiary enrollment; World Bank Document](#)

⁸⁴ [89 women graduated from the Makoi Women's Vocational Training Centre](#)

⁸⁵ <https://www.fijitimes.com.fj/fnu-and-graduate-women-fiji-hold-camp-to-boost-female-participation-in-stem/>

⁸⁶ Source:- PDF - Theresa Gigov May 2023. Tackling Gender Inequality in Fiji: Business Responsibilities & Opportunities. Ocean and Human Rights. (www.ocean-humanrights.org/resources)

⁸⁷ ILO (2024) Fiji country Factsheet 2024; [Fiji Female labor force participation - data, chart | TheGlobalEconomy.com](#);

the largest in the region—as well as the high concentration of women in low-paying, informal sector jobs, where they often lack access to maternity rights, legal protections, secure benefits, and representation. Additionally, workplace sexual harassment and other forms of sexual assault remains an issue. In response, Fiji became the second country to ratify ILO Convention C190 in 2020, aiming to address workplace violence and harassment. However, some women continue to face limited employment options, with some compelled to engage in sex work due to economic pressures and challenges with enforcing child support obligations⁸⁸.

Efforts to improve women's economic empowerment are ongoing, with the Fijian government and NGOs working on policies and action plans that target key barriers, such as lack of access to markets, limited training opportunities, and restrictive cultural norms. Additionally, Fiji's Women's Economic Empowerment National Action Plan aims to create a more supportive environment for women's equal participation in the economy, from microenterprise support to skills development and advocacy for fair policies⁸⁹.

Rural women play a significant role in the economic survival of their families. This is especially relevant in Fiji where agriculture is the main source of income and employment in rural areas. According to the FAO Country Gender Assessment of Agriculture and the Rural Sector in Fiji (FAO/SPC, 2019⁹⁰), employment opportunities are very limited in rural Fiji and informality prevails, accounting for two thirds of all informal workers in the country. Rural women earn a quarter less than their male peers and seem to be dedicated to physically strenuous and time-intensive agriculture tasks that add to their reproductive role responsibilities, linked to caring for her families, household chores and needs. Poor access to agricultural land is also a constrain to women's agricultural activities. Their inability to own land is a key factor that constrains women's ability to cultivate commercial crops and engage in agribusiness

Within agricultural households, women make decisions related to their domestic responsibilities and may share decisions about economic activities with their husbands/partners. However, due to patriarchal norms, men are generally considered the family decision makers. Women tend to have less voice in decisions about community-based management of natural resources, even though they rely on these resources for their subsistence and income-generating activities.

While women have full rights of inheritance and property ownership by law, local authorities often exclude them from decision-making processes on disposition of indigenous communal land. Limiting women's access to land is a central barrier to founding a formal business and puts pressure on them to marry. Women's dependency on men is a central challenge.⁹¹

Roles and Responsibilities of Women in Fiji's Economic Sectors

Women in Fiji play integral roles across various sectors in both rural and urban settings, contributing significantly to the economy and society. While their contributions are often essential to community well-being, they are frequently concentrated in sectors that may lack high visibility or tend to be lower-income and low-productivity fields. This section explores the diverse economic roles of Fijian women, from agriculture and food security to professional and technical fields, highlighting the challenges and opportunities they face⁹².

⁸⁸ ILO (2024) Fiji country Factsheet 2024

⁸⁹ [Urgent need to develop a Fiji Women's Economic Empowerment National Action Plan - Cabinet](#)

⁹⁰ <https://openknowledge.fao.org/items/bc192084-98fb-4efb-a1dc-8eec1d01dc90>

⁹¹ Theresa Gigov May 2023. Tackling Gender Inequality in Fiji: Business Responsibilities & Opportunities. Ocean and Human Rights. (www.ocean-humanrights.org/resources)

^{92a} FAO and SPC. 2019. Country gender assessment of agriculture and the rural sector in Fiji. Suva. <https://openknowledge.fao.org/items/bc192084-98fb-4efb-a1dc-8eec1d01dc90>

^{17b} Data and figures in section are taken from : Fiji Country Gender Assessment Deep Dive 2023. https://www.mwccsp.gov.fj/wp-content/uploads/2023/04/0227_DeepDive_SINGLE_PAGES.pdf

Food Security and Nutrition

Women are central to food security and nutrition across Fiji. In rural areas, women actively cultivate food crops, contributing to the community's food supply. In both rural and urban households, women are typically responsible for buying, preparing, and ensuring the availability of nutritious food for their families. This role is vital to the health and well-being of families and communities, as women ensure that family members have consistent access to essential nutrients.

Market Vendors and Economic Participation

Women account for 80% of the approximately 3,000 market vendors across Fiji's thirteen municipal markets. These vendors contribute significantly to local economies, providing a vital link between agricultural production and consumer markets. The presence of women in markets highlights their role as economic agents within their communities, enabling them to generate income, support their households, and participate in the local economy.

Farming, Forestry, and Fisheries

Though women represent only about 4% of the workforce in farming, forestry, and fisheries, their involvement in these sectors remains important. Women in these areas often engage in subsistence agriculture, fisheries, or small-scale farming to supplement household income and contribute to family food security.

Tourism Sector

The tourism industry employs a considerable portion of the female workforce in Fiji, with women comprising at least one-third of this sector. However, while women are prevalent in tourism, they are underrepresented in higher-paying professional or managerial positions, with only about one-quarter holding such roles.

Handicrafts and Artisan Value Chains

Women dominate the handicraft and artisan sector, where they often work as individual artisans, within family units, or in small collectives. These groups are sometimes self-formed but are also facilitated by civil society organizations (CSOs), international entities, and government agencies. By producing traditional Fijian crafts, these women preserve cultural heritage while also generating income through tourism and local markets.

Micro, Small, and Medium Enterprises (MSMEs)

Women are highly active in Fiji's informal economy, where most micro, small, and medium enterprises (MSMEs) operate. Approximately half of all MSMEs are owned by women, highlighting their entrepreneurial spirit and resilience. However, many of these women-owned businesses are concentrated in crowded, low-productivity areas like canteens, tailoring, beauty services, food processing, and second-hand clothing.

Professional and Technical Fields

Fijian women have an established and growing presence in professional and technical fields, supported by their strong participation in educational institutions. Increasingly, women are moving into leadership roles within various professional associations, indicating that their presence in professional fields will likely

continue to rise. This trend reflects the impact of education and professional networks on women's career growth and development.

Mining Sector

Although women's participation in the mining sector is minimal, their roles are often in skilled and technical positions. Women make up 12.4% of the workforce in mining, primarily holding scientific and technical roles such as laboratory technicians, geologists, and environmental management specialists. Their presence in skilled positions within mining demonstrates their capacity to succeed in traditionally male-dominated fields, though opportunities remain limited.

Global Outsourcing Services

Fiji's global outsourcing industry employs a substantial proportion of women, who make up approximately 65-70% of this workforce. This sector offers women employment opportunities that might not otherwise be available locally, contributing to both personal and national economic development.⁹³

Female Representation in Fiji's Public Sector and Governance

Women in Fiji are progressively entering traditionally male-dominated fields such as the judiciary, law enforcement, and senior civil service positions. However, only 5 out of 55 parliamentary seats (9.1%) are held by women⁹⁴. Similarly, women account for a low percentage of leadership roles in Fiji's management boards. A recent study by the Fiji Women's Rights Movement found that women comprise only 21% of board members (40 out of 192), with 10 out of 38 boards having less than 30% female participation⁹⁵. While the female share of senior and middle management roles in Fiji's workforce was 38.6% as of 2016, representation in ministerial-level positions was only 5.26% in 2022⁹⁶.

Gender relevant legal framework and institutional arrangements

Fiji has made significant strides in promoting gender equality through both international commitments and national legislation. As a signatory to the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) and through policies like the National Gender Policy and the Women's Economic Empowerment National Action Plan, Fiji demonstrates its commitment to gender equity. However, cultural norms and ingrained societal practices continue to challenge the pace of progress. This section provides an overview of some of Fiji's major gender-focused laws, policies, and initiatives.

National Gender Policy (2014)

In 2014, Fiji adopted the National Gender Policy, establishing a framework to promote gender equity, equality, social justice, and sustainable development. The policy aims to improve the quality of life for both men and women by:

- Promoting gender mainstreaming across government agencies
- Addressing and removing gender inequalities
- Advancing sustainable development through gender equity

⁹³ Fiji Country Gender Assessment Deep Dive 2023.

https://www.mwcsp.gov.fj/wp-content/uploads/2023/04/0227_DeepDive_SINGLE_PAGES.pdf

⁹⁴ Source: Inter-Parliamentary Union, 2023.

⁹⁵ Perception Study on Leadership in Fiji. Fiji Womens Rights Movement. Nov 2022.

http://www.fwrn.org.fj/images/fwrn2017/PDFs/research/FWRM_Gender_Diversity_Incl_GoF_Boards-Final_.pdf

⁹⁶ World Bank Gender Data Portal

Through this policy, Fiji prioritizes the integration of gender perspectives in all government policies, programs, and projects to reduce gender-based discrimination and support social progress.⁹⁷

Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) and Committee Oversight

As a CEDAW signatory, Fiji aligns its policies with global standards for gender equality. The CEDAW Committee monitors Fiji's progress and compliance, urging the government to enact measures that ensure women's rights across all spheres, including education, employment, and personal safety. Fiji's engagement with the CEDAW Committee highlights its ongoing commitment to these international standards.

National Action Plan to Prevent Violence Against All Women and Girls (VAWG) 2023-2028

This five-year action plan, launched in 2023, represents a comprehensive approach to tackling violence against women and girls in Fiji. The plan includes:

- Strategies to prevent violence through education and advocacy
- Initiatives that address social and cultural norms contributing to gender-based violence
- Partnerships with civil society, the private sector, and international organizations

The plan is critical to reducing violence and enhancing support for survivors, underscoring Fiji's dedication to safeguarding the rights and well-being of women and girls⁹⁸.

Family Law Act (2003)

The Family Law Act modernized Fiji's approach to family and domestic relations, introducing laws that promote gender equality within families. This includes:

- Rights regarding marriage, divorce, and custody that aim to protect both partners and children.
- Provisions to ensure that family law disputes are resolved fairly, with attention to the rights and needs of women and children.

The act is a step forward in addressing gender biases in family law and supporting equitable relationships within households.

Employment Regulations Act (2007)⁹⁹

The Employment Regulations Act introduced important provisions for workplace equality and protection, including:

- Ensuring fair treatment and protection from discrimination in employment
- Establishing standards for workplace safety, conditions, and benefits
- Protecting women's rights in employment, especially regarding maternity and equal pay

This law supports women's economic empowerment by providing legal recourse for gender discrimination in the workplace.

National Policy on Sexual Harassment in the Workplace (2007)

⁹⁷ National-Gender-Policy-Fiji-.pdf; <https://www.fiji.gov.fj/>

⁹⁸ Source - fiji_nap_2023-2028-digital-final.pdf

⁹⁹ [EMPLOYMENT RELATIONS ACT 2007 - Laws of Fiji](#)

To combat workplace harassment, Fiji introduced the National Policy on Sexual Harassment, which includes:

- Clear guidelines for employers and employees to prevent harassment
- Procedures for reporting and addressing complaints of harassment
- Protective measures to foster respectful and safe work environments

This policy is a critical measure for gender equality, ensuring that women feel secure and valued in their workplaces.

Fiji has enacted significant legislation and action plans to promote gender equality, protect women's rights, and empower women across all sectors of society. However, despite these advancements, cultural practices and longstanding social norms continue to pose challenges to achieving full gender equity. Nonetheless, Fiji's legislative framework sets a foundation for continued progress towards gender equality, creating pathways for women to thrive in all aspects of life.

Institutional arrangements

In Fiji, a diverse network of institutions works toward achieving gender equality and empowering women across all facets of society. This section examines key governmental bodies, such as the **Ministry of Women, Children, and Social Protection** and the **Ministry of iTaukei Affairs**, alongside Ministry of Rural, Maritime & Disaster Management and Ministry of Housing and Local governments which administers local councils, district and provinces, they drive policies and programs supporting women's rights, economic empowerment and social welfare.

Additionally, non-governmental organizations, international agencies, and faith-based groups play crucial roles in addressing issues like gender-based violence, economic empowerment, and inclusive leadership in Fiji. Organizations like the **Fiji Women's Crisis Center** and **Women's Fund Fiji** provide essential support and advocacy, especially for marginalized communities. Faith-based groups, including the **House of Sarah** and **Catholic Women's League**, further strengthen social networks for women through community engagement and support.

Together, these institutions form a comprehensive framework that promotes gender equity and helps build a more inclusive and resilient Fiji.

Government Ministries and Local Governments

Fiji's governance structure includes four divisions (Central/Western/Northern/Eastern) serving a total of 14 provinces, and various local councils with distinct functions.

Ministry of Women, Children, and Social Protection

The Ministry has three departments: **Department of Women**, **Department of Children**, and **Department of Social Welfare**. Each department implements various services and programs to support families, protect children, and empower women.

The **Department of Women** focuses on gender equality through initiatives like:

- **REACH Project** (Rights, Empowerment, and Cohesion of Rights for Rural and Urban Fijians): A mobile service delivery project prioritizing outreach to remote communities.
- **Women's Economic Empowerment National Action Plan (WEE NAP) 2024-2029**: Supports national development through economic empowerment of women.
- **Women's Plan of Action**: Focuses on gender-based violence.

Ministry of iTaukei Affairs

- Responsible for the welfare and governance of the iTaukei people, this Ministry oversees cultural mapping and integrates iTaukei development interests into national plans.
- Programs emphasize sustainable development, climate resilience, and partnerships with non-governmental organizations and community-based organizations.
- Provincial Offices and Soqosoqo vaka Marama: Represents iTaukei women in governance matters and supports their social and economic advancement.

Ministry for Rural Maritime Development and Disaster Management

The Ministry plays a critical role in working with other government agencies to better serve marginalised communities. The ministry has three main portfolio that is (i) rural development (ii) maritime development and (iii) disaster management. Under its five strategic priority areas are capital programs which includes:-

- Community access, roads, footpaths, footbridges
- Self-help programs
- Water programs
- Rural outer island programs
- Rural housing assistance
- Seafaring entrepreneurial assistance
- Sanitation programs

Projects under these capital programs have benefitted women, children, youths, aging people and persons living with disability

Ministry for Housing and Local Government

The ministry oversees two portfolios: Department of housing and local government. The department of Local Government supports & monitors the functions of the municipalities and is responsible for overseeing the administration and regulations of the councils, as well as overseeing the functions of the National Fire Authority. Local government plays a role in economic empowerment and development of rural women vendors in the municipal markets and licensing of informal business sectors such as street hawkers (for example jewellery sellers, florists, food sellers etc). It is responsible for infrastructural developments that address the safe keeping and protection of women and children within the market work space.

The National Fire Authority through their community awareness and education program are extending their training programs to build community capacity skills and knowledge by empowering abled men and women as well as youths as frontline responders during disasters. This information was gathered from the communities visited during the field interview.

Provincial Councils (supported by the Ministry of iTaukei Affairs): Empower and protect indigenous Fijians' land and resources.

Municipal Councils: Local governance bodies that oversee urban areas under the Ministry of Local Government.

Rural Local Authorities: Manage governance in non-municipal rural areas, providing additional support and services to rural populations.

Non-Governmental Organizations (NGOs)

Soqosoqo Vakamarama i-Taukei

Founded in 1924, this is one of Fiji's oldest Indigenous Women's Organizations, advocating for iTaukei women's rights and representation at the provincial level.

Fiji Women's Crisis Center (FWCC)

Provides essential services to survivors of gender-based violence, with counseling, legal, and medical support, along with a 24/7 toll-free hotline.

Fiji Women's Rights Movement (FWRM)

A feminist organization that champions women's human rights, focusing on legislative reform to remove discrimination and promote gender equality.

We Rise Coalition

A coalition of feminist organizations working toward gender equality and inclusive rights for diverse gender identities, supporting transformative change through collaboration.

Women's Fund Fiji

Established in 2017, the Fund provides financial support and capacity-building for women's organizations, especially those serving marginalized communities such as women with disabilities and LGBTQ+ groups.

FemLink Pacific

A feminist media organization with programs like **FEMTalk 89FM** and **Suitcase Radio** that give women a voice in media, disaster preparedness, and gender equality.

International and Multilateral Organizations

UN Women - Fiji Multi-Country Office

Works with governments and civil society to design and implement laws, policies, and programs aimed at achieving gender equality, in line with global standards.

UNDP Pacific Office – Fiji Multi-Country Office

UNDP Fiji MCO is committed to integrating gender equality and women empowerment into every facet of its programmatic activities. Its transformative gender-focussed efforts has led to the achievement of the Gender Equality Gold Seal. The accelerator lab has contributed immensely to finding innovative solutions to address the root causes of gender inequalities in the Pacific Islands. Furthermore, their work in Fiji has shown support and co-ownership of gender equality across programmes and operation. These includes the implementation of 2021-2022 Gender Equality Strategy, partnering with Fiji Police Force to support survivors of gender-based violence, supporting women's political participation in processes, in parliament and caucus membership, strengthening access to justice for women that survive violence, using sports to promote gender equality by establishing non-discriminating and teamwork values and analysing the laws and policies from a gender perspective.

Secretariat of the Pacific Community (SPC)

The Secretariat of the Pacific Community (SPC) has several initiatives to promote gender equality in the Pacific region, including: Pacific Platform for Action (a roadmap to achieve gender equality and improve the well-being of women and girls in the Pacific. It supports the commitments made by Pacific Island countries

and territories to gender equality), Pacific Gender Statistics Dashboard (a tool that provides a comprehensive overview of gender-related indicators and metrics in the Pacific. The dashboard organizes data into five thematic areas, including population demographics, economic participation, education, health, and safety), Pacific Partnership to End Violence Against Women (a program that includes working with Ministries of Education, teachers, and teacher training institutes to develop gender-responsive school policies. The program also partners with young people to build a pool of trainers to deliver community-based interventions, SPC Gender Flagship (a program that charts the pathway for gender equality in the Pacific).

It also highlighted other gender-related initiatives, including: training authorized justices to issue Interim Protection Orders for survivors of domestic violence, breaking barriers for LGBTQI human rights, International Women's Day and Micronesian Women's Conference.

International Women's Development Agency (IWDA)

IWDA is a feminist organization working towards resourcing women rights organization, promote systematic change, strengthens feminist movements, equality insights – the use of data to drive transformative change and resource empowerment through research to develop tools, frameworks and policies. In Fiji, IWDA works in partnership with FemLINKpacific and Fiji Women's Rights Movement (FWRM).

Faith-Based Organizations

House of Sarah (Association of Anglican Women)

Launched in 2009, House of Sarah provides counselling and support services for women affected by violence and builds coalitions within the Christian community.

Catholic Women's League, Fiji

Established in 1968, this organization has over 2,000 members across 36 parishes, promoting social and spiritual development for women.

Fiji Muslim Women's League

An organization affiliated with the National Council of Women in Fiji, addressing the needs of Muslim women, especially in areas like housing and youth support.

Methodist Women Fellowship in Fiji

Part of the World Federation of Methodist and Uniting Church Women, promoting leadership and development among Methodist women in Fiji.

Gender-Based Violence

Data reveals that Fiji has some of the highest rates of violence against women in the world. It is deemed that 72% of women in Fiji (aged between 18-64) have experienced physical, emotional, or sexual violence by their intimate partners in their lifetime²⁵ with 16% of adult women have reported being sexually abused before the age of 15 years²⁶. Director of Public Prosecution reported 94% of rape survivors were women with 68% of cases involving girls under the age of 18 years²⁷. Additionally, 73% of children directly or indirectly experience GBV, usually by male family members, 74% reported sexual violence cases of children under age

of 18 years.¹⁰⁰ The most common form of sexual violence in Fiji is child sexual abuse, primarily of girls and some boys under the age of 18 years¹⁰¹.

The trend of violence against women and girls in Fiji reveal 2 in 3 ever-partnered women in Fiji have experienced physical and/ or sexual violence from a male intimate partner in their lifetime, 1 in 3 ever-partnered women experienced marital rape, 1 in 4 women experienced economic abuse, 1 in 3 women have experienced violence from a man who is not their partner, and 1 in 5 women have experienced sexual harassment in the workplace in the formal sector, noting rates are unknown in the informal sector .

People of diverse sexual orientations, gender identities, and sex characteristics (including lesbian, bisexual and transgender women, gay men, transgender men and gender nonconforming people) experience higher rates, and more severe forms of violence from partners, family members and communities²⁶.

Women living with disability are particularly at risk for experiencing physical, economic, sexual and emotional violence from family members, partners, caregivers and institutions (health, social services, etc.) with more rigorous research required to understand the full extent of violence experienced²⁶.

Social Norms and Gender Roles in Fiji

In Fiji, social norms and cultural beliefs strongly shape women's roles, influencing their participation in decision-making and public life. These norms, rooted in Fiji's diverse ethnic traditions, often limit women's autonomy, particularly in community settings, where traditional hierarchies shape how women express their needs and concerns.

Women's Participation in Public Consultations

While women in Fiji do participate in public consultations, their involvement is often constrained by cultural hierarchies. When women have decision-making power, they are able to give voice to their interests, preferences and needs, either directly through holding positions of power in the public and private sectors or indirectly by lobbying and advocating through diverse individual and collective, formal and informal mechanisms and channels¹⁰².

In i-Taukei communities, women typically voice concerns indirectly through community women's groups or leaders, adhering to established protocols. In Indo-Fijian and settlement communities, similar structures exist but vary in form. Women's participation, therefore, is shaped by these social structures, which may limit their ability to openly and freely discuss issues affecting them.

Social Expectations and Empowerment

Fiji's patriarchal society reinforces a perception of male dominance, especially within households, and expectations around gender roles remain strong. Women are often seen as caregivers and supporters, while men hold primary decision-making roles. These expectations differ somewhat across ethnic groups and between rural and urban settings, with rural communities and older generations holding more traditional views. Younger Fijians, especially in urban areas, may be more open to shifting gender roles, though traditional beliefs continue to exert significant influence.

²⁵ Sexual and reproductive health and gender-based violence in Fiji: A review of policy and legislation.

https://pacific.unfpa.org/sites/default/files/pub-pdf/fiji_policy_and_legislative_review_290922.pdf

¹⁰¹ Fiji National Action Plan to Prevent Violence Against All Women and Girls (VAWG) 2023 – 2028.

https://asiapacific.unwomen.org/sites/default/files/2023-09/fiji_nap_2023-2028-digital-final.pdf

²⁷ Office of the Director of Public Prosecutions Rape and Sexual Offences Statistics for the years 2017, 2018, 2019, 2020.

<http://odpp.com.fj/mediaupdates-2/>

²⁸ Fiji Women's Crisis Centre (FWCC), 2013, 'Somebody's life, Everybody's business! National Research on Women's Health and Life Experiences in Fiji'. Available at: <http://www.fijiwomen.com/wp-content/uploads/2017/11/National-Survey-Summary.pdf>

Regional and Generational Differences

Gender expectations in Fiji vary widely based on ethnicity, region, and age. Rural areas and older generations tend to uphold traditional gender roles, while younger, urban Fijians are more likely to embrace progressive views on women's empowerment and participation. These differences reflect a gradual shift towards gender equity in some areas, though patriarchal norms persist, especially in rural and traditional communities.

Social norms in Fiji both shape and restrict women's roles in public life, with traditional beliefs reinforcing male dominance and limiting women's participation. As urbanization and generational change introduce more progressive perspectives, Fiji continues to navigate the balance between tradition and the push for gender equality.

Bridging Fiji's Digital Divide: Data Insights and Strategic Suggestions for Women's Inclusion

In Fiji, access to digital tools and information reveals notable gender disparities, especially among low-income communities and rural populations. Although 66% of Fijians were reported to have internet access in 2020, and mobile phone ownership exceeded the population at 139%¹⁰³, barriers remain for many women in terms of accessing and using technology on an equal footing with men. Digital adoption data is primarily general, but available research suggests that men and women in Fiji do not experience technology access in the same way.

Globally, data from GSMA's Women¹⁰⁴ project shows that women in low-income communities are 21% less likely to own mobile phones than men, with many women reporting comfort with basic phone functions like making and receiving calls but far less confidence in using advanced features such as texting. Additionally, fewer than half of the women surveyed felt proficient in sending text messages. Even so, many women expressed strong interest in learning about more advanced features, like mobile banking, online bill payments, and internet browsing—services that are increasingly essential for participating in modern digital economies. However, technical literacy and familiarity with these features are often limited, especially in rural or isolated areas. Local context also plays a significant role, as factors like gender norms, age, and geographic isolation can impact women's access and comfort with technology.

Internet access is also uneven between urban and rural areas. A 2022 survey by the United Nations Capital Development Fund (UNCDF) found that while 45% of Fijians aged 15–74 possess a payment card linked to a current account, only 16% of adults made an online purchase in the three months before the survey. This low adoption rate for digital transactions highlights several barriers, including limited eCommerce infrastructure, low financial literacy, and gaps in digital infrastructure, especially in rural and maritime regions. Fiji's dispersed islands and rugged geography complicate the development of consistent network coverage and technology access, and approximately 2% of the population remains outside of the reach of 2G, 3G, or 4G connectivity. Digital transformation in these rural and remote areas lags significantly behind that of urban centres¹⁰⁵.

In terms of accessing information, Fijian women rely on multiple sources, including radio, television, SMS, social media, and newspapers printed in English, iTaukei, and Urdu. Word-of-mouth remains a vital channel, especially in rural areas where community leaders, such as village heads, chiefs, and church leaders, play crucial roles in distributing important information. Although men and women generally use similar information sources, women from different demographics—especially those in rural areas, of different ethnicities, or varying age groups—experience these sources differently. For instance, older women and

¹⁰³ Country Gender Assessment report 2023 https://www.mwcsp.gov.fj/wp-content/uploads/2023/04/0227_DeepDive_SINGLE_PAGES.pdf
https://www.mwcsp.gov.fj/wp-content/uploads/2023/04/0227_DeepDive_SINGLE_PAGES.pdf

[Digital 2020: Fiji — DataReportal — Global Digital Insights](#)

¹⁰⁴ <https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/programme/connected-women/mwomen-bop-apps-challenge-closes-tomorrow/>

¹⁰⁵ <https://www.trade.gov/country-commercial-guides/fiji-information-and-communication-technology>

those in more remote communities often depend on local leaders or word-of-mouth rather than digital platforms, which may be more accessible to younger or urban women.

Nevertheless, technology access for Fijian women encompasses a broad spectrum, including smartphones, basic mobile phones, laptops, television, and radio. Digital access is becoming increasingly widespread and plays a pivotal role in empowering women's involvement in decision-making. As of 2023, an estimated 65% of Fijians (over 600,000 people) are Facebook users, underscoring the platform's reach and influence. Social media has even given rise to a form of digital activism among Fijian women, who leverage platforms like Facebook to share information, connect with community initiatives, and participate in decision-making processes. However, the extent of technology use can vary significantly across the country. Women in urban areas, especially those who are younger and more educated, are more likely to engage with digital resources, while those in rural areas or from lower socioeconomic backgrounds may lack similar access¹⁰⁶.

Significant challenges remain, however, in ensuring that women have equitable access to information, especially in emergency or disaster situations. Fiji is one of the most digitally developed nations in the Pacific Islands, with extensive submarine cables connecting it to neighbouring islands and major telecommunications hubs, yet the country still ranks only 107th out of 176 countries on the ITU ICT Development Index. In addition, approximately 5% of Fiji's rural population remains without mobile network coverage. As a result, remote communities often rely on alternative means of communication, such as community loudspeakers, sirens, and direct communication from local leaders¹⁰⁷. Studies and Post-Disaster Needs Assessments (PDNAs) in the Pacific have shown that these communities predominantly rely on interpersonal networks—village heads, church leaders, and women's groups—to receive warnings and critical information¹⁰⁸.

Given these challenges, engaging women effectively in digital communication strategies requires consulting with communities to understand their unique needs and preferences. Across the Pacific, communities tend to trust and rely on a combination of national authorities, local leaders, and religious institutions for information. This reliance underscores the importance of using culturally appropriate channels and trusted messengers when tailoring communication to reach women. Traditional and local knowledge also plays a vital role in how Fijian communities detect hazards, respond to emergencies, and share information, making these approaches indispensable for engaging women effectively.

Several strategies could enhance women's access to information, particularly in the context of climate hazards. Community-level interventions that involve social engagement, training, awareness programs, and consultations can play crucial roles in reaching women in rural and urban areas alike. For instance, local workshops on disaster preparedness, digital literacy training, and sessions on emergency response protocols can empower women to take proactive roles in their communities. Programs like FemLINKpacific's Young Women's Leadership Team exemplify how youth-centered initiatives can amplify young women's voices and strengthen community resilience. Adopting basic preparedness actions, such as creating emergency plans, participating in community drills, and assembling essential supplies, can also equip women to better prepare for emergencies.

In sum, while Fiji is advancing in digital access and ICT development, there are still significant gender-based barriers, particularly in rural and remote communities. Addressing these disparities requires concerted efforts to integrate gender-responsive approaches in digital and information strategies. By promoting digital literacy, fostering trust through local networks, and ensuring that women's unique needs are represented in emergency planning, Fiji can continue to bridge its digital divide and empower women across the nation.

Gender, Climate Change and Early Warning Systems and Disasters Response

¹⁰⁶ Country gender assessment report, 2023. https://www.mwccsp.gov.fj/wp-content/uploads/2023/04/0227_DeepDive_SINGLE_PAGES.pdf

¹⁰⁷ <https://www.trade.gov/country-commercial-guides/fiji-information-and-communication-technology>

¹⁰⁸ gender-responsive-and-disability-inclusive-early-warning-and-early-action-in-the-pacific-region_0.pdf

In Fiji, men, women, boys, and girls experience disasters differently, even within the same household. Social roles, expectations, and vulnerabilities vary significantly by gender, affecting how each group prepares for, reacts to, and recovers from natural hazards. Gender inequality in disaster contexts arises from these socially expected roles, which influence socioeconomic status, decision-making agency, and access to resources, ultimately shaping the diverse impacts of disasters on men and women¹⁰⁹. Additionally, certain groups, such as children, LGBTQ+ individuals and low-skilled migrant workers, face increased vulnerability across all ethnicities in Fiji due to intersecting social and economic factors¹¹⁰. Women often face differing risks during disaster preparedness activities, reflecting varied perceptions of vulnerability and required resources. Underlying drivers of vulnerability include living in high-risk disaster zones, poor housing conditions, poor health, and limited access to local institutions or preparedness measures.

Social and cultural norms in Fiji influence women's access to information and their capacity to act on early warnings. While traditional knowledge and past experiences are valuable, they can also create a false sense of security as recent weather events become more extreme than in previous generations.

Recommendations

Gender sensitive indicators and Capacity Building in Climate Resilience

Developing indicators related to women and climate change, particularly concerning natural hazards and emergencies, can provide actionable insights into gender-differentiated impacts. Empowering existing women's groups to take active roles in these initiatives is a promising strategy. Early Warning Systems (EWS) are essential tools that can benefit from gender-sensitive approaches, ensuring that women receive critical information and can respond effectively during crises.

Opportunities for Women's Participation in Disaster Preparedness

There are opportunities to build capacity among Fijian women to contribute to the development and implementation of Multi-Hazard Early Warning Systems (MHEWS). Women's church groups, active in many villages, have become vital to civil society and community resilience. Originally formed to support women's spiritual and domestic growth, these groups have evolved to raise awareness of gender-specific issues such as domestic violence and reproductive health. However, despite their impactful community role, they often face social pushback as they advocate for women's rights and broader community resilience¹¹¹.

These gender initiatives, built on existing structures, present an effective way to leverage local networks for disaster preparedness, thereby ensuring that gender dynamics are recognized and addressed in Fiji's approach to climate resilience.

Field work findings related to gender-specific vulnerabilities and needs in Hazard-Prone Communities of Viti Levu

The information collected are a result of interview of forty people living in hazard and disaster-prone communities using structured questions in a data collection tool. Respondents included an equal representation of male and female respondents from ten hazard-prone communities along Viti Levu's Queens Highway. Nine communities are Indigenous iTaukei communities and one is informal settlement with different ethnic groups settling on sugarcane tramline. Eight communities are located near river channels and flooding drains, one near the coast and one in drought area. Fieldwork results reveal that gender plays a significant role in shaping access to information, roles in community response, and specific needs in

¹⁰⁹ [Vulnerability and Vulnerable Populations \(https://wkc.who.int/\)](https://wkc.who.int/).

¹¹⁰ -PDF - Theresa Gigov May 2023. Tackling Gender Inequality in Fiji: Business Responsibilities & Opportunities. Ocean and Human Rights. (www.ocean-humanrights.org/resources)

¹¹¹ Fiji gender assessment report 2023, pdf

disaster-prone areas of Viti Levu. Women's needs are closely tied to household and caregiving roles, while men's needs align with community mobilization and protection efforts, underscoring the importance of gender-sensitive disaster preparedness and response strategies in these communities.

Leadership Roles: While men dominate leadership roles as community gatekeepers and youth leaders in iTaukei communities, women in leadership are generally involved in community or business-oriented women's groups. This division suggests that gender impacts the types of leadership roles accessible, with men more likely to handle broader community safety responsibilities and women focusing on social and economic aspects within female-oriented groups.

Awareness and Access to Early Warnings: Gendered access to early warnings varies, largely shaped by access to media and household responsibilities. Generally, those with access to media devices—like mobile phones, TVs, and radios—are well-informed about early warnings, regardless of gender. However, aging men and women (particularly those over 70) and people with disabilities often rely on family members or community gatekeepers for information. It was noted that there are cases where men do not take SMS alerts seriously, given that *"it is always too much work putting up shutters and putting it down when the warning is not really serious"*. In mixed-ethnicity settlements, the Fijians of Indian descent and iTaukei people approach disaster preparedness differently. In the mixed-ethnicity settlement of iTaukei and Fijians of Indian descent, social organization varies by group. The iTaukei community has a communal structure with strong kinship ties organized by religious or regional affiliations, fostering mutual support during emergencies. Fijians of Indian descent, however, have a more individualistic approach; while men tend to form social groups, women often stay focused on household roles with fewer community connections. Despite these structural differences, most households have access to early warning information through devices like TVs, radios, and mobile phones, allowing them to respond quickly to flood alerts and reach evacuation centers when needed.

Gender and Vulnerability-Specific Needs: Respondents unanimously agreed that early warning and disaster response needs differ by gender, age, and vulnerability group. Women's requirements during hazards focus on essentials for household maintenance, health, and personal hygiene, including cooking tools, food, sanitary products, and blankets. In contrast, men prioritize supplies and tools for community mobilization, such as ropes, torches, and boats, which reflect traditional roles focused on protection and infrastructure. For adolescent girls, personal hygiene products (e.g., sanitary pads, clean clothes) are necessary, highlighting the gender-specific needs that often accompany physical development. Boys, meanwhile, primarily need clean clothing. Vulnerable groups, including elderly and disabled people, face challenges that require specialized support, such as communication assistance or medical supplies.

Community Engagement and Participation: Women leaders and youth leaders note that vulnerable individuals, particularly the elderly, disabled, and isolated settlers, often depend on family or community members to receive critical information. Additionally, some community members from the iTaukei communities miss out on preparedness training, which is usually held at provincial centers and not directly accessible to all, particularly those who do not work outside the village. In settlements, social and cultural diversity complicates collective action, requiring focused campaigns to increase attendance and motivate participation across gender and ethnic lines.

Training and capacity building: Community training initiatives aim to build capacity for hazard response and early warning systems. These trainings, organized by provincial offices, the Red Cross, the National Fire Authority, and South Pacific Business Development (SPBD), cover topics such as climate change, emergency response, first aid, and small business empowerment for women. The training and capacity-building efforts in these communities show some gender inclusion but lack broader accessibility, especially for people with disabilities and the elderly. Both men and women have access to training sessions, although participation is limited and generally restricted to selected community leaders like women's leaders, youth leaders, and gatekeepers. Women in particular are engaged through initiatives like the South Pacific Business Development (SPBD), which focuses on empowering women in small businesses. However, training access remains uneven: while some youths, primarily males, receive specialized training (e.g., fire warden

certifications), women are generally less represented in these roles. People with disabilities face significant challenges in accessing these programs, largely due to physical barriers at venues and community attitudes that discourage their involvement.

Community members suggest a need for more practical and inclusive training on first aid, rescue techniques, interpreting warnings, and specific disaster responses. They also emphasize that these sessions should be accessible to all, especially for people with limited mobility, to ensure full preparedness across diverse needs and abilities within the community. Women and disabled individuals are often overlooked in emergency preparedness roles.

Recommendations for gender and vulnerability responsive actions: These recommendations made by the interviewed community members emphasize increasing gender inclusivity and accessibility in disaster preparedness and response through community-based, inclusive training and infrastructure improvements.

1. **Inclusive Training:** Ensure that training sessions include representatives from all community groups, with both men and women participating. Schools should also incorporate hazard awareness to reach children, as they are often more prone to ignoring warnings. Involve women in the National Disaster Management Office (NDMO) trainings to increase membership in the committee as they will have the skillset to qualify their involvement.
2. **Empowering Local Committees:** Strengthen all registered community committees, including youth, women, health, and disaster committees, by equipping members with training on early warning systems. This builds on traditional knowledge while enhancing response capabilities.
3. **Accessible Evacuation Centres:** Improve evacuation centres to be accessible for the elderly, people with disabilities, and women, particularly by addressing location, restroom accessibility, and flooring conditions. This would alleviate the burden felt by vulnerable individuals who currently face challenges reaching and using these centres.
4. **Enhanced Communication:** Use community women's groups and local sports teams for awareness programs, tapping into popular activities for effective communication about emergency preparedness. Additionally, improve communication channels like mobile networks in remote and disaster-prone areas to ensure timely warnings.
5. **Resource Stockpiling:** Establish a community charity committee to stockpile essential items (e.g., clothing, diapers) at community halls for emergency relief, which would address the needs of women, children, and vulnerable groups.
6. **Inclusive Leadership:** Increase women's involvement in disaster committees like the Disaster Management Committee (DISMAC), which are often male-dominated, by providing relevant training and mentorship. This helps ensure that women have the skillsets needed for active roles in early warning and emergency response.
7. **Consistent Engagement:** Maintain regular programs and outreach for women and vulnerable groups to sustain engagement and motivation in disaster preparedness efforts. This includes mentorship for young mothers and awareness programs targeting women and girls.
8. **Visible Support Structures:** Install billboards in vulnerable areas with instructions on responding to hazards, maps of evacuation centres, and preparedness guidance, ensuring clear and accessible information for all community members, including women and other vulnerable populations.

Based on the note provided at the end of the study it is also suggested to:

1. **Partner with Religious Groups:** Use existing networks (e.g., Christian denominations, Muslim Women's League, Hindu Women's Club) to deliver community programs, as residents align closely with these groups.
2. **Form Community Clusters within Religious Groups:** Organize smaller clusters within these groups for better engagement and cohesion.
3. **Improve Local Representation:** Appoint leaders from within the settlement or from each religious group to better represent resident needs and liaise with the District Office.

4. **Conduct Programs at Religious Venues:** Hold workshops and empowerment initiatives at religious sites, which are familiar and trusted by the community.

These findings and recommendations have been used to inform the Gender Action Plan for Fiji's EW4All proposal.

6 Gender Assessment for Chad

While the second and most recent General Population and Housing Census of 2009 (RGPH2) counted Chad's total population at 11,039,873, the World Bank estimated this population at 18,278,568 in 2023, compared with 18,675,547 according to recent INSEED projections for 2024. The proportion of forced migrants is significant, with 1,200,000 refugees from Sudan, the Central African Republic and Nigeria, plus 244,000 returnees and 215,900 IDPs in 2024¹¹². In 2021, the World Bank estimates a population density of 13.6 inhabitants per km², with a sex ratio of 1.008, i.e., 8,623,530 men (50.2%) for 8,556,210 women (49.8%). The sex ratio at birth was 1.042 for the same year. The average household size is 5.3 and the average age of the population is 19.3, while the median age is 13.0 in 2018¹¹³.

According to the ECOSIT4 survey conducted in 2018, Chad's population (15,503,179) is predominantly rural (76%). More than half (52%) of this working population in rural areas is made up of women. Their main activities are farming, livestock rearing and fishing. Most Chadian households derive most of their income from the rural sector. The urban population, which represents only 24% of the total population, is concentrated in N'Djamena and the departmental capitals, which are considered urban centres. The proportion of the nomadic population is relatively low, at 3.5% of the total population of Chad (RGPH2). Women represented 49.8% of Chad's total population in 2023¹¹⁴ and 22.1% of Chadian households were headed by women¹¹⁵ in 2015. Chad has a large rural population, with 76% of the population living in rural areas¹¹⁶ in 2023, the majority of whom work in the agricultural sector.

Chad's gender equality indices are the lowest in the world, in Africa and in the sub-region. In the UNDP Gender Development Index (GDI) 2023-24¹¹⁷, which compares the achievements of women and men in the UNDP Human Development Index (HDI) 2023-2024¹¹⁸, Chad is ranked 179^e out of 182 countries, with an index of 0.776. It is also ranked 145th out of 146 countries in the Global Gender Gap Report. In the UNDP Gender Inequality Index (GII) for 2022, which combines indicators of sexual and reproductive health, education and political and economic participation, the country is ranked 163rd out of 166 countries, with an index of 0.671. The Gender Global Gap (GGI) Index of the World Economic Forum (WEF, 2023), which takes into account four (4) areas (economic participation and opportunities, education outcomes, health and survival, and political empowerment), ranks Chad 145th out of 146 countries, with an index of 0.570.

According to the MICS 2019 survey based on the latest general population census (RGPH 2009), the disability rate for people aged 18 to 49 at national level is higher among women, at 5.8% (4.1% in urban areas compared with 6.2% in rural areas) than among men, at 3.0% (2.5% in urban areas compared with 3.1% in rural areas). Disabled women are doubly marginalised, both because of their gender and their disability. They have very limited access to education, health services and economic opportunities. Disabled women are also more exposed to gender-based violence (GBV) and social exclusion.

Chad's economy remains largely undiversified and heavily dominated by the primary sector, which accounted for 61.2% of national GDP in 2011, with oil accounting for 39.3%, agriculture for 10.5%, livestock farming for 9% and mining for 2.2%. As for the secondary sector, whose activities are becoming increasingly diversified, its contribution to GDP was limited to 7.7% in 2011. The tertiary sector contributed 20.7% to GDP in 2011, with commercial activities accounting for 13% and public administration for 10.7%¹¹⁹. The agricultural sector, which is limited to food crops such as cereals, and cash crops such as cotton, sugar cane and gum arabic, is highly vulnerable to the vagaries of the weather and is also dependent on the security situation. The private sector is made up of small and medium-sized enterprises, most of which operate in the informal economy.

¹¹² OCHA : Bulletin humanitaire Tchad mai 2024

¹¹³ ECOSIT4

¹¹⁴ World Bank: [Population, female \(% of total population\) - Chad | Data](#)

¹¹⁵ World Bank: [Female headed households \(% of households with a female head\) - Chad | Data](#)

¹¹⁶ World Bank : [Rural population \(% of total population\) - Chad | Data](#)

¹¹⁷ The GDI is the ratio of the HDI for men to that for women. Chad is ranked in group 5 of the GDI, where the absolute gender parity gap is greater than 10%.

¹¹⁸ [UNDP \(2024\). Human Development Report 2023-24.](#)

¹¹⁹ Rates assessed by INSEED and the World Bank

The rural sector plays a crucial role in the Chadian economy, contributing 42.5% of GDP in 2020¹²⁰.. Rural development therefore remains a government priority, with the aim of stimulating economic growth and reducing poverty in rural areas, while addressing cross-cutting issues such as adaptation to climate change and gender.

Education, vocational training and literacy

Education in Chad shows significant disparities between men and women aged 15 to 49. According to ECOSIT4, the literacy rate for those aged 15 and over is 41.5%. It is higher among men than among women (57.6% compared with 28.1%), and more so in urban than in rural areas (66.0% compared with 32.5%). There are major disparities between N'Djaména, the capital, and other urban centres (76.1% in N'Djaména compared with 58.8% in other urban centres). There are two categories of provinces: those with literacy rates well below the national level (Lac, Salamat, Batha, Hadjer-Lamis, Kanem, Barh El Gazal, Sila, Chari-Baguirmi, Guéra, Borkou/Tibesti, Logone Occidental, Wadi Fira, Ennedi Est/Ouest and Mandoul) and the other provinces with high literacy rates. The literacy rate varies from 8.1% in Le Lac to 76.1% in the capital. Looking only at the poorest households, it rises from 3.0% in the province of Le Lac to 60.7% in N'Djaména, while among the richest, it ranges from 17.4% in Le Lac to 83.5% in the city of N'Djaména. The disparities are caused by early marriage, with 55% and 18.9% of women married before the age of 18 and 15 respectively, as well as by economic constraints, encouraging women to stay at home after primary school (MICS 2019).

Educating girls and promoting gender equality in Chad requires a considerable amount of work, coordinated by all the players. The gender parity gap is observed throughout schooling and decreases considerably at the different levels of education. It is 0.9% at primary level, 0.7% at lower secondary level and 0.6% at upper secondary level (MICS 2019).

According to the MICS 2019 survey, the net primary school enrolment rate excluding pre-school attendance is 43% for the country as a whole. However, this rate varies according to gender, with 45.5% of boys attending school compared with 40.4% of girls, though this varies by area. For instance, in urban areas 59.1% girls attend school compared with 67.9% boys in urban areas, while in rural areas 36.3% of girls attend school compared with 40.8% of boys. These proportions also vary from one province to another, with the lowest rate recorded in Kanem (4.2% girls and 6.3% boys) and the highest in Moyen Chari (71.2% girls and 73% boys). Overall, the primary school completion rate is 27%, with a clear gender difference (30.3% boys and 23.3% girls), with girls more likely to leave school before completing their studies due to early marriage and domestic responsibilities. Depending on their place of residence, the primary completion rate varies significantly between urban areas (55.5%) and rural areas (19.4%), and also between provinces, with 2.4% in Kanem compared with 58.3% in N'djamena and 10% in Ouaddai.

In addition to the socio-cultural constraints that limit girls' attendance at school, there is the weight of religion, which is reflected, for example, in the low enrolment rate of children whose head of household is Muslim (23.6%) compared with children whose head of household is Catholic (68.5%) or Protestant (74.6%). There are also sexist prejudices in the education system, and this accentuates discrimination at professional level. Socio-cultural constraints such as women's and girls' very busy schedules, forced and early marriages, early and unwanted pregnancies, etc., limit girls' and women's access to education and vocational training. Girls and women find it difficult to obtain information, read, train and acquire the knowledge and skills they need to enter the labour market.

There are few young women in higher education or in technical and vocational education and training (TVET). In higher education, 2014 official sources say that the institutions were attended by more than 40,000 students, 16% of whom were young women. In 2012, the participation rate of young women in TVET and higher education was 28.5%, compared with 71.5% for young men.

The education sector is directly and indirectly affected by climate change, with extreme weather phenomena leading to child deaths and damage to school infrastructure, as well as shortages of water, sanitation and

¹²⁰ [Rapport analyse genre coopération Tchad 2021](#)

hygiene that compromise the quality of children's education. Chad is ranked as the second country in the world where children are most at risk from climate change according to UNICEF's Climate Risk Index for Children (CRIC).

The education sector plays a crucial role in reducing vulnerability to disasters, particularly for women and girls. Taking gender and climate change into account in education and communication will strengthen the knowledge and skills needed to deal with climate challenges at individual, institutional and community levels. In the context of climate change, the level of education influences the ability to carry out resilient livelihood activities. Unequal access to education can reduce the ability of women and girls to adapt to climate shocks through migration. Girls with limited access to education may be at a disadvantage in seeking alternative livelihoods, particularly when traditional livelihoods become less viable. This may have gender implications, as the skills traditionally acquired by children may be more or less adaptable to other social contexts. For example, as migration routes are closed to herders and livestock rearing becomes less viable, children voluntarily left out of school to help herd livestock may find themselves at a particular disadvantage later in life.

Employment and participation in the labour market

Chadian women face significant obstacles to their economic participation, including discriminatory regulations and restrictive social norms. Although they often hold informal and agricultural jobs, they remain under-represented in positions of responsibility and have difficulty accessing financial services. In terms of rural finance, women's access to financial services is limited, with only 5% having a bank account and 11.8% accessing credit. Initiatives such as the National Strategy for Inclusive Finance aim to improve access, but challenges remain, particularly in rural areas (ECOSIT 4).

In Chad 60% of the population aged 15 and over is economically active (49.3% of women compared with 72.8% of men) with a ratio of 67.7 women for every 100 men (World Bank, 2018)¹²¹. Although women make up a significant proportion of the working population in Chad, they remain under-represented in management positions and skilled jobs, which contributes to the persistence of gender pay gaps in other words, women earn less than men due to their position in the labour market. There are no statistics on wage differentials between men and women by sector of activity. However, women are in the minority in skilled and responsible jobs, i.e., 14% of senior managers, 10% of middle managers, 16% of skilled workers and 11% of employers (DAI, 2021)¹²². Thus, the traditional division of labour assigns domestic and craft tasks to women, relegating them to jobs that are less well paid and less valued than those of men (Ndoloum and Djimbaye, 2020).

Estimated gross annual income differs by gender. In 2017, a Chadian man earned an average of around \$2,088, while a Chadian woman earned an average of \$1,412. The agricultural sector employs the most women. Only 27% of working women work in another sector (FAO, 2021)¹²³.

In 2022, agriculture remains the main economic activity in rural areas, employing almost 69.1% of the working population¹²⁴. Women make up a large proportion of the agricultural workforce, 68.8% of them in employment, but are often confined to tasks that are less well paid than those performed by men (69.4% of men). Access to land is a major challenge for rural women, with only 15% of women owning land, compared with 58% of men¹²⁵.

Agriculture is a crucial economic pillar in Chad, contributing 23% of GDP in 2018 (PNA, 2021). Agricultural production in Chad involves both men and women, but the value of women's contributions is not fully recognised. In the agricultural sector, women account for a similar proportion of jobs to men, but are more

¹²¹ World Bank: [Labor force participation rate, total \(% of total population ages 15-64\) – Chad | Data](#)

¹²² UE : [Rapport analyse genre coopération Tchad 2021](#)

¹²³ FAO : [Profil national genre des secteurs de l'agriculture et du développement rural - République du Tchad](#)

¹²⁴ World Bank: [Employment in agriculture \(% of total employment\) \(modeled ILO estimate\) - Chad | Data](#)

¹²⁵ EDS-MICS 2015

involved in food processing activities. The differentiated roles of women and men reflect socio-cultural norms, with women often involved in activities considered secondary in the contribution to food.

Representation and participation in decision-making

Women are in the minority in almost all formal economic sectors. This situation is largely due to delays in school enrolment and the persistence of socio-cultural, religious and other stereotypes. In addition to these causes, school wastage is another factor, often due to early pregnancy and early marriage. This imbalance in formal employment can also be explained by the massive presence of women in the informal agricultural sector¹²⁶ in rural areas.

Women's involvement in rural development is marked by traditional production practices and a low level of product processing and conservation techniques, which limits added value. They spend an average of 63 hours a week on farming activities, in addition to domestic chores, but are under-represented in the sector's decision-making spheres, accounting for just 19% of decision-makers. Female representation in livestock umbrella organisations is also low, which limits women's voice in decision-making and the defence of their interests. Decision-making processes take place at different levels, from national to local, and involve a combination of formal and informal mechanisms.

At national level, women are mainly represented in these processes by civil society organisations, particularly umbrella organisations such as the Cellule de Liaison et d'Informations des Associations Féminines (CELI AF). However, this participation is often hampered by a lack of accountability and dissemination of information to rural women, which reduces their real impact on the decisions taken. At provincial and local level, women are often under-represented on provincial, departmental and local action committees, which reduces their ability to influence the choices made at these levels. This under-representation limits their access to resources and decision-making opportunities, even though they are increasingly present on water and natural resource management committees. Their role in sanitation management is also often secondary, focusing mainly on hygiene responsibilities.

Women are often excluded from land ownership and land management decision-making, which contributes to their economic marginalisation. Land conflicts, fuelled by these demographic and economic pressures, threaten family farms and access to land for the most disadvantaged.

Despite these challenges, women's organisations play a crucial role in promoting women's interests in the agroforestry and fisheries value chains, advocating for greater inclusion, equitable representation and greater access to resources and economic opportunities. As agents of change, these organisations continue to defend women's rights and work to improve their situation, helping to build a more equitable and sustainable society.

There have also been positive developments with the inclusion of a female component in bodies such as the Conseil national de concertation des producteurs ruraux au Tchad (CNCPR T). Nevertheless, this integration remains partial, with men often retaining a dominant position in these bodies, which highlights the need to continue efforts to achieve gender-balanced representation in decision-making processes relating to agriculture and rural development.

Over the past few years, the government has implemented a number of policies to increase the participation of women in decision-making, with a clear increase compared to previous years, but still far from equalling that of men.

With regards to the rate of women holding seats in the national parliament, it was only 12.8% in 2017, which represents a considerable improvement (2.4% in 1995) at national level, but places the country among the last in terms of women's representation in parliament worldwide (UNDP, 2018)¹²⁷.

¹²⁶ [FAO, National Gender Profile of the Agriculture and Rural Development Sectors, Chad.](#)

¹²⁷ [FAO, National Gender Profile of the Agriculture and Rural Development Sectors, Chad.](#)

In 2018, Chad adopted an ordinance-law aimed at introducing parity in nominative and elective functions, with a 30% quota granted to women from the outset. However, gender inequalities persist in access to civil and political rights. Women remain largely under-represented in decision-making bodies. After the death of President Idriss Déby in April 2021, the transitional government appointed on 2 May 2021 included only 9 women out of 40 ministerial posts and secretaries of state (22.5%), and women accounted for only 15% of the members of parliament designated as advisers to the National Transitional Council (CNT). There are no women on the Military Transition Council (CMT).

In the diplomatic and administrative sectors, the situation is similar, with only 2 women out of 24 ambassadors (8.33%), 6 women out of 30 at the Economic, Social and Cultural Council (20%), and 2 women secretaries-general out of 29 (6.89%). In the ministries, women hold 29% of department head posts, 14% of division head posts, 12% of director general posts and 13% of technical director posts.

According to data from the Direction de la Solde, women are in the minority in the civil service, with only 14% of women compared with 86% of men in 2021. The highest proportion of women is found among contract employees (46%), followed by middle management (22%), senior management (22.8%) and support staff (3.8%). A comparison between the sexes shows that of the 14% of women in the civil service, only 2% are in senior management. The highest proportion of women in any grade is in middle management (Grade B), which is the grade for executive assistants. The astonishing discrepancy in the first two grades is due to the very low level of education among women, which is now rising sharply as a result of policies in favour of women. The support staff category, which requires few qualifications, is the only category where the gap between the two is minimal. These figures show that women are rather marginalised in the field of employment, especially when it comes to access to positions of responsibility. The Ministry of Women's Affairs tops the list in terms of female employees, with 48%, compared with 5% for the Ministry of Territorial Administration and 7% for the Ministry of Justice.

At territorial command level, only 2 out of 56 prefects (3.6%) and none out of 23 governors are women. In the 2011 communal elections, 5 women out of 43 were elected mayors (12%), and only 13% were municipal councillors are women. Over and above these statistics, it should be noted that in Chad there are only 8 female notaries for 64 male notaries; there is only one female doctor specialising in immunology, and only one female bailiff. The difference between the provinces is also quite marked, with only 712 women working in the provinces, i.e., outside N'Djamena, which accounts for 95% of formal employment at national level. Lastly, 29% of the new government of the 5th Republic of 2 January 2024 (Republic of Chad 2024) is made up of women. This government, comprising a prime minister and 41 ministers, including 5 ministers of state and 11 secretaries of state, includes 12 women. Despite progress in involving women in grassroots organisations, their presence in governing bodies remains low. Women's participation in decision-making remains well below the 30% recommended by the Beijing Platform for Action.

Gender-based violence and social norms

In Chad, gender-based violence (GBV) is a major risk for women and girls, not only during conflicts but on a daily basis. The presence of refugees, displaced persons and humanitarian workers has an impact on the living conditions of local populations in a context of limited natural resources. Moreover, in eastern Chad, GBV accounted for 55% of reports of human rights violations (Solhjell et al., 2010). Domestic or conjugal violence is the type of violence most frequently reported among local populations, internally displaced persons and refugees in Chad (Global Protection Cluster, 2012).

According to Chad's latest multiple indicator demographic and health survey (EDS-MICS, 2014-2015), the structural and gender-based power differentials between men and women mean that women aged 15-49 are much more exposed to multiple forms of GBV, the most recurrent and particularly damaging of which are physical violence (29% of women), sexual violence (11.7% of women), physical or sexual violence (32.6% of women) and domestic violence (physical, sexual or emotional) (34.8% of women). In addition, among people aged between 20 and 24, child marriage before the age of 15 (28.5% of women compared with 4.4% of men), child marriage before the age of 18 (65.9% of women compared with 37.1% of men) and female genital

female genital mutilation (38.1%) are all types of violence that are difficult to eradicate because of communities' attachment to their cultural, social and religious customs and traditions.

On average, 74% of non-single women cite their husband or partner as the main perpetrator of violence (76.7% sexual violence and 71.7% physical violence), while others also report physical violence inflicted by their mother or father's wife. This last point concerns 12% of women in couples compared with 43% of single women. So the common idea that violence against women is necessarily perpetrated by men needs to be qualified. Furthermore, at any point in their lives, 6.5% of women who have experienced violence have also assaulted their spouse, compared with 1.7% of women who have never experienced violence who have nevertheless assaulted their husband (EDS-MICS, 2014-2015). Overall, 28% of women reported that their husband/partner had displayed at least three (3) controlling and domineering behaviours.

Gender equality policies and legal framework

Chad has specific legislation on gender equality and combating gender-based violence, as well as national programmes and committees dedicated to these issues. Here is an exhaustive list of the main legislation, strategies and structures in place:

Specific legislation

- Constitution of the 5th Republic (2023): This constitution reaffirms equal rights between the sexes, prohibiting all forms of discrimination (Articles 14 and 15) and encouraging the political participation of women (Article 34).
- Revision of the Penal Code (2017): Includes provisions on sexual and domestic violence, child marriage and female genital mutilation, taking into account gender-specific issues.
- Law n°022/PR/2018: Institutes parity in nominative and elective offices, requiring a 30% quota for women's participation.
- Law n°029/PR/2015: Prohibition of child marriage, protecting the rights of underage girls.
- Law n°007/PR/2007: Protection of the rights to education, training and socio-economic integration of people with disabilities, including women with disabilities.
- Law n°006/PR/2002: Combating female genital mutilation, early marriage and domestic violence.
- Law n°024/PR/2018: Ratification of the Convention on the Rights of Persons with Disabilities, which includes specific provisions for women with disabilities.
- Law n°021/PR/2019: Governs legal aid and legal assistance, including victims of gender-based violence.
- Law n°019/PR/95: Adoption of the Declaration on the policy of integrating women into development.
- Law n°038/PR/96: Labour Code guaranteeing gender equality in employment, with specific restrictions on night work for women.
- Law n°015/PR/2018: Creation of the Maison de la Femme, a centre dedicated to the promotion and protection of women's rights.
- Ordinance n°012/PR/2018: Institutes parity in nominative and elective functions, ratified by Law n°022/PR/2018.
- Order n°043/PR/2018: Orientation of the Agrosylvopastoral and Halieutic System, involving women in the governance bodies of development policies and strategies.

National programmes and strategies

- National Gender Policy (PNG 2017-2020): Drafted in 2011 and adopted in 2017, this policy accompanied by its five-year national action plan (PA-PNG 2019-2023) validated in December 2018, aims to promote gender equality and reduce physical and/or sexual violence against women and girls.

- National Strategy to Combat Gender-Based Violence (Stratégie Nationale de Lutte Contre les Violences Basées Genre - SNVBG): Drafted in 2015 and revised in 2022, this strategy aims to eliminate gender-based violence through prevention and by responding to the needs of victims.
- National Gender and Climate Change Strategy (SNGCC 2024): Designed to reciprocally integrate, the gender dimension into climate policies at national, provincial and sectoral levels. This strategy strengthens the resilience of women and vulnerable groups to climate change.
- National Social Protection Strategy (SNPS 2016-2020): Aims to establish a comprehensive social protection system to reduce vulnerabilities and ensure access to basic social services for all Chadians.
- National Strategy for Inclusive Finance in Chad (SNFIT 2017-2030): Promotes the financial inclusion of women, particularly in rural areas, through better organisation and use of financial services.
- Vision 2030 "The Chad we want": A global strategic framework that includes gender equality and social inclusion objectives in its various axes, in particular good governance and improving the quality of life for all citizens.
- National Development Plan (NDP): The 2017-2021 NDP and the future 2022-2026 NDP highlight the importance of gender equality for Chad's economic and social development.

Committees and coordination structures

- National Gender Policy (NGP) Monitoring and Coordination Committee: This committee is responsible for implementing and monitoring the NGP, in collaboration with the ministry responsible for gender promotion.
- Technical Committee for the National Strategy to Combat Gender-Based Violence or National GBV Coordination: Coordinates efforts to combat gender-based violence at national level. Its mission is to implement and evaluate the national strategy to combat GBV.

International commitments

Chad has ratified several international conventions aimed at promoting gender equality, including :

- The Charter of the United Nations (1945) ;
- The Universal Declaration of Human Rights (1948) ;
- The four World Conferences on Women, 1975-1995
 - Mexico City Conference (1975), which launched the "United Nations Decade for Women" by adopting a World Plan of Action focusing on equality, development and peace.
 - Copenhagen Conference (1980), which assessed progress since 1975 and highlighted the continuing challenges of equal access to education, employment and healthcare.
 - Nairobi Conference (1985), which marked the end of the Decade for Women, by adopting the "Forward-looking Strategies" for equality, development and peace to the year 2000.
 - Beijing Conference (1995), which is famous for adopting the "Beijing Declaration and Platform for Action", an ambitious plan to achieve gender equality in twelve critical areas.
- Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), ratified in 1996.
- Resolution 48/104 (1993): on the declaration on the elimination of violence against women;
- Resolution 54/134 (25 November 1999): on the International Day for the Elimination of Violence against Women;
- United Nations Resolution 13/25 on Women, Peace and Security (2000) and the ten (10) resolutions that followed (Resolution 18/20 adopted in 2008, Resolution 18/88 adopted in 2009, Resolution 18/89 adopted in 2009, Resolution 19/60 adopted in 2010, Resolution 21/06 adopted in 2013, Resolution 21/22 adopted in 2013, Resolution 22/42 adopted in 2015, Resolution 24/67 adopted in 2019, Resolution 24/93 adopted in 2019 and Resolution 25/38 adopted in 2020): Guide gender policies in conflict and post-conflict situations.

- African Charter on Human and Peoples' Rights (27 June 1981): Recommends guaranteeing women's rights in African states.
- Maputo Protocol (2003): Relating to women's rights in Africa.
- Solemn Declaration on Gender Equality in Africa (2004): Adopted by the African Union to promote gender equality.

Regulatory texts

- Decree no. 2035/PR/MFPPE/2017: Adoption of the National Gender Policy (PNG).
- Decree no. 0186/PR/MASF/2002: instituting SENAFET (National Chadian Women's Week).
- Decree n°0433/PR/MFPPE/2021: Application of Ordinance n°012/PR/2018 on parity.
- Decree no. 1521/PR/MFPPE/2019: Implementing rules for law no. 007/PR/2007 on the protection of people with disabilities.
- Order n°106/PR/PM/MASSNF/SE/DPFIG/2010: Creation and responsibilities of the national coordination for prevention, protection and response to gender-based violence

These laws and strategies demonstrate Chad's commitment to promoting gender equality and combating gender-based violence, in line with international and regional standards. However, challenges remain in the effective application of these measures, particularly in rural areas and with regard to traditional and religious practices.

Gender-related institutional arrangements

The institutional framework relating to gender issues and the promotion of women's rights in Chad is made up of several structures, including (i) national mechanisms or public institutions, (ii) specialised services responsible for the promotion of women or gender (gender focal points, the Observatoire de la Promotion de l'Egalité et de l'Equité de Genre (OPEG) and other public administrations), (iii) civil society organisations and technical and financial partners.

The Ministry for the Promotion of Gender, renamed the Ministry for Women, the Family and Child Protection (MFFPE) in 2021, is the national mechanism responsible for issues relating to the promotion of women and gender. Its mandate includes coordinating action on gender equality and equity, women's empowerment, advisory support, monitoring-evaluation and advocacy, as well as mobilising resources to finance the implementation of development policies and programmes (Final report, *Analyse Genre Tchad*, 2021/UE). This ministry is responsible for implementing the National Gender Policy (PNG) and its action plan (PA-PNG), but it suffers from budgetary constraints and a lack of technical and financial support, generally limiting its actions to improving women's living conditions rather than helping to change their position in society. This is reflected in the difficulties in implementing the PA-PNG, ensuring effective coordination and monitoring-evaluation (there is no monitoring and evaluation unit within the ministry) of the various and multifaceted actions implemented directly by the ministry or in partnership with the sectoral ministries, since the MFFPE has little contact with them and at central level those in charge do not understand the gender concept (Final report *Analyse Genre Tchad*, 2021/UE).

The gender focal point, appointed by each ministry at the request of the ministry responsible for gender issues, is responsible for ensuring that gender issues are integrated into the policies, strategies, programmes and projects of their administration. However, these officials often do not have the necessary capacity to really influence decisions within their administration. Their profile, their generally non-strategic position and their lack of institutionalisation, in particular the absence of this position in the ministry's organisation chart, are obstacles to their effectiveness. In addition, they do not always report to the structures responsible for gender issues in their administration, which further limits their action.

Gender focal points generally do not receive specific training on gender issues and do not have the necessary financial and technical resources to fully fulfil their role. This limits their ability to promote and advocate for gender mainstreaming. In addition, high staff turnover among gender focal points hampers the continuity of

initiatives and compromises communication channels. In addition, there is no mechanism for monitoring and evaluating the activities of these officials by the Ministry for Women or their own administration, which makes it difficult to assess their performance and improve their actions. Sometimes the new gender focal points are not even put in touch with the Ministry for Women, which hampers their integration and effectiveness in their role.

To overcome the lack of monitoring and evaluation, an Observatory for the Promotion of Gender Equality and Equity (OPEG) was set up in July 2022 in accordance with the provisions of the PNG. The Observatory's mission is to monitor, control and evaluate the progress made in promoting gender equality and women's rights. As such, it must collect, produce, centralise and disseminate qualitative and quantitative data through analyses, studies and research on the gender situation at local, national and continental level. It reports to the Prime Minister and is chaired by the Minister for Women. It is made up of a Steering Committee, an Executive Secretariat, technical committees and ad hoc committees. The Observatory's resources consist of State subsidies, donations and legacies.

The creation of the "Maison Nationale de la Femme" (National Women's Centre), inaugurated in 2014 and officially recognised as a structure for awareness-raising, training, supervision and psychosocial care for victims of violence, which offers low-cost computer and sewing courses to young women and girls to help them reintegrate into society and the workplace, is an effective way of combating violence and discrimination against women and girls.

There is also the Centre d'Étude et de Formation pour le Développement (CEFOD), set up in 1966. This research institute has studies and references on the status of women in Chad, as well as a collection of texts on women's rights. Although very active in achieving gender equality goals, civil society organisations (CSOs) often focus on women's practical needs rather than promoting their strategic interests. They face challenges such as lack of financial resources, technical skills and coordination, as well as socio-cultural barriers. Despite this, they play an important role in raising awareness and advocating for women's rights. They are essentially made up of NGOs and women's associations operating at national level (including women's groups and networks).

Technical and Financial Partners (TFPs) support the government's efforts to promote women's rights, but they encounter difficulties with harmonisation and coordination, which sometimes leads to inefficiency and wasted resources. They also identify challenges such as the lack of gender-disaggregated data, the lack of up-to-date information and the lack of financial resources.

Representation and participation of women in disaster risk management sectors

In Chad, women's participation in disaster risk management remains limited, despite their crucial role in community resilience. Specific data on their representation in this sector remains difficult to access due to a lack of national statistics and specific studies. However, initiatives to strengthen their involvement do exist. For example, the United Nations Development Programme (UNDP) has implemented projects aimed at improving ecosystem management¹²⁸ and strengthening the resilience of local communities¹²⁹, particularly women, to climate risks. In addition, as part of several initiatives, training has been given to women both in the provinces and at central level, to raise their awareness of hydrometeorological disasters and climate adaptation techniques, thereby increasing their participation in risk reduction actions.

The representation of women in disaster risk management, meteorology, hydrology and natural resource management remains very low. According to recent estimates, women account for around 15% of civil servants in these fields, but their presence in decision-making positions remains marginal, often below 10%. For example, among the civil servants in the provincial and national departments involved in risk management, very few women hold positions of responsibility such as department directors or heads of

¹²⁸ Lake Chad ecosystem restoration and rehabilitation project

¹²⁹ Community-based climate risk management project

technical services. According to a study carried out as part of the SWEED project, women account for 5% of civil servants at the Ministry of Territorial Administration¹³⁰ and Decentralisation (MATD), and 16% at the Ministry of Transport, Civil Aviation and National Meteorology (MTACMN). It is also important to note that the sector includes a director of civil protection in the MATD, a director of risks and disasters in the Ministry of Social Action, Solidarity and Humanitarian Affairs (MASSAH) and that both MASSAH and MTACMN are headed by women ministers.

In provincial and local disaster management committees (CPA, CDA, CLA), women are often under-represented, representing less than 15% of members, and their role is often limited to administrative or financial tasks. In addition, cultural norms and gendered perceptions, which associate technical skills and science with male roles, hinder women's access to influential positions in technical fields, particularly meteorology and hydrology. This low representation is also due to persistent disparities in access to education and professional opportunities, compounded by a lack of awareness and support mechanisms to promote female leadership in these strategic sectors.

Despite these efforts, it is essential to pursue initiatives to increase the representation of women in disaster risk management in Chad.

Gender in early warning and disaster risk response

Faced with increasing risks from floods and droughts, Chad is striving to strengthen its early warning systems and response mechanisms, notably through community-based EWS, Food Security and Early Warning Information System (SISSAP) at national level and the Platform for Real-time Impact Situation Monitoring (PRISM), supported by the World Food Program (WFP). However, a major challenge lies in the lack of data disaggregated by gender, age and other diversities, which limits a comprehensive understanding of the specific vulnerabilities of women, children, the elderly and people with disabilities in the context of climate-related disasters. This gap hinders the planning of inclusive and gender-sensitive interventions.

The PRISM platform, which uses satellite data, early warning systems and socio-economic assessments to prioritise aid, could become a more effective tool by integrating gender-disaggregated data and collaborating with national institutions such as the National Meteorological Agency (ANAM), the Water Resources Directorate (DRE) and the SISAAP. These institutions play a key role in data collection and analysis, but their systems often lack the capacity to capture and exploit information relating to gender and diversity.

For inclusive risk management, it is essential to strengthen the collection and analysis of data on the specific vulnerabilities of marginalised groups, including rural women, transhumant herders and people with disabilities. This could be achieved by expanding the capacity of PRISM and national institutions to include gender-sensitive indicators. Such an approach would not only improve understanding of the impacts of disasters, but also enable responses to be designed that are better adapted to the needs of the most vulnerable groups, while integrating national and local priorities into climate disaster risk management plans.

Vulnerability and findings from Consultations

During consultations with stakeholders in preparation for this project, several groups of people were identified as being particularly vulnerable to disasters, including women, the elderly, people with disabilities, children, farmers, nomadic herders, as well as refugees and Internally Displaced Persons (IDPs). These groups are at greater risk due to their limited access to resources, information and early warning systems, as well as their potential exclusion from decision-making processes and efforts. Early warning and risk management must take account of these specific vulnerabilities if they are to be truly inclusive and effective.

Although assistance theoretically depends on the targeting of the most vulnerable by the local authorities (administrative and traditional), in practice, food and non-food items are still not being distributed to the

¹³⁰ Study on women's skills in Chad (2021)

most vulnerable, even though this category is the most in need. This observation is reinforced by the insignificant amount of assistance, which does not cover the 10% of needs and which would therefore require better targeting of victims and other disaster victims. During our discussions, we were told that some vulnerable groups had not received the planned assistance, to the detriment of others. This is the case for disabled people's associations and certain women's organisations in Abéché and Bol. The exclusion of this category of people from formal and informal response mechanisms makes them more vulnerable to the impact of disasters.

Women and girls are often among the most vulnerable and exposed to disaster risks, due to their socio-economic position, their traditional role in society and their limited access to information and resources. In rural communities, women are traditionally responsible for managing water, food and resources for their families. In times of drought, when there is a shortage of water and food, they have to travel long distances to find these resources, exposing them to additional risks, particularly in terms of health and safety. In rural areas, they often have limited access to information and early warning systems, due to their limited exposure to the media, education and technology. Their multiple domestic responsibilities and social status keep them away from sources of information that are often the main means of disseminating warnings, such as community radio and community meetings traditionally aimed at a male audience. Women are often under-represented in decision-making processes relating to disaster risk management, which means that their specific needs are not always taken into account in the planning of interventions at local and provincial level, or even in contingency plans that do not reflect a real sensitivity to sex and gender.

The elderly are another vulnerable group in times of disaster, due to their reduced mobility, increased dependency and isolation. In the event of a disaster, the elderly may find it difficult to evacuate quickly or move to a safe location. In times of flooding or water erosion, their reduced mobility increases their risk of being isolated or left behind. Older people, particularly those living in remote areas, may have limited access to early warning systems and information in general, especially if this information is disseminated via modern channels, such as mobile phones or online platforms, to which they often do not have access. Older people are often dependent on family or community members for access to essential resources such as water and food. In times of crisis, this dependence makes them more vulnerable, particularly if their relatives are themselves affected by the disaster and unable to provide the necessary assistance.

Children and young people are particularly at risk during disasters, due to their dependency and vulnerability to physical and psychological impacts. Children are physically more vulnerable to the direct impacts of floods and droughts. Lack of access to drinking water and sufficient food can lead to health problems such as malnutrition and water-related diseases, which are particularly widespread after floods. For example, in Nokou, Lac et Kanem, there was a very high prevalence of water-borne disease and malaria during the rainy season this year. Droughts and floods can also disrupt access to education, as schools may be damaged, inaccessible or simply occupied by disaster victims. This directly affects children and young people, depriving these groups of essential education and increasing their vulnerability in the long term.

The national and local context lends itself to solidarity and stem or extended families rather than individualism and nuclear families, which gives rise to the presence of several generations within the same household, encouraging the sharing of information between young people and their elders. This remains true for the response and support of vulnerable people in times of crisis at household and even community level. It is very rare to find people living alone, let alone elderly or disabled people, without any assistance from a family member. However, there are more and more exceptions in certain areas affected by armed conflict, the influx of refugees, the displacement of people and drug trafficking, which has a devastating effect on young people (e.g. Lac). On the other hand, in times of crisis, young people, particularly those from poor households, are often forced to look for work to support their families, exposing them to dangerous environments, particularly when moving to urban areas or migrating to other countries considered to be prosperous. Young people used to migrate to Niger and Nigeria (in the Lac region), Libya (in the Kanem region) and Sudan (in the Ouaddaï region) in search of better opportunities. This trend is set to be reversed with the armed conflicts in these border countries.

People with disabilities are often among the most excluded and exposed during disasters. They face physical, social and communication barriers that limit their ability to protect themselves and access warning systems. People with reduced mobility or physical disabilities are particularly at risk in the event of flooding or water erosion. They may need special help to evacuate, but this is not always available, especially in rural areas where evacuation infrastructures are limited. The weakness of the legal framework was deplored, in particular law no. 007 of 09 May 2007 on the protection of people with disabilities, which is not being applied to subsidise transport for people with reduced mobility.

People with disabilities, particularly those with sensory impairments (hearing or visual), may have limited access to warning information. If warning systems are not adapted to their needs, they risk being excluded from early warning systems. During our interviews, people with disabilities and those living with HIV were keen to highlight the risk that they may be excluded from preparedness and response plans at community level, as they are not always consulted or involved in decision-making processes, which can lead to solutions that do not meet their specific needs.

Farmers and Nomadic herders are particularly vulnerable to disasters, due to their direct dependence on natural resources (water, pasture) and their increased exposure to climatic hazards. Depending on the amount of rainfall, urban development, the vulnerability of ecosystems and the province's hydrographic network, low rainfall leads to drought, while excess rainfall can cause flooding and water erosion. All these hazards contribute to food insecurity.

Farmers mainly grow rainfed crops (maize, sorghum, millet, etc.), market gardening in the ouadis (Kanem and Ouaddaï) and lowland polders (Lac), as well as off-season crops to a lesser extent. They are directly affected by droughts and floods, which destroy their crops through lack of water and excess water respectively, causing economic losses and exacerbating food insecurity. Farmers often have limited access to forecasts, where they exist, and cannot plan ahead sufficiently to protect their crops or store food.

Nomadic livestock farmers, who depend on water and pasture resources for their herds, are exposed to increased risks in the event of drought or flooding. In both cases, there are losses of livestock, income and increased food insecurity. Very often, animals are exposed to bad weather, parasites and other vectors of disease in the event of flooding, not to mention the destruction of pastures in some flood-prone areas. In the event of drought, they are exposed to a lack of water and grazing.

Movement is essential to the survival of nomadic herders, but in crisis situations they may be forced to move to areas with better water and grazing resources, leading to conflict with local communities living in these areas. Nomadic herders, in particular, may be more isolated from early warning systems, as they often move to remote areas with poor communications coverage. Their mobile lifestyle also makes regular access to weather information and early warnings difficult.

Refugees and internally displaced persons (IDPs), who are present in the Lac and Ouaddaï provinces as a result of armed conflict, are a particularly vulnerable group to disasters. These people often live in precarious conditions, with limited access to safe shelter, drinking water and basic services. Climatic disasters exacerbate their already fragile situation by destroying their temporary homes, reducing their access to essential resources and delaying humanitarian assistance when roads are impassable during periods of heavy rain. Refugees and IDPs are often excluded from early warning systems, as they live outside the usual community networks or in isolated camps and do not necessarily speak the same language as local communities. Their access to weather information is often restricted, and warning dissemination mechanisms are not always adapted to their specific needs. In Lac, for example, refugees mainly speak Hausa and English, while in Ouaddaï, they speak Arabic and English.

Gender and access to information and early warning technologies

Although 78% of Chadians live in rural areas, the country's two main mobile operators, Airtel and Moov, have increased 2G coverage to 86% of the population by 2020¹³¹. By 2020, 48.0% and 21.8% of the Chadian population will be served by 3G and 4G networks respectively. In terms of household ownership of a telephone, the differences between the proportions of households with a radio are smaller between urban and rural households with the same level of well-being (GSMA/Deloitte, 2016).

According to (ECOSIT4 2018), more than half of households (58.6%) own at least one mobile phone nationwide, 85.6% of urban households compared with 50.5% of rural households. In the capital, 97.3% of households own at least one mobile phone, compared with an average of 79% in other urban centres. Almost three people in ten (30%) own a mobile phone nationwide. This proportion is 56.9% in the capital. The provinces of Barh-El-Gazal (44.3%), Kanem (42.6%) and Lac (38.0%) have significant proportions, while Salamat, Moyen-Chari, Tandjilé and Sila, in descending order, remain below 20%. Ownership of other ICT (Information and communication Technologies) goods breaks down as follows: radio (20.0%), television (5.0%) and computers (1.4%).

Households with televisions are generally those that have access to electricity from various sources, such as the SNE electricity network, solar panels or generators (gensets). In Chad as a whole, 5% of households own a television set. By area of residence, this proportion is 40.9% in N'Djamena, 8.7% in other urban centres (20.3% for all urban centres) and only 0.4% in rural areas. The average for urban areas as a whole is driven by the capital, N'Djamena.

Nationally, only 1.4% of households have at least one computer, 12.2% in N'Djamena, 5.8% in urban areas as a whole and 0.1% in rural areas. Unlike mobile phones, computer ownership seems to be reserved to some extent, until now, for households with a higher level of prosperity.

Radio, a medium widely used to inform and communicate, is an asset that is used to a greater or lesser extent by household members. It is owned by 20% of households nationally. Compared with household ownership of a telephone, the differences between the proportions of households with a radio are smaller between urban and rural households with the same level of well-being. Among the poorest households, 14.3% of urban households have a radio, compared with 10.1% of rural households. Among households classified as wealthier, 38.8% of those living in urban areas and 23.1% in rural areas have at least one radio.

At 31 March 2023, it is estimated that more than 69.7% of the population¹³², i.e. 12,577,105 out of approximately 19,800,000 people, own a mobile phone based on SIM card users. Meanwhile, the fixed-line penetration rate is low¹³³ with 1,863 subscribers or 0.011%, mainly public institutions. The number of social media users in Chad is strongly linked to smartphone ownership, which determines internet access (3,598,419 people) and its penetration rate, which stood at 18% of the population at the start of 2023. The EW4All project will have to take this into account, in particular when developing approaches for disseminating early warnings.

In the provinces visited during the consultations, there are generally marked differences in access to information between men and women, particularly with regard to disaster monitoring, warning and management, irrespective of the channel (word of mouth, radio, television, mobile telephony and internet), the area (urban or rural) and the standard of living (household income).

Men generally have more direct access to information in general, as they are often more present in public spaces where information circulates more freely, they are more literate and they have more time and resources to access radio, television, mobile phones and the internet (social networks, online media, etc.).

¹³¹ ARCEP 2020

¹³² ARCEP 2023

¹³³ ARCEP 2022

On the other hand, women, particularly in rural areas, have limited access to these information channels because of their heavy involvement in domestic and agricultural tasks, which keep them away from the channels, but also because of socio-cultural constraints which mean that “a woman's value is measured by the number of children she has and her ability to look after the household, to the detriment of her free time, unlike a man whose value is measured by his bravery and the assets he owns, including the number and influence of the members of his family”.

In addition, women often report that they pay less attention to the information conveyed through these channels, either because the information is not perceived as directly relevant to them, or because it is less suited to their specific needs. Women are also more dependent on information provided at major events (weddings, births, etc.) and in collective settings, such as women's associations or groups, where they mainly obtain information through regular meetings and awareness-raising workshops organised by partners. These informal and formal meetings are becoming the preferred means of accessing information on climate risks and disasters, but this also means that women often receive information with a certain delay, compared with men.

Community radio stations are the main source of information for local people. Unlike the national radio and television stations of the Office National des Médias Audiovisuels (ONAMA), which are not picked up everywhere in rural areas, community radio stations enjoy greater coverage and the trust of local communities. These radios are particularly important for broadcasting early warnings of natural disasters or major climate changes. According to the testimonies gathered, the inhabitants of these provinces have more confidence in these radios, as they offer interactive programmes, tackle local issues and are more accessible in terms of language and content. The issues addressed include climate change and natural disasters, although weather forecasts are not really covered, unlike national television, which broadcasts daily global and provincial weather bulletins.

Agricultural extension agents and other rural development services (ANADER, SODELAC, SISSAP) also play an important role in disseminating information to local producers. They regularly receive information bulletins on changes in weather conditions and hydrological trends, and become essential channels for relaying this information to communities. Their role is crucial in popularising disaster risk management practices, particularly among farmers and livestock breeders. However, this transmission often depends on the availability of information and the qualifications and involvement of these agents in the areas concerned. Community meetings, word of mouth and town criers are also traditional channels that are still used to share information in areas where access to modern technology is fairly limited. Disaster risk information is often shared at these meetings through traditional or even religious leaders and chiefs. This process is slower, but it is still effective in close-knit communities, where information circulates well through these traditional social networks, whether formal or informal.

Increasingly, virtual platforms such as WhatsApp are being developed as modern social networks, enabling certain institutions, groups and associations to exchange and share information more generally and on climate risks and disaster management in particular. However, access to these platforms is still limited, particularly in rural areas where network coverage is poor and access to the internet is technically and financially restricted (internet costs remain relatively high). In rural areas, farmers and women are the most affected by this inaccessibility to new technologies, and their involvement in these platforms remains low compared with urban players and men respectively.

For example, during the discussion groups with the community of Bahboul in the Lac, it was observed that only one male out of around fifty people owns a smartphone with Internet connectivity, whereas the community's basic mobile phone owners are mainly men. This smartphone owner enjoys a certain notoriety, as the other members of the community are obliged to go to him for information and to communicate with the outside world.

Local communities in urban areas are not very well informed about weather forecasts, and those in rural areas are virtually unaware of them. Nevertheless, they express a greater need to improve information and

communication channels for forecasting and managing natural disasters. Men and women, although having different needs in terms of access to information, share the desire to have more reliable information through more effective information channels that are better adapted to their local realities.

Women, particularly those involved in groups and associations, express a strong need for training and awareness-raising adapted to their constraints and responsibilities. They want information channels that allow them to access information more easily, despite their multiple occupations. This includes community workshops, specific programmes broadcast on community radio stations at suitable times, and voice messages on mobile phones that are useful even when they are busy with housework. Training women in associations and groups is also a key channel for transmitting information on climate risk management and disaster resilience.

As far as men are concerned, community radio stations continue to be the most desired source for monitoring and warning, as they are already widely used and perceived as reliable. However, there is a desire to improve early warning systems, with greater involvement of extension workers and the creation of more formal mechanisms for disseminating alerts, notably via SMS or voice telephone notifications, where mobile coverage allows.

There is also a growing demand for real-time weather and climate information and tools that enable local communities, particularly farmers, to better anticipate periods of drought or flooding. Producers are also expressing the need to acquire work equipment and technological tools to improve their resilience (better production, better stock, better inputs, adapted seeds, etc.), which would enable them to overcome crises linked to natural disasters more quickly as part of anticipatory action.

Despite these expressed wishes, a number of barriers stand in the way of improving access to information, particularly for women. The first barrier is poverty and unequal access to modern technologies. Many rural families cannot afford mobile phones, radios, televisions or modern work equipment. This makes it difficult to disseminate information via digital or technological channels.

Low literacy, particularly among women, is another barrier, as many cannot read or interpret written or technical information. This requires appropriate communication strategies, such as the use of visual aids or simple audio messages in local languages.

The third barrier relates to access to energy, which is essential if less traditional channels are to be used, especially in rural areas where simply recharging a mobile phone can require a great deal of effort. Access to electricity in the provinces is far from sufficient, and even less permanent, in the same way as thermal generators, opening up a new market for renewable energies, particularly solar power, which is taking on significant proportions even in some of the most remote areas.

At community level, there is a lack of coordination between the various stakeholders (government, committees, local communities, the humanitarian community, community radio stations and extension workers), which can create gaps in the dissemination of coherent and reliable information on natural disasters. Greater synergy between these actors would be necessary to ensure more comprehensive coverage of risk information and early warnings, and to respond effectively to the needs of local communities, particularly those of women and vulnerable groups.

At national level, the country is facing a number of major challenges in terms of access to climate information and disaster risk reduction (DRR). Firstly, there is a lack of integration of DRR issues into school curricula at both primary and secondary levels. The lack of educational content adapted to the different school levels, combined with a lack of teacher training, limits the young generation's awareness of and preparation for climate risks and disasters. This shortcoming is exacerbated by the lack of documents and teaching materials dedicated to DRR, despite the potential efforts of Technical and Financial Partners (TFPs).

Chad must also overcome the challenge of informing and raising awareness among the general public about climate change adaptation and DRR. Although initiatives such as the celebration of International DRR Day and community awareness-raising programmes are envisaged, their implementation is hampered by a lack of coordination and resources. Awareness-raising programmes on climate change adaptation and disaster risk are less accessible to women, especially in rural areas. Unlike men, women, particularly those who are illiterate or monolingual, have limited access to the formal sources of information (radio, SMS, internet) used to disseminate these messages. They often rely on oral transmission, which reduces their ability to access crucial information in a timely manner. Men are more often involved in training and workshops organised by NGOs and local authorities.

Most awareness campaigns are conducted in French and Arabic, which could be a serious shortcoming for some of the country's provinces. The participation of the city of Ndjamen in international campaigns initiated by UNDRR, such as "Safe Cities", "Safe Schools" and "Safe Hospitals", remains limited, as does the production and dissemination of communication and information tools adapted through appropriate channels to the different population categories, particularly women and vulnerable groups. Finally, capacity-building for journalists and national NGOs through specific training remains insufficiently developed, which hinders the dissemination of relevant information and the mobilisation of civil society around DRR issues. However, although efforts have been made in recent years by the country, with the support of these TFPs, to raise awareness, train and inform local communities and national institutions about adaptation to climate change and disaster risks, women's participation remains low given their vulnerability to climate change. In the development and implementation of public policies to combat climate change and DRR, concrete efforts are currently focused on awareness-raising workshops to inform stakeholders of the risks involved and the importance of integrating gender issues, although the number of women generally attending these workshops is very limited (15-30%).

Preparation, early action and response

In Chad, preparedness, early action and response to climatic disasters such as floods and droughts are mainly coordinated by national and local institutions, including the Directorate General of Civil Protection (DGPC) and the Action Committees (CPA, CDA, CLA). These structures play a key role in drawing up and implementing contingency plans and emergency responses, although there are still shortcomings when it comes to including vulnerable groups.

Existing contingency plans do not sufficiently differentiate between the specific needs of vulnerable groups, such as women, people with disabilities, the elderly or children. Disabled people's organisations and community representatives report a low level of involvement of women and vulnerable people in consultation and response planning. This results in emergency responses that do not take their specific needs into account, particularly for people with functional limitations, who face increased barriers in accessing emergency services and resources.

Disaster management efforts are coordinated by the DGPC, provincial and local action committees, and partners such as ANAM (climate forecasting) and SISAAP (food security). These institutions are supported by international organisations such as the FAO, WFP and UNICEF to strengthen early warning systems and plan inclusive responses. However, greater efforts are needed to integrate the principles of gender and inclusion into their approaches, as well as to develop technical and logistical capacities.

Discussions with stakeholders reveal an urgent need to build institutional capacity for gender equality and the inclusion of people with disabilities. This includes raising community awareness, training those involved in disaster management, and integrating marginalised groups into consultation and planning processes. The development of contingency plans specific to the needs of vulnerable groups and their inclusive implementation are essential to ensure appropriate responses to climate shocks.

In the provinces visited during the consultations, anticipatory actions to deal with the risks of natural disasters such as floods and droughts are already in place in various forms, although they are often limited in scope

and effectiveness. Local communities rely on ancestral knowledge to anticipate the risks of natural disasters. They use natural indicators to anticipate floods and droughts. For example, they observe natural signs such as changes in the behaviour of animals and insects (spiders moving from south to north, the behaviour of certain birds, etc.), variations in vegetation, or atmospheric signs (stars, winds, the appearance of clouds in the east, the cycle of coolness and heat before and after the rainy season), to predict floods and droughts. In some areas, houses are built on higher ground to avoid flood damage, and dykes or small dams are built to channel water during the rainy season and protect inhabited areas and fields. This strategy, which is based on observations of the seasonal cycles of rainfall, high water and low water, helps to reduce the risk of flooding, although the expansion of inhabited areas and climate change are disrupting these traditional practices.

However, despite its effectiveness and reliability for communities, this knowledge, although valuable, is not sufficiently taken into account in modern forecasting processes and even less integrated into warning, preparedness and response systems, which represents a gap in risk management. For example, we were told that a dyke built by local communities on the lake using local materials was more resistant to the effects of weather and bad weather than the neighbouring dyke built by the humanitarian community using modern materials. An approach that combines modern methods with local knowledge could improve the accuracy of weather forecasts, impact predictions and community resilience.

Access to safe evacuation areas

In Chad, climatic disasters such as floods and droughts increase the vulnerability of women, children, the elderly and people with disabilities, particularly in terms of access to safe evacuation sites. Although public buildings such as schools, health centres and community facilities are often used as temporary evacuation sites, these spaces are rarely adapted to meet the specific needs of vulnerable groups. Basic sanitation facilities, such as separate toilets for men and women, access to drinking water and safe sleeping areas, are often lacking, making life difficult for evacuees, particularly women and people with disabilities.

Overcrowding, lack of privacy and insecurity expose women and children to an increased risk of physical and psychological violence in these places. The specific needs of women in terms of safety and dignity, as well as those of the elderly and disabled, are often not taken into account in the planning and management of these spaces. To respond to these shortcomings, it is essential to strengthen the capacity of evacuation sites to be inclusive and safe, by incorporating facilities that meet the specific needs of vulnerable groups, in order to reduce the risks and suffering during climatic crises.

Limited access to resources and services

In Chad, recurrent floods and droughts expose vulnerable groups, including rural women, the elderly, people with disabilities, households living in poverty and nomadic and transhumant indigenous peoples, to increased difficulties in preparing for climatic disasters.

Women and vulnerable groups remain the most affected by disasters, due to aggravating social factors such as gender inequalities, poverty and illiteracy, which reduce their ability to withstand the impacts. In Lac, for example, the CPA's needs assessment found that the majority of households in the province do not have sufficient food stocks to cope with crises, which makes them particularly vulnerable to climatic disasters, especially floods or droughts, and limits their ability to resist and recover quickly after a crisis.

These populations also face limited access to essential resources, such as drinking water, agricultural equipment, and the financial means to implement preventive measures. During periods of prolonged drought, water scarcity seriously affects agricultural and pastoral activities, reducing household income and limiting their ability to save to protect themselves against future crises. Many families have to buy water to meet their domestic needs, exacerbating their poverty and vulnerability.

The indigenous nomadic and transhumant groups have an economy based mainly on pastoral livestock farming. They face difficulties in accessing basic socio-economic services, as well as natural resources such as

water and grazing land, especially as climate change disrupts their seasonal movements and makes them more vulnerable (NAP 2021). Women are often marginalised in access to and management of land, even though they play an essential role in agricultural activities and rural livelihoods. Cultural norms limit their right to land ownership, often reserving this right for men. As a result, women depend mainly on family or marital relationships for access to land, making them vulnerable to family conflict or loss of access in the event of divorce or widowhood. Moreover, nomadic and transhumant communities are facing increasing restrictions on access to their traditional rangelands and pastoral resources. Demographic expansion, agricultural pressures and the absence of clear legal frameworks to protect their rights of access to transhumance lands are exacerbating tensions between farmers and herders. Conflicts over land frequently arise as a result of overlapping agricultural and pastoral activities. These limitations hamper mobility, which is essential for the resilience of pastoral systems in the face of drought and flooding, increasing their vulnerability to climatic and social crises.

The lack of inclusive policies and targeted support mechanisms, such as social safety nets adapted to climate shocks, also hampers people's ability to mitigate the impacts of disasters. A thorough understanding of the specific needs of women, people with disabilities and other marginalised groups is essential to designing programmes that respond effectively to climate risks. By integrating inclusive approaches into disaster risk reduction strategies, Chad could strengthen the resilience of the most vulnerable communities to the growing impacts of climate change.

Recommendations to improve the inclusion of women and vulnerable groups

The following are the main recommendations from the community consultations, which will help to identify areas for improving the inclusion of women and vulnerable groups in the development of the Gender Action Plan for the EW4All initiative:

- Strengthen the participation of women and vulnerable groups: Introduce quotas in local committees directly created by project and organise targeted training to include their perspectives in risk management.
- Integrating local knowledge and disaggregated data: Collecting and using data disaggregated by gender, age and disability to plan appropriate interventions, combining traditional knowledge and modern technologies.
- Include inclusive infrastructures in SOP, training and contingency plan: This to build adapted evacuation sites with specific facilities for women and vulnerable people (separate toilets, secure access).
- Promote equitable access to warnings: Develop appropriate alert systems (SMS, audio messages) in local languages, prioritising the needs of women and marginalised groups.
- Raising community awareness: Run campaigns on gender inequalities and the different impacts of disasters through community radio stations and local leaders.
- Create feedback mechanisms: Put in place accessible systems to collect complaints and needs of women and vulnerable groups in order to strengthen the relevance of interventions regarding SEP.

7 Gender Assessment for Ethiopia

As of July 1, 2024 the estimated population of Ethiopia is 132,059,767¹³⁴, of which 66,167,291 people are male and 65,892,477 (50.11%) are female (49.89%)¹³⁵. Ethiopia has a large rural population with 77% of the population living in rural areas¹³⁶. Twenty two percent (22%) of Ethiopian households are female-headed¹³⁷.

According to UN Women, people with disabilities in Ethiopia, make up 9.24% of the country's overall population. Unemployment is higher among women with disabilities compared to men, regardless of where they live.¹³⁸

Ethiopia has made strides toward gender equality, but challenges persist, particularly in rural regions. Significant progress has been achieved in areas such as maternal health and gender-responsive policies. However, social norms, patriarchal traditions, and gender-based violence (GBV) still heavily affect women's rights and opportunities. Ethiopia is ranked 176th out of 193 countries in the UNDP Gender Development Index (GII) and 79th of 146 countries in the Global Gender Gap Report 2024, with a score of 0.709¹³⁹.

Literacy, Education and Vocational Training

Despite numerous efforts to improve access to education and training, Ethiopia continues to face a high illiteracy rate, with women disproportionately affected. The gender gap is particularly striking, as women's lower enrolment and higher dropout rates at the secondary level contribute to their higher illiteracy rates. According to the 2016 EDHS¹⁴⁰, which has the most recent available data, the illiteracy rate among women stands at 57.1%, significantly higher than the 32.2% for men nationwide. In regions like Somali, the situation is even more dire, with 79.4% of women aged 15-49 unable to read or write. This educational gap severely restricts women's access to information and opportunities, limiting their ability to fully participate in society and contribute to the country's economic development.

In Ethiopia, there are gender disparities in education, with more girls than boys dropping out of school due to early marriage and domestic responsibilities. Efforts are being made to close the gap, but rural areas continue to lag behind urban regions in terms of access to education and literacy programs for women.

Pre-primary education

According to the Educational Statistics Annual Abstract (ESAA), the National Gross Enrolment Rate (GER)¹⁴¹ for pre-primary education in 2021 reached 36.7%, a sharp rise from 9% in 2010. Despite this improvement, enrolment remains low nationwide. Addis Ababa leads with a GER of 93.8%, while other regions vary from 67.4% in Gambella to just 3.2% in Somali. Male enrolment continues to surpass female enrolment, with a GER of 37.7% for boys compared to 35.6% for girls. Nationally, the Net

¹³⁴ [Population of Ethiopia 1950-2024 & Future Projections \(database.earth\)](#)

¹³⁵ [World Bank Gender Data Portal](#)

¹³⁶ [Ethiopia | Data \(worldbank.org\)](#)

¹³⁷ [Female headed households \(% of households with a female head\) - Ethiopia | Data \(worldbank.org\)](#)

¹³⁸ UN WOMEN (2024) Abridged version Ethiopia Country Gender Equality Profile Brief, p. 13

¹³⁹ World Economic Forum (2024) Global Gender Gap 2024, p.175

¹⁴⁰ Central Statistics Agency and the DHS Program, ICF (20016) Ethiopian Demographic Health Survey 2016. <https://www.dhsprogram.com/pubs/pdf/FR328/FR328.pdf>

¹⁴¹ Gross Enrolment Rate (GER) refers to the overall enrolment at a specific level or stage of education, irrespective of the students' ages. It is presented as a percentage—sometimes exceeding 100%—of the population within the officially recognized school-age group for that educational level or cycle. This indicator reflects how effectively the education system covers its target demographic. /

enrolment Rate (NER) was 21.8% for boys and 20.6% for girls, with the widest gender gap in Somali (GPI of 0.83). This indicates that girls are less likely to enroll at the official pre-primary age compared to boys.¹⁴²

Primary and middle school education

Primary education in Ethiopia covers grades 1-8, with a 2021 National Gross Enrollment Ratio (GER)¹⁴³ of 95.1%. Gender disparities are significant, particularly in pastoral regions like Somali and Afar, where the Gender Parity Index (GPI) is 0.76 and 0.83, respectively. Addis Ababa is the only region where female enrollment exceeds male enrollment. However, completion rates for grades 6 and 8 are lower for girls, who face higher dropout, delayed entry, and repetition rates compared to boys. In 2019/20, the national dropout rates for grades 1-8 was 15.2% for boys and 13.7% for girls.¹⁴⁴

Secondary education

In Ethiopia, secondary education covers grades 9-12, but enrollment remains low at 42.1% in 2020/21, compared to 95.1% for primary education. While this marks an improvement from 26.3% in 2014/15, access to secondary education is still limited for both girls and boys. However, girls continue to fall behind boys, with only 40.3% of girls enrolled in 2021, compared to 43.8% of boys.

Tertiary education

According to the Ethiopian Education Strategy Centre (ESC), of total students in 2020, women made up 35% of undergraduates, 22% of master's students, and 13% of PhD students in Ethiopian public universities. Though an improvement from 10 years ago, these numbers remain low, limiting women's access to higher education and better job prospects. Gender gaps are stark, with fields like engineering, having only 28% female undergraduates and just 9% at the PhD level.

Technical and Vocational training (TVET)

Female participation in TVET in Ethiopia has been comparable to that of males ranging between 51-54% over the past decade. However, women are more concentrated in lower levels (1 and 2) and underrepresented in higher levels (3 to 5). Enrolment is particularly low in regions like Afar, Gambella, Somali, and Harari, where access issues are more severe, especially for women. Many *woredas* lack TVET centres, making it difficult for women, particularly those from disadvantaged backgrounds, to attend. Despite significant government investment and a rise in institutions from 458 in 2013 to over 1,800 in 2023, access challenges continue to disproportionately affect women¹⁴⁵. In 2023, a policy shift was announced to enhance TVET quality and address the unique needs of various regions, with a particular focus on supporting female participation¹⁴⁶.

¹⁴² Federal Democratic Republic of Ethiopia, Ministry of Education. Education Statistics, Annual Abstract, 2020/21. <https://ecde.aau.edu.et/jspui/bitstream/123456789/276/1/MoE%2018%20ESAA%202013%20E.C%20%282020-2021%20G.C%29..pdf> ; UNICEF. 2022. Quality Pre-primary Education is the Basis of a Child's Journey. <https://www.unicef.org/ethiopia/stories/quality-pre-primary-education-basis-childs-journey#:~:text=In%20Ethiopia%2C%20pre%2Dprimary%20education,to%2044%20percent%20in%202022.> Federal Democratic Republic of Ethiopia, Ministry of Education. Education Statistics, Annual Abstract, 2020/21 <https://ecde.aau.edu.et/jspui/bitstream/123456789/276/1>

¹⁴³ Gross Enrolment Ratio (GER) measures the total number of children enrolled in a specific grade range, regardless of their age, expressed as a percentage of the school-age population. This indicator encompasses enrollment in both Alternative Basic Education (ABE) and traditional primary schools.

¹⁴⁴ Federal Democratic Republic of Ethiopia and UN WOMEN (2024) Ethiopia Country Gender Equality Profile 2024

¹⁴⁵ Ibid

¹⁴⁶ Cfr.: IPDR (2021) Hawassa University, Gender Audit of the Ethiopian Higher Education & Girum Abebe and Tsegay G Tekleselassie Chansathith Chaleu. (2022). "National TVET policies and systems in Ethiopia: Opportunities and issues in challenging times." International Labor Organization, 2022. https://www.ilo.org/wcmsp5/groups/public/---africa/---ro-abidjan/---sro-addis_ababa/documents/publication/wcms_863440.pdf

Education accessibility for people with disabilities

The Ethiopian Government has made strides toward equal educational access for students with disabilities, but challenges remain. Issues such as transportation difficulties, inaccessible school buildings, inadequate disability-friendly facilities, and insufficient learning materials persist. Consequently, the Gross Enrolment Rate (GER) for pre-primary students with disabilities is 1.5% for girls and 1.9% for boys. Across regions, enrolment rates for both sexes with disabilities are low, although boys tend to have slightly higher rates than girls. Addis Ababa shows better performance, with enrolment rates of 20.3% for boys and 12.8% for girls at the pre-primary level¹⁴⁷.

Control of Assets and Digital Divide

In Ethiopia, women's control over assets, including land, remains limited despite progressive legal frameworks promoting gender equality. While the Ethiopian Constitution and the Revised Family Code guarantee women equal rights to own and inherit land, traditional norms and practices often restrict women's access to and control over these resources. Women-headed households, particularly in rural areas, are more likely to face challenges in land registration and tenure security, which directly impacts their economic empowerment. Women's access to bank accounts is also less compared to men with 51% and 64% of women and men having access respectively (AFRO BAROMETER, 2024). Access to technologies like mobile phones is also marked by gender disparities, with fewer women owning or utilizing mobile devices and mobile internet services compared to men; 60% Vs 68 % as of 2023 creating a gender gap in mobile and internet usage (AFRO BAROMETER, 2024). However, relative progress in rural phone ownership with approximately two-thirds of rural households owning a mobile phone has been recorded (International Food Policy Research Institute, 2023). Despite this, sex disaggregated mobile phone ownership is not equally spread across the four major regions of Ethiopia. i.e Tigray, Oromia, Amhara and Southern Nations Nationalities and People's Region (SNNP) as per data as of 2019 according to IFPRI. Rates of ownership are growing quickly for heads of household in Tigray, Oromia and SNNP, with rates of male heads slightly lagging in the Amhara region. This digital divide hinders women's ability to access critical information, financial services, and opportunities for entrepreneurship. Furthermore, women's agency—their ability to make independent decisions and influence their socio-economic conditions—is constrained by societal norms that often prioritize male decision-making. While national policies and programs, such as gender-responsive agricultural and digital inclusion initiatives, aim to enhance women's agency and access to resources, significant gaps persist in bridging these inequalities.

Employment and Labor Participation

The 2021 Ethiopian Labor & Migration survey reveals significant gender disparities in labour market participation, with only 56.8% of women involved compared to 72.6% of men. When employed, women are more likely to work in the informal sector (28.6%) compared to men (15.7%), which limits their access to benefits like pensions and insurance¹⁴⁸. Additionally, a substantial portion of women (45.3%) engage in unpaid family work, compared to 30.4% of men. Women predominantly hold lower-level jobs, particularly in agriculture (41.3%) and elementary employment that consists of routine tasks typically requiring minimal skills, training, or education (35.5%)¹⁴⁹. Unemployment rates also highlight gender inequality, with 11.7% of women versus 5% of men unemployed in 2021. The COVID-19 pandemic disproportionately impacted female-dominated sectors.

¹⁴⁷Federal Democratic Republic of Ethiopia and UN WOMEN (2024) Abridged version Ethiopia Country Gender Equality Profile Brief

¹⁴⁸ Ethiopian Statistics Services. (2022). Statistical Report on the 2022, 2nd Round Urban Employment Survey

¹⁴⁹ Federal Democratic Republic of Ethiopia, Ministry of Planning and Development (2021) Statistical Report on the 2021 Labor Force and Migration Survey. Statistical Bulletin <http://www.statsethiopia.gov.et/wp-content/uploads/2022/01/THE-2021-LABOUR-FORCE-AND-MIGRATION-SURVEY-STATISTICAL-REPORT-1.pdf>

Representation and Participation in Decision Making

Women's representation in political and decision-making processes is improving, with recent efforts to ensure gender-balanced leadership in public offices. However, women still face cultural barriers that limit their participation at both local and national levels and significant leadership responsibilities continue to be dominated by men in both private and public sectors¹⁵⁰.

Meanwhile, progress in gender representation in Ethiopia's government has been notable, with women achieving parity in ministerial appointments in 2018 (AGI 1.021). Women's representation in line ministries rose from 13% in 2015 to 50% in 2018, highlighted by the appointment of a woman as Head of State, Supreme Court president, and Attorney General. However, by October 2021, women held only 36% of ministerial positions, a decline from 2018, with most now in less influential roles¹⁵¹. Systemic barriers continue to hinder women's participation in decision-making and leadership, with only 29% of state minister positions held by women. Regional disparities are significant, with female representation as low as 7% in some areas. Addressing systemic barriers and enhancing women's leadership in both public and private sectors remains crucial for fostering gender equality¹⁵².

Gender-Based Violence and harmful practices

Ethiopia has implemented various laws and policies to combat sexual violence, domestic abuse, and harmful practices. The Criminal Code addresses crimes like rape and domestic violence, while the country has ratified international agreements such as the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), the Protocol on the African Charter on Human and People's Rights on the Rights of Women in Africa (Maputo Protocol) and Beijing Declaration and Platform for Action that protect individuals from gender-based violence.

Despite these efforts, significant gaps remain in legal enforcement. For instance, the Revised Criminal Code does not recognize marital rape, and workplace harassment is inadequately addressed. Additionally, psychological violence in marital settings is not covered¹⁵³.

The 2016 Ethiopian Demographic and Health Survey (EDHS) found that 23% of women experienced physical violence and 10% experienced sexual violence in their lifetime¹⁵⁴. The 2022 humanitarian needs overview indicated an increase in individuals needing gender-based violence (GBV) support, rising from 3.5 million in 2021 to 5.8 million, primarily in conflict- and disaster-affected regions¹⁵⁵. Additionally, 63% of women surveyed aged 15-49 believe it's justifiable to beat a woman for violating social roles (such as burning food, arguing with her husband, leaving the house without permission, refusing intercourse and neglecting the children), demonstrating the need for awareness raising interventions to help women understand their rights¹⁵⁶. Violence in schools is the other facet of GBV where school children face at least one form of violence. One study of violence against girls in primary and secondary schools in nine regions of Ethiopia found that 46% of students spoke of sexual

¹⁵⁰The fact that significant leadership responsibilities continue to be dominated by men is reflected in Ethiopia's AGI score for empowerment and representation which stands at 0.247. The African Gender Index (AGI) in the area of Empowerment and Representation assesses the presence of women in leadership and decision-making positions within government and the private sector. It evaluates the comparative numbers of women and men serving in parliamentary and cabinet roles, as well as the percentage of businesses with female managers and the ratio of female to male managers, professionals, and technicians., African Development Bank (2020). Africa Gender Index report https://www.afdb.org/sites/default/files/documents/publications/africa_gender_index_report_2019_-_analytical_rep

¹⁵¹ Federal Democratic Republic of Ethiopia and UN Women. (2024) Ethiopia Country Gender Equality Profile 2024

¹⁵² Federal Democratic Republic of Ethiopia and UN Women. (2020) Status of Women in Leadership Positions in the Legislative, Executive and Judicial Branches of the Government of Ethiopia.

¹⁵³ Ibid.

¹⁵⁴ Federal Democratic Republic of Ethiopia and UN Women. (2024) Ethiopia Country Gender Equality Profile 2024

¹⁵⁵ Gender Based Violence AOR. (2022) Situation of GBV in Ethiopia. <https://reliefweb.int/report/ethiopia/gbv-aor-ethiopia-situation-gbv-ethiopia-september-2022>

¹⁵⁶ Federal Democratic Republic of Ethiopia and UN Women. (2020) Status of Women in Leadership Positions in the Legislative, Executive and Judicial Branches of the Government of Ethiopia, p.48

harassment, degrading treatment and sexual attacks, most frequently on the journey to and from school, but also in school compounds, with perpetrators including male students and older men (Save the Children et al., 2008).

In response, the government has made progress in working on a national policy for the prevention and response to Gender-Based Violence (GBV). This policy seeks to safeguard vulnerable groups and uphold their human rights. The policy will provide a comprehensive framework from which various laws, strategies, and guidelines can be developed.¹⁵⁷ The current development stands at MoWSA working with the Ministry of Planning and Development to merge Gender Equality GBV and Economic Empowerment in one Gender Equality and Women Empowerment Policy rather than being endorsed as separate Policies. The timeline for the endorsement of the Gender Equality and Empowerment Policy is however not clear.

Harmful Traditional Practices (HTPs) in Ethiopia mainly result from social, cultural and religious norms. Several national policies recognize Harmful Traditional Practices as detrimental to women's and girls' rights. The 1993 National Policy on Ethiopian Women aims to eliminate these practices, supported by the National Strategy for the Elimination of HTP and the roadmap to end Female Genital Mutilation (FGM) and child marriage (2020-2024)¹⁵⁸. In Ethiopia, 25 million women and girls have undergone FGM, the highest in Eastern and Southern Africa, with 65% of women aged 15-49 affected. While FGM prevalence has decreased from nearly 90% in 1970 to 50% in 2000 among adolescents, it remains a significant issue driven by entrenched social and cultural norms.¹⁵⁹ Social norms stigmatizing premarital sexual activity and cultural beliefs controlling female sexuality perpetuate child marriage and FGM¹⁶⁰.

Despite the government's commitment to end child marriage by 2025, the 2016 EDHS found that 40.3% of women aged 20-24 were married before 18, with 14.1% married before 15.

Gender Policies and Legal Framework

Over the past 30 years, the Ethiopian government has prioritized gender equality in its development efforts, introducing various laws, policies and reforms, such as the National Action Plan for Gender Equality, to support this goal. However, implementation remains a challenge, especially in remote areas, due to deep-rooted social norms hinder progress.

Ethiopia has advanced its international and regional gender commitments through national laws and policies:

- **National Policy on Women (NPW):** In 1993, Ethiopia implemented the National Policy on Women (NPW). It promotes gender-sensitive structures, ensuring equal political, economic, and social rights. Currently, the government is finalizing a new National Policy on Gender Equality and Women's Empowerment (NP-GEWE), which will eventually replace the NPW. This updated policy is being developed through a collaborative process led by the Ministry of Women and Social Affairs.

¹⁵⁷ Federal Democratic Republic of Ethiopia and UN Women. (2024) Ethiopia Country Gender Equality Profile 2024
UN Women (2022) UN supporting development of National Policy and Strategy on GBV Prevention and Response in Ethiopia.
<https://ethiopia.un.org/en/213638-un-supporting-development-national-policy-and-strategy-gbv-prevention-and-response-ethiopia>

¹⁵⁸ UN Women, UNAIDS, Federal HIV/AIDS Prevention and Control Office (FHAPCO) (2020) Gender Assessment of the National HIV Response in Ethiopia. <https://africa.unwomen.org/sites/default/files/2022-02/ethiopia%20gender%20assessment%20of%20the%20national%20hiv%20response.pdf>

¹⁵⁹ Federal Democratic Republic of Ethiopia and UN Women. (2024) Ethiopia Country Gender Equality Profile 2024.

¹⁶⁰ For more information regarding social norms that are also diving to harmful practices see also International Bank for reconstruction and Development (2019) Ethiopia Gender Diagnostic Report

- **National Action Plan for Gender Equality (NAP-GE):** Ethiopia's NAP-GE serves as a blueprint for integrating gender considerations across sectors, fostering gender-sensitive planning and implementation. The Plan aligns with Ethiopia's Ten-Year Development Plan and Growth and Transformation Plan (GTP), emphasizing the inclusion of women in economic and leadership roles.
- **Constitution (1994):** Guarantees gender equality, education, and affirmative action for women and girls.
- **Family Code (2000):** Establishes equality between men and women, raises the legal marriage age to 18, and grants women economic rights in long-term partnerships. However, some regions (Afar, Somali) have not fully aligned.
- **Criminal Code (2005):** Criminalizes domestic violence, rape, FGM, abduction, and early marriage.
- **Other Legal Reforms:**
 - Prohibition of sexual harassment (2017)
 - Protection for overseas workers (2016)
 - Free legal services for women (2016)
 - Gender budgeting in financial administration (2016)
- **Ten-Year Development Plan:** Promotes women's participation in education, leadership, and the economy.
- **Education Policies:** Aim to increase access for girls and women, including free education up to grade eight and gender-sensitive initiatives.
- **Health Policies:** Focus on maternal and child health, expanding reproductive health services.
- **Economic Programs:** Support women's entrepreneurship through microfinance and economic participation.
- **Disability Inclusion:** Ensures equal rights and employment for persons with disabilities, with a focus on women.¹⁶¹
- **Growth and Transformation Plan I and II:** GTP I aimed to mainstream women and youth agendas in the national development and governance strategy of the country. GTP II requires redoubling of efforts to address challenges around economic and social benefits for women.
- **Gender-responsive budgeting (GRB)** is a policy and planning approach introduced in Ethiopia in 2016 as part of broader reforms to ensure that national and local government budgets address gender disparities. GRB integrates gender considerations into the allocation of public resources, aiming to promote gender equality and women's empowerment through targeted investments in programs and services that benefit women and marginalized groups.

International Commitments

Ethiopia is a signatory to several international conventions promoting gender equality, including:

- CEDAW (Convention on the Elimination of All Forms of Discrimination Against Women)
- Beijing Platform for Action
- Maputo Protocol (African Charter on Human and People's Rights on Women's Rights) with reservations on marriage and inheritance
- Maya Declaration on economic rights
- Sustainable Development Goals (SDG 5) on gender equality

More specific to the context of disasters, Ethiopia has endorsed the **Sendai and the Hyogo Frameworks** both of which are applicable in disaster situations. These frameworks have addressed the importance of giving due recognition to the gender dimension of disasters and highlights on the need

¹⁶¹ For more info see: Federal Democratic Republic of Ethiopia and UN WOMEN (2024) Abridged version Ethiopia Country Gender Equality Profile Brief/ Federal Democratic Republic of Ethiopia and UN Women. (2024) Ethiopia Country Gender Equality Profile 2024

for a gender perspective to be integrated into all disaster risk management policies, plans and decision-making processes, including those related to risk assessment, early warning, information management, and education and training. Ethiopia monitors and reports on its compliance to the principles and stipulations in the frameworks. At a national level, the **Strategic Program Investment Framework (SPIF)** is a strategic framework for the prioritization and planning of investments that will drive Ethiopia's Disaster Risk Management (DRM) system. It recognizes the differential impacts of disaster on women and men and highlights gender-based risks and vulnerabilities to be identified as one of its objectives under disaster risk profile.

Ethiopia has aligned these commitments with its Ten-Year Development Plan (2021-2030), integrating gender equality across all sectors. Additionally, Ethiopia supports equal education rights through international treaties like the Universal Declaration of Human Rights, Beijing+, SDGs, AU 2063, and the African Charter on the Rights and Welfare of the Child. These frameworks ensure free and equal access to education, with special measures for girls.

Gender-Related Institutional Arrangements

Several institutions in Ethiopia work to promote gender equality and women's empowerment. The Ministry of Women and Social Affairs (MoWSA), established in 2014, leads these efforts, coordinating gender policies across various ministries through Gender Directorates. These offices operate from the federal level down to local Kebele units, reflecting the government's commitment to gender parity. MoWSA oversees the National Action Plan on Gender Equality and advocates for women's rights.

The Annual Women's Conference brings together stakeholders from government, UN agencies, NGOs, and the private sector to discuss progress and challenges in gender equality. It also includes a forum for women parliamentarians to address gender priorities and integrate them into national policies.

Various alliances and working groups, such as those focused on ending child marriage, FGM, and violence against women, collaborate to improve the status of women and girls. The Gender Working Group comprised of government representatives, development partners, and civil society organizations focuses on aligning national gender initiatives with international frameworks like CEDAW and the Sustainable Development Goals (SDGs). Community-level initiatives like the Women's Development Army (WDA) and professional women's associations play a crucial role in advocacy, training, and holding the government accountable for gender equality. Women's Associations and Cooperatives support capacity-building and economic empowerment initiatives. These groups provide training, financial literacy programs, and access to microfinance, enabling women to participate more actively in decision-making and disaster preparedness activities. Community-Based Organizations (CBOs) similarly, often in partnership with NGOs, facilitate targeted interventions for vulnerable women, including survivors of GBV, women-headed households, and persons with disabilities.

Gender, Climate Change and Early Warning Systems and disasters response

Ethiopia is highly vulnerable to climate change, facing frequent droughts, floods, crop failures, and locust infestations. These disasters disproportionately impact women and vulnerable groups. Women, especially in rural areas, face challenges in accessing resources, technology, and information, which limits their ability to adapt to climate change. They often rely on agriculture and forest products, both highly susceptible to climate shocks. Displaced women and girls face food insecurity, meaningful access to services including healthcare, shelter, water and sanitation and are also susceptible to risks of gender-based violence. Furthermore, women have limited access to agricultural inputs like seeds, fertilizers, and irrigation systems, which undermines their resilience. Male-headed households generally focus on farming adaptations, while female-headed households lean towards off-farm

income diversification. Women and girls also bear the burden of collecting firewood, often spending hundreds of hours annually, exacerbating deforestation and limiting their time for education or employment.

In addition, climate-related disasters increase health risks for women, particularly from waterborne diseases due to inadequate sanitation during floods. Traditional gender roles, coupled with the impact of disasters, lead to malnutrition, mental health issues, and physical strain on women and girls. Despite these challenges, Ethiopia has made progress with policies like the Climate Resilient Green Economy (CRGE) strategy, which aims to integrate gender-responsive climate action and move towards a low-carbon economy. However, women's involvement in decision-making remains limited due to persistent gender norms.

Community consultations undertaken within the framework of the EW4All project design also attest to the fact women are disproportionately impacted by disasters. The consultations conducted in Tigray, Oromia, Southern Nations, Nationalities, and Peoples' Region (SNNP), and Amhara, consistently highlighted the intersection of gender and climate change impacts. Women, particularly in rural settings, face heightened vulnerability due to their traditional roles in household management and agriculture. In regions like Tigray, the impacts of climate change—manifested through erratic rainfall patterns and severe droughts—have forced women to shift away from agriculture to alternative income-generating activities, often without adequate support. Their dependence on natural resources for household energy and food production further exacerbates their susceptibility to climate-induced risks.

Climate change also disproportionately affects women-headed households. The economic insecurity faced by these households is intensified by limited access to financial services, land, agricultural inputs and resources including climate-resilient technologies, making them particularly vulnerable to food insecurity. Platforms like the Emergency Coordination Center in Tigray offer some opportunities for gender-responsive planning; however, resource constraints still hinder comprehensive support.

In Oromia and SNNP, grassroots organizations like self-help groups (SHGs) and women's development networks provide some support to their members, but their reach and capacity remain constrained by socio-cultural barriers and resource limitations.

In relation to Gender-Responsive Early Warning Systems, across all the regions where consultations for the EW4All project took place, the importance of gender-responsive early warning systems was emphasized. Current systems often fail to consider the specific needs of women, persons with disabilities (PWDs), and the elderly, who face barriers to accessing early warning information. For instance, in Tigray, the destruction of community-based early warning networks due to conflict has left many women without critical information. Similarly, in Oromia, logistical challenges hinder the inclusivity and reach of these systems.

Efforts to integrate gender-sensitive practices into early warning systems have been made in SNNP and Amhara. For example, the revision of woreda risk profiles that includes gender-specific indicators is an opportunity to gather gender responsive data in all regions. Amhara's impact-based early warning roadmap aims to address the unique vulnerabilities of women and marginalized groups by mainstreaming gender in MHEWS. However, the absence of comprehensive data on gendered impacts of past disasters hampers the development of truly responsive systems.

Disaster response strategies in Ethiopia are gradually incorporating gender-sensitive frameworks. The Ethiopian Red Cross Society (ERCS), a key stakeholder, emphasizes gender diversity and inclusion in its operations, targeting women and children in 35% of its disaster response efforts. ERCS also conducts

vulnerability and capacity assessments that include gender considerations, enabling the development of targeted interventions for women and marginalized communities.

However, challenges persist. In Amhara, ongoing conflicts have disrupted traditional response mechanisms, disproportionately impacting women and the elderly among internally displaced persons (IDPs). In Hawzen Woreda of Tigray, the absence of reliable early warning and disaster response systems has left women vulnerable to gender-based violence (GBV) and economic destitution. This is attributed to the weakened collective resilience, community networks and limited women's access to resources. The compounded effect of all has direct impact on vulnerable women's exposure to GBV and exploitation. The youth in Tigray also face challenges related to climate change, including reduced agricultural productivity due to land degradation and irregular rainfall. These barriers discourage agricultural involvement, leading many young people to migrate to urban areas, onward migration or pursue less sustainable livelihoods, which negatively impacts both men and women. In Oromia, political conflicts further complicate disaster response efforts, leaving marginalized communities, including women, without adequate support. Additionally, the settlement of some communities in rural locations bordering rivers expose them to flooding and the resultant impacts. Their cultural beliefs however demand they stay on their lands because rivers overflowing is a symbol of abundance and prosperity. The government has not yet mapped out such settlements along river lines due to resource limitations.

In **SNNP** and **Amhara**, women, elderly individuals, and people with disabilities face particular challenges during recurring hazards like drought and floods. In SNNP, women's caregiving roles and social norms, such as eating last, increase their vulnerability in times of food shortages. SNNP also has high drought exposure, affecting 55% of woredas. Additionally, women in flood-prone areas are often unaware of early warning information, heightening their risk. Similarly, in Amhara, drought, floods, and ongoing conflict have intensified the hardships of women and other marginalized groups, especially among nearly one million internally displaced persons (IDPs), who are largely composed of women, elderly, and disabled individuals.

Some of the opportunities for gender-responsive and inclusive systems include capacity building programs targeting women and marginalized groups, which can empower them to participate actively in disaster preparedness and response. This can include technical skills training for managing early warning systems and leadership development initiatives. Furthermore, community-based mechanisms like self-help groups, women development associations, development agents and extension workers widely exist in rural settings. Strengthening these grassroots networks can enhance the flow of early warning information and foster economic resilience. Integrating these with formal disaster management frameworks would enhance gender responsive and inclusive early warning systems and disaster preparedness and response measures.

Women's Representation and Participation in DRM Sectors

Women are disproportionately affected by disasters, and their involvement in disaster risk management (DRM) is increasing. While Ethiopia is working on gender mainstreaming in disaster preparedness and response, capacity-building efforts are still needed to integrate gender into all levels of disaster risk management.

In Ethiopia, women's status within the society has made it challenging for them to acquire the necessary knowledge, information and resources to have agency. This affects their meaningful participation in community engagement, including disaster preparedness and mitigation plans and response and resilience processes. Women often bear the brunt of caregiving responsibilities, making them more vulnerable to displacement and loss of livelihood in crises. Without adequate representation, DRM strategies risk overlooking the unique needs and challenges that women face

during emergencies, such as access to reproductive health services or safe spaces for shelter. However, some changes have been recorded where women participate in DRM. Institutional reforms have ensured the participation of women in leadership positions of DRM undertakings through the institutionalization of gender departments that oversee the gender responsiveness of the different levels of project implementation. To this effect Pillar leads of the EW4All project have highlighted the importance for gender to be mainstreamed across the full cycle of DRM and response. While this is a good start into having a proper gender responsive process, there is the risk of missing the unique issues of gender and the vulnerabilities of women within the overall DRM cycle. This is because gender responsiveness may at times seem to only focus on numbers without actually understanding the differential impact of disasters and the tailored response to address that.

Women's perspectives are not included in DRM data collection, compilation, and analysis at the kebele or woreda levels. While there are some efforts in gender equity in data collection and several indicators, this does not necessarily influence if or how stakeholders prioritize gender needs or use gender-sensitive data in mitigation, action planning, or decision-making at the kebele or woreda levels. This is likely exacerbated by the limited number and voice of women in community planning and committees (USAID, 2019).

A comparative analysis of the location specific consultations for the EW4All project indicate that across all regions, women's representation in leadership roles within DRM structures remains limited. In Tigray, conflict and displacement have reversed many gains previously made in women's empowerment. Before the conflict, women in Tigray were active participants in development and decision-making platforms, including community-based disaster preparedness initiatives. However, the war has not only disrupted these structures but also led to a disproportionate burden on women, particularly as household heads managing economic and caregiving responsibilities. The number of female-headed households has substantially increased as a result of the war and they are left with little capacity to engage in decision-making processes.

In Oromia, cultural norms continue to restrict women's participation in DRM leadership roles. Women are often excluded from decision-making platforms, with leadership positions primarily occupied by men. However, there are promising examples of grassroots initiatives, such as women-led self-help groups (SHGs), which have started advocating for women's inclusion in local DRM committees. These groups offer a potential entry point for broader participation if adequately supported. In both Tigray and Oromia, women face significant resource limitations, including lack of access to transportation, information, and financial support, which restricts their ability to participate actively in DRM activities. Amhara region demonstrates a mixed picture. While the regional DRM offices have attempted to integrate women into local disaster preparedness committees, ongoing conflicts have marginalized their voices further. Ongoing conflicts in Amhara have disproportionately affected women, leading to their marginalization from DRM processes. Displaced women face additional challenges, such as lack of access to early warning information and vulnerability to gender-based violence (GBV), which further isolates them from DRM structures.

In SNNP, the engagement of women in DRM structures is relatively stronger in urban areas, where education and awareness levels are relatively higher. However, in rural areas, traditional social norms hinder their participation. Women often assume primary caregiving roles and are at greater risk during disasters, especially in drought-prone or flood-prone areas. Women in some kebeles in SNNP are the ones who cultivate land that is made fertile by rivers and flood. With limited access to early warning information women have been swept away by sudden and raging floods while working on their farms. Cultural norms such as women and children eating last in households, exacerbate their vulnerability during droughts food shortages also causing malnutrition in children.

Nevertheless, challenges persist in advancing women to leadership roles within DRM sectors. Leadership, coordination, and technical training opportunities are more accessible to men, often leaving women underrepresented at decision-making tables. Limited access to communication tools, such as mobile phones, TV and radios is the other significant challenge that contribute to the disproportionate impact of climate change on women in rural areas.

The ERCS, for instance, has acknowledged this imbalance and emphasized the need for targeted initiatives to empower women through leadership programs and tailored disaster management training. Overall, because of their roles within their households and communities, women have strong ties with the environment and natural resources. This should put them at the forefront in disaster preparedness, response and resilience discussions and engagements. The consultations identified that although DRM policies in Ethiopia promote gender equality, real-world execution is inconsistent. This gap suggests that existing frameworks and protocols, while theoretically inclusive and show the general intent, lack rigorous enforcement, especially at local levels which are attributed to logistical, resource and cultural barriers. Thus, women's representation in decision-making roles, within the broader governmental DRM structures, remains low, which restricts their influence in planning and response activities. By including women in community DRM teams, the EW4All initiative could tap into women's central role in community networks for the attainment of gender responsive early warning dissemination and higher responsiveness to alerts.

Gender in Early Warning Systems and Disaster Response

Ethiopia's DRM policies have evolved over the years, with the inclusion of gender as an important aspect in national frameworks such as the 2013 *National Policy and Strategy on Disaster Risk Management* ¹⁶², which serves as the key document guiding disaster risk management in Ethiopia. While it acknowledges the importance of gender sensitivity, it lacks specific measures to address gender-based violence (GBV) risks, offering no clear actions for prevention, mitigation, or response. The policy recognizes gender as a cross-cutting issue, noting that women and other marginalized groups are disproportionately affected by disasters. It emphasizes the need for DRM laws, programs, and plans to "give special attention to women" and other vulnerable groups. Additionally, it states that disaster response efforts should be accessible to pregnant and lactating women, along with other marginalized populations. A revision of this policy is currently in progress¹⁶³.

Ethiopian policies emphasize the importance of gender-sensitive approaches to DRM. The government and various international organizations are working towards enhancing women's participation, especially in decision-making roles within local communities. However, substantial work is still needed to ensure that women are actively involved in planning and implementing disaster preparedness and response measures. Although progress has been made in integrating gender into early warning systems, women still face barriers to accessing disaster-related information and technologies, especially in rural areas. Efforts are underway to ensure that women have equal access to early warning technologies. However, women's access to early warning information and technologies is hindered by several factors, including literacy barriers, limited access to technology, and socio-cultural norms. Many women, especially in rural areas, do not have access to mobile phones, radios, or other technology used for disseminating early warning messages. This limits their ability to receive timely and accurate information about impending disasters.

¹⁶²The National Policy and Strategy on Disaster Risk Management is a sectoral national strategic document applicable to the period of 2013-2023. Its main objective is to reduce disaster risks and potential damage caused by a disaster through establishing a comprehensive and coordinated disaster risk management system in the context of sustainable development. [Ethiopia: National Policy and Strategy on Disaster Risk Management | Refworld](#)

[national-policy-and-strategy-on-disaster-risk-management_9401f39d859678e1dfc6ea95493f2a01.pdf \(climatepolicyradar.org\)](#)

¹⁶³ IFRC and UNICEF (2021) Gender responsive disaster risk management-An analysis of gender integration in national and regional disaster risk management frameworks in Eastern and Southern Africa.

Challenges to women's participation in Early Warnings and DRM include:

- Limited access to education and training in DRM.
- Cultural and societal norms that restrict women's roles in decision-making.
- Lack of policies to enforce women's participation in leadership roles in DRM.
- Illiteracy
- Limited access to technology
- Limited access to financial services e.g. banks and credit

The status of women in early warning systems (EWS) across Ethiopia reveals significant gaps in representation, participation, and access to information. The consultation findings highlight a stark disparity in how women engage with EWS across Tigray, Oromia, SNNP, and Amhara, shaped by cultural, logistical, and structural barriers. While some progress has been made in specific regions, systemic challenges continue to undermine women's effective inclusion and leadership in early warning mechanisms.

In Tigray with many women overwhelmed by caregiving responsibilities, and many other in displacement, their engagement in early warning activities has dwindled. For instance, in Hawzen Woreda, the destruction of early warning networks and infrastructure has left women reliant on informal channels or indigenous knowledge systems, which are inconsistent and often exclude their perspectives. Prior to the conflict, women were active participants in community-based early warning structures. Women's development groups played a pivotal role in disseminating early warning information related to droughts and floods.

In **Oromia**, early warning dissemination is primarily carried out through government structures at the kebele level and local media. The Oromia BoWSA, although it exists down to the woreda level, the structure has not been operational at Kebele level until very recently where its functionality is being tested through piloting. This has its own limitations on early warning dissemination at community level. Cultural norms also restrict women's mobility and access to public spaces where early warning information is shared. The consultations also highlighted that women in pastoralist communities, where mobility is essential for survival, often miss early warnings disseminated through male-dominated forums. While self-help groups in Oromia have started integrating early warning education into their activities, these efforts are not yet widespread or well-resourced.

In SNNP, there are different groups that are functioning to enhance women's engagement in DRM with the support of governments. However, these networks are not as supportive as women's participation is limited because of cultural impositions. In rural areas, the lack of gender-sensitive communication tools—such as messages tailored to women's literacy levels or schedules—limits their access to timely information. The integration of women into woreda-level disaster risk committees has shown potential, but it is still nascent and faces resistance from traditional community leaders. There are different capacity building projects that are designed to empower women in DRM. But their effectiveness needs to be measured. Similarly, in Amhara, women's participation in early warning systems is constrained by ongoing conflicts.

The Ethiopian Red Cross Society (ERCS) has made strides in incorporating women into community-based DRM teams, where their insights contribute to better resource allocation and community engagement. Women on the ground often have a keen awareness of local resources and vulnerable individuals, and their involvement in disaster response and early warning can lead to quicker, more targeted relief efforts. Women's representation at community levels not only improves the reach and effectiveness of disaster warnings but also builds a sense of ownership and trust among community members.

Areas for Improvement in Early Warning, DRM and Gender Inclusiveness

Based on the information reviewed, and findings from consultations, the following general recommendations are made with regards to gender inclusion in EW and DRM. To the extent possible, this will be taken into consideration in the EW4All initiative based on its activities:

Enhance Gender-Sensitive Strategies: Develop and implement tailored engagement and training programs for women and vulnerable groups to ensure their active participation in disaster preparedness and response initiatives.

Improve Community Engagement: Enhance the role of community organizations in implementing culturally appropriate communication and dissemination strategies, ensuring that all community members receive timely and relevant information. It is therefore important to involve local governments, leaders, and women's organizations in the communities in the design and implementation of early warnings.

Finalize Legal Frameworks: Advocate for the swift endorsement of the DRM Policy Enforcement Document to solidify accountability mechanisms across all sectors involved in gender responsive disaster risk management.

Policy Review and Reform: Government should review national DRM policies to ensure they are gender-responsive and align with international frameworks such as the Sendai Framework for Disaster Risk Reduction (2015-2030). This includes strengthening gender-responsive disaster laws and regulations.

Gender Responsive Communication: Communication channels including media need to refine their approach, integrating gender-specific needs and stakeholder feedback into their communication strategy. Any measure to disseminate information must include facilitating access to the technology that will be used to receive alerts and real-time updates. This will enhance the capacity to disseminate early warning information as a nation that is not only accessible but also impactful.

Improve Access to Technologies: Improve access to technologies, such as mobile phones, for the most vulnerable and marginalized groups including women, elderly and persons with disabilities by providing subsidies. At the same time, invest in literacy and digital skills training to ensure comprehensive support. These efforts will complement and enhance each other, empowering vulnerable communities more effectively.

Revitalize or Strengthen Community Structures: Rebuilding/strengthening women's development groups, youth associations, and community-based organizations is crucial to strengthening social cohesion, information flow, and economic resilience.

Enhance Access to Financial Services: Providing credit, cash support, and micro-financing options can empower women and youth to re-engage in income-generating activities, from small businesses to livestock rearing. This will help women to become resilient to shocks and be in a better place to translate early warning information and take the necessary preparedness measures.

Enhance Gender-Responsive Early Warning Systems: Strengthening early warning systems to reach all communities, especially vulnerable groups, is essential. This includes expanding the logistical support and communication tools to improve access to remote areas. In line with this, needs assessments and the communication of alerts must be in clear and accessible language, including local languages and adapted to local culture. This may include the use of verbal messaging in parallel with the written based alerts to overcome literacy barriers.

Support Mental Health Services: Addressing psychological trauma and social isolation is essential for long-term recovery, especially for women and youth facing displacement, loss of livelihoods, and increased responsibilities.

Promote Gender-Specific Data Collection: Assessment of the needs of women, developing gender-specific indicators and reporting mechanisms within early warning systems will provide insights into the distinct needs of women, children, and other vulnerable groups, ensuring that responses are tailored and effective. Similarly, gender-disaggregated data, is crucial for effective planning, response and DRR capacity development

Foster Collaboration with Civil Society: Tap into existing local and organizational capacities to build upon already existing initiatives, understand local dynamics and vulnerabilities to hazards to build trust among communities and facilitate ownership of early warning and DRM processes.

Strengthen Community Feedback Mechanisms: Creating platforms for community feedback on disaster response initiatives can help authorities adjust their strategies based on real-time challenges and needs, promoting more inclusive and responsive systems.

Capacity Building for Local Organizations: Supporting women's associations, disability organizations, and youth groups with training and resources will increase their ability to contribute meaningfully to disaster risk reduction and climate adaptation initiatives. Groups of women who have active roles within their communities, such as heads of households, or those actors who want to participate actively in the development of project-related activities should be empowered.

Capacity Building for Government Institutions: Government staff must be provided with targeted capacity-building initiatives that focus on understanding and integrating gender and inclusive considerations in every aspect of EWS. Government should prioritize gender-sensitive training, inclusive decision making and strengthening gender equality in leadership.

Adequate Funding for Gender-Responsive Disaster Management: Government with the support of its partners to allocate dedicated funding stream for gender-responsive disaster risk reduction initiatives, including training, research, and program implementation that target both women and men equally.

Monitoring and accountability: Government with the support of its partners need to implement robust mechanisms to track and ensure that allocated funds are being used effectively to promote gender equality in disaster management.

Additionally, it is important to ensure the following are taken into account:

- Training and awareness workshops on interpreting early warnings and response measures should be adapted to the needs and available time of women for their participation.
- Comprehensive review and action plan for Emergency Warning (EW) systems should provide analysis and strategies to include inclusivity and to address the digital gender gaps.
- Provision of clear and actionable information tailored to different groups' needs.

8 Gender Assessment for Antigua and Barbuda

The estimated population of Antigua & Barbuda as of 2024 is 102,634¹⁶⁴, of which 48,311 people are male (47.07 %) and 54,323 are female (52.93 %). Approximately 76% of Antigua and Barbuda's population lives in rural areas and 97% lives in the island of Antigua; 60% of its population resides along the coast.

Antigua and Barbuda has made significant strides over the past decades on issues affecting women and girls and gender equality, particularly in education and health, where women outperform men. Both girls and boys in Antigua and Barbuda have a very high completion rate of lower secondary school¹⁶⁵. There has also been a positive trend in maternal health. In fact, the maternal mortality ratio in Antigua and Barbuda has significantly improved, decreasing from 51 in 2000 to 21 in 2020. This rate is notably lower than the regional average, which is 88¹⁶⁶.

However, challenges persist, including the prevalence of gender-based violence (GBV)¹⁶⁷ and a significant proportion of vulnerable single female-headed households, which reflects underlying economic vulnerabilities. In fact, approximately half of the households in Antigua and Barbuda are led by women, many of whom are single parents responsible for dependents such as the elderly and individuals with disabilities¹⁶⁸. This reflects a significant demographic where women often assume the role of primary household managers, especially in single-parent families, which are common in the country. The burden of providing for and caring for dependents, combined with challenges women face in accessing economic opportunities, can therefore increase the vulnerability of female-headed households (FHHs) in Antigua and Barbuda, especially during natural disasters¹⁶⁹. This was evident in 2017, during Hurricane Irma, when it was estimated that at least half of the affected households were FHHs, which were larger in size than male-headed households and may have experienced significant damage due to their particular circumstances.¹⁷⁰

In addition to these challenges, women continue to face inequalities in wages and access to economic opportunities, signaling the need for sustained efforts to address these gaps¹⁷¹. These issues are further exacerbated by the absence of legal provisions for parental leave or protections against the dismissal of pregnant workers. While specific data on time use is missing, unpaid caregiving responsibilities are largely borne by women¹⁷². Furthermore, despite a decline, Antigua and Barbuda's adolescent fertility rate in 2022 was 32.3, still notably higher than the average of 11.2 for high-income countries¹⁷³.

Turning to political participation, only 5.6% of seats in national parliament were held by women in 2023, marking a decline from 11.1% in 2022. The current rate is lower than the average rate in high income economies and the lowest in the country since 2004, when it was 10.5%¹⁷⁴.

An additional aspect that demands attention is the lack of gender-related data in several key areas, which hampers efforts to bridge the gaps between men and women, as well as boys and girls. For instance, the breakdown by sex of people with disabilities as well as the disability prevalence itself is

¹⁶⁴ [Antigua and Barbuda - The World Factbook \(cia.gov\)](https://data.worldbank.org/locations/locations?locations=LA)

¹⁶⁵ <https://genderdata.worldbank.org/en/economies/antigua-and-barbuda>

¹⁶⁶ Ibid.

¹⁶⁷ Recent data from 2023 shows that 16% of women reported having experienced intimate partner violence in the past year. Cfr: Commonwealth of Learning (2023). Gender Country Profile 2023. Antigua and Barbuda [content \(col.org\)](#)

¹⁶⁸ The 2014 CARIBANK Country Gender Assessment indicated that approximately 47.4% of households are headed by women.

¹⁶⁹ Directorate of Gender Affairs/DoGA 2019

¹⁷⁰ Gender and Disaster Risk Management (DRM)- Antigua and Barbuda Country Profile. [World Bank Document](#)

¹⁷¹ World Bank Group (2021) Antigua and Barbuda Country Gender Scorecard [World Bank Document](#)

¹⁷² World Bank Group (2023). Antigua and Barbuda Gender Scorecard 2023

¹⁷³ The Adolescent fertility rate is the number of births per 1000 women ages 15-19

¹⁷⁴ <https://genderdata.worldbank.org/en/economies/antigua-and-barbuda>

not readily available for Antigua and Barbuda, despite the fact that the country is actively working to enhance policies and frameworks to ensure better inclusion and representation of persons with disabilities in public policy and services¹⁷⁵.

Antigua and Barbuda ranks 54th out of 193 countries in the UNDP Human Development Index (HDI). Nevertheless, no value is indicated in the UNDP Human Development report for the Gender Inequality Index (GII)¹⁷⁶, likely due to the previously mentioned lack of available data. Interestingly and probably for the same reason, Antigua and Barbuda is not among the 146 countries included in the World Economic Forum Global Gender Gap Report 2024 and ranked according to their Global Gender Gap Index¹⁷⁷.

Literacy, Education and Vocational Training

Education plays a crucial role in shaping the gender landscape in Antigua and Barbuda. The country has achieved significant progress in terms of gender parity in education, with both boys and girls having relatively equal access to primary and secondary education. However, disparities exist at higher levels.

In Antigua and Barbuda, the literacy rate is approximately 99%. This high literacy rate applies to both men and women, though sex-disaggregated data is scarce. The literacy rate among adults (aged 15 and above) highlights the country's success in providing broad access to basic education¹⁷⁸. However, literacy gaps that affect specific groups, such as women in rural areas or those with fewer economic opportunities, might still exist, though the lack of available data makes it difficult to ascertain this.

In Antigua and Barbuda, education enrolment data shows distinct trends across the different levels. For primary education (ages 5–12), enrolment rates for both males and females are fairly balanced. In the 2017–2018 academic year, males represented 46% of the total primary school population, while females accounted for 54%, reflecting a slight female majority¹⁷⁹. At the secondary level (ages 13–18), the distribution becomes almost equal. For the same period, males made up 51% of the secondary school population, while females represented 49%, indicating balanced participation between the two sexes at this level¹⁸⁰. To further support this balance, the government has implemented measures to eliminate gender stereotypes within the school curriculum; for instance, subjects historically viewed as male-dominated, such as Technical Drawing, have now been included in the curriculum of the nation's sole government-run all-girls secondary school¹⁸¹.

When looking at tertiary education, Antigua and Barbuda shows a trend where females outnumber males. The gender parity index (GPI) for tertiary enrolment—measuring the ratio of females to males—has been above 1, indicating more females are enrolled in higher education institutions compared to males¹⁸². This aligns with global trends where women are increasingly accessing higher education. In this context, women and girls have benefited from tertiary education access through impartial scholarships, leading to more women receiving financial support despite the absence of affirmative action. Additionally, efforts to create a safe and harassment-free learning environment have been prioritized, exemplified by The Halo Foundation's campaign, which underscores the importance of

¹⁷⁵ [International Disability Alliance: https://www.internationaldisabilityalliance.org/countries/ATG](https://www.internationaldisabilityalliance.org/countries/ATG)

¹⁷⁶ UNDP (2024). Human Development Report 2023-24, pg.293.

¹⁷⁷ World Economic Forum (2024) Global Gender Gap 2024

¹⁷⁸ [UN Women Data Hub: https://data.unwomen.org/country/antigua-and-barbuda](https://data.unwomen.org/country/antigua-and-barbuda)

[World Bank Data: https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=AG](https://data.worldbank.org/indicator/SE.ADT.LITR.ZS?locations=AG)

[IndexMundi - Country Facts: https://www.indexmundi.com/facts/antigua-and-barbuda/indicator/SE.ADT.LITR.ZS](https://www.indexmundi.com/facts/antigua-and-barbuda/indicator/SE.ADT.LITR.ZS)

¹⁷⁹ Antigua and Barbuda Review of the Beijing Declaration and Platform for Action Report 2014-2019. p 8

¹⁸⁰ Ibid.

¹⁸¹ World Bank Group (2023). Antigua and Barbuda Gender Scorecard 2023

¹⁸² [Antigua and Barbuda - School enrollment, tertiary \(gross\), gender parity index \(GPI\) \(indexmundi.com\)](https://www.indexmundi.com/facts/antigua-and-barbuda/indicator/SE.ADT.LITR.ZS)

addressing bullying in schools and promoting inclusivity. The government also ensures that schools provide adequate sanitation facilities for girls and implements policies that allow pregnant adolescents to continue their education. However, it is important to highlight that so far, this educational advantage does not fully translate into labour market outcomes (discussed below)¹⁸³.

Employment and Labor Participation

While women excel in higher education, enrolling and graduating at higher rates than men, this has not translated into equal economic opportunities or leadership roles. Thus, despite progress in education, labour market participation in Antigua and Barbuda reflects clear gender disparities. Women have made strides in entering the workforce, yet they remain underrepresented in high-paying and leadership positions.

The 2018 Labor Force Survey revealed that women's participation in the labor market is lower than men's (68.9% compared to 76.1%), with the gap widening during reproductive years due to the greater burden of unpaid domestic and caregiving responsibilities¹⁸⁴. The employment-to-population ratio was higher for men (70.5%) than for women (62.1%), and the unemployment rate among women was higher than that of men, even if men were more likely to participate in the workforce. The higher unemployment rate for women suggests that a larger share of women within the workforce were unemployed and actively looking for work. In 2018, the unemployment rate for women stood at 9.9%, whereas for men it was 7.3%¹⁸⁵.

Similarly, the labor force participation rate for young men was higher than for young women. Around 52.8% of males aged 15 to 24 were either working or seeking employment, compared to 45.0% of females in the same age group. Similar to the trend seen in the overall labor force, even though young men had a greater participation rate, young women faced a higher unemployment rate, with 30.5% unemployed compared to 21.6% of young men¹⁸⁶. The same survey indicates that senior men were more likely to remain active in the labor force than senior women. Approximately 31.5% of men, versus 10% of women, aged 65 and older were still engaged in the workforce.

The impact of COVID-19 on women's employment has been particularly significant. Recent labor force survey data shows that about 60% of workers in the tourism industry are women, highlighting their vulnerability to the closures of hotels and related businesses during the pandemic, and potentially also during national disasters. Additionally, when accounting for the broader effects of the downturn on linked service industries, it's estimated that nearly one in three women may have been directly affected by this decline¹⁸⁷. This disproportionate impact on women is particularly concerning given that many single-parent households in Antigua and Barbuda are predominantly led by women, who often bear the responsibility of supporting larger families compared to men. As a result, any significant drop in women's income due to the effects of COVID-19 has had lasting repercussions, resulting in ongoing challenges for the well-being of the children in their care¹⁸⁸.

Gender-Based Violence (GBV)

According to UN Women, nearly one in three women in the Caribbean faces intimate partner violence or sexual violence at some point in their lives¹⁸⁹. A regional analysis of Women's Health Surveys

¹⁸³ World Bank Group (2023). Antigua and Barbuda Gender Scorecard 2023

¹⁸⁴ Government of Antigua and Barbuda- Statistics Division (2020) Antigua and Barbuda 2018 Labor Force Survey Report [2018-Antigua-and-Barbuda-LFS-Report.pdf \(gov.ag\)](#)

¹⁸⁵ Ibid.

¹⁸⁶ Ibid.

¹⁸⁷ ANTIGUA AND BARBUDA COVID-19 HEAT REPORT HUMAN AND ECONOMIC ASSESSMENT OF IMPACT, p 15 [Human and Economic Assessment of Impact \(HEAT\) Report - Antigua and Barbuda \(undp.org\)](#)

¹⁸⁸ Ibid

¹⁸⁹ [Caribbean GBV Law Portal: GBV In The Caribbean | UN Women – Multi-Country Office – Caribbean](#)

conducted between 2016 and 2019 in five Caribbean Community (CARICOM) Member States—Grenada, Guyana, Jamaica, Suriname, and Trinidad and Tobago—revealed that ever-partnered women aged 15–64 in relationships with men exhibiting attitudes and behaviors that reinforce male dominance and perpetuate gender inequality were more likely to experience both lifetime and current intimate partner violence.¹⁹⁰

In Antigua and Barbuda, Gender-Based Violence (GBV) disproportionately affects women and girls, reflecting broader regional trends. While there is no evidence of an increase in gender violence post-disaster¹⁹¹, high rates of domestic violence and a troubling prevalence of GBV remain significant concerns. National surveys and reports from Non-Governmental Organizations reveal that domestic violence, sexual harassment, and sexual assault are common, though often underreported due to stigma and fear of retaliation, further exacerbating the issue¹⁹². This is further underscored by recent statistics from 2023, which indicate that 16% of women¹⁹³ reported experiencing intimate partner violence in the previous year¹⁹⁴. This statistic illustrates the ongoing challenges many women face regarding safety and security in their personal relationships.

To address the issue, the Directorate of Gender Affairs (DoGA)¹⁹⁵ plays a key role in supporting victims and raising awareness through initiatives like the 16 Days of Activism against Gender-Based Violence. The 2023 campaign, under the theme "No Excuse: Changing the Narrative about Gender-based Violence," included events such as a candlelight vigil and educational programs aimed at preventing violence, particularly during public festivities¹⁹⁶. DoGA has also established the Support and Referral Centre (SARC), which offers services such as medical care, crisis counselling, and legal assistance for survivors. This initiative aims to provide comprehensive care and advocacy for victims, underscoring the government's commitment to tackling gender-based violence¹⁹⁷.

These efforts reflect a growing acknowledgment of the need to protect women and girls in Antigua and Barbuda, though more work remains to be done to eliminate GBV entirely. The government has established hotlines, shelters, and support services for survivors of GBV, but these services are often concentrated in urban areas, leaving rural women underserved. Training programs for law enforcement officials and healthcare workers are also critical to improving response mechanisms.

The legal framework protecting women and girls from Gender-Based Violence (GBV) in Antigua and Barbuda comprises the Constitution, Acts of Parliament, and common law. While domestic violence legislation offers protection against physical and sexual violence, there is currently no specific legislation addressing sexual harassment. Nonetheless, victims of sexual harassment in the workplace can seek remedies through common law, relying on tort law (duty of care) and contract law (breach of implied trust or constructive dismissal).

¹⁹⁰ UN Women (2021). Research brief: Intimate partner violence in five CARICOM countries: Findings from National Prevalence Surveys on violence against women.

¹⁹¹ Project Management Unit (PMU) of the Department of Environment, Ministry of Health, Wellness and the Environment – Antigua and Barbuda (2019) Gender Assessment and Gender and Social Inclusion Action Plan–Climate Change Programme Resilience To Hurricanes, Floods and Droughts in the Building Sector in Antigua and Barbuda, p.31

¹⁹² DoGA CONDEMNS ALL FORMS OF VIOLENCE & DISCRIMINATION AGAINST WOMEN & GIRLS – Directorate of Gender Affairs (gov.ag): Empowering Change: 16 Days of Activism Against Gender-Based Violence in Antigua & Barbuda - Antigua Observer Newspaper

¹⁹³ Proportion of ever-partnered women and girls subjected to physical and/or sexual violence by a current or former intimate partner in the previous 12 months.

¹⁹⁴ Data quoted in Commonwealth of Learning (2023). Gender Country Profile 2023. Antigua and Barbuda content (col.org)

¹⁹⁵ Directorate of Gender Affairs (gov.ag) Overview – Directorate of Gender Affairs (gov.ag)

¹⁹⁶ <https://antiguaobserver.com/empowering-change-16-days-of-activism-against-gender-based-violence-in-antigua-barbuda/>

¹⁹⁷ Support and Referral Centre – Directorate of Gender Affairs (gov.ag)

Gender relevant legal framework and institutional arrangements

The summary provided in this section highlights the key laws, policies, and international commitments relevant to the social, environmental, and economic framework guiding projects in Antigua and Barbuda, emphasizing gender considerations and sustainable development.

National Legislation

Antigua and Barbuda Labour Code (1975, with significant amendments): Prevents discrimination in employment based on various attributes, including race and gender. **Labour Code (Amendment) Act 2010** - This amendment focuses on the protection of workers' rights and introduced provisions related to the employment of young persons and women.

Constitution of Antigua and Barbuda (1981): Prohibits discrimination based on sex and outlines that discriminatory treatment cannot occur due to race, origin, political beliefs, and sex.

Sexual Offences Act (1995/amended in 2004): This act criminalizes various forms of sexual violence and abuse, including rape and sexual assault. It aims to protect the rights and dignity of victims and emphasizes the importance of consent.

Disaster Management Act (2002): Organizes preparedness and response for natural and man-made disasters.

Childcare and Protection Act (2004): Creates a Child Protection Agency and sets standards for child-care facilities, emphasizing child safety and the prohibition of child labor.

Domestic Violence Act (2015): This Act provides protection to individuals from domestic violence. It establishes legal mechanisms for the prevention and protection of victims, allowing them to seek restraining orders and providing for the care and support of those affected by domestic violence. The Act recognizes the importance of addressing domestic violence as a critical gender issue.

Disabilities and Equal Opportunities Bill (2017): Protects the rights of persons with disabilities, aiming to eliminate discrimination and ensure equal participation in society.

Environmental Protection and Management Act (2019): Serves as the principal framework for environmental governance, integrating Multilateral Environmental Agreements and promoting sustainable management.

National Policies

National Youth Policy (2007): Aims to empower youth across various domains such as education, health, employment, and gender equality.

National Poverty Strategy (2011-2015): Focuses on building resilience through environmental sustainability and integrating disaster risk reduction into planning processes.

Internal Policies

Department of Environment Gender Policy (2018): The Department of Environment in Antigua and Barbuda established its **Gender Policy** in 2018. This policy aims to integrate gender considerations into environmental management and projects and promotes gender equality in the context of sustainable development¹⁹⁸, it also designates gender officers to support in this work.

¹⁹⁸ Government of Antigua and Barbuda (2018) Department of Environment (DoE)(2018) Gender Policy

Environmental and Social Safeguard Policy (2018): In this policy the Department of Environment (DOE) in Antigua and Barbuda requires that all projects comply with standards set forth in Multilateral Environmental Agreements (MEAs) and the Environmental Protection Management Act of 2015, addressing issues like biodiversity and climate change. Additionally, the DOE emphasizes the importance of aligning projects with human rights, social protection, and gender conventions to which the government is a signatory¹⁹⁹.

Multilateral Agreements and International Conventions

Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) (1979): Protects women's rights and promotes equal representation and opportunities.

Convention on the Rights of the Child (1989): Ensures children's rights in civil, political, and social contexts.

United Nations Framework Convention on Climate Change (UNFCCC, 1992): Provides a framework for climate action, emphasizing gender balance in climate policy.

Beijing Declaration and Platform for Action (1995): Calls for women's active involvement in environmental decision-making.

UN Convention on the Rights of Persons with Disabilities (CRPD) (2006): Ensures rights for persons with disabilities, promoting equal access to rights and freedoms.

Rio+20 Outcome Document (2012): Affirms the need for gender-sensitive policies in sustainable development.

Sustainable Development Goals (SDGs) (2015): The 2030 Agenda for Sustainable Development, adopted by all United Nations Members States in 2015, created 17 world Sustainable Development Goals (SDGs), including Goal 5, which aims to achieve gender equality and empower all women and girls. The aim of these global goals is "peace and prosperity for people and the planet" – while tackling climate change, promoting gender equality, and working to preserve oceans and forests.

UNFCCC Gender Action Plan (2017): Aims to enhance women's participation in climate-related processes and policies.

Institutional arrangements

In Antigua and Barbuda, the Directorate of Gender Affairs (DoGa) within the Ministry of Social Transformation, Human Resource Development, Youth, and Gender Affairs spearheads the development and execution of gender policies.

Antigua and Barbuda does not have a specific national gender policy that has been formalized as of now²⁰⁰. However, there are ongoing efforts to develop a comprehensive National Gender Equality Action Plan and policy. This initiative is being undertaken by the DoGa with the aim of aligning gender considerations with the national development strategy. The process involves national consultations with civil society and various government levels, ensuring that the proposed plan incorporates monitoring and evaluation frameworks for accountability

¹⁹⁹ Government of Antigua and Barbuda (2018) Department of Environment (DoE)(2018) Environmental and Social Safeguards Policy

²⁰⁰ [Antigua and Barbuda | UN Women – Multi-Country Office – Caribbean: https://caribbean.unwomen.org/en/caribbean-gender-portal/antigua-and-barbuda](https://caribbean.unwomen.org/en/caribbean-gender-portal/antigua-and-barbuda)

Meanwhile, the Department of the Environment (DOE) developed in 2018 a gender policy²⁰¹ that focuses on the impacts of climate change and environmental policies on gender. This strategy also aims to create specific policies and project designs to address gender disparities. By implementing this approach, the DOE seeks to enhance project effectiveness, reduce risks, and increase the likelihood of securing support from primary beneficiaries²⁰².

In 2018, the DOE has also developed the **Environmental and Social Safeguards Policy** of Antigua and Barbuda, establishing guidelines for managing environmental and social risks in development projects, particularly those funded by international sources like the Green Climate Fund (GCF). It emphasizes **gender responsiveness** by integrating gender considerations into all programs, ensuring equal access to resources, justice, and environmental information for both women and men. In this policy, the DOE promotes a gender-sensitive workplace that influences project development and implementation. Continuous gender sensitization training is provided to staff, partners, and community groups. Additionally, all partners and contractors must either have a gender policy or collaborate with the Directorate of Gender Affairs to develop one, reinforcing the commitment to gender equality within environmental framework²⁰³.

Gender, Climate Change and Early Warning Systems and Disasters Response

In 2020 and 2021, UN Women conducted studies highlighting the gendered impact of climate change and disasters in the Caribbean, including Antigua and Barbuda²⁰⁴. These studies revealed significant gender differences in how men and women experience the effects of natural hazards, with women, particularly those heading households, being disproportionately affected. For instance, as already illustrated above, women in female-headed households face financial instability when sectors like tourism, on which they heavily rely, are disrupted by disasters. This was evident during the COVID-19 pandemic, where the absence of income support exposed many women to financial hardship.

Men, who dominate sectors like fisheries, also face challenges, but the loss of equipment like boats mainly impacts their financial stability. On the other hand, women's caregiving roles, especially for children and the elderly, also leave them vulnerable when infrastructure, such as roads and healthcare, is compromised by disasters. Pregnant women and those with disabilities are particularly at risk due to inadequate access to medical care during emergencies. Women's health, especially maternal health, is at heightened risk from post-disaster issues like disease outbreaks, while mental health services remain underutilized by men.

Moreover, female-headed households (that account for nearly half of all households in the country) often struggle to access disaster information and resources due to economic limitations, such as the inability to afford data plans for mobile devices, smartphones or transportation to social services. This lack of access exacerbates their vulnerability during and after disasters. The studies also underscored gaps in policies addressing gender inequality in disaster risk and climate change, with a lack of gender-focused indicators or policies in place.

²⁰¹ [Microsoft Word - Approved DOE Gender Policy.docx \(adaptation-fund.org\): https://caribbean.unwomen.org/en/caribbean-gender-portal/antigua-and-barbuda](https://caribbean.unwomen.org/en/caribbean-gender-portal/antigua-and-barbuda)

²⁰² GCF document (October 2020). Gender Assessment FP133: Resilience to hurricanes in the building sector in Antigua and Barbuda | DOE_ATG | B.26/02, p 10

²⁰³ Government of Antigua and Barbuda (2018) Department of Environment (DoE)(2018) Environmental and Social Safeguards Policy [Microsoft Word - Approved DOE Environmental and Social Safeguard Policy.docx \(adaptation-fund.org\): https://www.adaptation-fund.org/wp-content/uploads/2016/12/Approved-DOE-Environmental-and-Social-Safeguard-Policy.pdf](https://www.adaptation-fund.org/wp-content/uploads/2016/12/Approved-DOE-Environmental-and-Social-Safeguard-Policy.pdf)

²⁰⁴ UN Women Multi-Country Office (MCO) Caribbean (November 2020). Gender Inequality of Climate and Disaster Risk and Cost of Inaction (GICDR) Study & UN Women Multi-Country Office (MCO) Caribbean & International Institute for Sustainable Development (IISD) (2021) and (UN Women) Gender-Responsive Resilience Building in the Caribbean: Understanding the Role of Knowledge, Attitudes, Behaviours, and Practices in Coordination Mechanisms for Climate Change and Disaster Risk Reduction.

Overall, the need for gender-responsive policies and improved infrastructure to assist women, especially those leading households, in tackling climate change and disaster risks is evident²⁰⁵.

Women's Representation and Participation in climate and DRM Sectors

Women in Antigua and Barbuda face ongoing challenges in political leadership, holding just 5.6% of seats in the national parliament, which reflects broader gender disparities in decision-making roles. These disparities extend into key sectors like infrastructure and energy, where women are underrepresented in technical fields but dominate administrative roles. Childcare and household responsibilities further limit women's ability to engage in disaster recovery, particularly for single-parent households. This vulnerability weakens their resilience to climate change and delays recovery efforts in key public services²⁰⁶.

Despite these challenges, women play significant roles in disaster management and climate change. The National Office of Disaster Services (NODS) has women in nine (9) of the seventeen (17) district disaster coordinator positions²⁰⁷. Meanwhile, in the Meteorological Services, responsible for issuing warnings and forecasts and managing the early warning system (EWS), out of eight (8) meteorologists and forecasters, two (2) are women, and among thirteen (13) meteorology observers, five (5) are women (2018). Additionally, both the Directorate of Gender Affairs (DoGA) and the Department of Environment (DoE) are led by female directors. The Government of Antigua and Barbuda has made efforts to incorporate a gender perspective into disaster risk management (DRM) at the national, sectoral, and agency levels. Similarly in areas related to climate change, women are strongly present primarily through the Department of Environment as the government's negotiator for climate change and main implementing body of the Paris Agreement. As of 2021, the staffing of the Department of Environment (including administrative and technical) is primarily female²⁰⁸.

Following Hurricane Irma in 2017, women played a key role in the creation and distribution of dignity kits—containing essential health and hygiene items—to affected communities, as well as in recovery efforts through their involvement in the Barbuda Council, where over half of its 11 members are women²⁰⁹. Nevertheless, despite the presence of women in leadership positions within DRM-related organizations, this has not necessarily resulted in gender-responsive policies or practices within the sector²¹⁰. Women's contributions, although critical, are still hindered by broader systemic gender imbalances that reduce their influence in technical, political, and climate-related fields.

Gender in Early Warning Systems and Disaster Preparedness

In Antigua and Barbuda, men and women reportedly prepare for and respond to climate-related events in different ways, highlighting the role of gender in shaping resilience, as shown by the data and reports in this section.

²⁰⁵ Cfr. UN Women/EnGenDER (November 2021) Gendered Impacts of climate change and disasters risk in Antigua and Barbuda- policy brief [EnGenDER Gender Inequality CC DRR Brief Antigua and BarbudaF 20220203.pdf \(unwomen.org\)](#)

²⁰⁶ ESS Officer within the Project Management Unit (PMU) of the Department of Environment, Ministry of Health, Wellness and the Environment (October, 2019) Gender Assessment and Gender and Social Inclusion Action Plan- Climate Change program- Resilience to hurricanes, floods and droughts in the building sector in Antigua and Barbuda

²⁰⁷ World Meteorological Organization (WMO). 2018. Caribbean 2017 Hurricane Season. An evidence-based Assessment of the Early Warning System. https://library.wmo.int/doc_num.php?explnum_id=5459

²⁰⁸ UNFCCC (2022) Antigua and Barbuda National Gender Submission. <https://genderclimatetracker.org/resource/23227>

²⁰⁹ Local Government - Barbudaful: <https://barbudaful.net/the-barbudaful-community/barbudaful-politics/local-government/>

World Bank (2021) Antigua and Barbuda Country Gender Scorecard.

https://worldbankgroup.sharepoint.com/mcas.ms/sites/LCR/Documents/Gender/Country%20Scorecards/LCR%20Country%20Gender%20Scorecards%202021/Antigua_barbuda_SCORECARD-OK-linked.pdf?McasTsId=20892

Directorate of Gender Affairs (DoGA) (2019). Antigua and Barbuda Review of the Beijing Declaration and Platform for Action: 2014–2019.

Antigua and Barbuda: Directorate of Gender Affairs. https://www.cepal.org/sites/default/files/informe_beijing25_antigua_y_barbuda.pdf.

²¹⁰ Canada Caribbean Resilience Facility(CRF); Canada,Global facility for Disaster Reduction and Recovery (GFDRR), the World Bank (2022) Gender and Disaster Risk Management. Antigua and Barbuda Country Profile.

Evacuation and Shelter Access: Women in rural areas of Antigua and Barbuda face specific challenges related to mobility, primarily due to limited transportation options. During climate events, pregnant women, elderly women, and persons with disabilities (PWDs) may be particularly vulnerable as roads become inaccessible and coastal areas are prone to flooding. To address the risks of gender-based violence (GBV) in shelters, the Directorate of Gender Affairs (DoGA), in collaboration with the United Nations Population Fund (UNFPA), UN Women Caribbean Multi-Country Office, and the International Organization for Migration (IOM), developed guidelines to prevent sexual assault and GBV in shelter settings. They also provided training for shelter managers on implementing these guidelines²¹¹.

Early Warning Systems (EWS): As noted in a 2018 report by the World Meteorological Organization (WMO)²¹², radio remains a key source of hazard information, though social media is gaining influence in the region as evidenced through surveys taken during project preparation. Platforms like WhatsApp and Facebook, accessed via mobile phones, are becoming increasingly relevant for early warnings. While the internet serves as a valuable tool for disseminating warnings, it may inadvertently create barriers related to gender, location, and ethnicity, even if there is limited data on the digital divide between men and women in Antigua and Barbuda. UN Women in its 2021 policy brief²¹³ notes that most households in Antigua and Barbuda have at least one mobile phone during disasters, allowing them to access critical information. Nevertheless, low-income female-headed households may lack access to data plans that are required by mobile phones for receiving disaster alerts.

Additionally, the 2018 WMO assessment which included Antigua and Barbuda in its review of EWS systems, identified the following key gender considerations: (i) women and men may respond differently to EWS based on how they spend their time; (ii) differences in household decision-making power and family structures; (iii) the type of economic activities they engage in; and (iv) how they perceive risk. The review recommended improvements, such as collecting and analysing gender-specific risk and vulnerability data, establishing feedback mechanisms to ensure warnings reach key stakeholders (especially women and vulnerable groups), and tailoring communication systems to address the different needs of urban and rural populations, men and women, young and elderly, and people with disabilities. It also suggested greater use of amateur radio and community alerts via social media²¹⁴.

Disaster Preparedness: UN Women in its 2021 policy brief²¹⁵ reveal that women in Antigua and Barbuda tend to spend more on disaster preparation, stocking up on essentials when alerted to an upcoming disaster. However, this can increase their financial vulnerability if recovery takes longer than expected. This is supported by the Government of Antigua and Barbuda and Adaptation Fund report (2020), which found that women in the country spend an average of US\$800 more than men on disaster preparedness, including supplies such as shutters, tape for windows, food, and clothing. While this finding warrants further analysis to understand the driving factors and impacts, global evidence suggests that women's household responsibilities and family roles may influence their disaster

²¹¹ Directorate of Gender Affairs (DoGA) (2019). Antigua and Barbuda Review of the Beijing Declaration and Platform for Action: 2014–2019. Antigua and Barbuda: Directorate of Gender Affairs. https://www.cepal.org/sites/default/files/informe_beijing25_antigua_y_barbuda.pdf.

²¹² World Meteorological Organization (WMO). 2018. Caribbean 2017 Hurricane Season. An evidence-based Assessment of the Early Warning System. https://library.wmo.int/doc_num.php?explnum_id=5459

²¹³ UN Women/EnGenDER (November 2021) Gendered Impacts of climate change and disasters risk in Antigua and Barbuda- policy brief [EnGenDER Gender Inequality CC DRR Brief Antigua and BarbudaF_20220203.pdf](https://unwomen.org/en/gen/der/gender-inequality/cc-drr-brief-antigua-and-barbuda-f-20220203.pdf) (unwomen.org)

²¹⁴ National Office of Disaster Services (NODS) (2018). Multi-Hazard Early Warning Systems Report for Antigua and Barbuda. St. John's, Antigua and Barbuda: NODS.

²¹⁵ UN Women/EnGenDER (November 2021) Gendered Impacts of climate change and disasters risk in Antigua and Barbuda- policy brief [EnGenDER Gender Inequality CC DRR Brief Antigua and BarbudaF_20220203.pdf](https://unwomen.org/en/gen/der/gender-inequality/cc-drr-brief-antigua-and-barbuda-f-20220203.pdf) (unwomen.org)

preparedness²¹⁶. Notably, the Government of Antigua and Barbuda launched a social housing program in 2018 targeting vulnerable urban populations, with a focus on women, particularly single mothers. This initiative has improved access to safe, affordable, and climate-resilient housing for women, acknowledging their vulnerability and often disadvantaged status²¹⁷.

Opportunities for the project to support: Effective access to communication on early warnings and hazard response present an opportunity for the project to support women in making informed decisions that may reduce their impact to climate related hazards. In this regard the project must look to enable communications solutions that are accessible regardless of data plans and access to smart phones. In addition surveys taken in project development demonstrated the relevance of leveraging community organizations and social structures to enhance capacity building to not only disseminate information regarding hazards but also on potential preventive responses that allow for early access. Finally, increasing the lead time of early alerts to allow for planning and decision making as well as potential preventive actions would allow women headed households to ensure measures to reduce vulnerability.

DRM Policies and Gender

The Government of Antigua and Barbuda is making progress in incorporating gender considerations into disaster risk management (DRM) across national, sectoral, and agency levels. The National Office of Disaster Services (NODS) collaborates with the Caribbean Disaster Emergency Management Agency (CDEMA) to integrate gender into disaster risk reduction efforts. However, most national disaster policies still lack specific gender content²¹⁸.

The Environmental Protection and Management Act of 2019, outlines the Department of Environment's (DoE) role in leading climate resilience efforts. Through the Sustainable Island Resource Framework Fund (SIRF Fund), the government provides financial support for vulnerable groups, including loans for female-headed households to boost climate resilience. The Act also requires periodic gender impact assessments of the Fund's projects. Meanwhile the National Strategic Biodiversity Action Plan aims to ensure sustainable use of biodiversity in Antigua and Barbuda. It aligns with international goals like the Aichi Biodiversity Target 14²¹⁹, focusing on restoring ecosystems that benefit women, indigenous groups, and vulnerable communities.

The DoE's Gender Policy promotes gender mainstreaming in environmental management, emphasizing the active involvement of both men and women. The 2018 Environmental Social Safeguard Policy reinforces this by mandating environmental, social, and gender screening of projects, and ensuring vulnerable groups are consulted and included in decision-making²²⁰.

However, a review of Antigua and Barbuda's Post-Disaster Public Financial Management found that gender equality is not factored into resource allocation for disaster response. A UN Women survey

²¹⁶ Erman A., De Vries Robbé S. A., Thies S. F., Kabir K., and Maruo M. 2021. Gender Dimensions of Disaster Risk and Resilience. Existing Evidence.

²¹⁷ Directorate of Gender Affairs (DoGA) (2019). Antigua and Barbuda Review of the Beijing Declaration and Platform for Action: 2014–2019. Antigua and Barbuda: Directorate of Gender Affairs.

²¹⁸ World Bank. (2021) Gender-Responsive Disaster Preparedness and Recovery in the Caribbean: Desk Review. Canada Caribbean Resilience Facility. <https://openknowledge.worldbank.org/handle/10986/35215>

²¹⁹ **The Aichi Biodiversity Targets** were established during the CBD's 10th Conference of the Parties in 2010 in Aichi, Japan. These 20 targets aim to halt the loss of biodiversity by 2020 and provide a framework for action by governments, NGOs, and other stakeholders. They cover various aspects of biodiversity, including habitat protection, species conservation, and the integration of biodiversity into national policies and planning. Each target is designed to address specific drivers of biodiversity loss, promote sustainable practices, and enhance awareness and participation in conservation efforts. **Target 14:** By 2020, ecosystems that provide essential services, including services related to water, and contribute to health, livelihoods and well-being, are restored and safeguarded, taking into account the needs of women, indigenous and local communities, and the poor and vulnerable. [Aichi Biodiversity Targets \(cbd.int\)](#)

²²⁰ Canada Caribbean Resilience Facility (CRF); Canada, Global facility for Disaster Reduction and Recovery (GFDRR), the World Bank (2022) Gender and Disaster Risk Management. Antigua and Barbuda Country Profile

identified gaps in understanding gender concepts and biases, such as the expectation that women prioritize family over career, as well as a lack of gender-disaggregated data in the sector²²¹.

Potential Opportunities for the project to support

- In the preparation of climate information ensure that consultations with end users incorporate women-only consultations and Focus groups, or discussions with women leaders and organizations at rural and urban level, particularly accounting for coastal populations that are most affected. This will allow a better assessment of needs and preferred communication channels for women, ensuring that content and structure of message and timing to relay message as well gender communicator communicating the message.
- For the planning of activities related to the design and dissemination of alert messages, the project will work to enable support networks in the form of local cooperation groups. Additionally, the project will look to ensure that training and awareness workshops are organized on interpreting early warnings and response measures and are adapted to the needs and available time of women for their participation, taking into account the double burden placed on women.
- The project will look to support risk management education programs at a community level including in schools and social structures, specifically aimed at women and girls. This would allow to train and equip community members, including women and persons with disabilities, for early warning actions.
- Measures to disseminate information must take into account the existing gender based digital gaps that may become a barrier in receiving early warning alerts and real-time updates. In this regard the project will look to identify technological solutions that are accessible to both men and women, for example looking to identify SMS systems that do not require data plan nor access to smart phone technology.
- Ensure that Comprehensive review and action plan for Emergency Warning (EW) systems provide analysis and strategies to include inclusivity and to address the digital gender gaps, while working to promote the representation and participation of vulnerable groups in preparedness and response activities in community emergency groups.

Gender-disaggregated data, is crucial for effective planning and response. It is therefore essential to integrate gender-responsive data collection and analysis in DRR capacity development. In this regard the government of Antigua and Barbuda has made important advancement through their NDCs and Climate Change communication in including gender disaggregated data. This best practice will be promoted within the DRR space by working with NODS.

²²¹ UN Women/EnGenDER (November 2021) Gendered Impacts of climate change and disasters risk in Antigua and Barbuda- policy brief [EnGenDER Gender Inequality CC DRR Brief Antigua and BarbudaF_20220203.pdf \(unwomen.org\)](#)

9 Gender Assessment for Somalia

Somalia is ranked 'medium' on the UNDP Gender Development Index 2023-24, comparing Human Development Index achievements for women and men²²². Women consist of 50.5% of the total population in Somalia²²³ and 31.9% of Somali households are female-headed²²⁴. Somalia has a large rural population with 74% of the population living in rural areas, and most of them work in the agriculture sector²²⁵. About 24.4% of the population, or 3.4 million people, are living with disabilities in Somalia, including 20.3% with mild or moderate degrees of disability and 4.1% with a severe disability²²⁶. Of people living with a disability, women are represented more (57%) than men (43%)²²⁷. Meanwhile, Somalia's gender equality-related indexes are ranked the lowest among the neighbouring countries in the Horn of Africa region²²⁸. Somalia is ranked 122nd out of 193 countries in the Gender Inequality Index (GII) and 102nd out of 146 countries in the Global Gender Gap Report 2023²²⁹. On the other hand, Somalia's rapid economic growth from 2009 to 2019, combined with structural change, helped to increase labour earnings and reduce poverty²³⁰. Living standards have improved, helping Somalia to narrow urban-rural gaps²³¹. However, low-income and rural households still lag in access to basic services and earning opportunities²³². Access to healthcare services in Somalia, particularly maternal health, remains critically inadequate. The country has one of the highest maternal mortality rates globally, with 732 deaths per 100,000 live births as of 2015. Decades of civil war, ongoing instability, and limited healthcare infrastructure contribute to this dire situation. Efforts by the Somali government and international organizations like UNICEF and UNFPA aim to improve healthcare services, but challenges persist, especially in rural areas²³³.

Education, Vocational Training and Literacy

Somalia has made significant progress in achieving gender parity in education, though enrolments are still generally low. The net enrolment in primary education is around 57.6% for female students and 58.6% for male students for the academic year 2023-2024²³⁴. The enrolment rate at lower secondary schools is about 23.9% for female students and 24.8% for male students, while the enrolment rate for both sexes is lower for upper secondary schools, at 11.5% for women and 14.1% for men²³⁵. However, the proportion of female students in tertiary education and technical vocational training is lower than that of male students, particularly for non-traditional areas of study such as science, technology, engineering, and math. The Ministry of Education (MoE) estimates that only 10% of female students, compared to 22% of male students, pursued a Bachelor's Degree in Science, Technology, Engineering, Arts, and Mathematics (STEAM)-related subjects²³⁶. Literacy has improved steadily for both women and men but remains higher for men at 52.3% compared to 45.4% for women²³⁷. There is also a literacy gap between urban and rural areas, with urban areas like Mogadishu having higher literacy rates. For

²²² [Country Insights | Human Development Reports](#)

²²³ [Somali National Bureau of Statistics](#)

²²⁴ <https://nbs.gov.so/>

²²⁵ [MOAI](#)

²²⁶ mogfa.gov.so

²²⁷ mogfa.gov.so

²²⁸ [Global Gender Gap Report 2023 | World Economic Forum](#)

²²⁹ [Country Insights | Human Development Reports](#)

²³⁰ <https://www.worldbank.org/en/country/somalia/publication/somalia-economic-update>

²³¹ [Somalia | United Nations Development Programme](#)

²³² [Somalia | United Nations Development Programme](#)

²³³ <https://somalia.unfpa.org/en/somalia-health-and-demographic-survey-2020>

²³⁴ [Somalia | UNESCO](#)

²³⁵ [Somali National Bureau of Statistics](#)

²³⁶ [Wasaaradda Waxbarashada, Hiddaha Iyo Tacliinta Sare – Ministry Of Education, Culture and Higher Education](#)

²³⁷ [Somalia \(SOM\) - Demographics, Health & Infant Mortality - UNICEF DATA](#)

example, in cities like Mogadishu, about 60% of adults (15 years and older) can read and write, but in rural areas, only about 40% of adults are literate²³⁸. This gap shows how much harder it is for people in rural areas to get an education. Girls and women in rural areas, especially those from ethnic minorities and people with disabilities, face even more challenges. Over 72% of women in rural areas have never gone to school, which is much higher compared to urban areas²³⁹. For children, only about 30% are enrolled in primary school, but this drops to 22% in rural areas, and even fewer girls attend school²⁴⁰. In rural areas, the gender gap in education is wider, with less than half of the 22% of pastoralist children in school being girls²⁴¹. This means girls in these communities have fewer chances to get an education and improve their lives.

Additionally, gender gaps in access to education are particularly evident among ethnic minorities and people with disabilities²⁴². Indigenous women in Somalia have higher illiteracy rates and fewer educational opportunities compared to their male counterparts²⁴³. The dropout rate for Indigenous adolescents aged 14-18 is significant, with girls being more affected due to the higher demand for them to perform household chores. This results in a widening educational disparity for Indigenous groups, people with disabilities, and those living in poverty²⁴⁴.

The education-related obstacles faced by rural girls and women in Somalia, such as higher illiteracy rates and fewer opportunities, combined with other factors like gender stereotypes and patriarchy, hamper their ability to access informed decision-making, hold jobs outside the home, understand relevant laws and regulations, and access technologies, extension services, and finance²⁴⁵. Access to early warning information is also affected²⁴⁶. Men are more likely to receive early warning information from formal sources, while women often rely on informal and social sources²⁴⁷. This disparity means that women are less likely to receive timely and accurate warnings, impacting their ability to take appropriate action during disasters²⁴⁸. EW4ALL services need to be gender-responsive, ensuring equal access to early warning information for both men and women²⁴⁹. Engaging women in the design and implementation of early warning systems can help ensure these systems are more effective and inclusive.

Employment and Labor Participation

In Somalia, women's labour participation in 2023 was 32.7%, compared to men's 52.5%²⁵⁰. According to the Somalia Labour Force Survey, 57% of women aged 15 to 64 were employed in agriculture, forestry, and fishing sectors; 10% in the industry sector; and 33% in services like hospitality, finance, healthcare, and education²⁵¹. However, just 30% of women are wage and salaried workers, while a significant 70% are in vulnerable employment, often in the informal sector with poor working

²³⁸ <https://www.unicef.org/somalia/education>

²³⁹ <https://www.worldbank.org/en/country/somalia/overview>

²⁴⁰ <https://reliefweb.int/report/somalia/somalia-education-sector-analysis-2021>

²⁴¹ <https://www.unicef.org/somalia/gender-equality>

²⁴² <https://www.worldbank.org/en/country/somalia/overview>

²⁴³ www.worldbank.org/en/country/somalia/overview

²⁴⁴ <https://data.worldbank.org/indicator/SE.PRM.ENRR?locations=SO>

²⁴⁵ <https://www.iicba.unesco.org/en/somalia>

²⁴⁶ <https://www.citiesalliance.org/newsroom/news/results/somalia-early-warning-systems-climate-adaptation>

²⁴⁷ <https://www.citiesalliance.org/newsroom/news/results/somalia-early-warning-systems-climate-adaptation>

²⁴⁸ <https://www.citiesalliance.org/newsroom/news/results/somalia-early-warning-systems-climate-adaptation>

²⁴⁹ <https://www.citiesalliance.org/newsroom/news/results/somalia-early-warning-systems-climate-adaptation>

²⁵⁰ [UN Women – Headquarters](#)

²⁵¹ [Somali National Bureau of Statistics](#)

conditions, low pay, and no social protection²⁵². Gender disparities are stark in Somalia. Women earn about 62% of what men earn, own fewer assets, and face more barriers to accessing better-paying jobs²⁵³. The gender wage gap stands at around 38%²⁵⁴, influenced by lower educational attainment for women, labour market discrimination, and societal norms assigning women to primary domestic roles²⁵⁵. These inequalities hinder women's ability to participate in formal employment, access skills training, and advance their careers²⁵⁶.

Representation and Participation in Decision Making

Despite a robust gender equality perspective reflected in national policies and legal frameworks, Somalia still faces significant challenges in achieving equal representation and participation of women and men in decision-making positions. Women are under-represented at all levels of legislative and executive bodies. In 2024, women hold 24% of Federal Parliament seats, up from previous years but below the 30% target²⁵⁷. In the executive body, women occupy 10% of senior ministerial positions and 16% of under-secretary roles²⁵⁸. At the federal member state level, women hold 27% of leadership roles in civil service and constitute 42% of the total civil service workforce²⁵⁹. However, at the sub-national level, representation is low: 8% in governor positions, 3.43% at the district level, 10.65% of chief positions in local councils, and 13% of deputy governor positions at the regional level²⁶⁰. Women make up 20.3% of deputy governor positions at the district level²⁶¹.

Gender-Based Violence, Social Norms and FGM in Somalia

Gender-based violence (GBV) remains a significant concern in Somalia. Over 60% of women face physical abuse, denial of education, forced marriage, rape, and sexual harassment²⁶². Harmful gender norms, such as the belief that men are entitled to sex regardless of consent, contribute significantly to GBV²⁶³. Marginalized women and girls, especially, face more challenges in accessing legal advice, safe shelter, and mental support²⁶⁴.

Cultural practices and social norms continue to contribute to the unequal division of labour in households and communities. Women and girls are involved in productive activities such as farming and generating income, but they are also primarily responsible for household and childcare duties²⁶⁵²⁶⁶. Much of their time is devoted to household chores, childcare, cooking, and cleaning. Indigenous women, in particular, carry a heavy burden in terms of household chores, including housework, farming, raising animals, and collecting forest foods, especially during food shortages and disasters²⁶⁷²⁶⁸.

²⁵² [Somali National Bureau of Statistics](#)

²⁵³ [Somalia | United Nations Development Programme](#)

²⁵⁴ <https://www.worldbank.org/>

²⁵⁵ mogfa.gov.so

²⁵⁶ [Somalia | United Nations Development Programme](#)

²⁵⁷ <https://iknowpolitics.org/en/region/somalia-0>

²⁵⁸ <https://nbs.gov.so/wp-content/uploads/2024/03/Gender-Statistics-Booklet.pdf>

²⁵⁹ <https://nbs.gov.so/wp-content/uploads/2024/03/Gender-Statistics-Booklet.pdf>

²⁶⁰ <https://nbs.gov.so/wp-content/uploads/2024/03/Gender-Statistics-Booklet.pdf>

²⁶¹ <https://nbs.gov.so/wp-content/uploads/2024/03/Gender-Statistics-Booklet.pdf>

²⁶² <https://somalia.unfpa.org/en/somalia-health-and-demographic-survey-2020>

²⁶³ <https://somalia.unfpa.org/en/somalia-health-and-demographic-survey-2020>

²⁶⁴ <https://somalia.unfpa.org/en/somalia-health-and-demographic-survey-2020>

²⁶⁵ https://www.unwomen.org/sites/default/files/2024-09/b30_report_somalia_en.pdf

²⁶⁶ <https://sidrainstitute.org/european-union-somalia-gender-analysis-study/>

²⁶⁷ https://www.unwomen.org/sites/default/files/2024-09/b30_report_somalia_en.pdf

²⁶⁸ <https://sidrainstitute.org/european-union-somalia-gender-analysis-study/>

Women are still expected to follow social norms and beliefs that endorse men's privilege while undervaluing women's capacity and potential²⁶⁹. This impacts the roles of women and men in decision-making in households and society as a whole²⁷⁰. Women are involved in managing household budgets, but men are the primary decision-makers for major expenditures²⁷¹. Additionally, social norms and low levels of education and literacy are major barriers preventing women and girls from accessing information, technical skills, and training, having their voices heard, and enjoying preference in their families and society²⁷².

Female Genital Mutilation (FGM) is a widespread practice in Somalia. According to the Somalia Health and Demographic Survey (SHDS, 2020), 99% of women and girls aged 15-49 have undergone some form of FGM²⁷³. This practice is deeply rooted in cultural and social norms and continues despite efforts to eliminate it²⁷⁴.

Gender Policies and Legal Framework

The Federal Government of Somalia is committed to promoting gender equality and social inclusion in disaster risk reduction and management, aligning with the 2030 Agenda for Sustainable Development. Key national legislation and policies support this commitment:

The **Provisional Constitution of 2012** guarantees equal rights for women and men, including human rights, equality before the law, and active participation in all aspects of national life²⁷⁵. The **Sexual Offences Bill** addresses sexual violence and promotes gender equality, though it has faced challenges in implementation²⁷⁶. The **Somalia National Disaster Management Policy** focuses on the needs of women, children, the elderly, and people with disabilities during disasters²⁷⁷. The **National Gender Policy** promotes gender equality and women's leadership, reduces vulnerability to gender-based violence, and improves social protection for the most vulnerable²⁷⁸. The **Gender Action Plan (2021-2025)** emphasizes gender equality and women's empowerment across various sectors. Somalia's **Nationally Determined Contributions (NDCs)** include gender-focused measures for climate change adaptation²⁷⁹. The **National Action Plan for Disaster Risk Reduction (2021-2025)** emphasizes gender-responsive approaches²⁸⁰. The **National Gender Mainstreaming Guidelines** provide practical guidance for integrating gender perspectives into disaster management policies²⁸¹. The **Guideline for Gender Mainstreaming in Disaster Management** provides detailed instructions for incorporating gender perspectives in disaster risk reduction and management efforts²⁸².

²⁶⁹ https://www.unwomen.org/sites/default/files/2024-09/b30_report_somalia_en.pdf

²⁷⁰ https://www.unwomen.org/sites/default/files/2024-09/b30_report_somalia_en.pdf

²⁷¹ https://www.unwomen.org/sites/default/files/2024-09/b30_report_somalia_en.pdf

²⁷² https://sidrainstitute.org/wp-content/uploads/2018/04/Report_EU_Gender_Analysis_Somalia_Study-SIDRA_Final_2016.pdf

²⁷³ <https://somalia.unfpa.org/en/somalia-health-and-demographic-survey-2020>

²⁷⁴ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

²⁷⁵ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

²⁷⁶ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

²⁷⁷ <https://sodma.gov.so/wp-content/uploads/2024/08/National-Disaster-Risk-Management-Policy-Revised-Draft-Edits10-12-2020.pdf>

²⁷⁸ <http://nwm.unescwa.org/resources/312>

²⁷⁹ https://www.undp.org/sites/g/files/zskgke326/files/2023-06/undp_somalia_gender_equality_strategy_2023-2026_summary.pdf

²⁸⁰ <https://africa.unwomen.org/en/stories/news/2023/01/somalia-launches-national-action-plan-on-unsr-13-for-women-and-security>

²⁸¹ <https://www.undp.org/somalia/publications/undp-somalia-gender-equality-strategy-2023-2026>

²⁸² chrome-extension://efaidnbmnnnibpcjpcglclefindmkaj/https://www.arc.int/gender-platform/IMG/pdf/gender_analysis_on_drm_somalia.pdf

Additional gender-sensitive policies include the **National Climate Change Policy**²⁸³, the gender-sensitive **National Water Resource Management Policy**²⁸⁴, the **National Disaster Management Strategy**²⁸⁵, and the **GCF Somalia NAP Climate Change Adaptation Gender toolkit**²⁸⁶. These policies aim to integrate gender perspectives into climate change, water resource management, and disaster risk reduction efforts²⁸⁷.

Somalia is also committed to international frameworks like Convention on the Elimination of all forms of Discrimination Against Women (CEDAW), United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP), United Nations Convention on the Rights of Persons with Disabilities (UNCRPD), the Sendai Framework for Disaster Risk Reduction, and the African Union Agenda 2063²⁸⁸. Despite these efforts, challenges include limited gender analysis skills within ministries and insufficient funding for gender-focused interventions^{289 290}.

Gender-Related Institutional Arrangements

In Somalia, gender-related institutional arrangements have evolved significantly in recent years, reflecting both progress and ongoing challenges in promoting gender equality. The institutional framework for gender equality is shaped by various policies, legal frameworks, and the roles of key institutions. The Provisional Federal Constitution, adopted in 2012, includes provisions for gender equality and women's rights²⁹¹. However, the implementation of these provisions remains inconsistent. Efforts are ongoing to integrate a 30% quota for women's representation in political institutions, although this has yet to be fully realized²⁹². Somalia has established mechanisms at both federal and federal member state levels to advance gender equality and women's empowerment, coordinated by the Ministry of Women and Families (MoWF) in collaboration with sectoral ministries, local governments, and development partners²⁹³.

The Ministry of Women and Families (MoWF) plays a key role in promoting gender equality by coordinating efforts across public institutions, civil society, and the private sector. It works closely with the Technical Working Group on Gender (TWG-G) and supports the Somali National Committee for Women (SNCW) in implementing CEDAW and ensuring legal protection for women²⁹⁴. MoWF also collaborates with sectoral ministries through Gender Mainstreaming Action Groups (GMAGs), which are responsible for developing, implementing, and monitoring gender mainstreaming action plans (GMAPs)²⁹⁵. At the federal member state level, each state has its own Ministry of Women and Human Rights Development or equivalent, which works alongside MoWF and other sectoral ministries to promote gender equality. These ministries partner with local governments and development organizations to ensure the inclusion and participation of women in decision-making processes²⁹⁶. In addition, The Somali Disaster Management Agency (SoDMA), as per its Gender Strategy & Road Map

²⁸³ <https://napglobalnetwork.org/wp-content/uploads/2022/11/napgn-en-2022-somalia-nap-framework.pdf>

²⁸⁴ <https://drive.google.com/file/d/11ZpInqCSTz53RYYLHxP0IjvYxyQ9pIHe/view?usp=sharing>

²⁸⁵ <https://www.preventionweb.net/media/97400/download>

²⁸⁶ <https://drive.google.com/file/d/1byegURxSW39hfdOg31HcsQIn9rAlOl6b/view?usp=sharing>

²⁸⁷ <https://napglobalnetwork.org/wp-content/uploads/2022/11/napgn-en-2022-somalia-nap-framework.pdf>

²⁸⁸ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

²⁸⁹ <https://www.arc.int/gender-platform/key-gender-issues-priorities-in-drm-somalia-policy-brief.html?lang=fr>

²⁹⁰ <https://nwm.unescwa.org/resources/312>

²⁹¹ [adopted constitution eng final for printing 19sept12 - 1.pdf](#)

²⁹² [Ministry of Women & Human Rights Development –](#)

²⁹³ [UNDP Somalia Gender Equality Strategy \(2023-2026\) | United Nations Development Programme](#)

²⁹⁴ [Gender-in-Somalia--Progress-Report-2012.pdf](#)

²⁹⁵ [UNDP Somalia Gender Equality Strategy \(2023-2026\) | United Nations Development Programme](#)

²⁹⁶ [Country Fact Sheet | UN Women Data Hub](#)

2024 – 2027²⁹⁷, will establish a Gender-DRM Working Group. This group aims to bring together key stakeholders—including government entities, civil society organizations, and the private sector—to promote sustained innovation and cross-sector collaboration. The focus will be on building capacity through knowledge exchange and gender-focused research, ensuring that DRM policies and programs are informed by gender-disaggregated data.

Meanwhile, the National Gender Commission oversees the implementation of gender policies and ensures compliance with international gender equality standards²⁹⁸. Numerous Civil Society Organizations (CSOs) actively advocate for women's rights and provide essential services, such as legal aid and support for survivors of gender-based violence²⁹⁹.

Disaster management at the sub-national, district, and village levels involves a multi-faceted approach. The Ministry of Interior, Federal Affairs & Reconciliation (MoIFAR) collaborates with local governments to develop disaster risk management plans and strategies. District Disaster Risk Management Committees (DDRMCs) coordinate disaster response and preparedness activities. Community-based disaster risk management (CBDRM) initiatives build local capacity and resilience³⁰⁰.

Several inter-sectoral organizations are involved in disaster management, working closely with government entities, civil society organizations, and the private sector. The Somali Disaster Management Agency (SoDMA) leads emergency planning and response efforts, including establishing early warning systems. The United Nations Development Programme (UNDP) enhances risk management and builds national capacity in disaster tracking, data management, and early warning systems. The United Nations Office for Disaster Risk Reduction (UNDRR) supports the National Platform for Disaster Risk Reduction, facilitating multi-sectoral and multi-disciplinary mechanisms for resilience-building³⁰¹.

Additionally, the World Food Programme (WFP) collaborates with SoDMA to develop anticipatory action plans for floods and other disasters, while the International Organization for Migration (IOM) leverages data tools for humanitarian response. Organizations such as UN Women, FAO, Danish Refugee Council (DRC), Action for Women and Children Concern (AWCC), and Norwegian Refugee Council (NRC) also collaborate with local governments, NGOs, and community-based organizations to enhance disaster preparedness and response at all levels³⁰². These institutions ensure gender-responsive approaches in all phases of disaster management activities^{303 304}.

Despite these advancements, several challenges persist. Deep-rooted cultural norms and practices, such as female genital mutilation (FGM) and early marriage, continue to hinder gender equality efforts³⁰⁵. Ongoing conflict and instability in certain regions exacerbate gender-based violence and

²⁹⁷ <https://docs.google.com/document/d/1pJOgplRMDOfR247h0yEcZYbnwYrP-JyO/edit?usp=sharing&ouid=101183427218989265382&rtopof=true&sd=true>

²⁹⁸ [Advancing Generation Equality Forum Commitment in Somalia | UN Women – Africa](#)

²⁹⁹ [UN Women Somalia trains civil society on gender in humanitarian Action | UN Women – Africa](#)

³⁰⁰ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

³⁰¹ <https://napglobalnetwork.org/wp-content/uploads/2022/11/napgn-en-2022-somalia-nap-framework.pdf>

³⁰² <https://napglobalnetwork.org/wp-content/uploads/2022/11/napgn-en-2022-somalia-nap-framework.pdf>

³⁰³ <https://drive.google.com/file/d/1byegURxSW39hfdOg31HcsQIn9rAlOl6b/view?usp=sharing://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

³⁰⁴ <https://napglobalnetwork.org/wp-content/uploads/2022/11/napgn-en-2022-somalia-nap-framework.pdf>

³⁰⁵ [Where we are : Eastern and Southern Africa : Somalia | UN Women – Africa](#)

limit the effectiveness of institutional arrangements³⁰⁶. Limited financial and human resources within key institutions impede the full implementation of gender policies and programs³⁰⁷. Nevertheless, the institutional framework in Somalia provides a foundation for continued efforts to promote gender equality and empower women across the country³⁰⁸.

Women's Representation and Participation in DRM Sectors

Women's representation in Somalia's government remains low, especially in sectors like water resource management, meteorology, hydrology, and disaster management. As of 2024, women hold about 24% of parliamentary seats nationally³⁰⁹. In specific fields, 15% of senior positions in water resource management are held by women, 10% in meteorology and hydrology, and around 20% in the National Disaster Management Committee^{310 311}. At the district level, their involvement decreases significantly to about 5% in disaster management committees³¹². These disparities reflect broader societal norms and systemic barriers that limit women's participation in governance³¹³. Efforts are needed to improve training and resources for women to enhance their leadership in these critical areas.

Table 1: Estimates of Female officials in Disaster Management in selected provinces³¹⁴

Regions/Committee's Roles	Total	Women	Percentage
Banadir (Mogadishu)	50	10	20%
Provincial Committee for Disaster Management	30	6	20%
Explore, Rescue, and Security Team	15	0	0%
Response and Recovery Team	25	4	16%
Secretariat	20	2	10%
Total	120	22	18.3%
Puntland	40	8	20%
Provincial Committee for Disaster Management	25	5	20%
Explore, Rescue, and Security Team	10	0	0%
Response and Recovery Team	15	1	13%
Secretariat	10	1	10%
Total	100	16	16%
Somaliland	45	7	15.5%
Provincial Committee for Disaster Management	20	4	20%
Explore, Rescue, and Security Team	8	0	0%
Response and Recovery Team	10	2	20%
Secretariat	7	1	14%
Total	90	14	15.5%

³⁰⁶ [Overview of Gender-Based Violence in Somalia - Advocacy Brief, 2021 | United Nations in Somalia](#)

³⁰⁷ [UNDP Somalia Gender Equality Strategy \(2023-2026\) | United Nations Development Programme](#)

³⁰⁸ [COUNTRY GENDER PROFILE: SOMALIA | Publications | UN Women – Africa](#)

³⁰⁹ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

³¹⁰ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

³¹¹ <https://moewr.gov.so/>

³¹² <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

³¹³ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

³¹⁴ [Home - Somali National Bureau of Statistics](#)

Notes:

- The numbers and percentages are an estimate only and will be updated as needed during project implementation.
- Key regions with the estimate data include Banadir, Puntland, and Somaliland
- The roles listed correspond to those typically found in disaster management committees.

(Source: Somali National Bureau of Statistics, African Development Bank. (2024). AfDB Gender in Somalia, World Bank. (2023). World Bank Somalia Overview, UN Women. (2024). UN Women Somalia, International Organization for Migration (IOM). (2023). IOM Somalia, UNDP Somalia. (2023). UNDP Somalia Projects).

Gender in Early Warnings and Disaster Risk Response

In Somalia, the integration of gender considerations in early warning systems and disaster risk response is essential to effectively address the vulnerabilities faced by different demographic groups. Recent efforts by various organizations and governmental bodies have highlighted the need for comprehensive data collection and analysis to inform disaster management strategies.

As part of the Key Platforms and Initiatives, The United Nations Office for Disaster Risk Reduction (UNDRR) supports Somalia in implementing the Sendai Framework for Disaster Risk Reduction, promoting gender-inclusive disaster risk management practices that emphasize understanding vulnerabilities through a gender lens³¹⁵. The Somali Disaster Management Agency (SODMA) plays a crucial role in coordinating disaster response efforts, aiming to strengthen community resilience and incorporate gender perspectives in disaster preparedness and response planning, although challenges in data collection and women's representation in decision-making persist³¹⁶.

The United Nations Development Programme (UNDP) has been active in enhancing disaster resilience, focusing on integrating gender into disaster risk management through capacity-building programs for local authorities and communities, highlighting the need for gender-disaggregated data³¹⁷. Similarly, the Food and Agriculture Organization (FAO) works to improve food security and resilience to climate shocks, employing gender-sensitive approaches to support women farmers and advocate for inclusive data collection³¹⁸.

The Ministry of Energy and Water Resources is involved in disaster risk reduction related to water management, creating flood risk maps and integrating gender considerations into water resource management, which is crucial for affected communities³¹⁹. Additionally, the World Meteorological Organization (WMO) collaborates with Somali authorities to enhance weather forecasting and early warning systems, which need to incorporate gender-sensitive data to ensure effective communication with all community members³²⁰.

Despite the efforts of various organizations, the availability of gender-disaggregated data remains limited, hampering the understanding of specific vulnerabilities and the impacts of disasters on

³¹⁵ <https://www.undrr.org/news/enhancing-risk-management-somalia-building-national-capacity-disaster-tracking-data-management>

³¹⁶ <https://sodma.gov.so/som/annual-reporting/>

³¹⁷ <https://www.undp.org/geneva/blog/advancing-resilience-building-through-gender-equality-and-womens-leadership>

³¹⁸ <https://www.fao.org/somalia/fao-in-somalia/en/>

³¹⁹ <https://www.fao.org/somalia/fao-in-somalia/en/>

³²⁰ <https://community.wmo.int/en/members/som>

different groups, particularly women, children, and those with disabilities³²¹. Furthermore, women's representation in disaster management roles is often insufficient, limiting their input in planning and decision-making processes, which is essential for inclusive and responsive disaster risk management strategies³²². Additionally, an improved coordination among various agencies, including UN bodies, local government, and NGOs, is necessary to streamline data collection and ensure that gender perspectives are integrated across all disaster management initiatives³²³. Establishing robust mechanisms for collecting gender-disaggregated data across all disaster risk management platforms is essential, including incorporating gender analysis into existing databases³²⁴. Additionally, building the capacity of local authorities and communities to understand and address gender issues in disaster risk management can improve resilience; training programs should emphasize gender-sensitive disaster management practices³²⁵. Engaging women's groups and community organizations in disaster preparedness and response planning can provide critical insights into local vulnerabilities, enhancing the effectiveness of interventions³²⁶. Finally, advocating for policies that mandate the inclusion of gender considerations in disaster management at all levels will help create a more equitable approach to risk reduction and response³²⁷.

Gender and Access to Early Warning Information and Technologies

In Somalia, recent efforts by various organizations and governmental bodies have highlighted the need for comprehensive data collection and analysis to inform disaster management strategies and ensure inclusivity. The implementation of EWS in Somalia is primarily managed by the Federal Government of Somalia (FGS) through the Ministry of Energy and Water Resources (MoEWR) and the Somalia Water and Land Information Management (SWALIM) project³²⁸. These systems are designed to monitor and provide alerts for various natural hazards, including floods and droughts. As of 2023-2024, Somalia has several centralized and decentralized EWS in place³²⁹.

Many international and local partners support the Federal and Federal Member States governments in implementing and maintaining EWS. Key partners include United Nations agencies such as the Food and Agriculture Organization (FAO) and the United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA)³³⁰, international NGOs like ActionAid, Danish Refugee Council, Mercy Cops, Oxfam, Save the Children, Norwegian Refugee Council, and World Vision³³¹, and bilateral partners including the United States, European Union, and various other countries³³². The United Nations Development Programme (UNDP), the United Nations Office for Disaster Risk Reduction (UNDRR), and the World Meteorological Organization (WMO) also play critical roles in enhancing disaster risk management and early warning systems in Somalia³³³. For flood and drought alerts, organizations like ActionAid, Danish Refugee Council, Mercy Cops, Oxfam, Save the Children, Norwegian Refugee Council, and World Vision

³²¹ <https://africa.unwomen.org/en/stories/news/2024/03/un-women-somalia-trains-civil-society-on-gender-in-humanitarian-action>

³²² <https://www.afdb.org/en/documents/somalia-country-gender-profile>

³²³ <https://somalia.iom.int/>

³²⁴ <https://www.undp.org/geneva/blog/advancing-resilience-building-through-gender-equality-and-womens-leadership>

³²⁵ https://www.wfp.org/operations/annual-country-report?operation_id=SO02&year=2023#/27256

³²⁶ <https://www.arc.int/gender-platform/key-gender-issues-priorities-in-drm-somalia-policy-brief.html?lang=fr>

³²⁷ <https://nbs.gov.so/wp-content/uploads/2024/05/SNBS-2023-Institutional-Annual-Report.pdf>

³²⁸ [Ministry of Energy and Water Resources, Comprehensive Assessment of Existing EWS](#)

³³⁰ [International Organization for Migration, Disaster Risk Assessment](#)

³³¹ [Acted Somalia, Integrated Support for IDPs and Communities](#)

³³² [U.S. Embassy in Somalia, Partnership with Somalia](#)

³³³ [Action Against Hunger, Flood Response in Somalia](#)

provide integrated emergency and early recovery support, including the dissemination of alerts and information to improve preparedness and mitigation³³⁴. Leading telecom companies such as Hormuud Telecom, Golis Telecom Group, and Somtel Network also play a crucial role in disseminating these alerts.

Receiving warnings about natural hazards through EWS in Somalia does not necessarily require users to have a registered mobile phone. However, mobile phones are a common medium for disseminating alerts, especially in urban areas³³⁵. The primary beneficiaries of EWS in Somalia are vulnerable communities, including internally displaced persons (IDPs), pastoralists, and agro-pastoralists. These systems help them prepare for and mitigate the impacts of natural disasters by providing timely alerts and information³³⁶. However, awareness and usage of EWS are generally low, particularly among marginalized groups such as people with disabilities, the elderly, women, and Indigenous people³³⁷. Many community members are not aware of the existence of these systems or do not understand how to utilize them effectively³³⁸. Women, in particular, often receive information indirectly from local authorities, which contributes to lower awareness and usage among them³³⁹.

The most preferred media for early warning alerts in Somalia include radio, mobile phones, and community meetings. Radio remains the most popular platform, with over half of the population listening to it weekly³⁴⁰. However, men tend to use all media platforms more than women³⁴¹. According to the Somalia Integrated Household Budget Survey (SIHBS) 2022, 85.6% of males and 84.5% of females over 15 years old own mobile phones³⁴². This data highlights a relatively small gender gap in mobile ownership. Access to mass media in Somalia shows significant gender disparities. Men are more likely to use all media platforms compared to women. For instance, 46% of men use the internet weekly compared to 37% of women³⁴³.

Water level management in Somalia is overseen by the Ministry of Energy and Water Resources and the Somalia Water and Land Information Management (SWALIM). These institutions monitor and manage water resources, including groundwater levels, to ensure sustainable usage³⁴⁴.

Focus group discussions for the Early Warning For All (EW4ALL) at the National Stakeholders consultation and national surveys have highlighted the gender disparities in access to early warning information and technologies. Women, especially in rural areas, have less access to information compared to men³⁴⁵. These findings emphasize the need for targeted interventions to improve awareness and usage of early warning systems (EWS) among women and other marginalized groups³⁴⁶.

Key recommendations from project-level national stakeholders consultations include, but are not limited to, the need for:

³³⁴ [Acted Somalia, Integrated Support for IDPs and Communities](#)

³³⁵ [Central Bank of Somalia, Mobile Money Regulations 2020](#)

³³⁶ [Ministry of Energy and Water Resources, Comprehensive Assessment of Existing EWS](#)

³³⁷ [Country Policy and Information Note on Somalia](#)

³³⁸ [Country Policy and Information Note on Somalia](#)

³³⁹ [World Bank Document on Early Warning Systems for Improving Food Security](#)

³⁴⁰ <https://drive.google.com/file/d/1byegURxSW39hfdOg31HcsQIn9rAlOl6b/view?usp=sharing>

³⁴¹ <https://drive.google.com/file/d/1byegURxSW39hfdOg31HcsQIn9rAlOl6b/view?usp=sharing>

³⁴² <https://somalia.unfpa.org/en/publications/somalia-integrated-household-budget-survey-2022>

³⁴³ <https://somalia.unfpa.org/en/publications/somalia-integrated-household-budget-survey-2022>

³⁴⁴ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

³⁴⁵ <https://drive.google.com/file/d/1byegURxSW39hfdOg31HcsQIn9rAlOl6b/view?usp=sharing>

³⁴⁶ <https://drive.google.com/file/d/1byegURxSW39hfdOg31HcsQIn9rAlOl6b/view?usp=sharing>

- Increased awareness through targeted campaigns to educate communities, especially marginalized groups, about the existence and usage of EWS;
- Improved accessibility by ensuring that EWS are accessible through multiple channels, including mobile phones, radios, and community meetings;
- Implementing a gender-sensitive approaches to ensure that women have equal access to early warning information;
- Strengthening partnerships with international and local organizations to enhance the effectiveness of EWS;
- Investment on capacity building for local institutions to manage and disseminate early warning information effectively.

By addressing these areas, Somalia can improve its disaster preparedness and resilience, ensuring that all communities benefit from early warning systems at all levels.

Preparedness, Early Actions, and Responses

Observations from the Early Warning for All (EW4ALL) project national level consultations have revealed that contingency and emergency plans in Somalia often overlook the specific needs of women and vulnerable groups, such as persons with disabilities, living in affected areas. Despite the importance of community engagement, these groups are frequently unaware of these processes and are less likely to be involved^{347 348}.

In regions like Banadir, Galmudug, HirShabelle, Southwest, Jubaland, and Somaliland, community meetings have shown that many women and people with disability have not been invited to participate in the consultation process for local contingency and emergency plans or development plans. Consequently, they remain unaware of the specific needs and interventions included in these plans³⁴⁹. Focus group discussions (FGDs) with the Somalia National Disability Agency (NDA) and representatives from organizations for people with disabilities (OPDs) revealed that disaster management interventions often fail to address the specific needs of persons with disabilities, particularly women and girls. This includes considerations for different types of disabilities and ensuring access to necessary services.

Stakeholders emphasized the need to enhance the capacity and skills of individuals involved in disaster management, including training on gender equality and disability inclusion to ensure that interventions effectively address the needs of all affected groups³⁵⁰. Consultations for the Gender Strategy & Road Map 2024–2027³⁵¹ by SoDMA revealed that women and girls face unique vulnerabilities during disasters. These include health risks, economic hardships, and psychosocial stress. Their needs encompass access to reproductive health services, protection from gender-based violence, privacy and safety, economic empowerment, education, and leadership opportunities³⁵².

³⁴⁷ <https://africa.unwomen.org/en/digital-library/publications/2023/11/country-gender-profile-somalia>

³⁴⁸ <https://undrr.org/ew4all-nairobi-workshops-2024>

³⁴⁹ <https://undrr.org/ew4all-nairobi-workshops-2024>

³⁵⁰ <https://undrr.org/ew4all-nairobi-workshops-2024>

³⁵¹ <https://docs.google.com/document/d/1pJOgplRMDOFr247h0yEczYbnwYrP-JyO/edit?usp=sharing&oid=101183427218989265382&rtpof=true&sd=true>

³⁵² <https://drive.google.com/file/d/1byegURxSW39hfdOg31HcsQIn9rAlOI6b/view?usp=sharing>

To effectively address the gendered impacts of disasters, it is crucial to incorporate gender considerations into all DRM initiatives. This involves identifying and responding to gender-specific vulnerabilities, ensuring equitable access to resources and decision-making, and promoting women's leadership in disaster preparedness, response, and recovery efforts^{353 354}.

Key recommendations from project-level consultations include, but are not limited to, the need for:

- Comprehensive data collection, with gender-disaggregated data;
- Gender analysis of past disasters, to understand challenges and opportunities;
- Integrating gender considerations into all stages of disaster management, including planning, preparedness, response, and recovery;
- Capacity building for SoDMA staff;
- Disaster preparedness and response efforts should focus on equipping communities with the knowledge, skills, and resources needed to actively participate in disaster management. This should include the creation of local disaster response committees with gender-balanced leadership structures and the development of inclusive early warning systems;
- Promoting women's participation and leadership in decision-making processes related to disaster risk reduction, response, and recovery is essential for ensuring inclusive and effective strategies that address their needs and priorities;
- Engage diverse stakeholders such as women's organisations, CSOs, and marginalised communities to ensure that their perspectives and needs are included in decision-making processes;
- Develop and distribute gender-sensitive DRM toolkits and guidelines to standardise gender integration across all sectors in alignment with the UN Women's strategy on gender-responsive disaster risk reduction and supports SDG 5, promoting gender equality through inclusive DRM processes.

Mainstreaming gender considerations into policies, procedures, and guidelines relating to disaster management and reduction involves integrating gender perspectives at all levels of decision-making and across all aspects of disaster preparedness, response, recovery, and risk reduction.

Access to Safe Evacuation Spaces

Efforts to improve access to safe evacuation sites in Somalia have been made through collaboration between the Federal Government, UN agencies, donors, INGOs, and CSOs. However, significant gaps remain in ensuring that these facilities meet basic standards for safety, hygiene, and security, especially during natural disasters such as flooding and drought. Across the Federal Member States (FMS), including Puntland, Somaliland, Galmudug, Southwest, Hirshabelle, and Jubaland, disparities exist in the availability and quality of evacuation sites.

The Federal Government, in collaboration with the UN and humanitarian partners, has set national minimum standards for safe evacuation sites. These standards include the provision of separate sanitation facilities for men and women, access to clean water, and secure shelter for vulnerable groups such as women, children, the elderly, and people with disabilities. Despite these standards, many evacuation sites remain under-equipped due to resource constraints, insecurity, and infrastructure challenges. The Federal Government's National Disaster Management Agency (SODMA) has been leading efforts to coordinate with the Federal Member States to ensure preparedness, but challenges remain in implementation and monitoring at the state level³⁵⁵.

³⁵³ <https://undrr.org/ew4all-nairobi-workshops-2024>

³⁵⁴ <https://drive.google.com/file/d/1byegURxSW39hfdOg31HcsQIn9rAlOl6b/view?usp=sharing>

³⁵⁵ [Ministry of Energy and Water Resources, Comprehensive Assessment of Existing EWS](#)

In Puntland, coastal and flood-prone areas such as Bosaso and Garowe face frequent displacement during flooding and cyclones. However, only 8 out of 70 identified evacuation sites meet the national minimum standards. Public spaces such as schools and mosques are often used as temporary shelters, but overcrowding is a persistent issue, especially in urban areas like Bosaso. Women and children are particularly vulnerable to gender-based violence (GBV) in these evacuation sites, and reports from local NGOs indicate that women lack access to separate toilets, increasing their vulnerability to abuse. Many displaced families are forced to rely on unsafe water sources due to insufficient clean water supply³⁵⁶.

In Somaliland, the government has made some strides in establishing safe evacuation sites, particularly in areas prone to drought and flooding, such as Hargeisa and Borama. However, only 6 out of 50 sites meet the minimum standards. Challenges include limited access to clean water, poor sanitation facilities, and insufficient shelter for displaced populations. Women in focus group discussions reported concerns about personal safety, especially during periods of extended stay in these sites. Reports from international NGOs indicate that women and girls are at heightened risk of physical and emotional abuse in overcrowded shelters³⁵⁷.

In Galmudug, regions like Galkayo and Hobyo frequently experience flooding, leading to displacement. Unfortunately, only 4 out of 60 designated evacuation sites meet the basic needs of displaced families. Communities in rural areas face additional challenges, with many relying on makeshift shelters that offer little protection from the elements. Women and girls, in particular, lack access to adequate sanitation facilities, often sharing toilets with men, which increases their vulnerability to violence. The elderly and persons with disabilities are also disproportionately affected, as evacuation sites are often not accessible for those with mobility issues³⁵⁸.

In Southwest State, areas like Baidoa and Afgooye are heavily impacted by seasonal flooding, leading to frequent displacements. Of the 100 evacuation sites identified, only 10 meet the national minimum standards. Key concerns raised by displaced families include overcrowding, lack of clean water, and poor sanitation. Women in these areas reported to CSOs that they often face the risk of sexual harassment when using shared sanitation facilities. Additionally, families in rural areas face challenges in accessing evacuation sites due to poor road infrastructure, making it difficult for them to reach safer areas during floods³⁵⁹.

Hirshabelle, particularly in regions such as Beledweyne and Jowhar, is highly vulnerable to riverine flooding along the Shabelle River. Only 5 out of 80 identified evacuation sites meet the minimum standards for safety and hygiene. During focus group discussions for the Somalia EW4ALL national level consultations women in Beledweyne expressed concerns about the lack of clean water and the need to rely on river water for drinking and cooking, which poses serious health risks. Overcrowding in shelters is also a significant issue, with many families sharing limited space. The elderly and persons with disabilities are also disproportionately affected, as evacuation sites are often not accessible for those with mobility issues³⁶⁰.

The Federal Government, in partnership with the UN, donors, and INGOs, has recognized the need for urgent action to improve the safety and accessibility of evacuation sites across all Federal Member States. Key recommendations from project-level consultations include:

³⁵⁶ [International Organization for Migration](#)

³⁵⁷ [Acted Somalia](#).

³⁵⁸ [U.S. Embassy in Somalia, Partnership with Somalia](#)

³⁵⁹ [Action Against Hunger, Flood Response in Somalia](#)

³⁶⁰ [Somalia National Bureau of Statistics](#).

- Improving infrastructure in evacuation sites to ensure they meet the national minimum standards, with a focus on providing separate sanitation facilities for men and women.
- Ensuring clean water is available at all evacuation sites to reduce reliance

Limited Access to Resources and Services

Vulnerable groups in Somalia, including the poor, rural women, the elderly, persons with disabilities, and nomadic communities, face significant challenges in preparing for disasters due to restricted access to resources and services. According to focus group discussions conducted for various projects, community members in disaster-prone areas reported not having sufficient savings to buy extra food, boats, or household supplies for emergencies, as they barely earn enough to survive each day³⁶¹.

Both women and men in drought-prone areas noted that during dry seasons, especially with increasing high temperatures and heat, their crops, agricultural activities, and livelihoods were severely affected. This negatively impacted their income, leaving them with inadequate means to support their families and no savings for emergencies. They also experienced a severe lack of water for household use, including cooking, washing clothes, and bathing, as well as for safe drinking water, due to insufficient supplies from wells, water pumps, and rainwater storage. Some families had to purchase water for drinking and household use, further exacerbating their vulnerability and poverty³⁶².

Limited access to resources significantly hampers the capacity of the poor, women, elderly, and persons with disabilities to better prepare for and take early actions to mitigate disaster risks and impacts on themselves and their households. The Somali government, supported by various international donors and organizations, has expanded its social protection policy to include 'shock-responsive social protection' for at-risk households. This initiative aims to build resilience and mitigate disaster impacts³⁶³.

Recurring droughts in Somalia have had devastating effects on agriculture, livestock, and water resources, exacerbating food insecurity and malnutrition and pushing vulnerable populations toward famine and displacement. Additionally, periodic floods, especially in riverine and coastal areas, have disrupted livelihoods, destroyed infrastructure, and increased the spread of waterborne diseases, further complicating recovery and reconstruction efforts. Environmental degradation, such as desertification and deforestation, worsens poverty and food insecurity, intensifying the country's vulnerability to disasters³⁶⁴.

The impact of disasters on vulnerable populations—particularly women and girls—is a critical issue requiring urgent attention. The intersection of gender and disaster risk management (DRM) is increasingly recognized as a vital consideration in addressing the disproportionate effects of disasters on marginalized groups. Women and girls are often more vulnerable during disasters due to preexisting social, economic, and cultural inequalities. These include limited access to resources, lower socio-economic status, restricted mobility, and lack of decision-making power. As a result, women and girls

³⁶¹ [IOM Somalia](#)

³⁶³ [Somalia - Acted](#)

³⁶⁴ [Homepage - U.S. Embassy in Somalia](#)

are more likely to suffer higher rates of mortality, injury, displacement, and loss of livelihoods compared to men³⁶⁵.

Community-Led Early Warning and Disaster Risk Reduction Initiatives

The Federal Government of Somalia has implemented several community-led early warning and disaster risk reduction initiatives in collaboration with international donors and organizations. These include the Building Resilient Communities in Somalia (BRCiS), which is a humanitarian consortium led by the Norwegian Refugee Council (NRC) that supports Somali communities in developing resilience to shocks and moving out of poverty. The program focuses on integrated programming models, flexible multi-year funding streams, and community structures to generate systemic change³⁶⁶.

Meanwhile, the Somalia Recovery and Resilience Framework (RRF), developed by the Federal Government of Somalia's Ministry of Planning, Investment and Economic Development, in partnership with the Ministry of Humanitarian Affairs & Disaster Management, aims to break the cycle of recurrent disasters and move towards medium-term recovery and long-term resilience³⁶⁷.

Additionally, the Federal Member States have implemented various community-led early warning and disaster risk reduction initiatives tailored to their specific needs. For instance, the Hirshabelle State Flood Response Plan was launched in response to severe flooding in 2019. With support from the Federal Government and international partners, the plan provides critical lifesaving assistance and invests in longer-term solutions³⁶⁸. Additionally, the Integrated Support for Internally Displaced Persons (IDPs) and Communities program, implemented by the International Rescue Committee (IRC) and other partners, supports IDPs and host communities in regions including Gedo, Lower Shabelle, and Hiraan³⁶⁹.

International organizations and NGOs play critical roles in supporting these initiatives. UNDP supports various disaster risk reduction initiatives in Somalia, including capacity building and resilience programs³⁷⁰. SODMA (Somali Disaster Management Agency) is the central body for disaster management in Somalia, coordinating efforts at federal and state levels to improve early warning systems and response capabilities³⁷¹. FAO (Food and Agriculture Organization) works on improving food security and resilience to climate shocks, supporting community-based disaster risk reduction and early warning systems³⁷². Organizations such as World Vision, Oxfam, Danish Refugee Council (DRC), and Norwegian Refugee Council (NRC) are also involved in various community-led initiatives to enhance resilience and disaster preparedness³⁷³. These initiatives aim to enhance the resilience of vulnerable communities to natural disasters and climate change.

³⁶⁵ [Somalia - Hunger Relief in Africa | Action Against Hunger](#)

³⁶⁶ [BRCiS Consortium - Building Resilient Communities in Somalia | NRC](#)

³⁶⁷ [Somalia-RRF-Summary-Report_final_layout6July2018-2.pdf](#)

³⁶⁸ [Somalia Flood Response Plan - November 2019 - January 2020 - Somalia | ReliefWeb](#)

³⁶⁹ [Somalia | International Rescue Committee \(IRC\)](#)

³⁷⁰ [Somalia | United Nations Development Programme](#)

³⁷¹ [SoDMA | Hay'adda Maareynta Masiimoobiya Soomaaliyeed](#)

³⁷² [FAO in Somalia | Food and Agriculture Organization of the United Nations](#)

³⁷³ [FAO in Somalia | Food and Agriculture Organization of the United Nations](#)

Areas for Improvement in Early Warning and Gender Inclusiveness

In Somalia, several areas require improvement to enhance early warning systems (EWS) and gender inclusiveness. These improvements are crucial for ensuring that EWS effectively serve all community members, particularly women and vulnerable groups. Below are the key areas for improvement and recommendations.

- **Enhancing GEDSI Analysis:** Integrate Gender Equality, Disability, and Social Inclusion (GEDSI) analysis into disaster risk assessments to capture the needs of diverse groups. Improving platforms like PRISM and CamDi to include disaggregated data and make them publicly available is essential³⁷⁴.
- **Improving Early Warning Dissemination:** A comprehensive review and action plan for Emergency Warning (EW) systems should provide analysis and strategies to include inclusivity and address digital gender gaps. Multiple communication channels, including cell broadcasting, should be used for wider reach. Community-based interventions, including outreach activities targeting women and persons with disabilities who have no access to mobile phones, should be implemented. Providing clear, actionable information tailored to different groups' needs and conducting training and awareness campaigns specifically targeting women, ensuring gender balance and youth participation, is crucial³⁷⁵.
- **Preparedness and Early Action:** Ensure evacuation sites meet minimum standards and are safe for all, particularly women and girls, and are designed inclusively. Train and equip community members, including women and persons with disabilities, for early warning actions. Building evidence for shock-responsive social protection schemes for preparedness and anticipatory action in target provinces, especially to include the poor, women, and persons with disabilities, is necessary³⁷⁶.
- **Cross-Pillar Coordination:** Create GEDSI mainstreaming guidelines and training for emergency warning systems. Increase the representation and participation of vulnerable groups in preparedness and response activities. Incorporate GEDSI aspects into standard operating procedures and monitoring frameworks, collecting and analyzing disaggregated data³⁷⁷.
- **Enhancing Gender-Responsive Strategies:** Developing and implementing gender-responsive strategies in crisis prevention, preparedness, and recovery is essential. Early warning systems should address the specific needs and vulnerabilities of women and girls by incorporating gender perspectives into all aspects of disaster management³⁷⁸.
- **Removing Structural Barriers:** Addressing structural barriers that limit women's economic autonomy and unpaid work is crucial. Aligning public and private finance through integrated national financing frameworks and investing in gender-responsive infrastructure can help

³⁷⁴ [BRCIS Consortium - Building Resilient Communities in Somalia | NRC](#)

³⁷⁵ [Somalia-RRF-Summary-Report_final_layout6July2018-2.pdf](#)

³⁷⁶ [Gender equality, disability and social inclusion analysis – Good practice note](#)

³⁷⁷ [The Role of Climate information and Early Warning Systems in Supporting Disaster Risk Reduction in Somalia - Somalia | ReliefWeb](#)

³⁷⁸ [Investment in early warning – early actions reaps massive reward against El Niño flood threats | United Nations in Somalia](#)

remove these barriers, enabling women to better participate in and benefit from early warning systems³⁷⁹.

- Preventing and Responding to Gender-Based Violence: Improving the skills and knowledge of police officers, judges, prosecutors, and other officials to ensure justice for survivors of sexual and gender-based violence (SGBV) is essential. Providing medical, legal, and psychosocial support to SGBV survivors ensures their needs are met in disaster situations³⁸⁰.
- Promoting Women's Participation and Leadership: Encouraging women's participation and leadership in all forms of decision-making is vital. This includes promoting women's political participation and leadership and ensuring their involvement in disaster risk reduction initiatives. Women's leadership can lead to more inclusive and effective disaster management strategies³⁸¹.
- Strengthening Data Collection and Analysis: Improving data collection and analysis on gender-specific impacts of disasters and early warning systems is vital. This data can inform more effective policies and programs that address the unique needs of women and girls³⁸².

Building Capacity and Awareness: Enhancing the capacity of communities, especially women and persons with disabilities, to understand and respond to early warning information is essential. Conducting awareness campaigns and training programs can help build this capacity, enabling communities to better prepare for and respond to disasters³⁸³.

³⁷⁹ <https://docs.google.com/document/d/1pJOgplRMDOfR247h0yEczYbnwYrP-JyO/edit?usp=sharing&ouid=101183427218989265382&rtpof=true&sd=true>

³⁸⁰ [SIDA 2015 Preventing and Responding to GBV .pdf](#)

³⁸¹ [UNDP Somalia Gender Equality Strategy \(2023-2026\) | United Nations Development Programme](#)

³⁸² [UNDP Somalia Gender Equality Strategy \(2023-2026\) | United Nations Development Programme](#)

³⁸³ [SOMALI DISABILITY NETWORK](#)

10 Gender Action Plan MCP EW4All

This GAP is a consolidated document for Outputs 1 and 2, and as such it is considered the global GAP. Output 2 countries have country-specific GAPs as part of the country funding proposals which provide a further breakdown for each country. Each country's gender staff will report to the global PMU on their country-specific GAP in order to inform this global GAP. It will be the responsibility of the global PMU to report to GCF on this global GAP. Since the project activities are mainstreamed, the main cost of the GAP will be the hiring of country-level and global-level gender staff. Nevertheless, an estimate for each country has been provided for each activity based on a proportion of the activity budget. This GAP is considered a flexible document and will be further updated during project implementation.

Principal Objective of the GAP: To enhance early warning systems and disaster preparedness through gender-sensitive strategies that accommodate to the unique needs of women, men, girls, and boys, ensuring inclusive resilience and timely response to climate-induced and other natural disasters.				
Output 1				
Activities	Targets/Indicators	Timeline	Responsibility	Estimated Cost/Resources
Development of guidelines, training materials, and strategic orientation documents	All those developed	Years 1-3	UNDP, WMO, UNDRR	\$50,000 total
Design and implementation of analytic frameworks and models for national adoption take into account gender considerations	All those developed	Years 2-3	UNDP, WMO, UNDRR	\$50,000 total
M&E framework for EW4All take into account gender considerations	Gender-specific targets included in the M&E framework	Years 2-3	UNDP, WMO, UNDRR	\$30,000 total
UN global knowledge bank and knowledge sharing products include lessons relating to gender aspects	At least 50% of knowledge sharing products include gender-related lessons	Years 3-5	UNDP, WMO, UNDRR	\$15,000 total
Global and technical expert working groups include female representatives	Efforts made to achieve equity in female and male participation, aiming for 50% of females whenever possible	Year 1 to Year 5	UNDP, WMO, UNDRR	none
Output 2				
Specific Objective for Output 2 countries:				
1. Strengthen disaster risk knowledge by developing a comprehensive risk assessment framework, databases, and capacity-building initiatives for national and local actors. 2. Improve detection, monitoring, and forecasting capabilities through harmonized data collection, forecasting tools, and interoperability across sectors.				

3. Enhance early warning dissemination and communication by establishing functional networks and communication protocols, ensuring last-mile connectivity to vulnerable populations.
4. Boost preparedness for early response and recovery by developing participatory and gender-responsive disaster preparedness measures and enhancing local and national capacities for anticipatory action.

Pillar 1: Improved Disaster Risk Knowledge and Capacities

Results: Sector based project plans and implementations in disaster early warning, preparedness and response are inclusive and gender sensitive

Activities	Targets/Indicators	Timeline	Responsibility	Estimated Cost/Resources
1. Integrate gender-responsive data collection and analysis in DRR capacity development, particularly on disaster risk data, that includes vulnerability, capacity and exposure of populations. Develop a centralized disaster impact database that tracks and addresses gender-specific disaster impacts. 2. Develop and deploy a centralized database that also tracks gender-specific disaster impacts. 3. Train DRM staff on gender-sensitive data analysis techniques.	• Number of hazard risk assessments that include gender considerations (target: 3-5). • Availability of a gender-sensitive disaster impact database (target: 1). • % reports include sex- disaggregated data (target: 100%) • 70% of DRM staff trained by Year 3, 100% by Year 5. • 2 trainings in each country annually targeting 50% female participants.	Year 1 to Year 5, with most of the activities expected in years 2-4.	UNDP and implementing agencies in each country	\$250,000 per country

Pillar 2: Integrated Actions for Detection, Monitoring, and Forecasting

Results: Increased awareness and preparedness among women and vulnerable groups in response to hazard warnings

Activities	Targets/Indicators	Timeline	Responsibility	Estimated Cost/Resources
1. Conduct community awareness campaigns targeting women, elderly, and persons with disabilities on impact-based forecasting 2. Capacity development trainings related to impact-based forecasting (ensure that the training content includes information on how women and men of different ages	• Percentage of early warning systems that provide gender-sensitive information (target: 100%). • % of campaigns conducted in rural areas (target: 80%) • % increase in awareness of DRM strategies among women by Year 5 (target: 40-50%)	Year 1 to year 5, with most of the activities	UNDP and implementing agencies in each country, relevant NGOs and partners	\$200,000 per country

may be affected by hazards and how to communicate warnings effectively to diverse groups, including children, women, the elderly, persons with disabilities as well as other vulnerable populations). 3. Establish women-led community DRM teams to support awareness and preparedness. 4. Develop and distribute accessible early warning messages in local languages and formats.	• % of females who have participated in capacity development sessions (target 40-50%) • % of training sessions include gender-sensitive content on the differential impacts of hazards and effective communication strategies for diverse groups (target: 100%) • % of target areas receiving warnings in accessible formats by Year 3 (70%), by Year 5 (100%)	expected in years 3-5.		
Pillar 3: Nation-wide Warning Dissemination and Communication Result: Increased gender equality and inclusion in early warning systems and disaster risk communication				
Activities	Targets/Indicators	Timeline	Responsibility	Estimated Cost/Resources
1. Develop inclusive communication strategies with specific outreach efforts for women, girls, and female-headed households. 2. Strengthen communication networks to ensure last-mile connectivity to vulnerable groups, especially women living in remote and rural areas. 3. Integration of inclusion and gender aspects into technical assistance on warning dissemination channels including Common Alerting Protocols (CAP) 4. Design and implement inclusive and gender sensitive Early Warning Systems	• Percentage of early warning messages that specifically target women and girls (target: 60%). • Number of SOPs developed with a gender lens (target: 3-5). • % of simulation exercises that engage women as leaders (target: 50%). • % of areas with updated gender-sensitive early warning dissemination plans by the end of the project (target: 70% by year 3, 100% by year 5) • % of project beneficiaries receiving news or improved early warning information disaggregated by sex, age, and diversities compared to baseline (target: 80-90%) • % of community early warning systems incorporated gender-responsive elements and met	Year 1 to year 5, with most of the activities expected in years 3-5.	UNDP and implementing agencies in each country, relevant NGOs and partners	\$130,000 per country

	established criteria for gender inclusivity compared to baseline (target: 80-90%) • % of Common Alerting protocols (CAP) and other technical assistance integrated inclusion and gender aspects in warning dissemination channels by year 3 (target: 90-100%), by year 5 (100%) • % of individuals from vulnerable groups targeted report receiving timely and actionable early warning messages via targeted SMS alerts (target: 60-80%) • % of alerts developed/used are non-written, to overcome literacy barriers (target: 30-50%)			
Pillar 4: Enhanced Preparedness for Timely Responses and Early Actions				
Result: Gender-Responsive Capacity Building and Leadership of women and marginalized groups in national and community early action response initiatives				
Activities	Targets/Indicators	Timeline	Responsibility	Estimated Cost/Resources
1. Develop disaster preparedness plans that incorporate the unique needs of women, men, and vulnerable groups, ensuring gender balance in decision-making. 2. Build community capacity with a focus on empowering women in leadership roles in disaster response teams. 3. Early Warning & Early Action champions training for community members including women, persons with disability, children, and youth. 4. Leadership training for women in the development and dissemination of early warning systems.	• Percentage of disaster preparedness plans developed with gender considerations (target: 100%). • % of women's groups that participate in preparedness training annually (target: 50%) • % men and women community members who have received training on EW and EA across all regions (target: 50:50) • % of women assume leadership roles in the development and dissemination of early warning systems as a result of the training (target: 10-20%) • % of women participants report increased confidence and skills in participating early warning	Year 1 to year 5, with most of the activities expected in years 2-4.	UNDP and implementing agencies in each country, relevant NGOs and partners, women's organizations	\$350,000 per country

	system development and dissemination, as measured by pre-and post-training surveys (target: 70-80%)			
General: Enhanced GEDSI mainstreaming in EW4All project management and disaster preparedness and response.				
Activities	Targets/Indicators	Timeline	Responsibility	Estimated Cost/Resources
1. Integration of GEDSI aspects into SOP and relevant developed protocols on EW 2. Design monitoring, evaluation, and learning frameworks that are GEDSI-responsive 3. Collect and analyze disaggregated data, and GEDSI-related case studies, and progress updates on GEDSI implementation of all relevant project reports 4. Ensure the GRM within the framework of EW4All is gender responsive.	• GEDSI mainstreaming guidelines developed and adopted for early warning (EW) systems (Guidelines on risk communication, inclusive preparedness, regional adaptation guideline, feedback mechanism and data collection). (Target: 3-5) • Number of SOPs and protocols on early warning that explicitly include GEDSI aspects (target: 3-5) • % increase in vulnerable groups (disaggregated by sex, age, and diversities) participation in early warning, preparedness, and response activities compared to baseline (target: 30-50%) • Monitoring, evaluation, and learning frameworks that include specific GEDSI-related indicators and considerations (participation, impact analysis, gender disaggregated data and feedback integration) developed (target: 3-4) • % of all relevant project reports include disaggregated data and GEDSI-focused analysis by the end of the program timeline (target: 100%) • At least 1 case studies per year per country focusing on successful GEDSI implementation. • % of grievance reports received are sex-disaggregated annually (target: 100%)	Year 1 to year 5, with most of the activities expected in years 2-4.	UNDP and implementing agencies in each country	\$150,000 per country
Total				\$1,080,000 per country
